

THEORY ON DEMAND

PARITANCE
A PHILOSOPHICAL
INVESTIGATION BEHIND
COGNITION AND
SIMULATION
HENRY WARWICK

**THEORY
ON
DEMAND**

A SERIES OF READERS
PUBLISHED BY THE
INSTITUTE OF NETWORK CULTURES
ISSUE NO.:

59

PARITANCE: A PHILOSOPHICAL INVESTIGATION BEHIND COGNITION AND SIMULATION

BY HENRY WARWICK

COLOPHON

Theory on Demand #59

Paritance: A Philosophical Investigation Behind Cognition and Simulation

Author: Henry Warwick

Edited by: Tripta Chandola

Cover Design: Katja van Stiphout

Design and EPUB development: Klaudia Orczykowska

Published by the Institute of Network Cultures, Amsterdam, 2025

ISBN: 978-90-83520-98-8

Contact: Institute of Network Cultures Amsterdam University of Applied Sciences (HvA)
Email: info@networkcultures.org Web: www.networkcultures.org

Order a copy or download this publication for free at: www.networkcultures.org/publications

This publication is licensed under the Creative Commons Attribution NonCommercial ShareAlike 4.0 Unported (CC BY-NC-SA 4.0). To view a copy of this license, visit www.creative-commons.org/licences/by-nc-sa/4.0/

Institute of
network cultures

ACKNOWLEDGEMENTS

Firstly, I wish to thank Geert Lovink for this opportunity to again publish with INC. Geert is a great man, INC is a fabulous and important publisher, and my words here, already meagre, falter in expressing my gratitude.

Secondly, I wish to (again) thank Tripta Chandola for her editing, support, and clarity. This document is a widely and weirdly ranging thing, and it is only as readable as it is thanks to her. Tripta - you're amazing. Thank you.

Thirdly, I wish to thank my wife and life partner, Beth Scannell, without whose love and support, none of this would have been possible. And I wish to thank our big grown-up kid, Desmond, for being such a lovely and fabulous person—you inspire me. I love you both so very much.

TABLE OF CONTENTS

INTRODUCTION	5
PART 0: FORM IS EMPTINESS, EMPTINESS IS FORM	7
PART 1: PARITANCE: A PHILOSOPHICAL INVESTIGATION	27
PART 2: THE MELLOTRON VS. THE SYNTHESIZER	49
PART 3: DIGITAL NECROMANCY	69
PART 4: DIGITAL EVOCATION AND DEMONOLOGY	103
PART 5: AI: RESURRECTING A DEAD GOD	125
PART 6: CODA: POSSIBILITIES	157
BIBLIOGRAPHY	169

INTRODUCTION

This volume exists to explain and explore some ideas that are now more relevant and directly important than ever in our hypermediated world of Internet interconnection and LLM-based Artificial Communications. I discuss a number of different ideas here, and a few of them are almost adequately argued. The purpose of this book is to provide a kind of outline or act as a shelf for these ideas. The purpose will be to use this as a springboard to other statements in media which are presently under development. That should keep me very busy for a very long time.

The book's schematic follows the path of my thoughts and inquiry. *I don't think in a straight line, and I usually don't think in words.* As a consequence, writing was a difficult task for me to do. However, it has led to a better appreciation of this way of working for me. I'm not trying to frustrate anyone—I'm just trying to get the ADHD fuelled arguments in my head to shut up long enough to get a few ideas typed up into this laptop. Normally, I think in shapes, colours, textures, and effects, filled with a variety of emotional positions that inform and propel my efforts. Also, I enjoy improvisation, riffing, digressing, and following where something goes. I am capable of many things; however, being able to effectively plan a work of writing is not one of them. So, I do what I can, and call on help.

Initially, I saw how much our technical culture is emulating, if not directly recapitulating, occult practices. I had read Sconce's book *Haunted Media*¹, and while I found it very insightful, I feel it missed the point I was looking at. He saw how people were resorting to seeing the occult in media as a response to new media. *And it's a fair take.* What I'm looking at is a similar condition, only from the other direction—it's not that people appeal to the occult to explain new media technology; more that new media technology is actively working at emulating activities that were once the province of the occult, and doing so in a very direct and obvious way, and I found that fascinating especially from a kind of post-Nietzschean post-religious perspective.

The pattern for this volume is that I started with an abstracted yet material discussion of the differences between the Synthesizer and the Mellotron. The Mellotron was the ancestor of the Digital Sampler. I see the Synthesizer as the voice of modernity, the voice of Frankenstein's Creature, and the Mellotron as something else—something vampiric, like Dracula. Of course, Dracula wins. This discussion of Dracula led me to consider the undead in terms of technological opportunity, and so I began to research Necromancy and Evocation. This was a powerful experience, as during my time spent writing it, I lost a few dear friends, and one even *called me on the phone...*

This inquiry into how AI technology is recapitulating occult practices led me to consider why any of this is even sensible, and I saw that this is a property of human cognition I call 'paritance'—the acceptance of parity. It is such paritance that allows us to conceive of gods, devils, angels, computers, thought, intelligence, consciousness, cats, whatever. That what *is*,

1 Jeffrey Sconce, *Haunted Media: Electronic Presence from Telegraphy to Television*, Duke University Press, 2000.

is unknowable, and everything we know about it claims parity, which we more or less accept. This acceptance is quite nearly automatic for common perceptions—we haven't the time or energy to interrogate every bit of incoming data from our cognitive sensorium, and so our cognitive capacities simply 'work with it'. Cognition is a higher order of complexity, and even more distanced from what it is cogitating about, and examining it at an even greater distance, while we accept the parities our cognition derives. That is paritance. We can't really know anything, so we accept this lack as knowledge. Beyond that? We get creative—we can say it's God or we can say it's a howling void - whatever flips your crank and makes you happy - there's a range of possibilities there. These possibilities that we create will be subject to the same set of limitations and paritance. This can be a great source of meaning creation and joy—we know we can't get there, so we can make our travels through it as interesting, pleasurable, and meaningful as we make them. Also, the possibilities this opens up can be of great use, as we can now create new channels of ideas, and mix, modify, disrupt, any combinations of previous systems. As we do.

That this volume is readable or makes any sense is the work of my fine editor, Tripta Chandola, whose clarity, precision, and enthusiasm were critical. With her help, this volume was deeply restructured. *Remake / Remodel*.² It starts with where this journey began - in a Zen Centre in 1985 and 1986.

2 'Remake / Remodel' by Roxy Music, from the LP, *Roxy Music*. Island Records. 1972.
 'I tried but I could not find the way. / Looking back all I did was look away. / I could talk talk talk talk talk myself to death. / But I believe I would only waste my breath - Show ME!'

PART 0: FORM IS EMPTINESS, EMPTINESS IS FORM

In the latter part of 1985 and for most of 1986 I lived in a Zen Centre, Ka-Shin Zendo Genzo-Ji, in Washington, DC, near Takoma Park, Maryland. It was a profound and life-altering experience. For a variety of reasons, I was in a desperate psychological state—daily and massive panic attacks, everything seemed to be collapsing in on me. It was psychologically imperative for me to deal with this. The reasons for my misery are not important to my dialogue here; suffice to say, they were non-trivial.

There isn't much to say about Zen practice—you sit on a cushion facing a wall. With your eyes half closed, you count your breath and meditate. That's about it. Because of the simplicity and austerity of the practice, the focus and flair of Zen rhetoric is more well-known, with dramatic stories of monks being turned into foxes, Zen masters cutting cats in half, monks carrying beautiful women across rivers, and all manner of drama. In this section, I will tell you a series of some memories from my time there, and then I'll get into how they affected my life and practice and ideas after.

It was June 1985. I was interested in learning Zen meditation. I found out where the Zendo was located, and a friend gave me a lift there. They drove off, and I went to the address of a nondescript, if tidy, suburban brick house with an enclosed porch. The door was securely closed with a note that they would open in about 20 minutes. I took the 20 minutes as an opportunity to walk a few blocks over to a 7/11 and get a can of Coke—it was brutally hot (it was June in Washington, DC) and the cold drink and caffeine helped. As I came to the street where the Centre was located, this East Asian woman wobbled past me on a bike and fell. I trotted over to help her. 'Are you OK? Can I help you?' She was in tears. She was middle-aged, probably in her late 40s, and in a heavy accent, she said she had never learned to ride a bike and had decided she wanted to learn. She was terribly frustrated. She cried, from the pain of falling off her bike, saying how it's so hard, she is always falling over! She found it frustrating and painful. I, being a bit of a bicycle fanatic at the time, told her that *learning how to ride a bike without falling over is a bit like learning how to swim without getting wet*. It made sense to her—she tilted her head and thought about it, and smiled through her pain. Still, she was not happy. She was unsure if she could do it. I told her I would give her a big push to help her on her way, and that it's easier to go a little faster on a bike than slowly, as the rotation of the wheels helps with stability. She sat up on the seat, feet on the pedals, and I pushed and trotted alongside her for a bit. She pedalled on and sped up and soon outpaced me. She kept going. I smiled and shouted, 'Goodbye—good luck!' and off she went—wobbling yet steady, and then she turned left up a street and disappeared toward the suburban wilds of Takoma Park.

I went back to the Zen Centre and someone was there to let me in—an older woman, tall, slender, with bright blue eyes and grey hair. Her name was Martha. She greeted me, and I came in and set about sitting on a cushion in the main room. Soon, others arrived, including a senior member who gave us basic instructions. 'Sit on the zafu and zabuton. Keep your back straight. Face the wall and half close your eyes. Count your breath. That's about it, really.' Candles were lit, a bell was rung, and we sat. After 20 minutes, the bell rang again, and we turned around and introduced ourselves. Tea and cookies were served. The instructor asked me what I thought of the meditation.

I said, 'I counted my breath. I got to about 300 when I lost count.'

Everyone laughed. He said, 'No, no, you can just count to three and start over.'

Embarrassed, I replied, 'Oh. OK. Also, my knees hurt.'

With that, he smiled, and everyone nodded in agreement and sympathy. He said, 'Well, Henry, *learning to sit zazen without knee pain is like trying to learn how to swim without getting wet.*'

I was completely blown away.

I laughed and replied, 'Yeah—well, I know all about *THAT*.'

Little did he realize what I had just experienced earlier with the woman learning how to ride a bike. Also, while I knew what it took to learn how to swim, I didn't know how to swim, either.

...

In August, one of the residents, Linda LaMacchia, a scholar of Buddhist nuns, said she was going to go to India for 6 months and if anyone wants to sublet her place, let her know. I did so immediately—I so wanted out of the hellish downtown tenement I was living in at the time. I liked Linda—she even took me on a short road trip in Maryland, where I got to meet some of her family and friends. At the end of August, she went to India, and I moved in, September 1. She was in India for six months, which soon turned into a year. When she returned, she was a markedly different person. She had spent a year in rural Northern India with Buddhist nuns, and I had experienced a very different path. I liked Linda very much, even though we were very different sorts of people. However, I decided to move out after her return, having lived there for 13 months. She later went on to continue her studies in India and died of a heart attack while hiking in Dharmasala, India, on 31 July 2017. She was 73.

...

My other 'housemate' was Martha Kent. She was 64 and looking forward to retirement the following year. She was going to move to San Diego to study under Charlotte Joko Beck. Martha and I got along great. She was delightful. While there, she gave me an album by Watazumido Shuso (aka Watazumi Doso) who played a special Japanese flute, similar to a shakuhachi, called a hotchiku, and established his own Zen school based around the flute. He's one of my musical heroes—I took his idea of 'meditation *THROUGH* music' very closely, and it has influenced my musical output in many, many ways. It is still essential to me and my creativity to this very day.

...

There was a third bedroom, and that was reserved for guests, usually monks from the Zen Centre in the Catskills, Dai Bosatsu Zendo. Once we had a visit from Eido Shimano, roshi of

the New York Zen Centre, with which the Zen Centre I was living at, Ka-Shin Zendo Genzo Ji, was affiliated. Years later, it was revealed that Shimano was having affairs with his students. He was forced to resign. He died in Japan in 2018.¹ All of his transgressions were unknown to me back in 1985 / 86. I was just a noob in DC.

One of the guests we had was a resident from Dai Bosatsu. He was smart, funny, and a real wit. I don't remember his name, so I'll call him Mike. Prior to moving to the Zen monastery, he was a pharmacist in Manhattan for several years and started sitting at the New York City Zen Centre. One day, I was having a very bad time of it—multiple panic attacks. Mike asked me why I seemed so out of it that day, and I explained—panic attacks, massive depression, the litany. He said he was sorry I was so miserable, and that being in a Zen centre is a perfect place, as the whole point of Buddhist practice was around the problem of suffering, something I seemed wrapped up in. I said something to the effect of 'Well, it doesn't seem to work very quickly, does it?' and I will never forget the rest of the conversation, almost verbatim:

He said, 'Well, don't worry—everything changes in 70 days. It might not be better, but it will be different.'

I asked, 'Why 70?'

He said, 'Because it's the smallest weird number.'

At that, I laughed; he smiled and walked away.

For years, I just thought Mike was being his obtuse self—weird numbers—Mike was being weird—what on earth was he going on about?

A few months later, Mike was back in town and asked how I was doing. I said I was feeling better. He smiled and said, 'See? 70 days. It works.'

Decades later, in the 2010s, I started getting into maths. Not to be a mathematician, or even formally study the subject, I saw it more as a collection of conceptual points and values. I was especially interested in the concept of number itself, and read a number of books about this. One I found very interesting in the early 2000s was *The Number Sense* by Stanislas Dehaene² and I worked from there, where number is not so much a natural thing that exists outside consciousness, as much as it is a product of cognition. I investigated aspects of numbers and number theory, and I found that Mike was correct! *70 actually is the smallest weird number*. 'Weird' is actually a mathematically technical term. A weird number is one where the sum of the proper divisors of a number is greater than the number, and no subset of the divisors sums to the number itself. So, for example, the number 12 has the proper divisors 1, 2, 3, 4, and 6, and these sum up to 16; however, three of them will add up to 12, e.g.: $2 + 4 + 6 =$

1 Neil Genzlinger, 'Eido Shimano, Buddhist Leader Who Resigned in Scandal, Dies at 85', *New York Times*, 26.02.2018.

2 Stanislas Dehaene, *The Number Sense*, Oxford University Press, 1999.

12. For the number 70, its proper divisors are 1, 2, 5, 7, 10, 14, and 35; they all sum up to 74, however, *none of them sums to 70*. And 70 is the smallest number that does this. Mike was correct. *70 really is the smallest weird number.*

...

I remember Mike was there for a party we had at the Zen centre. I had my music collection and stereo, so we were listening to a variety of ambient music albums while we sat on cushions around a low Japanese table. As the evening went on, people wanted more active music, so I complied with what I and Martha had in our collections. Then Mike stood up and stepped up onto the table, grabbed some chopsticks like they were a microphone. He started talking like Elvis Presley, 'Here's a little number called Heartbreak Hotel' and lit into it with the rest of us sitting on our cushions, clapping or slapping the table in response,

'Well, since my baby left me'

CLAP CLAP

'I found a new place to dwell'

CLAP CLAP

'Down at the end of Lonely Street at Heartbreak Hotel...' ¹³

Which made sense, because in many ways, it was.

...

A few months later, I was talking with a friend, Bob, and he asked me what I was thinking of, creatively. I told him my housemate, Martha, had turned me on to Watazumido-Shuso, and I thought he was brilliant. I told him I'd like to make music, but unfortunately, I had no instruments or outlet for it. I had been in a prog-rock cover band in High School—we played all kinds of crazy stuff—Pink Floyd, Yes, Genesis, King Crimson, Beatles, Renaissance. In University, I had some classes in Electronic Music, and while I never followed up on it, I found it fascinating—back then, even crappy synthesizers weren't cheap. Bob said he had an old Arp Odyssey he could lend me, and I was over the moon. That winter I spent hours and hours playing that synth (on headphones of course) and while I was a pretty awful keyboard player, I was able to get some cool sounds out of it, and began pursuing drone textures, like Eliane Radigue; only instead of her massive Arp 2500 and its rich noisy complex textures, I had this buzzy little Arp Odyssey. A few months later, I returned the synth to Bob, and the seed had been planted—I knew I had to make electronic music, abstract, spirited, evocative, ambient music. It had to come from within.

And that's what I proceeded to do in the following years.

...

3 Elvis Presley, *Heartbreak Hotel*, 45 RPM single, RCA Victor, 1956.

I have found that Zen, in its purest form, is fairly simple. People, for the most part, are not. And, of course, philosophy intrudes. One thing that attracted me was that one could characterize Zen as a kind of 'anti-philosophy', at least in a Western cultural sense, where Logic and Reason are paramount. This is to be very clearly distinguished from Western 'anti-philosophy' philosophers, as they approach anti-philosophy from the same sources and predispositions of logic and reason as philosophy. Zen doesn't care. It simply proceeds from its principles and has no interest in the language or structures presented by Western Philosophy or Anti-Philosophy.

Western philosophy often starts from the position of the individual, the ego. Philosophers from Descartes to Wittgenstein have approached the subject from the context of the supremacy of Reason, matters of Personality, and questions about Subjectivity. Zen basically doesn't accept these premises. A basic Zen position is that logic and the analytical, while extremely powerful and useful, cannot fully grasp the nature of reality. Personality is just an accident of personal development and growth. In Zen, everything must be approached through direct, lived experience that moves beyond the ego and the individual. In some ways, I've always felt that Zen (and Buddhism in general) is, in one narrow way, a path similar in disposition to pre-Socratics like Heraclitus.

It's also important to understand that Zen doesn't punt rationality as a waste of time. On the contrary, a fundamental Buddhist precept is that there is no escape from the laws of cause and effect. While Reason and Logic and its material practice, Science, are extremely useful; they have shortcomings, especially in their dependence on binary logic. There's a classic old Zen phrase, first written by the Zen Master Seigen (660–740ce), later made into a folk-pop song by Donovan, 'First there is a mountain, then there is no mountain, then there is.'⁴ It's not a specific repudiation of logic and reason, as much as it's a call to move beyond dualism and dialectical either/or logical structures.

Zen literature abounds with statements, questions, stories, and koans that defy logical categorization as a path of evading the limitations of conventional thought. At the same time, koans themselves can have internal logics. To say koans and zen stories lack structure as they point at realization would be to denigrate them as a provocative exercise in nonsense is overly simplistic. There's a koan for that—'don't confuse pointing at the moon for the moon.' For Zen, its purpose is not to explain the universe, but much as it is to bring an awakening to the practitioner in order to reduce human suffering. From this methodological foundation, Zen has developed a profoundly different understanding of reality that departs sharply from the Western philosophical tradition's reliance on ego-consciousness and discursive reason as the foundation of knowledge. In this sense, Zen is 'anti-philosophical'. It's not in opposition to inquiry per se, as much as it actively resists any systematic formulation grounded in dualistic epistemological certainty or rational analysis.

At the same time, Zen is also necessarily grounded in literature, and thus, a philosophy, one rooted in embodied practice and always evading attachment to a fixed position. This refusal to cling, *even to insight itself*, is central to Zen's understanding of freedom. To grasp and to hold is, in Zen literature, to bind oneself, a subtle yet powerful form of attachment that obstructs liberation.

4 Donovan, 'There Is a Mountain' from *Mellow Yellow*, Epic Records, 1967.

Thus, we get anti-Zen Zen statements like, 'If you meet the Buddha on the road, *kill him*.'

My first introduction to Zen was in High School, when I was studying the music of John Cage. From his influence, I read the *Tao Te Ching*⁵ and the *I-Ching*.⁶ I read three of his books back then and still own two of them to this day (*M* and *A Year From Saturday*). I had purchased a few albums of his music, and I was very intrigued by his music, as it is a kind of anti-music. A poem he wrote clearly sums up his work in a very tidy way from his book, *M*:⁷

the soundS
of the buGle
were out of my Control,
tHough without
my hAving
Made the effort
they wouldn't have been Produced.

I always found that mesostic poem insightful, inspiring, and prophetic, especially lately, as it generally describes the creative process of working with today's so-called Artificial Intelligence systems. You can tell it what you want it to do; however, you're not going to actually control what it does. At the same time, none of it would happen without your making the effort for its production. It was this 'do something to make something else happen', this juncture between the rigid laws of cause and effect and the stochastics of experience that resonated for me then, and continues to guide my creative processes. Note: matters of Artificial Intelligence will be foregrounded later in this volume.

...

A typical evening for me would be my arrival at Zendo after work. The air was pungent with incense, the room spotlessly clean, and the late afternoon light slanted through the windows. I would take up a position on my black zafu and zabuton cushions. The zafu is thick and round, and it's where I planted my bum. Below the zafu is a large square flat cushion called a zabuton. Thanks to serious injuries sustained at a younger age, my knees are not very flexible, and I was never able to get into a 'proper' lotus position. The best I could muster was a 'half lotus' at best, and I usually just adopted a simpler Burmese position— more of what one would think of as a tight cross-legged position. At my cushions, I would get into position, clasp hands and bow to the wall from my seated position to demonstrate respect for the process and act of meditation. Then I would put my hands in my lap in a particular position, where my left-hand fingers are below my right-hand fingers, and my thumbs would gently touch. These hand positions are called Mudras, and this is called the Dhyana Mudra. I would then start my meditation, albeit lightly, as others would arrive, and do the same.

5 Lao Tzu, *Tao Te Ching*, Written ca: 420bce. https://www.with.org/tao_te_ching_en.pdf

6 Anon, *I-Ching*. Written ca. 900 bce, China. <https://www.ichingonline.net/>

7 John Cage, *M: Writings* '67—'72. Wesleyan University Press. 1973. '36 Mesostics re and not re Marcel Duchamp, for Shigeo Kubota.', p. 30.

The leader of our group was usually a more advanced lay practitioner who was more of a hall assistant, a position called the Doan. They would light the candles and get things going. To start the meditation, the Doan would ring a large brass bowl, and the group would begin the first meditation session. On beginner nights, usually Wednesday evenings, these would only be for 20 to 25 minutes—enough for beginners to get a taste of what meditation's about. The other meditations for the more experienced were 40 minutes. In these longer sittings, after about 30 minutes into it, the Doan would get up and very slowly walk behind us with the *keisaku* stick. Usually, no one asked for it. If I was slouching too much, the Doan would gently put a finger in the middle of my back and then touch the front of my shoulder and gently push my shoulders back until I was sitting up straight. Posture was always important, and I came to see why—if one slouched, one could doze off, or even snore. While I never fell asleep while meditating, I have been to meditations where that did happen, and it was always embarrassing for the person who dozed off. This was especially common on meditation retreats called *sesshin*, where we would meditate for hours and hours for days and days, getting up before dawn, and sitting well into the evening—exhaustion was common. I remember at one *sesshin*, a fellow not only fell asleep, he just passed out—fell right over from exhaustion!

After 40 minutes, the bell would ring and we would bow to the wall, turn around on our cushions, stand up and then perform walking meditation, known as *kinhin*. This was done by maintaining enough of a view of the person ahead of me so I wouldn't bump into them, and changing from the Dhyana Mudra to a different hand position, the Shashu Mudra, where I would grasp my right thumb with my left fist and place this Mudra at my solar plexus. Then we would all silently and very slowly walk in a path, usually clockwise, around the meditation room for about 5 minutes. Eventually, we would arrive back at our cushions, bow, and resume sitting. The bell would ring, and we would sink/sync into meditation for another 40 minutes.

At the end, there was chanting. To do that, we would turn around on our cushions, facing each other / centre, and open our Sutra Books for the chanting materials. The Sutra Books themselves were of interest. Having been raised Catholic, I was used to massive liturgical tomes—thousands of pages of worship material, Bibles, and hymnals. In contrast, the entire Sutra Book was only 48 pages. In twelve-point Optima. That was it. I approved of its concision.

There were two chants that I found especially interesting. One was very simple—the General Vows, and the other is the more complex and profound *Heart Sutra*. These ideas I found while living at the Zen Centre set me up for my creative existence going forward, as well as the ideas expressed in this volume. First, I will examine the General Vows. The General Vows are just that, general; however, they point to many of the inherent theories native to the Zen discipline.

The General Vows

However innumerable all beings are, I vow to save them all.

However inexhaustible delusions are, I vow to extinguish them all.

However immeasurable Dharma teachings are, I vow to master them all.

However endless the Buddha's way is, I vow to follow it.⁸

8 Dai Bosatsu Zendo, *Sutra Book*, 2010, General Vows, p.16.

I found these four simple declarations to be powerful and important, if impossible, goals. I came to the conclusion that this kind of rhetoric is necessary for the maintenance of a particular hierarchy. The sentiments are such that they inspire a kind of esprit de corps in the practitioners. I remember having an amusing conversation with a monk who said that of the different schools of Buddhism, Zen was like the Special Forces of the lot. I thought that was an odd angle on it; however, I could see how it gave practitioners a sense of specialness and value. It's not that Zen looks down on other practices, as much as Zen is cognizant of the great rigour of its discipline. This gives a sense of 'I'm doing Zen, and it's hard and amazing.' There's a sense of 'to do this special hard thing, special hard rules must be followed.'

This validates the rigour of discipline to the point where I heard someone say that if you can't sit in full lotus, you can't experience enlightenment. This made me laugh, as it struck me as one of the single most 'unenlightened' things I had ever heard anyone say. Still, this vow to save all beings is a generous, if impossible, one.

Zen's strict rules of operation are there to maintain the Zen practice and structure. At the same time, they don't care if you're a Buddhist or not. That's *your* business. If you express interest in Zen, they will cheerfully help. At the same time, I've never seen anyone insist that people be Zen Buddhists. *It just doesn't work that way.* Obviously, those involved with the hierarchy of the Zen organisations are dedicated Buddhists, which is hardly surprising as systems of power and organization require lines of demonstrable fealty. You're not going to have a Jewish pope, or an Islamic Dalai Lama. Zen, when examined and experienced in a Buddhist context, has a rich and deep literature expressing a variety of related ideas; however, one needn't be a Buddhist to find value in them. This brings me to *The Heart Sutra*.

The Heart Sutra

(My notes and comments in parens and *italics*)

The Bodhisattva of Compassion, When he meditated deeply, Saw the emptiness of all five skandhas And sundered the bonds that caused him suffering. (*The Bodhisattva of Compassion was Avalokiteshvara. He is a very popular and widely revered figure, symbolizing the infinite compassion and mercy of all Buddhas. The figure of Avalokiteshvara is transformed in East Asian societies to a female form, in China, Guangyin, in Japan, Kannon. There was a figurine of Kannon on the altar in the Zen Centre. It reminded me of the Virgin Mary statues from Catholicism.*)

Here then,
Form is no other than emptiness,
Emptiness no other than form.
Form is only emptiness,
Emptiness only form.

(*This is the essential message, and requires unpacking later.*)

Feeling, thought, and choice,
 Consciousness itself,
 Are the same as this.

All things are by nature void
 They are not born or destroyed
 Nor are they stained or pure
 Nor do they wax or wane

So, in emptiness, no form,
 No feeling, thought, or choice,
 Nor is there consciousness.

No eye, ear, nose, tongue, body, mind;
 No colour, sound, smell, taste, touch,
 Or what the mind takes hold of,
 Nor even act of sensing.
 No ignorance or end of ignorance,
 Nor all that comes of ignorance;
 No withering, no death,
 No end of them.

(The stanza here is essential, as it dovetails with the Vow discussed earlier. If one is to save all beings while ignorance is eternal, then the work can never end. At the same time, it doesn't exist, so there is no work to do...)

Nor is there pain, or cause of pain,
 Or cease in pain, or noble path
 To lead from pain;
 Not even wisdom to attain:
 Attainment too is emptiness.

(If there is no ignorance and no end to ignorance, then there is no opposite of ignorance, nor any end of it. Thus there is "no wisdom to attain". At the same time, there is no end of wisdom and no end of "attainment.")

So know that the Bodhisattva
 Holding to nothing whatever,
 But dwelling in Prajna wisdom,
 Is freed of delusive hindrance,
 Rid of the fear bred by it,
 And reaches clearest Nirvana.

("Prajna wisdom" is the profound understanding of the true nature of reality and phenomena—that form is emptiness and emptiness, form—achieved through practice. This stanza sets up the next few, which form the "coda" of the Sutra)

All Buddhas of past and present,
Buddhas of future time,
Using this Prajna wisdom,
Come to full and perfect vision.

Hear then the great dharani,
The radiant peerless mantra,
The Prajnaparamita
Whose words allay all pain;
Hear and believe its truth!

Gate Gate
Paragate
Parasamgate
Bodhi Svaha
Gate Gate
Paragate
Parasamgate
Bodhi Svaha
Gate Gate
Paragate
Parasamgate
Bodhi Svaha

*(Gone, gone, gone to the other shore beyond. O what an awakening, all hail!)*⁹

Of the entire Sutra, the essential point, as I noted above, is:

Here then, Form is no other than emptiness, Emptiness no other than form. Form is only emptiness, Emptiness only form.

This is contradictory to much Western Enlightenment¹⁰ thinking, which insists on self-identity as a fundamental axiom in both logic and mathematics. $X \text{ must equal } X$. Zen disputes these notions. 'Form is no other than emptiness, Emptiness no other than form' is a key philosophical concept in Zen, as it focuses on the interconnectedness and interdependence of all things. We 'let X equal X '.¹¹ Form refers to the physical and experiential aspects of reality, and emptiness refers to the lack of inherent existence or self. They are inseparable and ultimately essentially the same. This concept suggests that everything is empty of inherent self because it is constantly changing and

9 Dai Bosatsu Zendo, *Sutra Book*, 'The Heart Sutra', p.5.

10 A term whose irony I find delightful.

11 Laurie Anderson, *Let $X = X$* , from *Big Science*, Warner Brothers, 1983.

dependent on other factors. At the same time, this emptiness is not an endless nihilistic black hole of a void—it is a *dynamic potentiality from which form arises*. Form arises from emptiness, and emptiness is revealed through form. The emptiness of one thing is related to the emptiness of all other things, and the arising of one thing is related to the arising of all other things. To repeat, this is a vision where Emptiness is not a void or nothingness; it is a dynamic potentiality from which all things arise or emanate. It is a state of open possibility and creativity. This is an essential Zen Buddhist theory. It is also an important part of this volume.

Speaking of 'Buddhist Theory', I always found it amusing. The fundamentals of Buddhism/Buddhist Theory are worthy of examination. For that, allow a short, potted history/story of the founder.

The story of Siddhartha Gautama's life has been subject to a great deal of amplification, distortion, and mythology. The basic narrative is that he was born into a royal family, raised in luxury and shielded from the harsh realities of life. He was completely protected from exposure to suffering, disease, and death. He wanted to learn more about the world, and so he would sneak out of his golden cage of a palace. On one outing, he saw an old man. Then he saw a sick man on another excursion. Then, on another covert outing, he came across a dead body. At that point, Siddhartha realized the impermanence and suffering inherent in life, leading him to renounce his princely existence.

He left his wife and children, his wealth and family, his possessions and palace, and embarked on a journey to find a path to liberation. He practiced strict asceticism and meditation for years. After intense and lengthy meditation under a Bodhi tree, he attained enlightenment, becoming the historical 'Buddha' or 'Awakened One'. He spent the rest of his life traveling and teaching his philosophy and established a monastic order. According to Buddhist tradition, he passed away at the age of 80, achieving Parinirvana, a state of final enlightenment and the cessation of the cycle of rebirth.

Whether any of this actually happened isn't as important as what it means as a guiding narrative. It is also very important to our discussion here. We can talk about Buddhist Theory as much as we want, however, deep down at the core of Buddhist Practice, is the simple fact that it is deeply anti-theoretical. The core point of Buddhism and Siddhartha's message and practice is one for the alleviation of human suffering. Period. Siddhartha's life and experience is largely yet not entirely fictive. This is fine—here, we can simply carry on and take it at face value. Siddhartha observed that in human existence one is born, one grows old, one suffers, and one dies. Fundamentally, this is true for everyone to greater or lesser degrees. One cannot control being born, or aging, or dying, however, *one can control to some extent how much one suffers in the process*, and... that's about it. The Buddhist practice is developed specifically for this purpose—to alleviate the mental anguish, *the suffering*, brought about by the realisation of the contingency of existence and the existential dread it can bring. By meditating, one learns to control one's thoughts and feelings, and thus, how much suffering one experiences. Not necessarily somatic suffering like broken limbs, massive burns, and similar injury—it's not specifically designed for that. It's the mental anguish brought by the awareness of being born without choice in the matter, only to grow old and infirm, and eventually shuffle off this brief mortal coil that can have some remedy in Zen practice.

There is an aspect of this in Zen 'Theory' that I found especially useful in this regard, the concept of Maya as discussed in the Heart Sutra and other Prajnaparamita sutras. Maya is the illusory in life, and when applied to the mind itself, the mind, the self, and all ideas: *all are Maya*.

This has been met with some opposition from some important Western thinkers, such as Hegel, Nietzsche, and Hume. Nietzsche criticized Buddhism for being 'life-denying' and nihilistic, especially its concept of Nirvana. He also saw Buddhism as promoting a 'self-deception of moral concepts'.^{12 13} In *Philosophy of History* Hegel suggested that the religious systems of China and the Far East were characterized by a lack of individual freedom and a dependence on nature, which he associated with a certain type of Buddhist thought.¹⁴ Hume, ever the skeptic, doubted all religious miracles, including those attributed to the Buddha.¹⁵ Hume's critique is one with which even Zen would agree. Miracles are essentially Maya. Nietzsche and Hegel clearly 'don't get' Buddhism, and this makes sense given their philosophical proclivities and backgrounds.

The Buddhist rejection of Maya isn't simply asserting that reality is false or meaningless. It's more of a shift in how perception, cognition, and existential identity are understood and experienced. Saying these things are an illusion in this context is not a blanket denial of existence. It's more a radical critique of reified concepts and the habitual clinging to appearances *as if* they were intrinsically real. This is not nihilism—it's a kind of freedom from illusion *and* the abyss of negation.

In the Zen view and associated doctrines, the sensual world is seen as more of a dream, a mirage, or, as Dogen described, 'like seeing the moon reflected in water'¹⁶: beautiful, vivid, seemingly tangible, yet ultimately lacking any enduring or independent essence. Disturbing the water disturbs the reflection, not the moon. This insight is not a rejection of experience, as much as it is an appreciation of its nature. When Prajnaparamita texts like the Heart Sutra say that 'all dharmas are like an illusion,' this includes not just external objects, it also internal phenomena: thoughts, emotions, the self, *even enlightenment itself and nirvana*. Nothing stands outside the reach of this analysis. All are empty (sunyata). This emptiness is *not* a nihilistic void; it is a refusal to assign any fixed essence to any phenomena.

The understanding that all things are Maya, illusion, can lead some to despair, and that's what many westerners, such as Hegel and Nietzsche, picked up on. However, it is meant (and as I

12 Benjamin A. Elman, *Nietzsche and Buddhism*, *Journal of the History of Ideas*, Vol. 44, No. 4. (1983), p. 671–686.

13 Friedrich Nietzsche, *On the Genealogy of Morality*, (1887) Dover Publishing (2003), II 21 'a nihilistic turning-away from existence, the desire for nothingness or desire for the 'antithesis', to be other, Buddhism and such like'.

14 G.W.F. Hegel, *The Philosophy of History*, (1837), 'Part One, the Oriental World, Section Two, (continued) India, Buddhism.' Batoche Publishing, 2001, p.185-191.

15 Nathan Rockwood, *Religious Studies*, 'Locke and Hume on competing miracles', (December 2023), 59 (4): 603–617.

16 Dogen, *Actualizing the Fundamental Point (Genjo-koan)*, 1233, Translated by Robert Aitken and Kazuaki Tanahashii, in *Enlightenment Unfolds*, Penguin Books, 2000.

take it) a release. Illusion is not inherently bad; it becomes problematic only when it is perceived as something that's inherently real. Just as in a dream, the problem is not the dream itself, as much as waking and assuming the contents and narratives of the dream were real. This can be extended to existence itself: to see the dreamlike nature of reality is to wake up. Some might say, 'This is just psychological dissociation'. I would disagree, as this metaphorical waking up is not into some higher metaphysical truth—it's more of a realization of the thusness or suchness (tathata) of existence—an ungraspable, dynamic, and interdependent flow of phenomena.

From this, there is no intrinsic identity, no self-sustaining core to anything, not to objects or the self. The self is not a permanent soul or ego directing experience, as much as a convenient designation for a stream of causally connected events and conditions. The self, like a rainbow, depends on these conditions; the Mind appears to us as a vividly real thing, yet cannot be grasped. This is not a claim of non-existence; it is a claim of dependent co-arising (pratityasa-mutpada). Things exist through momentary convention, not as ultimate objects.

The rejection of Maya and delusion is not an argument for annihilation or nihilism. It is an invitation, *like jazz*. We live in an all-pervasive illusion, while what we do in that illusion has very real consequences. Ethical action matters. Compassion arises naturally when one sees that others are caught in illusion—not as fools to be scorned, more as beings to be liberated, and this is why Buddhism is not nihilistic, even as it is clear that the vows of Buddhism to bring all to enlightenment are impossible to fulfil. Nihilism claims there is no meaning, no value, and it drags us into an existential paralysis. In contrast, Buddhism affirms that while 'ultimate reality' is empty of fixed form, it is not meaningless. 'Emptiness is Form', the Heart Sutra says; the illusion itself is the display of ultimate reality; and thus, Form is Emptiness. To see the world as Maya is not to negate it—it is to see through it all to the luminous, interdependent (metaphorically speaking) 'scalar field' from which all appearances arise. The world becomes less of a prison and more like a dance or musical performance.

To live with the awareness of Maya is to live lucidly, like a dreamer who knows he dreams... and, in knowing, becomes free in all dreams.

{

I did that a lot. For years, when I would dream, I would dream about the same place. A town I later called Wislenberg. I started dreaming about Wislenberg when I was in the Zen Centre. I had a life there. It was great. I knew all the streets, the businesses, the people. Every detail was etched like a memory. I would wake up fine. Usually. By 1987 I was almost always dreaming in Wislenberg.

I even met a lover in Wislenberg. She was quiet yet hardly shy. Her name was Emily. Her introverted demeanour was aloof yet curious, snarky and sweet. I remember her apartment up Elm Street towards Water Street. It was up a hill and had a bit of a view of downtown's building towers. This went on for several months, almost a year. Then one day, I got to Wislenberg to see her at her shop at the corner of Main and Elm—across the street from the church and the graveyard, right on the roundabout - it was empty. Gone. I went up Elm St to her place. I got

there and the apartment was empty. Everything was gone. I went back to Main St and into the basement level saloon/art gallery/music space next to where her shop was, and there were two of her friends, Max and Alan.

I liked Max; he often had amazing parties. Anyway, I asked them about Emily and they were as puzzled as I was, especially Alan, because he's known her since High School and they always kept up as neighbours. (He lives on Water Street, a few long blocks beyond Emily's place.) We agreed to keep in touch about this. Alan's girlfriend, Frieda, and Emily were good friends. Everyone was sad and weirded out. Even though Emily was gone, I would still go to Wislenberg. On the west side of the River, I took flying lessons. It was great. I sucked at flying. It was always fun, although I couldn't fly very high because I'd get vertigo. Still, I'd zip around at high speed at low altitude. At the end, the class would be hanging out in the large back room of the leader of the group, a paraplegic named Carter. Someone turned on the television, and the news show blared at us. The shaky, vibrating anxiety-fed newscaster said 'You can only fly in your dreams!' And people started disappearing. Carter yelled, 'NO IT'S NOT TRUE!' and that's when I woke up. I still dream of Wislenberg, but only a few times a year now.

}

While all this is well and good, you have to dream in the first place. You need an inner dialogue to reject an inner dialogue, and even this is a feature not universally found in the human species - apparently a huge percentage of people don't always have an inner dialogue.¹⁷ That's why I enjoy making music so much. *It's thinking*. It's thoughtful, it's emotional, it's conceptual, and yet it's none of that at all. I can just be in the sound. It is from this that I have come to reject Wittgenstein's dictum 'Whereof one cannot speak, thereof one must be silent.'¹⁸ In my view, whereof one cannot speak, THEN SING. And thus, *we arrive at Watazumi Doso*.

The saxophonist, Steve Lacy, said of Watazumi, 'He was the most modern improviser I've ever heard in my life. He surpassed anybody I could think of, including Braxton or Derek Bailey. Doso, to me, was just... whew, outside all of that, really.'¹⁹

From my perspective, this is a bit of an overstatement—I don't see music as competitive, so there's no 'surpassing' to do. However, it does show Watazumi's level of work and commitment to his practice. Watazumi himself deserves a bit of an unpacking here.

Born Tanaka Masaru in 1911, he passed away in 1992. He studied Rinzai Zen, and attained the title of Roshi, or Master, and at some point became Kanjo, the unifying head of the Fuke

17 Alex Soloducha, 'What it's like living without an inner monologue.' *CBC News*, 6 March 2020, 'According to Hulburt, not many people have an inner monologue 100 percent of the time, but most do sometimes. He estimates that inner monologue is a frequent thing for 30 to 50 percent of people.'

<https://www.cbc.ca/news/canada/saskatchewan/inner-monologue-experience-science-1.5486969#:~:text=According%20to%20Hulburt%2C%20not%20many,close%20to%20100%20per%20cent.%22>.

18 Ludwig Wittgenstein, *Tractatus Logico Philosophicus*, Routledge, 2001, p.89.

19 Christopher Cox, 'Invisible Jukebox: Steve Lacy.' *The Wire*, #225, November 2002.

sect of Zen Buddhism. He walked away from standard Zen practice in the late 1940s and developed his own, a practice centred on breath training and exercise. A focus on breath control and awareness is, as discussed earlier, central to the Zen Buddhist practice. In a presentation to musicians in Woodstock, NY, he said,

It's fine that you are all deep into music. But there's something deeper and if you would go deeper, if you go to the source of where the music is being made, you'll find something even more interesting. At the source, everyone's individual music is made. If you ask what the deep place is, it's your own life and it's knowing your own life, that own way that you live.²⁰

Also, his approach is international and inviting:

The Way of Watazumi has come from an old Japanese word. I was born in Japan and the way I live is through Japanese customs, but just because my way was born with me in Japan does not mean that it can only be transmitted or learned in Japan. If I had been born in this country, or any other country, I would still be teaching you the Way of Watazumi, but maybe in a different form.

And

When you hear some music or hear some sound, if for some reason you like it very well, the reason is that the sound is in balance or in harmony with your pulse. And so, making a sound, you try to make various different sounds that imitate various different sounds of the universe, but what you are finally making is your own sound, the sound of yourself.²¹

The results of his efforts and ideas for our experience is an amazing body of work, spread across a few recordings, starting with the album Martha gave me *The Mysterious Sounds of the Japanese Bamboo Flute*.²² I found it fascinating that he could basically form his own school of Zen, centred around playing music as *meditation*. Bob lending me the Arp Odyssey at that time was crucial for me to make that connection, because I saw that this is a legitimate form of creativity and personal development. After I moved out of the Zen Centre, in the following year, 1987, I took out a loan and built an electronic music studio.

This way, I could work with my machines and make music directly from myself, making sounds I design or sample and then process. The Samplers I procured produced audio snapshots, representations of different sounds in the universe. I would program synthesizers, engage in a practice of sound design, and then process them into something different, making the sound my own, and thus, the sound of myself.

20 Watazumi Doso, *The Way of Watazumi*, Lecture at the Creative Music Studio, Woodstock, N.Y., 1981. <https://www.komuso.com/people/people.pl?person=1222>.

21 Ibid.

22 Watazumido Shuso, *The Mysterious Sounds of the Japanese Bamboo Flute*, Everest Records, 1971.

Having a keyboard to play was a revelation, and I immediately set about developing a number of studies for the synths, recording into my old Atari computer as MIDI data, and then record directly to a Cassette. In 1987, I quickly put together an album, *Electronic Etudes*.²³

This album, and the following album, *Metawind*²⁴, are very similar, as they are for me primarily investigations in keyboard playing and learning subtractive and FM synthesis. In fact, I would suggest that the two, *Electronic Etudes* and *Metawind*, are actually very much companion pieces. Unfortunately, the sound quality by today's standards is not the greatest—in fact, a number of etudes are missing because the master cassette tape was so worn and had degraded so badly that by the time I decided to digitize these albums in the early 2000s, I found some of the tracks damaged and unlistenable. *Electronic Etudes* was made very quickly—I was in a very excited and obsessed state.

Metawind was recorded and released in 1988. It is very similar to *Electronic Etudes*, and shares a number of remixed tracks with it. Like *Electronic Etudes* is also a very 'keyboard' album, as I was still discovering things I could do by playing my keyboards. This was important to me at the time—Watazumi was a master of his instrument, even though he wasn't the 'best' player by traditional shakuhachi standards. His playing is considered very rough and has a very contrastive capacity between aggressive, loud parts and gentle, quiet parts.

Overall, *Metawind* is a bit more emotional and sophisticated recording than *Electronic Etudes*, and still very much one of my 'early' works. The title, *Metawind*, was invented by my friend, the poet Paul Grant. We worked together at Phil's Photolettering as Typographers. He very much liked this record. We would always listen to music or the news on our headphones while we worked, and we would trade cassettes. This record is as close to 'New Age' music that I ever got. I still like some of it; however, I find much of it now a bit tedious and overdone.

I began looking into other music by solo artists. I was well familiar with the works of Brian Eno and other electronic musicians who put out *their own* music. At the same time, I was also listening closely to music by Keith Jarrett. I was especially struck by his 1986 album *Book of Ways*.^{25 26} It has 19 recordings of him playing a clavichord. The clavichord is similar to a harpsichord—press a key and it plucks a string. However, it is much smaller, and the strings are shorter, so it is a much quieter and more intimate instrument.

Jarrett noted:

A Note on *Book of Ways*: To my knowledge, this recording is unique in several ways. We had three clavichords in the studio, two of which were angled together so that I could play them both simultaneously, and the third off to the side. Also we miked the

23 Henry Warwick, *Electronic Etudes*, Kether Records, 1987, <https://henrywarwick.bandcamp.com/album/electronic-etudes>.

24 Henry Warwick, *Metawind*, Kether Records, 1988, <https://henrywarwick.bandcamp.com/album/metawind>.

25 Keith Jarrett, *Book of Ways*, LP. ECM, 1986.

26 Note: 'The Book of Ways' is a rough translation of 'Tao Te Ching'.

instruments very closely so that the full range of dynamics could be used (clavichords are very quiet and cannot be heard more than a few feet away). The two CDs were made on an off day between concerts with my Trio, and no material was organized beforehand. Everything was spontaneous. The recording was done in four hours.²⁷

The entire record of 19 songs was recorded in a single day, in an afternoon, instantly improvised on the spot. One might respond with 'So?' That would be a suboptimal response, as the music on *Book of Ways* is stunning. He frequently has multiple melodies crossing between his hands—this is deeply woven music, and he sometimes hums along with it—you can hear him living inside the music. For me, this was confirmation that I was looking in the right place, and Martha had pointed me in a good direction with her gift of Watazumi Doso's music.

Beyond Jarrett's amazing work, I was also listening to others who stood solo in their music, ranging across styles. Acoustic off-kilter guitar from John Fahey to Derek Bailey; Keith Jarrett's keyboard playing and other ECM artists; Jon Hassell's trumpet improvisations to the avant-garde electronic artists like Hafler Trio and :zoviet*france:, and another one of my musical heroes, Laraaji.

My interest in Laraaji was also powerful. I had discovered his music in 1980, with the release of *Ambient 3: Day of Radiance*.²⁸ At that time, I went to New York City with a friend, and we wandered about the Lower East Side, came to Wiser's Occult Bookstore and wandered in. I could hear Laraaji's music playing. But it wasn't anything I recognized from the album. Shortly, we came upon Laraaji himself. He was seated on a cushion, between two zithers. The zithers had pickups that went into some electric guitar processor 'stomp boxes' (from what I could gather, a phase shifter and a delay) and from there into a small amplifier. He gently played his extraordinary and beautiful music. We stood there spellbound for quite some time. That moment stuck with me and eventually I made an homage record, *Zither Studies (after Laraaji)*.²⁹ I found a zither software instrument online and approached the sounds of Laraaji, only from my own electronic angle. It's more than a simple homage, as my computer-based approach is, in some important ways, very different from Laraaji's practice. He played every note (albeit processed) while I played much of the note data, but not all—I used some MIDI processors either in conjunction with or as a processor of manually played data on my keyboard. These studies used a plethora of effects, in order to create a unique atmosphere, and I used effects familiar to Laraaji and Brian Eno, who was also involved with his album, *Day of Radiance*. Consequently, with this musical interpretation of Laraaji, it may 'sound like' Laraaji, but it's really something else altogether, a music of alliance.

In 1989, I made my third album, *Breathless*. This was an important record for me, as I went directly into it with the insights from Watazumi, Laraaji, Jarrett, and others, especially Watazumi. Not his playing, per se, I had no skills at the flute, nor any background in Japanese music, and frankly, I had little interest in learning. It was the idea of meditation *in* music, how

27 Keith Jarrett, *Selected recordings*, Liner notes, ECM Records, April 2002.

28 Laraaji, *Ambient 3: Day of Radiance*, EG Records, 1980.

29 Henry Warwick, *Zither Studies (after Laraaji)*, Kether Records, 2021.

it can be more than just a catalyst for meditation, it can actually BE the meditation—away from 'keyboard playing' towards direct digital composition. Not that I was tired of playing keyboards—it is more of a shift to a methodology of playing combined with digital composition and sound design as a unified practice. Sonically, *Breathless* is very ambient and stripped down to a very few essential sounds. Everything about this record was nuance and subtlety. The slow swirling of modulating oscillators just ever so slightly detuned to cause slow tonal shifts that are carried through echoes into a deep reverberation.

One of the works on this album, 'indusTrial', was also released as part of Gargoyle Magazine's audio release.³⁰ The final piece, *Breathing*, is a short re-approachment of the title track, *Breathless*. In hindsight, I could have left off a few of the works on *Breathless*. Still, I consider it my first mature musical work. At the age of 30, one would hope so. With *Breathless*, I felt like I finally knew roughly what I was doing, and had found the beginnings of a practice, much like Watazumi had his Hotchiko flute in the 1940s.

I wasn't interested in 'enlightenment'. The Heart Sutra had wrapped that up for me—'*Attainment too is emptiness*.' From what I had learnt, there is just the practice, breathing in, breathing out, *and it works*. My response? *Breathless*. Which ends in *Breathing*. Did this make my musical practice a spiritual practice? From my perspective, no, because spirits and ghosts have nothing to do with it, even though spirits and ghosts are an extensive part of cultures from around the world. Spirit is sometimes seen as breath. God breathed life into the world is something not uncommon, and is specified in Genesis 2:7.³¹ When we stop breathing, life comes to an end and 'the spirit is no longer present' and we may find ourselves in a world of ghosts. In Buddhist mythology, there is a discussion of people who die with great desires, outsized desires, that prevent them from engaging in the karmic cycle of rebirth. Per ancient Buddhist writings, there are six realms, six different stages of existence. They are: hell, the world of *hungry ghosts*, the world of beasts, the world of fighting spirits, the world of human beings, and the world of gods and devas.³² It's interesting that they place humans just below gods, and hungry ghosts below that of beasts. It's a classic tension in Buddhist practice, where the purpose is to remove, or at least attenuate, human suffering. This requires a human, a specific type of animal. And all the complexities and difficulties native to human existence are only open to other human animals. Humans imagine something beyond this world of flesh, some even divide it into six realms, 'hell, the world of *hungry ghosts*, the world of beasts, the world of fighting spirits, the world of human beings, and the world of gods and devas', others divide it into hell, earth, and heaven, and others simply remove hell and heaven, and posit only the earth as the only real existence. This runs into a problem.

30 Henry Warwick, 'IndusTrial' instrumental, *Gargoyle Magazine* #36, Paycock Press, 23 April 1989.

31 The Bible, Genesis 2:7, 'Then the LORD God formed a man from the dust of the ground and breathed into his nostrils the breath of life, and the man became a living being.'

32 Koun Yamada, *The Gateless Gate*, Center Publications, 1979, p. 5.

The 'earth' isn't either, i.e., this material existence is as much of an illusion as anything else.

No eye, ear, nose, tongue, body, mind;
No colour, sound, smell, taste, touch,
Or what the mind takes hold of,
Nor even act of sensing.
No mind.
No thought.
No attainment.
Indeed, there is nothing to attain.³³

Yet, we continue with the discussions, poems, arguments, lyrics, songs, images, visions, stories, narratives of our lives, we are driven by our senses and bodily requirements imposed by cause and effect to conclude the Material as a minimum, as a base that cannot be questioned. Yet again, we must accept a parity between what we understand as a world 'out there' and our representations of it, even though we know it's wrong, that we're only experiencing what our brain affords us to understand. And our brain is a very weak and easily restructured system. Some micrograms of LSD or 5-MEO DMT can easily rewire one's experience to come to different conclusions. We know that what our brain cooks up as cognition and world, as myth and philosophy, as truth and fiction, is isomorphic and not actual. It is paritance.

33 Heart Sutra, *ibid.*

PART 1: PARITANCE: A PHILOSOPHICAL INVESTIGATION

A term essential to this volume is *paritance*. It's a portmanteau of parity and acceptance. Tupac's resurrection at Coachella was paritant. It sought parity with the memory of the actual Tupac and met with the acceptance of the audience. Paritance is not a Baudrillardian simulacrum. A simulacrum is part of a hyperreality where the original is lost in a world of *mediated* copies that are indistinguishable from the real, such that the real fades under the precession of the simulacra. *A Baudrillardian simulacrum is natively a media object—no media—no simulacra.*

Paritance is the affective material condition or object where a representation is accepted as sufficient. Paritance is not trying to replace the real. It is a real condition of cognition. It doesn't care, nor do its users. It's *good enough as the acceptance of the real, because our notion of the real is itself a paritance*. There are two basic forms of paritance and three basic traits of paritance, and one overall attenuation value.

Forms of Paritance

There are two basic forms of paritance—cognitive and social. Cognitive paritance is the paritance of our sensorium—we are forced to accept what our sensorium tells us. This leads us to believe in things like Reality. We don't have any options. However, because we are the kinds of creatures we are, we can see them as 'Maya' or paritance. In fact, the idea of Maya itself is a term based in cogitation, and thus paritant. We can also find things like optical illusions that examine the limits of human perception. Cognitive paritance is a feature of our human condition.

Social paritance is one induced by higher orders of thought conditioned by human interaction. For example, someone engages in a dastardly deed done dirt cheap. Whatever it is, it is one humans would commonly condemn as being 'bad'. Is the person who did this a Bad Person, or are they a person who is just bad at being a human? People would say 'Hitler was a bad person'. Was he? Or was he just a person who was just really really shitty at being a person by being so wilfully vicious? Paritance explains the former, and permits the latter. It explains the former because we accept that Hitler was a bad person. It permits the latter, because we don't really know what anything is, however, we are compelled into acceptance by our sensorium of Hitler's human behaviours, and judge them as profoundly suboptimal—therefore he is a person who's really bad at being a person, as our sensorium also indicates people are social animals and require cooperation. There are other political fallouts of this, which we shall arrive at later. Here, we will drill down a bit on the Traits of Paritance.

The Traits of Paritance: Acceptance, Parity, Intensity

Acceptance

Acceptance is affective and essential, as affect is primary. For example, something like mathematics. $2 + 2 = 4$. We can say there is no affectivity about this—it's just math. This is not true. (Before rationalists get their knickers in a twist, yes, $2 + 2 = 4$ in mathematics—that detail *is as true as it is irrelevant*) We have to *trust* that $2 + 2 = 4$. Without that, we won't believe that $2 + 2 = 4$, and for us, it won't be true. This effect is most apparent with people who have Capgras Syndrome.

Capgras Syndrome is a rare and complex psychological condition where an individual believes that a person they know, such as a family member, friend, or spouse, has been replaced by an impostor or double. This delusional belief is so strong that the subject is convinced that the impostor may be identical to the original person in appearance, however, the Capgras sufferer lacks the emotional connection or bond—they don't *trust* that the person presenting themselves as someone they know actually *is* the someone they know. Even though the sensorial evidence is plainly true, they can show identification cards, line up witnesses, show DNA tests—it doesn't matter. To the person with Capgras Syndrome—they are not the person they say they are—they are impostors. The subject has to *trust* them to be who they say they are, or they refuse to believe it. Whether what's presented is your Uncle Dave or $2 + 2 = 4$ or the planet Mars or a Studebaker Golden Hawk is irrelevant. It's how one *accepts and feels* about the information that *permits* the information to be true. Thus, we Let $X = X$.

We have evolved to *trust* our senses, as easily fooled as they can be, because they're the only 'window to the world' we have, so we must *accept* the information from our sensoria as real. We don't have to use a chromatograph to average the colours of the sky together to come out with the frequency range that we understand as blue is approximately 620-680 terahertz.

We have a cultural linguistic code of colour relations that allows us to quickly ascertain whether the sky is blue, and the regulation of this is affective, not data driven. We can use devices to take measurements, yes, however, they are not always necessary to tell if the sky is blue. However, we also *have to trust our instruments*.

The metaphysics of mathematics aside, this is a difficult problem when notions of truth and organising human behaviour (private or public) are performed and practiced through the paritance factory of media, when the paritance is thought itself. *We have to trust that our ideas conform to what we trust is a representation of reality*. Without that trust, without that emotional regulation, we engage madness—our emotions become unmoored and our ability to function or cope is profoundly impaired, or, at minimum, we spiral into a severe case of Capgras Syndrome. Which would be suboptimal.

Without acceptance, there is no paritance as acceptance is affective.

Giordano Bruno was burned at the stake for ideas that we consider nearly self-evidently true—that the stars were distant suns surrounded by their own planets like Earth or Mars, the possibility that these planets might foster life of their own, and that the universe is infinite and has no centre. He also held a variety of religious propositions that were at variance with the dogma of the Church at the time. All of these ideas challenged the ideas held by the orthodoxy. As their adherence to these ideas was deeply affective, engaging emotions of devotion, sublimity, comfort (in the maintenance of social hierarchy), defensiveness (impelled by centuries of conflict against people who also have similar emotional attachment to very different, if not oppositional social hierarchies and preferences), the orthodox were compelled to kill him to prevent affective friction and protect the comforts of their orthodoxy. One might argue that Bruno was correct and if sufficient technology was available at the time, his position would have been vindicated. This would be a false conclusion, as the technology would be sufficiently advanced as to appear as magic, again provoking the affective hysteria of the orthodoxy. Whether he was right or not is irrelevant here—what's important is the primacy of acceptance in the configuration of paritance, especially social paritance.

Parity

Parity, the second factor, is functional equality, the claim to identity that the paritance holds. It's what the paritance points at in a network of information relative to the noumena that impel description. Logic is built on parity and is a clear example. Logic features a few basic laws; the *Law of Identity*, where a thing is identical with itself, the *Law of Non-Contradiction*, which says it is impossible for a proposition and its negation to be true at the same time, and the *Law of the Excluded Middle*, which states that a proposition is either true or false, with no middle ground.

Whether these laws are true or not is also not of interest here. The 'Law of the Excluded Middle' is especially questionable here, as the Paritance argument excludes the truth or falsity of any statement—it's all the attachment of truth or falsity to a statement is subject to paritance. What is important here is that parity lurks underneath these laws, and that parity is perceptual and affectively regulated.

Arguments will be made. These arguments are products of paritance, *as is this volume itself*. From a paritant perspective, nothing can be completely identical to something else, not even itself, as context is time based, and context is necessary for meaning. Something cannot be equal to itself, as that which was stated as the original one second, and then said to be itself the next second has missed a second of change to the thing, and in Planck Time, that's 5×10^{44} possible states of the thing in time, and therefore, cannot exactly be that thing. From one Planck moment to the next, things change, and therefore are not the same. Self-identity is paritant. Once the self is perceived, this perception is now in another moment and thus different, and thus beyond itself. Thus, any self-image is a paritant image. Thus, there is no self-identity.

We are compelled to let $X = X$. From this, *all is haunted, all is paritance*. We can have parity.

We can have self-identity. We can have noncontradiction. We can have all of that with an innate, preformed, and built in sense of acceptance, thus, *paritance*.

Intensity

The third factor is **Intensity**, and it's more of a general attenuation value.

Intensity would be the relation of how much of a paritance a paritance is. For example, a paritance of zero intensity would be a paritance that has never existed. A paritance of maximum value would be the kinds of paritance that compelled the orthodoxy to kill Giordano Bruno and imprison Galileo or the Khmer Rouge to slaughter their population, or Nazis to gas Jews to list a few maximal examples. Maximum paritance intensity need not be violent. $2 + 2 = 4$ is a paritance. It just happens to be accurate in *accepted* mathematics, in *paritance*. Just because something is a paritance doesn't mean it's *necessarily* wrong.

It is through this paritance that human thought is possible as it is necessary for signification—signification itself is a paritance. The classic semiotic illustrates this—the signifier claims a level of parity with the signified, and this is met with a level of acceptance. When the signifier no longer has this acceptance, it loses intensity—a failed paritance.

Paritance: Beneath Simulation

We can see how paritance operates underneath even Baudrillard's simulation theory. A fun example of Baudrillardian logic is well illustrated by Eugene Wolters in his essay, 'Understanding Jean Baudrillard with pumpkin spice lattes'.¹ Pumpkin Spice lattes are a beverage this author finds unworthy of consumption², so it's a useful abstract example. The process of Baudrillard's precession of simulacra is outlined here:

1. Images are reflection of a profound reality: A picture of pumpkin is like the real thing.
2. Images mask and denature a profound reality. A picture of the pumpkin is like a shitty version of the real thing.
3. Images masks the absence of a profound reality, the cake pumpkin was a lie.
4. The image has lost all connection to reality. It is pure simulation. Pumpkins, as a natural phenomenon, are a lie and this picture I just handed you is just a napkin that I peed on.³

1 Eugene Wolters, 'UNDERSTANDING JEAN BAUDRILLARD WITH PUMPKIN SPICE LATTES.' *Criticaltheory.org*, Blog entry, 24 September 2014, <https://www.critical-theory.com/understanding-jean-baudrillard-with-pumpkin-spice-lattes/>

2 I drink my coffee black, no sugar.

3 Wolters Ibid.

Paritance is prior to this, as it is essentially held within and exists prior to, the first step, (emphasis mine):

1. Images are a reflection of a profound reality: A picture of pumpkin is **like** the real thing.

There is an immediate claim to a level of parity—the picture of the pumpkin is *like* the real thing. Everything immediately goes off into the weeds of paritance. Everything after is a logical follow on in paritance. The picture is a claim to some parity, it is accepted as such—at whatever level it claims. You are handed a picture of a pumpkin. It is not a pumpkin. It is not nutritious. It is paper and ink. Or, on screen, a field of glowing pixels.

Everything one might use to discuss pumpkins, spice, or lattes, are all going to be symbols, signifiers, concepts, and ideas: social constructs. Paritance is how we let it be that way. At the same time, one might watch an expensive blockbuster film and be wholly unconvinced by the special effects, even though the effects are incredibly realistic. It would fall into the uncanny valley, and the uncanny valley is evidence of paritance. We permit signifiers of 'realism'—paritance. The film tries to fulfil realism with these 'realist' techniques. Insofar as they succeed, we accept the proposed parity. When they don't we have the uncanny valley, and the domain of paritance.

Politics of Paritance

A common complaint would be, 'well, so what? We all know a picture of a pumpkin is not a pumpkin.' However, this is not how images are treated, as paritance is part of the deal, and parity is an adjustable parameter.

Ask someone of the Islamic faith about a picture of Mohammed. Is it a picture of Mohammed? Of course not. No one alive today really knows what he looked like, so making a picture of Mohammed can't possibly resemble Mohammed except in the most rudimentary of terms of a depiction of a medieval Arabian male.

Islam has a long history of aniconism, which is the prohibition of creating or displaying images of living beings. This is based on the Quranic verse that says, '*And they have taken besides Allah gods, which do not create anything, but are themselves created, and possess no harm or benefit to themselves, nor can they control death or life, nor to the resurrection.*'⁴ This prohibition originally applied to all forms of visual representation, including paintings, sculptures, and photographs. The stringency of this ban has varied over the centuries, and today, there is no shortage of visual media in Islamic nations—television, the internet, and advertising are as prevalent there as anywhere else, and the aniconism of Islam is more about images of Islamic religious figures, especially that of Mohammed. This demonstrates the paritance of the Islamic prohibition—the image is not Mohammed or Allah or anything. *It is an image saying it is.*

4 (Quran 21:24)

The image is a paritance, and Islam is **very** sensitive to this particular vector of paritance. Islam is not alone in this—there are a number of Calvinist Christian fringe churches, such as the Amish and Mennonite traditions, who agree with such bans of visualising their own god or prophets / saviours / revered figures. For them, also, the parity and intensity parameters are extremely high. A recent example of this kind of sensitivity happened in the Paris Olympics, where a Bacchanalian feast of Olympian gods was presented in the media representing the bounty and glory of the Olympic games from whence the Olympics got their name. However, the way it was presented was as a tableau of these gods arranged around a feasting table. This reminded some people of the painting, *The Last Supper* by Leonardo Da Vinci, an Italian.

The people who were most offended were Christians, who saw this as a cruelty to their faith, never mind that Leonardo lived and died about 1500 years after their saviour ruined his weekend with a crucifixion. In actuality, the image more closely resembled another painting by Jan Harmensz van Biljert, *The Feast of the Gods* from 1635, which is presently in the Collection of the Magnin Museum, in Dijon, France.⁵ This didn't stop said Christians from flying into the high dudgeon of a moral panic, because their *intensity of parity* is exceptionally high.⁶

2000 years ago a rabble rouser was killed by the Romans. A religion sprang up around his memory. 1500 years later, an Italian makes a fabulous painting of the last meal with his cohort. A hundred or so years later, a Dutchman makes another fine painting depicting a feast of Olympian gods. One does normally feast at a table, so there is a superficial resemblance to the Leonardo painting, simply based around European eating habits. 400 years later the image of the Leonardo painting is widely reproduced in print and a century or so later in electronic media, grounding the image with social weight in history. Then in 2024, an image of the Olympian gods feasting, as represented by a diverse variety of people gather at a table in similar rhetoric to the van Biljert painting is produced, and rather than do the research (As the people at Artnet did) they immediately took offence to the Olympic work and flew into a moral panic.

To reframe this, in paritance, they accepted that depictions of their saviour retain a sacred value—there is some accepted level of parity between their saviour and a painting of their saviour. Anything that *claims resemblance* must therefore either be in support of their ideology/religion or against it. They accepted parities between the representation of something and what they imagine as the thing itself. This precedes simulation—simulation is not possible without paritance, as paritance is beneath or behind it. A worldwide kerfuffle then blew through the mediasphere—a kerfuffle quickly blown off the screen by other

5 Sarah Cascone, 'Controversy Erupts Over Paris Olympics Take on Leonardo's "Last Supper"' The controversial scene at the opening ceremony actually more closely matches Jan Harmensz van Biljert's 'The Feast of the Gods.' *Artnet*, 29 July 2024. <https://news.artnet.com/art-world/paris-olympics-last-supper-controversy-2517433>

6 Anthea Butler, 'The ridiculous moral panic over the Olympics' opening ceremony.' *MSNBC*, 29 July 2024, <https://www.msnbc.com/opinion/msnbc-opinion/olympics-2024-apology-christians-last-supper-elon-musk-mike-johnson-rcna164008>

more pressing concerns, such as that of the US Election entrance of Kamala Harris and Tim Walz—more of the hypernormalisation of digital media and the Baudrillardian critique, which are aftereffects of paritance.

The Mundanity That Enables the Ubiquity & Insidiousness of Paritance

Paritance, with its vectors of Parity, Acceptance, and Intensity lies at the centre of a cross-roads between cognition, reason, perception, and affect. Depending on their settings for each vector, relative to the subject of the paritance in question, the success of the paritance in its simulative career and precession are evaluated. Cognition and Perception permit the possibility of parity. This is not, as pointed at earlier, a Signifier/Signified relation.

Paritance is more originary than the Signifier.

A Signifier can only signify if it is already accepted as signifying the signified. *Paritance is what allows this to happen.* We are constantly forced to accept paritance, just to get out of bed in the morning. One wakes up, turns off the alarm. All of this is accomplished through perceptions which are always already in the past. Perception, indeed consciousness is paritant. It only exists as a smear in time facilitated by the thermodynamics of our brain's activity in acquiring, storing, and burning energy in order to create brain patterns we can use as a memory activity for example. The inherent thermodynamic nature of life is well explored by Sagan and Schneider in their book *Into The Cool*.⁷

Life and Cognition as Thermodynamic Processes

A TLDR of Schneider and Sagan's thesis is: energy flows not only power evolution, they also have determined said evolution. Life operates from the second law of thermodynamics. The 'standard view' of the second law of thermodynamics is that the total entropy of a closed system always increases over time. Entropy is a measure of disorder or randomness. In other words, as energy is transferred or transformed, more and more of it becomes unavailable to do useful work because it becomes randomly distributed and dispersed.

What is key to Sagan and Schneider is that they show that this isn't completely true in practice. A part of an entropic system can devolve into, for example, a crystalline condition. For example, take something like liquid water. Water has a highly disordered (i.e. liquid), if relatively warm (compared to, say, the average temperature of space), system. Taking liquid water and dumping it in a cold environment results in ice. Thanks to the molecular and atomic structures of H_2O , it easily forms hexagonal forms when frozen. Hardly chaotic. This extends to other structures in the universe. Another example could be a large star. Eventually it burns through its nuclear fuel and continues its thermodynamic journey. For our example, we'll stipulate that it's such a big star it collapses into a black

7 Schneider and Sagan, *Into the Cool: Energy Flow, Thermodynamics, and Life*, University of Chicago Press, 2006.

hole singularity—the most orderly system possible, possessed of only mass, spin, and charge—it occupies no space or time and does exactly one thing: hoovers up anything that falls within its event horizon.

However, according to physicists like Stephen Hawking,⁸ this is not the thermodynamic end of the road for black holes. A greatly simplified description of why would go like this: a black hole is defined by its event horizon, a spherical space from which nothing, not even light, can escape due to the enormous gravitational density, hence 'black hole'. Throughout the universe, virtual particles are created in opposed pairs and destroy themselves constantly. However, if one appears right at the event horizon, one of the virtual pair will fall into the black hole, while the other is thrown away from it, wicking away a tiny amount of energy from the black hole.

Since, per Einstein, energy = mass, this means that some tiny bit of mass is removed from the black hole. This energy is so small it would be lost amidst the background noise of the contemporary universe. However, as the universe continues to cool, because of thermodynamics, eventually the relative temperature of the 'Hawking Radiation' will exceed the local temperature, and the black hole will begin to lose mass. Eventually this becomes a runaway process and the black hole evaporates. The amount of time this will take is enormous. Hawking Radiation becomes the thermodynamic end of the huge star, and the universe itself as it enters what is predicted by some cosmologists as a cosmological 'heat death'—a final thermodynamic collapse of the universe.

However, between the peculiar contemporary *Now* of computers, bowties, and plastic corn forks, and the Frozen Future of evaporating black holes, we have *structures*, not random noise. Structures are afforded by thermodynamics, one of which is biological structure itself. Sagan and Schneider spend much time discussing Bénard cells. Bénard cells are a type of cellular structure that forms in a planar horizontal layer of fluid heated from below, characterized by a regular pattern of convection cells. Bénard cells form from the combination of surface tension-driven convection and variations in the temperature gradient-driven surface tension. Bénard cells are quite familiar. For example, if you paint a wall in your house, it will look different when wet than dry—the colour itself shifts slightly. This is because as it dries, Bénard cells form on the surface that are characteristic of that paint's chemical formulation, changing the reflectivity at various wavelengths of light, thus changing the apparent colour of the paint. These structures come for 'free' in any thermodynamic process. Sagan and Schneider take this material property as an indication of the thermodynamic nature of cellular formation—chemicals just '*do that*'. Thus, a biological system (creature) depends on thermodynamic systems to exist.

A schematic of this would be that a biological system stores and releases energy in order to replicate more biological systems that store and release energy. In this way, life itself can be seen as a thermodynamic process.

8 Stephen Hawking, *A Brief History of Time*, Bantam, 1988.

Brain functions are this way as well. It stores energy (sugars, carbohydrates, ATP, etc.) in its cells. The cells burn some of this energy in their functioning, one function being 'consciousness', however one chooses to define it. Thus, it is impossible to 'live in the moment' i.e., for consciousness to be present to the most infinitesimal span of time. The smallest possible length of time that we as physical things can measure is Planck Time, which is equal to about 5×10^{-44} second. This is an incredibly short period of time—there are 50,000,000,000,000,000,000,000,000,000,000,000,000,000,000 in every second. Considering human vision is fooled by a cycle of images appearing to be in motion in as few as a couple dozen per second, being conscious of a moment is simply not possible. The flow of energy, the release of the sugars, ATP, etc. in the brain which produces consciousness is that: a thermodynamic flow, a *blur* from one state to another. We are never in the moment. We are only in the blur, and the blur is impelled by our physical limitations as biological organisms and the thermodynamic underpinnings of our operation. Biological organisms are entropy engines. Life operates to store and burn energy in order to reproduce more entropy engines, increasing the efficiency of the entropic nature of the universe. Thus, we are forced by the very structure of our brain to *accept paritance*. This is the material nature of paritance.

Existence Itself Is Paritant

The past does not exist—we have memories, which are structures formed with the entropy of biological operation. The future does not exist. It is simply a conjecture we entertain; a conjecture made possible by said thermodynamics. The science communicator and astrophysicist, Neil deGrasse Tyson pointed out *that the Universe is under no obligation to make any sense to us at all*.⁹ Proceeding from there, we can also conclude that the universe is also under no obligation to exist at all. It could easily wink out of existence from some kind of a Big Rip or a vacuum collapse that would expand at the speed of light if that happened -- *we would never even see it coming*. We would simply *cease. to. exist*. We have no past, no present, no future. Our existence in this universe is completely contingent. We *accept* existence, even its contingency—we have the capacity for the brain configuration afforded by thermodynamics that we can do this. And, this awareness isn't subject to linguistic understanding—we could write a million books about it, and even with such scholarship we would only have the linguistic understanding, which is not primary, per Fedorenko et al.¹⁰ Thus, *existence itself is a paritance*.

A paritance isn't what it 'is', and for human existence to be a paritance, it creates an inherent contradiction and we're off to the dialectical races—another exercise in paritance. And we *accept* it, because it feels right to do so—'it has to'—we have evolved for this capacity over the past few billion years.

9 Neil deGrasse Tyson on *Real Time with Bill Maher*, Episode 14, 3 June 2016 'the universe is under no obligation to make any sense to [you]'

10 E. Fedorenko, S.T. Piantadosi, E.A.F Gibson, 'Language is primarily a tool for communication rather than thought.' *Nature*, 2024, 630, pp. 575-586.

Structural Paritance

Paritance applies across technologies and markets.

Houses in North America (and elsewhere, certainly, however especially so in North America) are no longer built to last for centuries. If they last as long as the average term of residency, that is sufficient. In the USA over 47 percent of residents live in their current homes for between six and ten years.¹¹ With a *paritant* strategy new homes can be built to last just long enough for the buyer to not notice the inherent faults—never mind the length of the mortgage. This was made personally obvious to me when travelling in Paris. We stayed in the Marais, near the Marché des Enfants Rouges. The building was constructed in 1609, made of stone, easily half a meter thick, and ancient heavy wooden beams supporting the thick flooring planks. Two floors below us was a gay S&M bar that had loud thudding music going **UN TSA UN TSA** all night long. *We never noticed*. And it wasn't the jetlag. The building was solid as the stone it was made of. Barring direct exposure to a nuclear detonation, it will likely last another 400 years.

In contrast, many if not most North American homes are built cheaply; often 11mm plywood is sold as 'half-inch plywood.' A half inch is 12.7mm. That is nailed to studs made of 2' x 4' (51mm x 102mm) wood, which is actually 1.5 inches by 3.5 inches (38mm x 89mm) and then the place is wrapped up in Tyvek plastic and foam and sprayed with cheap stucco or fake brick, an acrylic top-coat, and painted, or just covered over with aluminum siding. A hilarious examination of these horrible building practices and the hideous houses that have resulted can be found on the McMansion Hell blog.¹² These are not mansions—they are weak *paritances* of mansions. They claim the space of mansions of old through the signifiers of what constitutes a mansion: vast floor space, turrets, es-cutcheons, 'stained glass', multiple roof peaks, stone facing to emulate stone construction, decorative elements made of hardened plastic foam instead of stone or concrete, etc. We accept them as mansions, rather than the cheaply built and horrifically designed exurban architectural blight they actually are. That we can critique these buildings by calling them 'McMansions' shows they are weak paritances with a reduced intensity of parity, one so obvious it is a common subject of ridicule.

There are other results that feed into the inherently paritant 'good enough' principle.¹³ Where consumers will use products that are *good enough* for immediate requirements, even when better technology comes to market—they illustrate the acceptance parameter of paritance.

11 Joint Center for Housing Studies - Harvard University, *America's Rental Housing*, Report, Harvard, 2024, https://www.jchs.harvard.edu/sites/default/files/reports/files/Harvard_JCHS_Americas_Rental_Housing_2024.pdf

12 *McMansion Hell*, Blog, <https://mcmansionhell.com/>

13 Mark Wilson, 'The New Mantra of Tech: It's Good Enough', *Gizmodo*, 27 April 2009.

This has had an effect on Moore's Law. Classically, in Moore's Law, the power of computers doubled every 18 months or so. This was true in the 1980s and 1990s. It began to run out of steam in the 2000s. However, thanks to **good enough** computing, it's no longer relevant—it doesn't matter. How powerful does a computer have to be to run Notepad? Not very. MS Word? More than Notepad, and yet not as powerful as it needs to be in order to run, say, an advanced audio workstation like Ableton Live. And while Ableton Live may need significant resources, it won't be nearly as resource-hungry as an advanced video editor like Adobe Premiere or DaVinci Resolve. It is now common for people to maintain computers for many years, running software that is *good enough* for their present needs. For example: this part of this text is being typed on an Apple MacBook Air from 2015—a ten-year-old computer. This would have been difficult in the 1980s and 1990s when computers and software were doubling their abilities every year or two.

Applying this to AI is obvious. It's not that AI actually needs to pass the Turing Test. It's not that it's even intelligent. It's just that its imitation of intelligence is *good enough for most purposes*. If it fools some of the people all of the time and all of the people some of the time, that's good enough for most purposes.¹⁴ AI is a paritance. It doesn't have to be God. It merely needs to be god-like *enough* to suit local needs and definitions. We let it fool us, because it is convenient. Because it is *good enough*. It's part of the blur.

Political and Cultural Paritance

Politically, paritance is a continuous state built into the compromises of democracy and the fabrications of authority. In our present state (wildly overpopulated industrial urbanism) even using a crude measure like a utilitarian argument, socialism is the obvious and logical system to insure the greatest good for the greatest number. Given our present conditions, it would be **good enough**. Paritance comes into play as 'national health', or 'free university', or 'universal basic income', or similar ideas are introduced. While these policies and projects may be aspects of a more socialised system, *they are not socialism*. They are paritants of low parity. This doesn't mean such policies and projects should be abandoned—on the contrary—they should be expanded as they have demonstrably positive effects on the lives of millions of people. At the same time, their use doesn't make them less paritant, nor does their paritance make them less useful. Also, an unfortunate side effect is that the discussion of socialism itself becomes a paritant discussion between society's material needs and the means of acquisition of the materials needed to satisfy said material needs.

The abilities of systems that operate outside of the verbal or text based (music, video, imaging, etc.) to be imitated by machines indicates another 'good enough-ness'. That the meteoric rise in popularity of a virtual idol like the vocaloid Hatsune Miku may have been something peculiar to Japanese pop culture, is no impediment to making something

14 Tom Weller, *Minims: Or, Man Is The Only Animal That Wears Bow Ties*, Houghton Mifflin, 1982.

'You can fool some of the people all of the time, and all of the people some of the time, and that's good enough for most purposes.'

similar happen in other genres beyond music. American film production, especially given recent advancements in technology, is also ripe for such. Insofar as a film adheres to, say, Joseph Campbell's story of the Hero, or at least Vogler's interpretation of it,¹⁵ whatever film one is watching is a paritance of the actual story—the so-called *Journey of the Hero*, which in itself is a 20th Century assemblage borne of the relations of a wide variety of myths and parables.

Nietzsche discusses the potential for Art to replace the void left in the human imagination when one evicts God from society's imagination in 'The Birth of Tragedy'.¹⁶ There, he argued that art can provide a similar kind of metaphysical consolation and affirmation of life that religion once did. In the absence of a shared religious belief, Nietzsche suggested that art can serve as a powerful force for meaning-making and cultural renewal. This substitution precludes a number of possibilities of the imagination. However, these preclusions can also be reversed into *invitations*. This was picked up by Roger Scruton,

A high culture is a tradition, in which objects made for aesthetic contemplation renew through their allusive power the experience of membership. Religion may wither and festivals decline without destroying high culture, which creates its own "imagined community", and which offers, through the aesthetic experience, a "rite of passage".¹⁷

We suggest that Art or 'High Culture' was and is still not capable or robust enough to shoulder such responsibility. It lacks the social institutional frameworks and power. Also, it's not designed for such responsibilities. Ellen Dissanayake argues that art was central to human evolutionary adaptation and that the aesthetic is a fundamental psychological faculty. For Dissanayake, art's intentional decoration of materials is often tied to spiritual and social practices and rituals regarding birth, death, and the transcendental or sublime in its power to take an ordinary thing and say 'this is special', requiring a distinct affective state and approach.¹⁸ This 'specialness' is an attention directive—this line isn't just any line, this line *matters* and requires a special kind of attention. It is an instrument. From this, we suggest that Art helps us 'get there' - it points the direction, as it doesn't have the capacity to be *the destination*. It is a device, an instrument, not the music itself. As Frank Zappa noted,

...a composer's job is the decoration of fragments of time. Because without music to decorate it, time is just a bunch of boring production deadlines or dates by which bills must be paid.¹⁹

15 Christopher Vogler, *A Practical Guide to The Hero with a Thousand Faces*, Memo, Disney, 1985.

16 Friedrich Nietzsche, *The Birth of Tragedy and Other Writings*. Edited by Raymond Geuss and Ronald Speirs; translated by Ronald Speirs, Cambridge, 1999.

17 Roger Scruton, *The Roger Scruton Reader*. Bloomsbury Academic, 2009, p. 144.

18 Ellen Dissanayake, *Homo Aestheticus: Where Art Comes From and Why*, University of Washington Press, 1995.

19 Frank Zappa, *The Real Frank Zappa Book*, Poseidon Press, 1989, p. 107.

Zappa's point brings in the obvious problem: Art under capital is a precious commodity factory, despiritualized and only left with empty symbols of mysticism. The mystical and occult are essential to any religion, for even as the orthodox often rail against it, it is the struggle of the mystics attempting to put their experiences into words and the occultists trying to make the spiritually immanent materially manifest that forms the backbone, the core, and the systemic glue that gives religion weight and vigour. For example, even in a supposed religion of mercy, like Christianity or Islam, practices like astrology, necromancy, demonology, and angelic evocation, herbal medicines, witchcraft, shamanic states of consciousness and much else have been endorsed or condemned over the centuries given the exigencies of rhetorical goals in the acquisition of social or political power by religious orthodoxies. These practices, when condemned by said orthodoxies, were and are often put down with brutal and murderous results.

The Hauntological as the Paritant Mirror of Contemporary Cultural Production

Here it is useful to take another angle on this, the Hauntological.

Hauntology

For those wondering what hauntology is, a quick description:

Hauntology is a philosophical concept that explores the idea of the present being haunted by lost futures. It is a portmanteau of haunting and ontology, coined by French philosopher Jacques Derrida. It is also a pun, as in French, Hauntology and Ontology are fairly homophonic, similar to how the name Mary and the word Merry are nearly homophonic in various regions of the USA. Derrida used the term in his book, *Spectres of Marx*²⁰, where he took up where Marx pointed in the *Communist Manifesto*, how 'there is a spectre haunting Europe, the spectre of communism'.²¹ The idea of the Hauntological was later taken up with equal seriousness and rigour by Mark Fisher, most especially in two of his finest works, *Capitalist Realism*²² and *Ghosts of My Life*.²³

Mark Fisher's position on hauntology focuses on the idea of 'lost futures' and how they haunt the present. He argues that neoliberalism's dominance has led to a cultural and political landscape where the future is perceived as something already predetermined, causing a sense of melancholic nostalgia for what could have been. This feeling of being haunted by unrealised possibilities, according to Fisher, is a defining characteristic of contemporary culture.

Hauntology is a complex and multifaceted concept that challenges understandings of time, memory, and the human experience. As a philosophical framework, it encourages us to think about the ways in which the past continues to shape our present and future, and how

20 Jacques Derrida, *Spectres of Marx*, Routledge, 1994.

21 Karl Marx, Friedrich Engels, *The Communist Manifesto*, 1848, p.1.

22 Mark Fisher, *Capitalist Realism: Is there no alternative?* Zero Books, 2009.

23 Mark Fisher, *Ghosts of My Life: Writings on Depression, Hauntology and Lost Futures*, Zero Books, 2014.

we can find meaning and significance in the remnants of lost futures.

Hauntology itself is not directly about ghosts or supernatural entities. However, it is an important component of our discussion, for while Hauntology is more about the lingering presence of 'the past inside the present'²⁴ and lost futures, it does so using some of the structure and terms of spiritualism and the occult—it's right there in the label on the tin: *Hauntology*. The Hauntological is the paritant mirror of contemporary cultural production. The idea of Lost Futures is one of the more potent aspects of the Hauntological project. Examining Lost Futures is to see that certain futures have been lost or failed to materialize, leaving behind a sense of longing and nostalgia. With Marx, it was the spectre of communism that was haunting Europe, a spectre that failed to materially manifest where it was expected—in advanced capitalist societies like Germany and UK. Whether it was a practical possibility is irrelevant. Whether the Soviet model was useful or not is also irrelevant. It is sufficient that it provided an alternative direction and set of possibilities counter to the demands and interests of industrial capitalism, and in the process, failed, for better or worse.

Music Is Like a Canary in a Coal Mine

The creative space that has been most effective in Hauntological terms has been music. Because music is always readily at hand, its production requirements are low, and its history is beyond ancient, music is an innate practice that can review social norms and systems, or used as a means of investigation into a society and its creative flow. Jacques Attali noted that music is the harbinger of the future.²⁵ He's held that position since, and most recently in 2023 in his essay, *When music, once again, reveals the future*.²⁶ This is extremely useful, as music is like a Canary In A Coal Mine—by observing its health (or lack thereof) one can make determinations about the mine it's in. As a consequence, music and the material and economic lives of musicians can be a leading voice in a culture. Towards this, Jacques Attali noted, 'music is a foretoken of evolution on the basis of behaviour in the human world'²⁷ and made an argument for Noise in music as an example of the harbinger function in music:

What practice of music should be read as the real harbinger of the future? The pseudo-new proliferates today, making it difficult to choose. Musicology always situates this essential fracture back at the entry of noise into music. That was indeed when provocation and blasphemy, the cry and the body, first entered the spectacle. Their entry was imperative in a world in which brutal noise was omnipresent; it did not, however, translate into a real rupture of the xexisting networks. As early as 1913, Russolo was talking about "the crashing down of metal shop blinds, slamming doors, the hubbub of crowds, the variety of din, from stations, railways, iron foundries,

24 Boards of Canada, 'Music is Math' from *Geogaddi*, Warp Records, 2002, Lyric: 'The past...inside the present...'

25 Jacques Attali, *Noise: the political economy of music*, University of Minnesota Press, 1985.

26 Jacques Attali, *When music, once again, reveals the future*, Blog Post, 19 April 2023, <https://www.attali.com/en/society/when-music-once-again-reveals-the-future/>

27 Jacques Attali, 1985, p. 136.

spinning mills, printing works, electric power stations and underground railways and the absolutely new noises of modern war." 121 He invented an orchestra of vibrators, screechers, whistles. Honegger wrote a work, *Pacific 231* (1924), which reproduces the rhythm of the wheels of a train, and Antheil wrote a *Ballet mecanique* (1926), which calls for airplane propellers. In 1929, Prokofiev wrote *Pas d'acier*; Mossolov wrote *The Iron Foundry*; and Carlos Chavez, *HP*²⁸

This line of analysis by Attali recaps the important shifts in the Western Art Tradition of music making, especially as it relates to the rise of Modernist avant-garde art music of the first half of the twentieth century. What happens a century later, when the 'new' isn't, when the New becomes a re-presentation of new ideas as a new idea? We will see how this happened with the digital sampler—where the technology of reproduction made manifest the Futurist dreams of Russolo with a digital device that instrumentalises the reproduction of sounds.

In short: if you want to see what the future holds, look at what musicians are contending with today. Today, most musicians make little or nothing from their musical labour, and their very livelihood is under threat of evacuation or redundancy in the face of Artificial Intelligence.

Musicking

Music comes to operate differently, as in our present condition the magic abounds, as we have been able to substitute the instrument for the song, the recording paritance for the music, and then instrumentalising it, converting a recording from a means of information distribution to a form of information generation—the message becomes a medium. Music is thoroughly dependent on the thermodynamic blur of consciousness. There isn't anything particularly 'wrong' with this, paritance itself is non-moral, and at the same time there is a great gain in the expansion of the possibilities of information and expanded creativity. At the same time, there is a tremendous loss with the abandonment of mystical thought forms which can then (and have) inform(ed) musical thoughtforms.

In a musical context, we went from a limited and purely performative understanding of music, to an expanded notion where recordings of music are instrumentalised to make a more hybrid musical experience that can only be experienced in the context of an industrial society—a paritant 'Mellotronisation' which I will discuss in a later chapter.

What is lost is what Christopher Small points at this in his book, *Musicking: the meaning of performing and listening*.²⁹ Small suggests that music is a process, 'musicking,' which includes all musical activities; the consumption of musical recordings, experiencing a concert as a listener, experiencing a concert as a performer, singing in the shower, sitting at a desk and instrument and composing music - these are all aspects of 'musicking'.

28 Ibid.

29 Christopher Small, *Musicking: the meaning of performing and listening*, Wesleyan, 1998.

This approach of music as a process has common ground with Dissanayake, as Art is a way to say 'this is special' and a way to decorate our space. With a Musicking process, it is a way to say 'listen to *THIS*' and Art becomes a method or process for humans to 'decorate time' (per Zappa) in a synchronous fashion. This is not to say that the decorative is shallow—*quite to the contrary*—this is to elevate the possibilities in said decorative practices to a depth and meaning that is more useful, inviting, and liberating for us as a species.

Also, as discussed earlier, humans can use the arts for very practical and pragmatic purposes, such as hunter-gatherer songline map-making,³⁰ to visual map making, to Google Maps. The switch we are focused on here is that which occurs in the amplification of said maps so that in a Borgesian / Baudrillardian sense, the map *replaces* the territory. However, it only does so as we accept it to do so, an acceptance based in a cognitive fait accompli: paritance.

Paritant Glitchery

In his essay, *The Aesthetics of Failure*³¹ Kim Cascone wrote how the 'post-digital' tendencies in contemporary electronic music transformed into a product for data capture and marked a fundamental dividing line between the twentieth and twenty-first centuries in the history of music. The main interests in the post-digital are the defects in the digital. The glitches and errors of the digital became the *modus operandi* of the post-digital, resulting in a number of different artistic directions explored by the likes of Oval, Venetian Snares, Autechre, Matmos, BT, Secret Mommy, and others; investigating and or using glitch aesthetics for creative ends. This effort was furthered by software. As the advent of digital music subsumed noisy and crude analogue methods of musical recording and reproduction under a brittle digital veneer of desperate precision, forming the gateway for the vampirism of the Mellotron to consume the Modernist Synthetic (to be discussed in a future chapter).

For example, Oval's early works were exemplars of this—Oval (essentially the artist Markus Popp) 'recorded smooth music and samples on CDs, which they then scratched themselves, covered with paint and pieces of sellotape, drew on with felt-tip pens—in short, they did everything so that the CD reading laser would constantly stumble and give non-smooth random music through trial and error.'³² The interest of the post-digital being in the failures of the digital underlining Cascone's reasoning in *The Aesthetics of Failure*, was then quickly imitated into software from a variety of companies such as Glitchmachines.³³ This software provides the stuttering glitchy effects that Oval and others would have hand practiced with paint, knives, and markers to create. It is this sort of software that Oval/Markus Popp soon developed and adopted to make their music. BT (Brian Wayne Tran-

30 Lynne Malcolm, Olivia Willis. Ibid.

31 Kim Cascone, 'The Aesthetics of Failure.' *Computer Music Journal*, MIT Press, 24:4 Winter 2002.

32 George Pallidev, 'Oval – 94.Diskont. A brief story behind first glitch album.' Blog post. 11/11/2020. <https://blog.12edit.com/oval-94-diskont/>

33 Gitchmachines, Website, <https://glitchmachines.com/products/everything-bundle/>

seau) is another artist like this, only even more involved with glitch software development, and even holds a patent in this regard.³⁴ His company, Sonik Architects, released *Sonifi* which allowed people to do the kinds of stutter effects well known in his music. A prime example would be *The Antikythera Mechanism*,³⁵ which combines glitch, classical music, and EDM stylings.

This method of glitch production is not based on the failure of the digital, as much as it is manipulating the digital to emulate the glitch. What was previously a digital error, a glitch, *becomes a copy of an error or glitch*. We suggest it is a paritance of a glitch. And now one can hauntologically revisit the post-digital with software, and create significant results, for example my album *Covidian Meditations*³⁶, or *Phantom Power* by West End Electric,³⁷ where software is used to make the music sound *as if* it is from highly eroded analogue formats recorded through a variety of degraded noisy media. It becomes a stuttering spastic effect of glitch or the sonic rust of analogue systems, giving way to the morose complexities of the hauntological.

This hauntological music draws on cultural memories and aesthetics of the past, often incorporating elements of library music, film and TV soundtracks, psychedelia, and public information films, as well as the technological limits of analogue media platforms. 'Hauntological Music' (viz. music from GhostBox Label, Boards of Canada, Broadcast) abounds, and other styles such as Vaporwave and Lofi Hiphop have distinctively hauntological components, and beneath it all is the acceptance of parity—we accept the representations of glitch and noise as glitches and noise.

The Sounds That Resist: Hauntological Musics

Hauntological groups like Broadcast, Boards of Canada, Belbury Poly, The Focus Group, and the Advisory Circle share some similar concerns and formal similarities. I often describe it as if one was listening to the soundtrack to someone else's childhood of the 1970s or 1980s. Noisy recordings, simplistic friendly melodies, samples from television shows like Sesame Street or arrangements that sound like television show themes or background music for documentaries, game shows, mystery shows, and a variety of possible media, all are parts of the hauntological palette. The Boards of Canada are prime examples—the band consists of two brothers, Michael Sandison and Marcus Eoin, who took their band name from The Film Board of Canada. Born in Scotland, they spent their childhood in Calgary, Alberta, Canada, and grew up watching documentaries on CBC, dominated by the Film Board of Canada. They loved the electronic soundtracks of the

34 Brian Wayne Transeau / Sonik Architects, 'Time varying processing of repeated digital audio samples in accordance with a user defined effect.' Patent, US20090281793A1, applied: 25 May 2006, Granted: 12 November 2009.

35 BT, 'The Antikythera Mechanism', *Binary Universe*, Binary Acoustics / DTS, 2006.

36 Henry Warwick, *Covidian Meditations*, Kether Records, 2020, <https://henrywarwick.bandcamp.com/album/covidian-meditations>.

37 West End Electric, *Phantom Power*, Kether Records, 2022. <https://westendelectric.bandcamp.com/album/phantom-power>.

films and these recordings were a significant influence on Sandison and Eoin's creative vision. Another creative response to the cul-de-sac of contemporary capitalist realist culture is Vaporwave. It's a direct, ironic, and immediate result of Fisher's Hauntological examinations.

Much vaporwave is a type of plunderphonics, where vacuous pop music of the 1980s is slowed down, chopped up and hacked into a new work of music -- much fulfilling the ideas of Oswald explored in his Plunderphonic essay. Vaporwave production reached a kind of peak in the late 2010s. Vaporwave is well examined by Grafton Tanner³⁸. His book, *Babbling Corpse*, explores the intersection of Vaporwave music, politics, media theory, and the fascination with uncanny, cosmic horror under global capitalism. Tanner argues that vaporwave, a genre characterized by its appropriation and recontextualization of 80s and 90s popular culture, is a quintessential example of late capitalism's commodification of art and culture. Vaporwave was a rebellion against the commercial music establishment, while perpetuating capitalism's marketing and aesthetic tricks, such as endless repetition, rehashed *a e s t h e t i c s*, and other vague gestures toward the capitalist / realist/postmodern condition.

Vaporwave uses notions of nostalgia and retrofuturism as a means to critique the dominant cultural narrative of the late post collapse 2000s and 2010s. Vaporwave was often seen as a meme in search of a meaning— a kind of musical joke as it celebrated its lack of innovation and use of digital audio as little more than a sonic blender. With this critical air, vaporwave's connections to politics, media theory, and our fascination with uncanny, cosmic horror brought a focus on global capitalism's impact on art, and the changing relationship between art and commerce by raiding commercial music that is then instrumentalised in a paritant re-presentation of music as grist for a computer music mill. Tanner sees Vaporwave as a symptom of our postmodern, hyper-consumerist society, where the lines between authenticity and simulation, originality and appropriation, are increasingly blurred.

Lofi Hip Hop came about in the mid 2010s. A quiet and calming style of music, it shares some properties with ambient music, however, Ambient Music rarely has a trap kit tapping out a dusty rhythm like LoFi Hip Hop. Featured on Youtube channels like Chilled Cow (which switched its name to LoFi Girl in 2021), it became instantly popular with students as musical accompaniment for study and relaxation. Most LoFi Hip Hop is instrumental and thus conducive for those purposes.

The Hauntological musics described here, from the Boards of Canada to the Vaporwave of Chuck Person or Vektroid, to the lofi hip hop beats of eevee or Kupla, all rely on the instrumentalisation of samples and reproductions to render a pastiche of previous styles. The level of raw stylistic innovation in popular music peaked in the late 1960s and early 1970s. Most popular music styles came into being or are rooted in innovations of that time. Fisher notes that things were changing so quickly that sometimes you can not only pick out what year a song was released, you can say what month it was, as change in music was so rapid

38 Grafton Tanner, *Babbling Corpse: Vaporwave And The Commodification Of Ghosts*, Zero Books, 2020.

and periodic.³⁹ Fisher's position is one of a lament in the collapse of innovation in music under capitalist realism, the loss of the modern impulse to originality, where 'the 'cancellation of the long term', is invariably stagnation and conservatism, not innovation'⁴⁰, as he sees the problems in music as symptomatic of the broader cultural malaise of Capitalist Realism.

Fisher expressed his view on hauntological music in *Ghosts of My Life*.

When it was applied to music culture—in my own writing, and in that of other critics such as Simon Reynolds and Joseph Stannard—hauntology first of all named a confluence of artists. The word confluence is crucial here. For these artists—William Basinski, the Ghost Box label, The Caretaker, Burial, Mordant Music, Philip Jeck, amongst others—had converged on a certain terrain without actually influencing one another. What they shared was not a sound so much as a sensibility, an existential orientation. The artists that came to be labelled hauntological were suffused with an overwhelming melancholy; and they were preoccupied with the way in which technology materialised memory—hence a fascination with television, vinyl records, audiotape, and with the sounds of these technologies breaking down...

...And hovering behind much sonic hauntology is the difference between analogue and digital: so many hauntological tracks have been about revisiting the physicality of analogue media in the era of digital ether. MP3 files remain material, of course, but their materiality is occulted from us, by contrast with the tactile materiality of vinyl records and even compact discs.

No doubt a yearning for this older regime of materiality plays a part in the melancholia that saturates hauntological music. As to the deeper causes of this melancholia, we need look no further than the title of Leyland Kirby's album: 'Sadly, The Future Is No Longer What It Was.' In hauntological music there is an implicit acknowledgement that the hopes created by postwar electronica or by the euphoric dance music of the 1990s have evaporated—not only has the future not arrived, it no longer seems possible. Yet at the same time, the music constitutes a refusal to give up on the desire for the future. This refusal gives the melancholia a political dimension, because it amounts to a failure to accommodate to the closed horizons of capitalist realism.⁴¹

'I wonder if we're dead.'

Hauntology is also concerned with the nature of being and existence, exploring how our understanding of the present is shaped by the remnants and shadows of the past, thus affecting our ontological present. A fine and appropriate example of this is the song 'I won-

39 Mark Fisher, 'The Slow Cancellation of the Future', Youtube Video, *Pmilat*, 21 May 2014, 9:01. <https://www.youtube.com/watch?v=aCgkLICTskQ>

40 Mark Fisher, *Capitalist Realism - is there no alternative?*, Zero Books, 2009, p. 76.

41 Mark Fisher, *Ghosts of My Life*, Zero Books, 2013, p. 9.

der if we're dead' by the Dutch producer eevee from her album, *Seeds*.⁴² The haunting melody is derived / sampled from the show *The Legend of Korra*, a piece called 'Separate Ways.'⁴³ By taking a theme from a children's show from the 2000s and playing it over a dusty slow hip hop beat, enormous pathos are pulled from the generation for whom *The Legend of Korra* was a formative experience. *The Legend of Korra* was an American children's animation program that was executed in a style derivative of Japanese anime. Eveline Schuyff (eevee) skillfully and syncretically pulled a variety of sources together—the melody from an American TV show pretending to be a Japanese anime (Korra), put it over an American Hip Hop beat, and then filled in sections using standard music theory of counterpoint and harmony using a jazz vocabulary. The video for the song consists of edited sections of the Japanese anime *Dareka no Manazashi (Someone's Gaze)*.⁴⁴ *Someone's Gaze* takes place in the near future, and describes a coming of age story of a young woman and her changing relationship with her father. The story develops through nostalgic scenes of her childhood and the connection between her, her father, and their cat. When edited and pared down to work with *I wonder if we're dead*, the video and song form a deeply affective work bound in themes of nostalgia, loss, struggle, and independence.

The title, *I wonder if we're dead*, is directly hauntological as it implies a present and aware consciousness after death — an essential condition of a ghost.

'Asking a ghost whether he believes in ghosts.'

In 1983, Jacques Derrida appeared in a film, *Ghost Dance*,⁴⁵ where he discussed matters of a hauntological nature that are of some use here.⁴⁶

Pascale Ogier—I'd like to ask you something, do you believe in Ghosts?

Jacques Derrida—That's a difficult question. Firstly, you're asking a ghost whether he believes in ghosts. Here, the ghost is me. Since I've been asked to play myself in a film which is more or less improvised, I feel as if I'm letting a ghost speak for me. Curiously, instead of playing myself, without knowing it, I let a ghost ventriloquize my words or play my role, which is even more amusing. The cinema is the art of ghosts, a battle of phantoms. That's what I think the cinema's about, when it's not boring. It's the art of allowing ghosts to come back. That's what we're doing now. Therefore, if I'm a ghost, but believe I'm speaking with my own voice it's precisely because I believe it's my own voice that I allow to be taken over by another's voice. Not just any other voice, but that of my own ghosts. So, ghosts do exist. And it's the ghosts who will answer you. Perhaps they already have. All this, it seems to me, has to do with...with an exchange between the art of the cinema in its most original unedited form and an aspect of

42 eevee, 'I wonder if we're dead', from *Seeds*, Inner Oceans Records, 2017.

43 Jeremy Zuckerman, 'Afraid (Separate Ways)', *The Legend of Korra: Original Music From Book One*, 2013.

44 *Dareka no Manazashi (Someone's Gaze)*, Dir. Makoto Shinkai, Animation Film Short. CoMix Wave Films, 2013.

45 *Ghost Dance*, Dir. Ken McMullen, Film, 1983.

46 Ibid. 15:53

psychoanalysis. Cinema plus psychoanalysis equals the science of ghosts. You know that Freud...Freud had to deal all his life with ghosts.

(telephone rings)

Now the telephone is the ghost.

(answers the telephone)

Hello. Yes. There will be a small seminar tomorrow afternoon. It's a closed seminar but you can come if you want. At four and fifteen— fifteen past four PM. Saturday. And another seminar next Wednesday at five pm. Yes. Yes. OK. I will be very glad to meet you. Goodbye.

(hangs up the telephone receiver)

Well, that was the phantom voice of someone I don't know. He could have told me any old story. Someone who's arrived from the USA, and says he knows a friend of mine. Well, what Kafka says about correspondence, about letters, about epistolary communication, also applies to telephonic communication. And I believe that modern developments in technology and telecommunications, instead of diminishing the realm of ghosts, as does any scientific or technical thought, is leaving behind the age of ghosts as part of the feudal age, with its somewhat primitive technology, as a certain perinatal age. Whereas I believe that ghosts are part of the future. And that the modern technology of images, like cinematography and telecommunications enhances the power of ghosts and their ability to haunt us. In fact, it's because I wished to tempt the ghosts out that I agreed to appear in a film. It could perhaps offer both us and them the chance to evoke the ghosts: The ghost of Marx, the ghost of Freud, the ghost of Kafka. That American's ghost—even yours! I only met you this morning, but to me you're already permeated by all sorts of phantom figures. Whether I believe in ghosts or not, I say: "Long live the ghosts". And you, do you believe in ghosts?

Pascale Ogier—Yes. Certainly. Yes. Absolutely.

There are a few crucial points here relative to Paritance here.

'I feel as if I'm letting a ghost speak for me.'

'I feel as if I'm letting'

Acceptance

'a ghost speak for me'

Parity

Thanks, Jacques. That sums it up very nicely.

Humans find themselves betrayed by their senses, nothing is as it seems. Humans crave human or human forms of sociality—if sociality isn't immediate and consistent from childhood on, an array of psychological problems ensues—this has been documented in

Eastern European orphanages.⁴⁷ So, when faced with an implacably mysterious universe, one which, as we have seen is actually quite beyond human experience, and which can only be experienced by humans in paritance, there is a common desire for a reflection of the human person and personality in those things which are not human. Ideas of a 'personal God' are developed to fill that void.

As a consequence, rather than sit and work it out, we project our insecurities onto the world, and invent fictions of Personal Gods, angels, demons, and *ghosts*. These 'ghosts', impelled by our communication systems and software, are responses to the paritance beneath our cognition and reality. There's nothing there, but our brains, operating in a degraded thermodynamic flow and having evolved on a small planet between the interactions of a variety of hominens over hundreds of thousands of years, are compelled to fill it all in with narratives from human or human adjacent (gods, ghosts, AI) agents. We even have a meme for this:

YOU ARE A GHOST DRIVING A MEAT COATED SKELETON MADE FROM STARDUST RIDING A ROCK HURLING THROUGH SPACE. FEAR NOTHING.

Stripped of Futurity, we have a dead culture, a ghost culture, that is embodied in the very machines we use to enact our civilisation. Ideas, ideologies, cultural norms, public perceptions, all of these are parts of the dreams that our stuff is made of. This is a constant flow between necessity and creativity, desire and consumption, paritance and will. We will now see how we can examine two cultural artefacts that embody such ideals: The Synthesizer and The Mellotron.

47 Charles Nelson, *Romania's Abandoned Children: Deprivation, Brain Development, and the Struggle for Recovery*, Cambridge, MA and London, England: Harvard University Press, 2014, pp. 335-368.

PART 2: THE MELLOTRON VS. THE SYNTHESIZER

Encountering Telstar

This particular story begins in 1963. I was about four or almost five years old. My dad worked at the Johnson and Johnson Personal Products 'Modess' Factory. His job was to test tampons and pads on a machine. It wasn't a glamorous or particularly fun job. In fact, his boss was a bit of a dick, and he would often come home feeling very upset at how his day had gone in the tampon factory. Still, it was a job, and it was the early 1960s, and with the wages he made at the factory, we were able to own a home and a car, and my mom could stay at home and raise us kids. Our 'home' was a two-bedroom duplex hurriedly built in the 1940s for officers waiting to be shipped to Europe. It was clad in asbestos shingles and sat next to the New Jersey Turnpike and under the landing pattern for Newark Airport, with a spur of the Lehigh Valley Railroad a block away. Our neighbourhood was surrounded by the Rheem toilet factory, the Zeston fibreglass factory, the JT Spivey tool and die mill, and the Bienfang paper factory, and Exit 10 of the Turnpike for traffic to access local roads and Interstate 287. So, not only was it a tiny house clad in toxic asbestos next to the largest highway in New Jersey, stuck under the path of shrieking jets landing at Newark, it also smelled bad. Besides all the car exhaust, the fibreglass company smelled like glue, and we would frequently be subjected to horrible odours from the oil and gas refineries just north of us in Carteret, Rahway, and Linden, New Jersey. My parents, being 'good Catholics,' proceeded to have five more children and stuffed us all into that tiny little 875 square foot (81.3 square meters) duplex, and while it was truly a rather miserable existence, that particular misery is a different discussion for a different day. When I was small, we also had a car—a two-door robin's egg blue 1960 Rambler American station wagon. While it was not the most reliable car, it worked well enough for several years. I remember it was a beautiful day—the sky was a deep blue, the temperature was pleasant, and it was not especially humid. After playing outside with friends that beautiful afternoon, my mom said that it was time to go collect my father from work. My brother, Matt, was two years old at the time, and he came along as well.

Matt and I clambered into the back seat of the wagon. There were no 'child seats' back then—nor were there any seat belts, there was just the springy back bench seat of the Rambler covered in cloth and Naugahyde. In today's modern a-go-go world, we have seat belts, head rests, and child seats. Today, in these days of modern times, we don't have Naugahyde, because in a greedy fit that lasted for a few decades, the automobile companies drove all the poor little Naugas into extinction for their precious little hides. Poor little Naugas. Once my mom got the Rambler rambling, we sped down US1, past the giant Ford plant (which later built the Ford Pinto—a car notorious for exploding into flames when struck from behind), past the Mack Truck dealership, and past Edison Lanes, one of the largest bowling alleys ever built with 110 lanes. We went over the Morris Goodkind bridge across the Raritan River, made our way past Route 18, and approached the enormous Modess facility, across from the Permacel factory, which made sticky tape. We arrived a bit early, so we parked in the shade and turned on the radio. A variety of pop hits poured out from the AM radio built into the car's light blue dashboard—songs like 'I Can't Stop

Loving You' by Ray Charles,¹ 'The Lion Sleeps Tonight' by The Tokens,² and 'Big Bad John' by Jimmy Dean³ owned the airwaves in this pre-Beatlemania musical landscape. My mom read a book, Matt took a nap, and I listened to the music on the radio.

I soon heard something I had not heard before—a rumbling, slashing, rippling noise stuttering out of the radio's speaker—I was entranced. Then a fast repetitive drum beat began, and an odd-sounding instrument I'd not heard before—something like an oboe, something like a bagpipe, and something like a reed organ—started playing a cheerfully optimistic melody over some energetic and strummy guitars. The melody repeated and I hummed along to it—daaaa da daaa, da daa dada da daaaa... The song broke into a middle eight section with these beautiful harp sounds and a guitar playing a similar melody. Then the weird oboe/bagpipe sound came back, playing the familiar melody, and with a man singing 'la la la' along with it. It built to a crescendo, and then the thumpy slashy rippling sounds re-appeared and faded away, quickly interrupted with the station's audio logo—'DOUBLE U—A—B—C!!!' The DJ said the song was called 'Telstar' by a British group called The Tornados.⁴ Do listen to it—it's three minutes and twenty seconds of modernist pop bliss.

My father arrived momentarily after that, and we went home. As usual, my dad was in a foul mood—he usually was after work. I asked him about the song, 'Telstar'. What is a Telstar, daddy? He said it's a machine in outer space that flies around the planet letting rich people call each other on the phone, all over the world. It's called a Satellite. I was deeply impressed—it had its own song! I never forgot that song, or forgot that we shared a party-line for our phone service and long distance charges were insanely expensive.

Many years later I discovered a lot more about the song. For example, the swishy stuttering slashy woofy sounds in the beginning and end were made using recordings that were then processed and re-recorded using techniques of a musical style called *Musique Concrète*. I learned that the song was made by Joe Meek and while it was wildly successful charting in countries all over the globe, it caused him enormous grief, as the record company basically ripped him off. I also learned that a few years later, he eventually flipped out and murdered his landlady and then, himself, just a few months before the case was resolved in his favour. As a four-year-old, I didn't know any of the back story, and was more impressed by the wild sounds coming out of the radio that this song, 'Telstar', had presented as music.

This was the miracle of music recording. There was a history to this. And a consequence.

1 Ray Charles, *I Can't Stop Loving You*, Written by Don Gibson 1957, ABC Paramount, 1962.

2 The Tokens, *The Lion Sleeps Tonight*, Written by Solomon Linda, 1939, RCA Victor, 1961.

3 Jimmy Dean, *Big Bad John*, Columbia, 1961.

4 The Tornados, *Telstar*, RGM Sound, London, 1962.

What Happens Now?

The Synthesizer is the voice of modernity, the voice of Frankenstein's Creature, and the Mellotron is something else something vampiric: Dracula. Of course, in a fight between Frankenstein's Creature and Dracula. Dracula wins. Always. Here we will see how paritance has had a profound effect on human creativity, specifically in –the field of electronic music, a field I have some 40 years of expertise in. We will see how paritance is woven into the very material existence of electronic music, it's an essential characteristic. First, a brief discussion of the Synthesizer.

The Synthesizer: Frankenstein's Creature: Our Hero

Earlier drafts of this part included a lengthy discussion of the evolution and development of synthesizers and electronic music. After some consideration, it was seen as superfluous to the discussion. Here, I will discuss this from a very different angle.

Since synthesizers are more well known, I won't spend too much time discussing their workings. The basics are fairly simple—electrical current makes a speaker vibrate. By modulating that voltage, you can create different kinds of electrical impulses that make the speaker vibrate, making different kinds of sounds. The math and physics behind this sort of thing is extremely well defined, as there have been electrical methods of generating sound since they invented speakers and amplifiers. Early synthesizers were expensive. For example, a Theremin set, including vacuum tubes, cost about \$240 in 1930, which would be around \$4600 today. The most capable ones were incredibly expensive. The RCA/Columbia synthesizer cost \$500,000 in 1957, which would come to about \$5,720,000 today. It was the size of a room, and was housed in ten audio racks from floor to ceiling.

Over time with miniaturisation and economies of scale in industrial production, the prices collapsed and capabilities multiplied decade after decade. They were, and are, extremely popular devices for music making, and to this day, most musical keyboards sold today are some kind of synthesizer. They have sparked the imagination of composers for a century. In the first half of the twentieth century, classical composers wrote music for the Theremin⁵ and its keyboard based cousin the Ondes Martenot⁶, as well as for the more capable and avant garde Trautonium⁷. The recordings and sheet music of these are fairly simple to find. It is easy to understand why the synthesizer was met with some acceptance: it is a product of modernist science and technology. The waveforms, filters, modulators, and envelopes used by these machines are bound up with mathematics, physics, and engineering - *the sound itself is science made audible*.

5 Saint-Saëns: *The Swan* (Carnival of the Animals).

6 Messiaen: *Turangalila-Symphonie, Fête des belles eaux, and Trois petites liturgies de la présence divine*.

7 Hindemith: *Concertino for Trautonium and String Orchestra*.

As was Frankenstein's creature. Frankenstein's Creature was also a product of science, physics, and technology. *He was Science made flesh.*

'Each fashioned thing speaks of its change'⁸ And each change is conditioned, limited, and impelled by the ideas inherent to the material conditions of its production—thus each fashioned thing speaks of these ideas as well—we have to negotiate the dreams our stuff is made of. The synthesizer is the musical zenith of ideas from Newton, Maxwell, Gauss, Leibniz, and many other scientists who used mathematics and the physical sciences to describe the world, and the technological scientific worldview of that world enabled the Synthesizer to exist.

The Mellotron was a very different affair from the synthesizer, and while a vastly simpler device, the ideas it embodies are vastly more complex. It uses basic tape-recording technologies, modified and manipulated to produce sounds not of its making.

The Vampiric Mellotron

The Mellotron is a keyboard instrument. When you press one of its keys, a tape is set into motion, and as the tape moves against a playhead, an electrical impulse is sent to the amp and speaker. When you let go of the key, the tape snaps back into position by a spring. Essentially, that is all there is to a Mellotron. It's a box full of tapes that play when you play the keyboard.

The Mellotron itself makes no sound—its material form and configuration are inherently silent, save for the sound of the capstan spinning and the whirring sound of the electric motor that powers the capstan. It only sounds like whatever is recorded on the tapes that are in it. It doesn't have a 'native' voice. It always sounds like 'something else'. At the time, a synthesizer sounded like a synthesizer. (that it closely imitates other instruments today is a subject we will get into later) The Mellotron did not. Nor did the Mellotron have great works of classical music written for it. On the contrary, it was picketed by angry music union members afraid of this new technology taking away their jobs. It was often reviled by its users because of its expense, delicacy, and fragility. Because it ran tapes, any slowing of the capstan that pulled the tape would slow the tape playback, and that would drop the pitch. So, if you tune the Mellotron for a single note played at a time, this will become useless when you play a chord, as every key played at once is a multiplier of friction on the capstan, making it play more slowly, causing the pitch to drop. Because of this, Robert Fripp of the band King Crimson famously quipped, 'Tuning a Mellotron doesn't'.⁹

What is a good example of the Mellotron? Many point to the opening 'flutes' from *Strawberry Fields Forever* by the Beatles. Or the ghostly Strings one hears on a Moody Blues

8 Art Bears, Final lyrics to the song 'Slave' from the LP *Winter Songs*, Ralph Records, 1978.

9 Robert Fripp, 'The Night Watch', *dgmlive.com*, blog post, 2016.
<https://www.dgmlive.com/in-depth/the-night-watch-robert-fripp>

record. One of my favourite examples is 'Epsilon In Malaysian Pale' by Edgar Froese.¹⁰ The title track is a long (17-minute), moody and lushly beautiful solo on two Mellotrons, one set for flutes, the other for strings. It also has a synthesizer occasionally playing swooping noise parts. It's possibly the finest example of Mellotron playing as a lead solo instrument. The opinion by musicians regarding the Mellotron was mixed and often negative. Mike Pinder, of the Moody Blues, loved it, while Rick Wakeman hated it so much he worked with an engineer, David Biro, to come up with an alternative. They patented a project called the Birotron, which used 8 track tape loops instead of simple tape strips.¹¹ This tape cartridge system was much less prone to failure. I've heard rumours that Wakeman gleefully set his last Mellotron on fire in his garden, and given his disdain for the instrument and his comical and flamboyant personality, I wouldn't find it at all surprising if that were true. He, in a generous spirit, reflected, 'In the past 30 years I've owned and sold, loved and loathed, dozens of Mellotrons.'¹²

My friend John Hawken, who was the keyboard player for The Nashville Teens, Renaissance, and The Strawbs have, over the years, regaled me with tales of the Mellotrons he has used and his experiences of using the Mellotron. He told me over coffee that his introduction to the Mellotron was by way of David Cousins, the singer and key songwriter of The Strawbs. Cousins basically told him 'Well, John, this is our Mellotron, we use it a lot. Get acquainted.' John played it and was instantly appalled by the feel of the keyboard. It felt 'dead' – nothing like a piano, and without the springy playfulness of a quality organ keybed. Just... flat and lifeless. Also, there's a very slight delay between pressing the key and hearing a sound come out of the speaker, as it takes a very short moment of time for the tape playhead to engage and for the capstan to pull the tape through. As a professional keyboard player, well known for his piano and organ playing, his attention to such playability issues was sharp. So, his immediate impression of the Mellotron was one of instant disdain, which changed over time.

Then he started playing it for the songs the Strawbs had written with it, and the profound power of the Mellotron became manifest. John said, 'There was nothing like it. It was amazing. The roadies hated it because it was heavy and delicate. But the sound was incredible. It may only have had strings, choir, and flutes, but for the time, there was nothing like it. I came to love the damn thing.' I asked him, 'If you dig it so much, why don't you have one now?' He rubbed his fingers together and said, 'Money. To get one is expensive, and their upkeep is ridiculous.'¹³

Yes, Mellotrons were delicate, expensive, and fragile, yet bombastic, powerful, imperious, and mesmerizing. Just like Dracula, *it was vampiric*. If the voice of Frankenstein's Creature is the Synthesizer, the Mellotron is the voice of Dracula. And Dracula's *modus operandi*

10 Edgar Froese, *Epsilon In Malaysian Pale*, LP, Virgin Records, 1974.

11 US Patent 4,182,214: *Electronic Musical Instrument*, Richard C. Wakeman and Birotronics, Ltd. Filed: 27 January 1975, Issued: 8 January 1980.

12 Yes Source. Ibid.

13 From conversations I had with John in September 2023 and May 2024.

was the opposite of Frankenstein's. Frankenstein's Creature was erudite, if misunderstood, a misbegotten yet heroic product of a modernist scientific industrialism. The appeal of the Frankenstein/Synthesizer problem is intellectual and philosophical. Is it human/music? If it is, then we need to redefine what it means to be human and make music. If it isn't, then we have to ask some hard questions about what it means to be human and make music. The appeal of this electronic synthetic voice is undeniable, and the past century has seen its growth into every nook and cranny of musical production. From classical musicians having worked with and written for synthesizers to thousands of pop musicians cranking out hits in every style of music. Synthesizers are essential for contemporary popular music production. The synthesizer is paradigmatic of modernity and its technical wonders.

Not the Mellotron. The Mellotron is vampiric, and like Dracula, its appeal is to the emotions. It is a seductive instrument, beguiling the listener by being sweet and always slightly out of tune. Its sound quality is mediocre—it's a tape machine, and the tapes are narrow and not that fast, they stretch and wear out. Yet it does sound very much like other instruments, or whatever one chooses to put inside it. This '*very much like*' aspect is important, as it was the essence of the British Musicians Union complained about the instrument, that it was taking away work from string players, for example. Dave Cousins of the Strawbs responded to this,

Their argument was that you can't use a Mellotron on a track because you're replacing string players. Well, our argument was that's not correct, because they don't sound like strings, they don't sound like brass, the machine doesn't sound like anything it's meant to sound like. It is its own individual sound.¹⁴

Cousins is correct in that it does have its own unique sound, and one could, if in a cranky enough mood, argue that if a musician's union string player who is so poor at their instrument can't sound better than a warbly out-of-tune Mellotron, perhaps they should concentrate on another line of work. The disagreement revolves around this '*sounds like*' problem. Also, one need not put recordings of musical instruments in it.

Keith Richards, the guitarist of the Rolling Stones, recalled how, when he arrived at Tom Waits's studio, he thought: 'Hello! He had a Mellotron... which was loaded entirely with train noises.'¹⁵ This is especially interesting and historically ironic, given the so-called premiere work of *Musique Concrète* by Pierre Schaeffer in 1948 was *Étude aux Chemins de Fer*. (*Study at the Railways*). Tom Waits is taking train sounds and instrumentalizing them with a Mellotron *materialized the promise of Musique Concrète*, as the Mellotron was the device that came of the material production of the aesthetic inherent to *Musique Concrète*, premiered by Schaeffer's work, made of recordings of train sounds.

¹⁴ Awde. p. 233.

¹⁵ Tim Adams, 'All these bulletproof songs, one after another': remembering Tom Waits's extraordinary mid-career trilogy', *The Guardian*. 20 August 2023

The popularity of the Mellotron peaked in the mid-1970s. Soon, polyphonic string synthesizers were developed by ARP, Solina, and Freeman, which were weak synthetic approximations of a 'String' sound. They sounded vaguely like strings, especially when buried in reverb and echo—similar to how a Mellotron sounds like a diminished replica of an orchestra. Or a flute. *Or whatever is on the tape.* These String Synthesizers weighed less than half that of a Mellotron, were vastly more dependable, *and they were able to stay in tune when you played a chord.* This cut the value of using a Mellotron dramatically. In the late 1970s/early 1980s, the children of the Mellotron, devices like the Fairlight CMI and the Synclavier were on the horizon, using digital recording and sampling technologies. The Fairlight and Synclavier were eye-wateringly expensive, making a Mellotron seem cheap in comparison.¹⁶ The base model of the Fairlight IIx, the most popular model by Fairlight, cost \$27,500 in 1980 dollars, which is over \$106,000 in 2025 dollars.

Mellotron went out of business in 1986, the same year Korg came out with the DSS-1 sampling synthesizer, which I used for some time. Fairlight ceased production of its samplers a few years later, in 1989. Fairlight failed for much the same reason Mellotron did: the march of electronic technology's penchant for increased speed and miniaturization, and the falling rate of profit in a capitalist commodity market. By 1989, inexpensive samplers, such as the Korg DSS-1, which I procured in 1987, were widely available and came with huge libraries of sounds and instruments—including samples of sounds from the Fairlight.

Frankenstein vs Dracula

One might think this was the end of the road for the Mellotron. However, we have to remember—the Mellotron is a vampire. It can't die—it's *undead*. It lives on through the Sampler—it's not so much that the Fairlight, Korg, Yamaha, E-Mu, Sequential Circuits, Akai, Ensoniq, and Roland samplers killed the Mellotron, as much as they were possessed of the vampiric nature of the Mellotron by the innate nature of their recording and playback technology. So, even as they collectively buried Streetly Electronics and their Mellotron, then were operating under the same vampiric aesthetic of the Mellotron. In this way, even as the Mellotron died in the battles of the marketplace; it won the war of ideas.

In 1985, John Oswald presented his paper, *Plunderphonics, Or Audio Piracy as a Compositional Prerogative*, to the Wired Society Electro-Acoustic Conference in Toronto, Ontario. This seminal work was made in the ascendancy of the digital sampler (at the time dominated by Fairlight) and noted:

A sampler, in essence a recording, transforming instrument, is simultaneously a documenting device and a creative device, in effect reducing a distinction manifested by copyright.¹⁷

16 In 1965, a Mellotron cost \$4000, or \$38,692.32 in 2023 dollars.

17 John Oswald, *Plunderphonics, or Audio Piracy as a Compositional Prerogative*, 1985, <http://www.bitwisemusic.com/wp-content/uploads/2015/08/Plunderphonics-or-Audio-Piracy-as-a-Compositional-Prerogative.pdf>

This ability to use recordings as an instrument was pioneered by Chamberlin and the Mellotron some decades earlier. The digital sampler made it all faster, higher quality, and more explicit. The Mellotron is generally and obviously considered the spiritual parent of the Sampler as their fundamental function is nearly identical: *recordings are played at the touch of a key*. Whatever the recordings may be *are not innate to the sampler itself*. The instrumentalization of recorded sound that is native to the Mellotron, and all the histrionics surrounding it, naturally played into the reception of the Sampler. However, due to scaling issues (duration of sample, loopability, processing, and recording quality) the sampler soon ran afoul of the sonic Turing Test that informs copyright law. A telling work in this regard is *Pump Up The Volume* by MIAIRIRIS¹⁸ (essentially the group Colourbox¹⁹), which was largely composed of samples of other songs.

Instead of a chorus, there's just the title phrase, sampled from Rakim, over a beat that cribs from James Brown, the Bar-Kays, the Last Poets and the Montana Sextet, not to mention Public Enemy, Wolfman Jack, and a fire engine.²⁰

To make that song today would be financially prohibitive to clear all those samples. The perfidy of contemporary copyright is something I have discussed elsewhere,²¹ and is well outside this volume's purview. It is sufficient to say that intellectual property rights and copyright are the central gating issue with the development of Plunderphonics. Some artists, such as myself,²² Girl Talk,²³ Buttress O'Kneel²⁴ and the entire related Vaporwave movement, and many of the hauntological musicians in general, have simply chosen to operate and create works outside and around these legal limitations.

A key issue is charging people to procure our music files. By having it available for free, this removes economics from the discussion. Secondly, we hack the music up a great deal. For me, I (usually) process the samples to such an extent that they are nearly unrecognizable from the original source, as my interests are textural, not necessarily mimetic. Others are much more blatant in their sampling practices. Oswald's original use of the term, Plunderphonics, was that it referred to a musical work that was created from samples of a single artist and no other material. In this way, リサフランク420—現代のコンピュー by Macintosh Plus²⁵ is much more strictly Plunderphonic, as it is composed of a chopped up and slowed down rendition

18 MIAIRIRIS, *Pump Up The Volume*, 4AD, 1987.

19 MIAIRIRIS was a combination of two bands on the 4AD label – Colourbox and A.R. Kane, at the behest of the label. Unfortunately the two groups came to dislike each other, and so the release of the single had *Pump Up The Volume* on one side, and was the work of Colourbox, and the other side was *Anitina* by A.R. Kane.

20 Michelangelo Matos, 'How M/A/R/R/S' 'Pump Up the Volume' Became Dance Music's First Pop Hit', *Rolling Stone*, 14 July 2016.

21 Henry Warwick, *The Radical Tactics of the Offline Library*, Institute of Network Culture, Amsterdam, 2014.

22 My music is available here: <https://henrywarwick.bandcamp.com/music>

23 Girl Talk's music is available here: <https://illegalart.net/girltalk/shop/>

24 Buttress O'Kneel's music is available here: <https://buttressokneel.bandcamp.com/>

25 Macintosh Plus, 'リサフランク420—現代のコンピュー' (Lisa Frank 420 / Modern Computing)', from *Floral Shoppe*, Beer on the Rug label, 2011

of *It's Your Move* by Diana Ross.²⁶ Girl Talk, Butters O'Kneel, and most others working in this vein tend to sample from a variety of sources, as was found in *Pump Up The Volume* decades earlier, which set a kind of template for this way of working.

Pump Up The Volume could have been produced with a couple of Mellotrons with custom tape sets—a few keys with the rhythm, another few keys with the bassline, and the rest of the keys set to tapes of all the samples. However, the song was not made with Mellotrons—it was made using samplers. John Fryer, the producer, noted the equipment used 'In this case, I recall him listening to it for a couple of days, coming out of an Akai S1000 sampler or an Emulator...'²⁷

The sampler, as noted by John Oswald, is more than just a recorder—it is an instrument. And the instrumentalization of pure sound is the fulfillment of the aspirations laid out at the beginning of the twentieth century in the Futurist Manifesto,

1—We must enlarge and enrich more and more the domain of musical sounds. Our sensibility requires it. In fact, it can be noticed that all contemporary composers of genius tend to stress the most complex dissonances. Moving away from pure sound, they nearly reach noise-sound. This need and this tendency can be totally realized only through the joining and substituting of noises to and for musical sounds.²⁸

Clearly, the sampler is the technological fulfillment of these aspirations, and as we will see in the second part of this work, the technological fulfillment of atavistic and ancient interests is essential to our present condition.

This fulfilment was first accomplished by way of the conceptual shift inherent to the Mellotron, where a 'sound' is available at the touch of a key. One might argue that this is also inherent to Musique Concrete, where the tape recorder itself is used as an instrument. This is fair enough; however, the Mellotron, as a development of the Chamberlin from the early 1950s, is not a follow-on from Musique Concrete, as much as it is a co-evolving application of recording technology in an overtly instrumentalized form. Reel-to-Reel tape decks found greatest popularity as part of a fairly high-end stereo kit or in recording studios. The amount of labour and effort of accessing the recording, cutting it out from its source tape, then placing and editing it into a work of musique concrete and then copying the original edit to another tape is, frankly, a right pain and a lot of work. Conversely, one could take a song that is less than 864 seconds long and cut it into 36 sections and 'play' the song section by section on the Mellotron or Sampling Keyboard. It would be monstrously difficult, yet possible. This was actually done by Roxy Music for their song 'A Really Good Time'²⁹ where they hired a string quartet for the album, and then chopped it up

26 Diana Ross, 'It's Your Move', from *Swept Away*. written by Steve Kipner and Terry Shaddick. RCA, 1984

27 Richard Buskin, 'Classic Tracks: Pump Up The Volume', *Sound On Sound Magazine*, August 2012. <https://www.soundonsound.com/techniques/classic-tracks-marrs-pump-volume>

28 Luigi Russolo; Robert Filliou (trans), *The Art of Noise, futurist manifesto, 1913*, UBU classics, 2004.

29 Roxy Music, 'A Really Good Time', written by Bryan Ferry, From *Country Life*, Island Records, 1974.

to go on a Mellotron, so they could rehearse and play the song without having to bring the quartet with them. These tapes are now available in a digital form with the G-Force *M-Tron* Mellotron emulator VST instrument³⁰ as a preset, 'Roxy Tapes'.

This memetic imitation of sound is beguiling and alluring; there is sensuousness to this—like any vampire's promise. Frankenstein's Creature, while he was an artificial being, was still a being made of flesh and blood, and the Frankenstein of synthesis fell under the spell of Dracula's Mellotronics and became vampiric. This was made possible with the miniaturisation of the integrated circuit—the same computer chips that made miniature computers possible, also permitted the sampler, and at a vastly reduced and ever reducing cost. A classic part of the vampire mythos is that a vampire cannot enter a home unless invited, and these vampiric samplers were quickly invited into studios and homes all over the world.

The Frankensteinian, indeed, *promethean*, world of the modernist synthesizer and its inherent universalism could not permit the paritant Mellotronic sampler its own special alternative sonic development space. Out of that opposition, Synthesis itself fell under the spell of the Vampiric Mellotronic Sampler A E S T H E T I C with Virtual Analogue (VA) and Physical Modelling (PM) synthesis in the 1990s. Samplers had been sampling synthesizers, or like the Fairlight and DSS-1 had the ability to draw single waveforms, or mold a waveform using additive synthesis techniques. Samplers were immediately sampling synthesizers - synths are just another sound, like any other instrument. A synthetic response was soon fielded.

A Physical Modelling synthesizer makes a virtual model of a sound making object or instrument, and then allows a set of parameters that go into making that model subject to modification. For example, let's say you're modelling a flute—a metal tube with holes at specific places to permit pitches at specific frequencies. A high-quality PM synth would have parameters to be modified, such as a flute made of wood, or is 100 meters long and 1cm wide, or made of platinum,³¹ or the holes placed wherever you want. It would certainly make for a very different sounding 'flute'. The important angle on this is that it's mimetic—it starts with a non-synthetic sound which is then re-presented on the plane of representation. It's not that the sound coming out of the speaker IS the sound of a flute, or ever was. *We let it be the sound of a flute.* To appropriate a quote from Laurie Anderson, 'Let X=X'.³²

You hear someone play a flute, at, say, the pitch of C4. You hear someone play the note C4 on their PM synth set to the Flute preset, and out of the speaker comes something that sounds very much like a flute playing the note C4. It's not the same as listening to an actual flautist play C4 on their flute. It's an imitation. However, over the decades of enculturation into the simulative re-presentative plane which had been exercised in the

30 G-Force, *M-Tron Pro IV*, <https://www.gforcesoftware.com/products/m-tron-pro-iv/>

31 The flute solo work, *Density 21.5*, by Edgard Varese (1936), was written specifically for a platinum flute, hence the density of 21.5, which is the density of platinum.

32 Laurie Anderson, 1982, *Ibid*.

1970s with the development of the String Synthesizer, people have naturally glossed over the difference. This is not a new phenomenon. Remember, the British Musicians Union would picket Moody Blues shows in the 1960s because they used a Mellotron. They found the paritance in the warbly third-rate copy of strings in economic and practical parity to that of live players.

'It is a question of accepting the signs of the real as the real.'

Baudrillard made a career out of his simulation and hyperreality theory that came out of the re-presentation illusion. This illusion is actively defeated when one simply refuses to accept that the recording is the thing it has recorded. Paritance ends with saying 'No.' At the same time, it's clear that refusing to admit the instrumentalisation of recording as a musical practice is inherently reactionary, if not ludicrously luddite. Our position here is a good bit more nuanced. However, the re-presentation of music as instrument results in the re-presentation of instruments as information subject to manipulation and arrangement.

With VA, the scope of re-presentation is narrowed. In a VA synth, the computer inside the synthesizer simulates the behavior of electronic circuitry in order to digitally replicate the tone of a given synthesizer. So, just as a Mellotron doesn't make the sound of (instrument) and plays a simulation of it on an analogue substrate (tape), the VA synth operates as a simulation of a synth and its waveforms and filters. The important point of this is the abstractive simulation principle inherent to the Mellotron is made manifest in material synthesizer design with VA synthesis, and a shift from material analogical processes to ones based in information and data representations. It is this informational data representation that is essential to the vampirism, in the musical mockingbird of the Mellotronic and the Paritant. 'Parts Is Parts.' Data is data. Ones and Zeros are Ones and Zeros. The essential abstraction of the modernist Frankensteinian industrial machine created its opposite, the virtualised postmodernist vampiric, in its own re-presentative field: from the synth to the Mellotron and its digital offspring, the sampler.

As the Mellotron materially demonstrated, it isn't anything. *It's just a box of tapes.* (sonic information) The sampler isn't anything. It's just a machine that needs data, samples. (sonic information) The VA synth continues this Mellotronic practice: it's not what it sounds like, it is simply a computer that is programmed to behave and sound and even look like a synthesizer. Its Mellotronic and paritant nature is abstracted into its operative programming. With the VA synth, the vampirism of the Mellotronic becomes flesh, Dracula becomes Frankenstein by Frankenstein operating on the terms of Dracula's vampirism. What is most important is this: Dracula wins. The vampirism spreads through the living, and the paritant commodity of musical creativity is thus opened to every vector of imitation.

Software synthesizers (commonly using the VST format) of the past few decades underline this even more—there are very few synthesizers of the past that haven't been

virtualised into software. Even the Fairlight CMI is now available as an iPhone app³³ or as software in Arturia's V collection³⁴, and the Mellotron is also virtual software instrument for electronic musicians as G-Force M-Tron³⁵, Arturia Mellotron V³⁶, or even as an iPhone app³⁷. And now, with AI music, the commodity of the song itself is a paritant vehicle of wealth extraction.

Even the gigantic synthesizers of the past, like the Moog Series III (viz Wendy Carlos or Keith Emerson) are now virtualised. One can get a version from Arturia in its V collection of soft synths, or one can download Cardinal or VCV Rack, and build an arbitrarily large modular synthesizer, only limited by one's screen real estate, memory, and processing power. This vampirism is now also in the analogue realm itself, through the shenanigans of the Behringer corporation.³⁸ Behringer (named after its founder, Uli Behringer) has been building copies of synthesizers from the past. Some manufacturers don't appreciate Behringer cloning their old synths. Behringer's response is that creating clones of older hardware is legal where the patents have expired.³⁹ The difference between what is legal and what is ethical is an ongoing discussion outside of this one. Here it is sufficient to show that the paritant vampirism of the Mellotronic is now manifest in the material production of the Frankensteinian synthetics of these new 'analogue' retrosynths. It is also important to note that this retro-revivification of old synths is also something the manufacturers themselves are engaged with. For example, in 2015, Moog revived the Moog System 55, System 35, and Model 15 synthesizers, which were originally created and manufactured in 1973. And in 2022, Moog revived the Model D (aka 'the Minimoog').⁴⁰

The self-vampirism of the Mellotronic and virtual was inherent to Frankenstein's creature, even bound inside the text of Frankenstein by Shelley, as noted by Sarah Fuller,

The Creature clearly unsettles another distinction as well, that between good and evil. While Victor, with his numerous designations and descriptions of the Creature as "vampire, fiend . . . spectre" clearly places it on the side of evil, the novel urges its readers to be highly suspect of these designations and Victor's own ability to assign them.⁴¹

33 Peter Vogel Instruments, *CMI App for iPad and iPhone*, <https://petervogelinstruments.com.au/ios/>

34 Arturia Instruments V Collection X, <https://www.arturia.com/products/software-instruments/v-collection/overview>

35 G-Force, *Ibid.*

36 Arturia Instruments, *Ibid.*

37 Insideout Ltd. *Manetron iPhone App*, Apple Store, <https://apps.apple.com/us/app/the-manetron/id1641254309>

38 For purposes of full disclosure, I don't really have any problem with Behringer copying synths of the past. If these synths were so valuable and important, then the manufacturers should have kept building them. I do have problems with Behringer's litigiousness and labour practices, although, that's another discussion for another time.

39 Scott Wilson, 'Curtis chip company speaks out against vintage synth cloning', *Fact Magazine*, 22 March 2017.

40 Note: with Moog's recent purchase and closing of its factory these have all been discontinued.

41 Sarah Canfield Fuller, 'Reading the Cyborg in Mary Shelley's 'Frankenstein'', *Journal of the Fantastic in the Arts*, Summer 2003, Vol. 14, No. 2 (54) (Summer 2003), pp. 217-227

Indeed, with the capture of Frankenstein's Creature by Dracula, discussions of 'good or evil' are superfluous and profoundly decontextualised.

People have been 'doing remakes' of cultural objects as long as cultural objects have existed. The mechanical reproduction of cultural objects has also been going on for centuries (printing presses, etchings, lithographics, engravings, etc.) and have had a great effect on both the exchange of these objects as well as the development of cultural, economic, social, aesthetic, and media theories, from Marx to Weber to Benjamin to Baudrillard to Haraway. That is not the novelty. What is valuable and new here is the materialisation and ascension of cultural vampirism and its instrumentalization by the aesthetic driven by the Mellotron and its digital child, the Sampler. With the example of these musical instruments we can see how *paritance is prior to hyperreality or simulation*.

*I'm after rebellion—I'll settle for lies.*⁴²

'The Precession of the Simulacra' and the Condition(ing) of Parity: Acceptance

This mimetic aspect of synthesis was 'built in' to the synthesizer as commodity project itself, as synthesizer programmers did their best to mimic non-electronic sound sources as well as create new sounds. One of the earliest and most successful of these was, of course, Wendy Carlos, who spent countless hours using her analogue Moog synthesizers in the 1970s and her Crumar GDS Synergy additive synthesizer and software in the 1980s, built imitations/replicas/interpretations of classical music instruments, anticipating the goals of VA, Physical Modelling, and later synthesis modes. Especially noteworthy are such moments as the horns at the beginning of 'The William Tell Overture'⁴³ and her suite of works about moons in the solar system, *Digital Moonscapes*, 'Written for orchestra (or orchestra replica), it is inspired by several astronomical subjects.'⁴⁴ In between both, interestingly titled, was her work on the soundtrack for the movie, *TRON*.⁴⁵

TRON was a sci-fi fantasy film about a computer programmer and video game developer who is somehow physically transported into the software of a mainframe computer, where he interacts with various malevolent programs in his attempt to escape this digital virtual world. He eventually defeats these systems and reappears in the real world in front of his computer console. His victory has completely opened all access to the computer, and the evidence that our hero's work had been plagiarized/stolen is printed out. Our hero, Flynn, is then promoted to CEO of the company where he works, ENCOM. When he is inside the computer, he is known as *TRON*.

So, a film about plagiarism is based on a sense of mimetics—Flynn becomes *TRON*, an

42 Blue Oyster Cult, 'Flaming Telepaths', from *Secret Treaties*, Columbia Records, 1974.

43 Wendy Carlos, 'William Tell Overture', *A Clockwork Orange (soundtrack)*, Columbia Records, 1972.

44 Wendy Carlos, *Digital Moonscapes*, East Side Digital, 1984.

45 Wendy Carlos, *Tron*, Original Motion Picture Soundtrack, CBS Records, 1982.

electronic version of humanity. A human in paritance, only not as a tape-based replica or sampling technology, as much as a digital re-creation. This mimetic and paritant approach was evident from early on in synthesizer construction.

In 1973, Roland built what was, for the time, an affordable, simple synthesizer, the SH1000. It only had one oscillator and one LFO, one filter, and one envelope generator. A very basic synth that proved to be very popular—many popular musicians used it.⁴⁶ It had 'presets'—flip a switch and the synth would make that sound.

It featured Tuba, Trumpet, Saxophone, Flute, Clarinet, Oboe, Violin, Bass Guitar, Harpsichord, and Piano presets. These presets sounded about as 'exactly like' what they're named for as some might see the face of Jesus in a piece of burnt toast. It's audio pareidolia. However, that is enough, as this pareidolia is also conceptual, and that is where this comes to a significant place.

Baudrillard noted in this age of hyperreality and simulation, 'It is no longer a question of imitation, nor duplication, nor even parody. It is a question of substituting the signs of the real for the real.'⁴⁷ When we listen to a recording of, say, a symphony by Mozart, we can say 'we are listening to classical music'. Can we say 'we are listening to an orchestra'? Can we say 'we are listening to violins, cellos, bass viols, trumpets, French horns, trombones, timpani, flutes, clarinets, oboes, and many other instruments'?

Because we are not. We are listening to a recording of these things. We are not listening to those things.

Can we say 'we are listening to classical music'? Can we say 'we are listening to folk music'? Can we say 'we are listening to jazz'? Or, are we merely listening to a recording of people performing these musical styles? We are not listening to them. We are listening to recordings of them. The parity of 'I'm listening to this Musician' is accepted—pure paritance. Given the importance of improvisation to jazz, this especially problematizes jazz recordings—is a recording of Jazz actual live jazz, in all of its live synchronic participation? Obviously, not. The Mellotron was a material manifestation, an instantiation, of cultural Paritance prior to Baudrillardian simulation theory as it was devised (in the form of its predecessor, The Chamberlin) when Baudrillard was still an undergraduate at the university. By the time Baudrillard wrote *Simulacra and Simulation*, the Mellotron corporation was twenty years old and in deep trouble, and Fairlight was already selling the Fairlight CMI sampler. A few years later, John Oswald would lecture on Plunderphonics. The material world and practice of music production were decades ahead of Baudrillard's theoretical workings.

Baudrillard's ideas have been deeply criticized by many people. For example, Susan Son-tag thought Baudrillard was 'condescending', and that he was 'ignorant and cynical' as

46 Wikipedia, *Notable SH1000 users*, https://en.wikipedia.org/wiki/Roland_SH-1000

47 Jean Baudrillard, *Simulations*, Semiotext(e), 1984. p. 4.

well as 'a political idiot'.⁴⁸ Regardless of these critiques surrounding him and his work, however ill or well founded, I should like to centre an observation of his I noted earlier: 'It is no longer a question of imitation, nor duplication, nor even parody. It is a question of substituting the signs of the real for the real.'⁴⁹

The histrionics around, and the drama within, his overstatement and presentation of his theories of simulation and hyperreality, such as the kerfuffle was inspired when he challenged whether or not the Iraq War existed and similarly provocative ideas, those things are not of interest here. I would suggest that the situation he points to is actually much deeper and much more profound than his ideas regarding electronic media, because Paritance precedes simulation.

I would suggest that the core distinctions pointed out in the statement '*substituting the signs of the real for the real*' are much deeper than what Baudrillard discusses, and actually go to the core of human cognition in Paritance. This paritance, due to it being part and parcel of human cognition, is completely hypostatic to the human condition. Here, I simply want to narrow this to (1) Paritance, where the normalization and naturalization of accepting parity is materialized and instrumentalized in the development of electronic music instrumentation, and (2) tying this to a narrative fiction-based struggle between the modernist heroism of Frankenstein's Creature and a postmodernesque/Capitalist Realist form of Vampirism.

Baudrillard's overstatements and extreme claims have not served his insights well, and there is a great deal of critique and rejection of his ideas. At the same time, his insights in Simulacra and Simulation have informed and are seemingly prophetic with recent advancements in Virtual Reality (VR) and Augmented Reality (AR) technologies. I would suggest that such materialities/mate realities are now in a precession against their simulation. However, in the particular plane of a given simulation, the Precession of the Simulacra maintains, while they are simultaneously circumscribed and horizoned by these Mate realities. A simple example is something like AR/VR. The mate reality of Information and Communication Technology (ICT) completely encompasses AR/VR. There is no AR/VR without it. Thus, whatever materials that compose ICT give AR/VR the ability to exist at all. For example, we can look at an iPhone.⁵⁰

Silicon, as a metal, is rather brittle and hard, and for it to be used in a computer requires an incredible amount of toxic chemicals and processing. Engineering.com has a fairly detailed description of how all of this is made, and I would recommend it for more details.⁵¹

48 Robert Antonio, *The Passing of Jean Baudrillard*, University of Texas at Arlington, 2007.

49 Baudrillard, *Ibid*.

50 <https://www.engineering.com/story/what-raw-materials-are-used-to-make-hardware-in-computing-devices>

51 Andrew Wheeler, 'What Raw Materials Are Used to Make Hardware in Computing Devices?', *Engineering.com*, 29 September 2018. <https://www.engineering.com/story/what-raw-materials-are-used-to-make-hardware-in-computing-devices>. '...high purity silicon is bathed in hydrogen peroxide and sulfuric acid first to clean the wafers. Next, it is bathed in deionized water and sulfuric acid to remove any particulate matter. After the second

Suffice to say, it is complicated; with enormous amounts of energy and resources required for an intensive process of hardware development and construction. It continues from there. Without materials like tantalum, aluminum, hafnium, indium, copper, tungsten, and many, many others, some of which are rarer than gold, *ICT is simply not possible*. Everything about AR/VR—any claims to any 'reality', any claims to value or importance, is completely encompassed and horizoned and limited by the material requirements of the ICT that permits AR/VR to exist. Furthermore, the very experience of AR/VR is paritant. It is 'like' being in a space, and your sensorium has autonomous reactions to such an experience. The result is how often AR/VR makes people vomit from motion sickness.

These claims, including some made by Baudrillard, are conditioned by a pretense, and, I would suggest, are products of themselves—creators and creatures. As Virilio noted, 'The invention of the airplane was the invention of the plane crash'⁵²—the belief in an all-encompassing hyperreality can be seen as a cause and a creature of itself. This happens because it is a historical object that has material precedence and history and is circumscribed by the development of the technology that permits it. The invention of electronic media, starting with the telegraph, the 'material/matereal', created the effective 'space' for the 'hyperreal'.

However, the 'space' created by electrical or electronic media is purely *fictive* and based upon a much more basic delusion—the one which Baudrillard noted—the *acceptance* 'of substituting the signs of the real for the real'. This shows that his simulation theory is a function of paritance and human cognition itself. Fiction tells us stories. Myths tell us stories. All of our communication systems are reductive and exist as something other than what they purport to represent. What they actually represent, the 'thing in itself', is far outside this discussion. What gives Baudrillard's insight power is when we exchange one for the other within or between media.

We listen to someone playing a violin. We record their performance. As discussed earlier, we are not listening to a violin. We are listening to a recording—sound produced by a vibrating cone inside an enclosure. It 'sounds like' a violin. The invention of the recording was the invention of the original performance, and points at the essential conditions of the problem as enframed by electronic media. Is It Live or is it Memorex? In the early 1980s, the Memorex brand of video and audio tape ran a series of commercials that played off this situation.

bath, the oxide layer is removed with deionized water and hydrofluoric acid. The wafer is finally bathed one last time in hydrochloric acid, hydrogen peroxide and deionized water.'

'To speed up the performance of their CPUs, IBM and Intel began using hafnium, a rare metal... GPUs are silicon layered with tantalum and palladium transistors and capacitors for better storage on a smaller chip, which is perhaps the most interesting of the materials used for a GPU and RAM cards are made from a mind-boggling array of chemicals and copper, boron, cobalt, tungsten, for starters.'

'(to make NAND chips)...Hundreds of trillions of transistors in the circuit design are etched onto the circuit board using potassium hydroxide (also known as caustic potash), layered with a photoresist chemical and cured part-by-part. The caustic potash reacts with the silicon, forming an oxide layer of silicon dioxide, which forms a layer of non-conductive material which sets apart other layers of the circuit. The chip then receives more conductive layers of copper and aluminum.'

52 Paul Virilio and Sylvère Lotringer, *Pure War: Twenty-Five Years Later*. Semiotext(e), 2008, p. 46.

"Look at me, do you like what you see? Good, because it's not me" (zoom out to see her sitting next to a video screen showing what we just saw prior to the zoom out) "It's a recording of me—on Memorex video tape..."⁵³

The commercial pulls this trick a few times as it continues, and it was effective in selling Memorex video and audio tape. The matereality of this conceit is that it was all on videotape. Within the plane or field of videotape, on the plane or field of simulation, everything is exchangeable and everything is a simulation—Baudrillard's fourth level of simulation. However, it only acquires meaning as the field of videotape simulation exists within the confines of a particular materiality or 'matereality'—video—a field which is not the meat space that people live in, as it is the dissonance between the imaginary space of the video and the real space of the video observer that creates the uncanniness of the commercial's presentation for the observer. By merely observing or participating in a media field, uncanniness comes for free.

There was another Memorex commercial where Nelson Riddle, who produced much of Ella Fitzgerald's music, was asked to listen to some sound. The sound: Ella Fitzgerald's singing.

When asked, 'Can you tell if that's Ella live or a recording on Memorex cassette tape with MRX2 oxide?' he listened for a few seconds, smiled, and replied, 'No, I can't'. Immediately, there's a cut to Ella and the band saying in unison, 'It's Memorex!' and Riddle replies, 'Sounds live to me!'⁵⁴

Again, this illustrates that the uncanny valley is *inherent* to the mediated field of its own existence. Again, Virilio: *the invention of the airplane invented the plane crash*.

The mimetic created the uncanny. However, there is also a further point, a more basic and unacknowledged reification involved, wherein the imitation is banally accepted as the real. Again, as Baudrillard noted: 'It is a question of substituting the signs of the real for the real.'⁵⁵ This is not the place to argue the metaphysics of 'what exactly is the real'. That said, I would like to modify Baudrillard's statement to something more useful for this context: 'It is a question of accepting the signs of the real as the real.' And we immediately see that paritance precedes simulation.

The reason is that with 'substitution', there's a certain lack of agency — one thing is withdrawn and something else takes its place. This can happen involuntarily, or even accidentally. It's a beautiful day. The lovely weather is then substituted with a tornado an hour later. (It could happen.) No one's perceptual apparatus had anything to do with this. One type of weather substituted for another over time. It wasn't like one day the Real disappeared and was replaced with a cardboard cut-out of itself and we said, 'yeah—that's the Real.' No, the cardboard cut-out of the real arrived, and we *accepted* it as the real, as consciousness itself is a paritant structure.

53 Memorex, 'Ella Fitzgerald for Memorex 1974, 1979', *GarfieldFco channel*, Youtube video of TV Commercial, 31 December 11, <https://www.youtube.com/watch?v=1-lvTF0xUxM>.

54 Memorex, *Ibid*. Time: 0:15.

55 Baudrillard, *Ibid*.

We can choose to accept many different things for things they are not, insofar as we decide that they are worthy of such acceptance. We can choose to accept that a giant three-ton 4x4 pickup truck is a car. We can choose to accept that people can be resurrected from the dead. We can choose to accept that capitalism is the perfect and ultimate stage of civilization. We can choose to accept that the proletariat exists, and has always existed as a creation of the bourgeoisie—again: invention of the airplane/plane crash. We can choose to accept that things that have no material existence outside being a set collection of electrical or magnetic charges in an SSD or hard drive hidden away in some server farm on some other continent far away are somehow 'property' like one's underwear or socks. We can choose to accept that certain experiences are 'real' and other experiences aren't. Of course, due to circumstances and historical particularities, one's 'choices' may be limited to varying degrees. However, those limitations are not germane to the general structure of what obtains here. What we see in the Mellotron and Synthesizer are manifestations of ideologies and sensibilities impelled and made complex by the cognitive dissonances that are a natural product of an awareness of Paritance.

Baudrillard invented an entire and elaborate position around this with his ideas of hyper-reality. While he overstated his position, at the same time *he didn't carry it far enough*. Also, there's this kind of fatalism involved with his hyperreality, as if the hyperreal condition is somehow uniquely unavoidable and contemporary. This inevitability, this implacable loss of the real is not (really) true, and it is through this musical lens of the Mellotron and synthesizer that we will see how our media isn't just a fictive struggle between Frankenstein and Dracula, between the cold, mathematical modernist precision of the synthesizer and the cloying warbling and grandiose emotional effects, and affects of the Mellotron; it is much more ancient, atavistic, and occult.

Dracula Won

The synthesizer is the voice of Frankenstein's Creature. Frankenstein's creature is a modernist industrial replica/interpretation of a human. The Mellotron is vampiric as it is no voice of anything—it is dependent on the tapes and samples it receives. Synthesizers are the voice of modernist industrialism, as unique heroic voices of our age - their appeal is *intellectual*. Mellotrons are the voice of postmodernist postindustrialism. Their voices are unique in the quality of their musical diction or accent impelled by the shortcomings of their playback system, although they are not unique in their native timbre, as these are recordings and samples of musical instruments and sound effects. Their appeal is *affective, alluring, and evocative*. Synthesizers acquired sampling technology, and in the process, the synthesizer inhabited the aesthetic of the sampler, and at the same stroke, in the plane of digital synthesis (being an exchange of symbolic information) the synthesizer was then abstracted into the ideology and material practice of the Mellotronic. Frankenstein's Creature then becomes another vampire—Dracula won. Frankenstein's Creature was made of flesh and blood, and therefore subject to the wiles and pathologies of the vampire, and succumbed to the allure of the 'immortality' of the undead—beings who are and aren't alive, or dead.

However, this is where we start; the problem is older and much more dramatic, as we find in the sources of Necromancy, Demonology, and beneath it all, the underpinnings of the invention of metaphysics itself as a significant philosophical pursuit which permits the possibility of these practices, and presently surrounds and outlines the possibilities of human experience, all because of the cognitive 'trick' and flattening condition of paritance.

PART 3: DIGITAL NECROMANCY

The Necromantic

The subject of Digital Necromancy (DM) has been discussed at some length in a variety of forums for a variety of purposes, both scholarly and popular and creative. Much of it predictably revolves around questions of property rights and relations. It will be the subject of this section to mostly go around the mundane questions of property to more interesting questions and problems of Digital Necromancy, in order to move the discussion to Digital Demonology and Evocation.

Necromancy is an occult practice where one summons the dead for purpose of communicating with them.¹ Stories of necromantic pursuits go far back into antiquity to Medieval and Renaissance sorcerers and necromancers. They also flourished with Victorian spiritualism in the late 19th century. These tales are frequently about deception and grifting on the part of the necromancers and séance conductors. Much of the literature surrounding this with the Victorians owes a big nod to John Dee in the Renaissance. Note: John Dee owned a copy of *The Sworn Book of Honorius of Thebes*.

We are not alone—other mammals also demonstrate profound attachment to dead loved ones—other primates², whales and dolphins³, elephants⁴, even dogs and cats, have exhibited sadness and stress with the passing of a family member or associate of their own kind. Some have also acted with concern towards the release and deposition of the . . . body. This has been widely observed and understood. Thus, it comes as absolutely no surprise that we, modern humans, also experience profound shifts in affect and enact social rituals at the passing of a loved one. And, given our long history of trying to bring loved ones back from the dead through various necromantic practices, it only makes sense we would also make use of digital means to do so.

I am of deeply mixed feelings regarding Digital Necromancy, as it goes to matters of mortality and existential matters that have been around me my entire life—premature and miserable deaths of people I cared about were a part of my upbringing.

...

-
- 1 A. Kapcár, "The origins of necromancy on how we learned to speak to the dead." *Sacra*; 13(2) pp. 30–58, 2015.
 - 2 A. Gonçalves, S. Carvalho, "Death among primates: a critical review of non-human primate interactions towards their dead and dying." *Biological Reviews of the Cambridge Philosophical Society*.10:15, 2019.
 - 3 G. Bearzi, L. Eddy, S. Piwet, M. A.L. Reggente, B. Cozzi, "Cetacean behavior toward the dead and dying." In: Vonk J, Shackelford TK (eds) *Encyclopedia of animal cognition and behavior*. Springer. pp. 1-30, 2017.
 - 4 I. Douglas-Hamilton, S. Bhalla, G. Wittemyer, F. Vollrath, "Behavioural reactions of elephants towards a dying and deceased matriarch." *Applied Animal Behaviour Science*. 100:87–102, 2006.

One committed suicide when he was 14. He lived across the street from me, and we played together often, especially when I was 8 or 9 years old. He was really into outer space shows like *Lost In Space* and *Star Trek*. One day, a few years later, he 'woke up' in a mental health facility and realized he was never going to get better. He spiraled down into a massive depression and checked out as soon as he could. My mother knew what had happened as she was friends with his mother. My mother didn't tell me about his demise until years later.

...

Another friend, we were best buddies in grade 7, died with his dad when he was 20 because his father fell asleep and rolled the truck they were driving. They were driving it overtime to pay for the investigators and expenses surrounding the murder of my friend's younger sister. I knew her, too. She didn't deserve to be raped, carved up, and dumped in a reservoir at the age of 17.

...

Another dear friend died when she was 19 from a freak heart attack, causing her to pass out and drown while rinsing her hair in the tub. As my bandmate's (drummer) girlfriend, we saw each other frequently. She affectionately and playfully filled most of a page in my high school yearbook—she was funny and sweet.

...

Another childhood friend, he and I set fire to the Raritan River when we were 10. He drove his motorcycle into a bridge abutment in an alcoholic rage of self-loathing at the age of 19.

There were others—*those are just a few off the top of my head*. People age, get sick, and die—this is part of the Four Noble Truths. Dying is tragic enough; however, the untimely deaths of young people, so full of promise with their road ahead for them cut short, whether accidentally or intentionally, is something else altogether. Half of my grandparents passed before the age of 56, and another one left at 71 from a lifetime of smoking. Growing up, I honestly didn't have great expectations of a long life. Yet here I am, aged 67, and I have lived longer than so many dear friends and relatives. Would I want to talk with them? Perhaps—why not? Can I? No, of course not. Can they speak to me? Can they appear before me? That's a different sort of question. Again, would I want to talk with them? Perhaps—why not? *Or would I?*

Encountering the Dead with a Call from Beyond

I took a break from writing the sentence above. I was at 'Claudia's Café' in The Junction here in Toronto, and went to the counter to order (yet another) coffee from Abby, the delightful barista. I got my coffee, took a few sips and started typing out some ideas about Digital Necromancy. Fully engrossed, I was reading for this section on Digital Necromancy, and reading

actual Necromantic and Demonological practices in *The Sworn Book of Honorius of Thebes*.⁵ In short, *Very Weird Stuff*. And just as I'm really deep into it and reading *The Sworn Book* on how to raise the dead, I got a phone call. My phone was buried in my computer bag, and I didn't get to the call in time to answer. By the time I pulled it out, the phone stopped ringing. When I saw the name of the caller, *my flabbers were most thoroughly and completely gasted*.

It was a call from my friend Jodi Patterson.

*Jodi died of Pancreatic Cancer in 2021.*⁶

Stunned, I momentarily didn't know what to do. However, it left a number in my 'Recent Calls' list. I promptly called the number—it rang once and was instantly declined. The disembodied, automated robot angel voice of the phone company said,

We're sorry, but the number you have dialled is no longer in service.

I was completely shocked and unnerved by the uncanny coincidence of reviewing medieval Necromantic rituals and receiving a phone call from a dear departed friend. Upset, I thought, 'WHAT in the actual *fuck* was that all about? Just...like... WHAT? So weird...'

I was perplexed. Shocked, I talked with a fellow café denizen at the adjacent table about what happened, as she saw and heard my exclamations of horror and surprise. Thankfully, she was kind and sympathetic, and suggested that perhaps this meant something more than mere coincidence. Also, I found out she's also an artist, a painter, like Jodi. I soon packed up my stuff and made my way home, flooded with memories of my departed friend, and then typed these few paragraphs. How could this have happened? I honestly and truly have no proof of what actually transpired. My educated guess is that someone spoofed her phone number some time ago and used it to get people to pick up and deal with whatever nefarious plan or grift the caller has cooked up. This is not uncommon. I have no proof of such; it simply makes sense. In any case, it was very disconcerting and uncanny as it provoked a flood of difficult emotions. The phone call from Jodi was, obviously, and most assuredly, NOT HER. She's quite dead, and I can speak for everyone who knew her, her husband, her family, and all her friends, *we all truly miss her deeply*.

Walking the Dead: The Sworn Book of Honorius of Thebes

What struck me for a loop was the context of the call—I was deep in reading *The Sworn Book of Honorius of Thebes*. And while I am a strong agnostic in such matters, that sort of reading is intense and fraught—the endless lists of angels, the hundred names of God, the endless lists of demons, the repetitive prayers and impassioned incantations are, even for a modern reader, profound and moving, and the tone is exacting and demanding. In its original medieval

5 Joseph Peterson (trans., ed.) *The Sworn Book of Honorius: Liber Inratus Honorii, by Honorius of Thebes*, Ibis Press. 2016.

6 Jodi Ann Patterson, Obituary, <https://www.wswfh.com/obituaries/Jodi-Ann-Patterson?obId=23411326>.

context, it's even more profound, as this book was actively persecuted for centuries—merely possessing it could result in an extremely painful death at the hands of an Inquisitor. As I read the book, I was drawn into the frame of reference inside the book, and even through translation (from what I can gather, the translation efforts are excellent), I was in an emotionally heightened state. The odds of that happening—that I would get a (spoofed) call from 'Jodi' over the few hours I spent reading *The Sworn Book* — are absurdly small. Yet real.

It is typical for humans to make associations between personal activities and events. The stove is hot, you put your hand near it, and you associate the heat with possible danger, and refrain from touching it. If you don't, you might touch it and burn yourself. This sort of experience scales up into associations between image/event/ perception and object/result in the perceiver, and thus, with our language facility, these symbols, semiotics, media theory, philosophy, etc., act within language and judgment to create ever more elaborate framings of these perceptions. This becomes especially directed and powerful when these associations and perceptions are around the people we love. And the people we love, being people, **always die**.

The reactions among the living toward the dead can be dramatic. Hallucinations can be common—people might see a vision of a loved one, or hear their voice calling them from another room, or have powerful and realistic dreams of them at night. These feelings of loss can be so powerful and overwhelming that people will desire to continue some contact with their loved one.

Over the next few days, I thought a great deal about Jodi, and I thought about Necromancy. I mused that Jodi would probably be VERY amused by all this. If she were alive, I would have missed her call, pressed the Return Call button, called her back, and had a nice chat. We could have caught up on recent events and had a lovely conversation. I would have found out why she was calling, and we would have had a really lovely time — as usual. Unfortunately, she is dead, and none of these options exist. What if one of them was, what if I *could* speak to her again? What would I say?

Hey Jodi, I miss you a lot. Too bad we didn't get to do that "Artist Union" digital community thing, because that would be so relevant today in the face of AI imaging capabilities and practices which are based in the sampling and analysis of "other people's" artwork. The initial idea we discussed didn't directly anticipate it, however, it would have been easily extensible, and your idea was and is still well ahead of its time. As usual. It would have been great to do that with you. I remember our wonderful conversations that would go far into the evening and all the lovely experiences we shared in grad school. Miss you, lots.

Another conversation with her would have been delightful. I think she would have laughed at the spoofed call. She would enjoy the conversation — we always had a fine creative rapport. She was a real friend. Would it be worth going through the procedures of *The Sworn Book* to contact her? Nope. Not for me.

Media: An Onramp to Our Present Hauntological Condition

The Sworn Book of Honorius is an amazing document, a fundamental text and manual, on the invocation and summoning of angels and demons. The book was largely ignored by historians until recently, which is unfortunate, as it shows the relations inherent to Kabbalah and Jewish mysticism and its importation into European Hermeticism and the evolving Western Esoteric Tradition. If mysticism was a sport, *The Sworn Book* is very 'inside baseball'.

To that end, there are the instructions by Eliphas Levi, in his book *The Doctrine and Ritual of High Magic*, where, in Chapter 13,⁷ Levi also discusses the process of necromancy. This aspect of the book is well discussed in the email I received from John Greer, one of the translators:

In Lévi's view, the evocation of the dead is justifiable only for two motives, love or knowledge: to communicate with a dead person the mage loved, or to seek instruction from a dead person who knew something the mage needs to know.

His description of these two methods is precise and practical. In both cases a period of intensive preparation involving a restricted diet, meditation, and prayer builds up to a final evocation, after which the desired spirit appears briefly to the mage. The period of preparation helps the mage attune himself or herself to the consciousness of the desired spirit and increases his or her sensitivity to the astral light. The prayers to use in this working, Lévi stresses, should always be based on the concept of the Divine accepted by the dead person.

No other conjurations, invocations, or other magical actions are necessary or appropriate; the mage simply enters into a state of expanded consciousness in which the astral light reveals the forms that are sought. For the evocation of love, the potent emotional bond between lover and beloved does most of the work. For the evocation of knowledge, a simple technical method Lévi has described before—directing a ray of light into a billowing cloud of incense, which the mind of the mage can then use as a sort of ever-changing Rohrschach blot—becomes the medium through which forms are projected from the Unseen.⁸

There are a number of similarities between Honorius's thirteenth century tome and Levi's volumes from 1856. Dietary restrictions, constant prayer and meditation, and an informed focus on the subject to be evoked, to name but a few. Levi's prescriptions are far less complex than Honorius's long and detailed grimoire. The similarities are greater, especially as they seek the same aesthetic ends: to bring about vivid visions of the dearly departed.

Just as the Synthesizer was overwhelmed by the *A E S T H E T I C* of the Mellotron, Frankenstein's Creature would pose little threat to Dracula, as the living, pose little threat to the

7 Levi Eliphas, *The Doctrine and Ritual of High Magic: A New Translation*. Translated by John Michael Greer and Mark Anthony Mikituk, TarcherPerigee, 2017.

8 John Michael Greer, email, 12 June 2024.

undead. It's very difficult to 'kill' something that isn't alive in the first place. This structure formed a lead-in to the Necromantic, which has become a significant point in the ongoing integration of the digital, this peculiar Imitation Game, into human social practices.

It took the Mellotron *AESTHETIC* as made materially manifest in the paritant Digital Sampler to permit the possibility of Digital Necromancy. The Necromantic possibilities of electronic and even electrical media cannot be ignored in this regard, as they supply the onramp to our present hauntological condition. The onramp to the Digital Sampler was, as we saw in part one, the invention of the Chamberlin and its descendant, the Mellotron, which flowed from the invention of the tape recorder, and previous audio replication systems of the wire recorder and the phonograph. These media were recordings of people performing, singing, speaking, acting—Derrida noted earlier that the film industry is a ghost factory for the hauntological. That was in 1983. It is now more than 40 years later, and much has changed. Our media can now raise the dead.

The Landscape of Digital Necromancy

Much has been written about Digital Necromancy, and here we will only be using a few examples. An excellent book that discusses the ramifications of this pre-digital space time is the book *Haunted Media* by Jeffrey Sconce.⁹ It gives an excellent history of the evolution of the affect and reactions to the then new and very spooky electronic media, and how it developed around angles of the occult and necromantic chicanery. Sconce also discusses aspects of our hauntological culture and why new technologies are seen as supernatural rather than simply natural technological developments.

As valuable as Sconce's research is, it's just a few years out of date, as it largely ends with television and similar televisual broadcasts media. Here, today, we have a completely different approach, where rather than viewing new technology through an occult lens, technology is simply fulfilling a far more ancient scope of knowledge, and is based on fundamental features (or bugs, depending on your perspective) of human cognition itself.

Another is a sociological paper, *Digital necromancy: why bringing people back from the dead with AI is just an extension of our grieving practices*, by Michael Mair, et al.¹⁰ which explores the more 'mundane' aspects of Digital Necromancy as it is being integrated into human grieving over the death of loved ones.

Leah Hendrickson, in her paper, *Chatting with the dead: The hermeneutics of thanabots*, examines the performativity of, and discourse surrounding, chatbots that resurrect the dead, often ostensibly to facilitate grieving. Hendrickson quotes Dan Crowe, editor of the book *Dead*

9 Jeffrey Sconce, *Haunted Media: electronic presence from telegraphy to television*, Duke University Press, 2000.

10 Michael Mair, Phillip Brooker, Dipanjan Saha, Terence Heng, "Digital necromancy: why bringing people back from the dead with AI is just an extension of our grieving practices." *The Conversation*, 2023. <https://theconversation.com/digital-necromancy-why-bringing-people-back-from-the-dead-with-ai-is-just-an-extension-of-our-grieving-practices-213396>

Interviews, who points out essential aspects of necromancy that, if these words weren't written so recently, they could be translations of the situation from the ancients:

"So what does it mean to interview someone who is dead?" Crowe (2013: 4)¹¹ asks: We want the dead to give us information, provide revelations, tell us something they didn't know when they were alive. We want to believe that they know more than we do, can advise us, help us. Where are they? Is there a God? How scared should be we? We want to ask the dead why they did what they did. We want them to say sorry. We want to say sorry to them. Above all, I think, we want to believe it is possible to talk after we die.¹²

Hendrickson notes:

The deceased loved one is remediated in a way that is simultaneously immediate—one can chat with a version of the loved one in real time—and hypermediated—one is acutely aware that this is an artificial representation of that loved one.¹³

We would suggest that the awareness of artifice is more a function of the uncanny valley and not hypermediation per se. An extensive examination of digital necromancy is *Digital Afterlife and the Spiritual Realm*, by Maggi Savin-Baden.¹⁴ Her work examines how digital media and the creation of digital immortals may affect religious understandings of death and the afterlife within a religious context, examining the impact of digital media on bereavement and digital memorialization and how this can change our theological understandings of death. Savin-Baden's stance (emphasis mine)

...is that although people may talk to the deceased on social media sites, they do not expect them to respond and the practice of doing this is not religious. Yet this stance seems rather stark and seems to suggest that religion is clear and bounded. Yet many people who do not attend a church or temple, as well as those who position themselves as atheists, still have a belief in an afterlife of some kind. For example, Walter et al. (2012) argue that the online dead are always accessible and often spoken about as **angels**. There is a sense then from Walter et al.'s perspective that angels have some kind of digital embodiment: **Angels are messengers, traveling from heaven to earth and back, and cyberspace is an unseen medium for the transfer of messages through unseen realms, so there may well be a resonance between how some people imagine online messaging and how they imagine angels.**¹⁵

11 Daniel Crowe (ed.), *Dead Interviews: Living Writers Meet Dead Icons*, London: Granta, 2013.

12 Leah Hendrickson, "Chatting with the dead: The hermeneutics of thanabots", *Media, Culture, and Society*. Vol. 45 (5), 2023, p. 949

13 Ibid. p. 950

14 Maggi Savin-Baden, *Digital Afterlife and the Spiritual Realm*, CRC Press, 2022.

15 Ibid, p. 68., quoted from: Walter, T., Hourizi, R., Moncur, W., & Pitsillides, S. "Does the internet change how we die and mourn? An overview." *OMEGA: Journal of Death and Dying*, 64(4), 275–302, 2012. Emphasis by author.

This seeming 'embodiment' of the angelic in the technological is part of our discussion here—just as the Synthesizer embodied the spirit of modernity, science, and progress, and the Mellotron, the spirit of post-modernity, simulation, and paritance. Unlike Sconce, who sees the supernatural as a refuge from the shock of new media and ICT, the 'seemingly' part of an apparent 'resonance'—these fleeting sidelong glimpses are what we have experienced for millennia—recent technology is simply amplifying these ancient practices of necromancy, evocation, and deification, and in such amplification that the sensorial coverage of new media becomes a quality of its own, where our new media is not so new.

Our dreams of talking with the dead or angels or a deity are materialized with an acceptance of some parity between the virtual and the dead. So, after all that, would it be worth going through the procedures of *The Sworn Book* for me to 'contact' Jodi? Again, *no*. What if she could be resurrected digitally?

Hallucinating the Dead: Haunted Media

If I went through the hundreds of pages of the book and memorised the hundreds of names of angel and the countless prayers, psalms, and other rituals, and if I was even able to find a Catholic priest willing to do this with me, (which is highly unlikely as it would invite excommunication upon discovery) the only real effect would be me hallucinating her presence. If I did enough LSD or DMT, I suppose I could probably get the same effect. For example, when I was in High School, I did some LSD at a party. I had a conversation with God there. He was sitting at the kitchen table drinking a beer. I knew it was God because he had a big gold G on his t-shirt. He was pretty cool. I don't remember much of the conversation. I do remember being impressed by his T Shirt. Anything *The Sworn Book* could do, could not be much more effective than that. So, no, it would not be worth it. It would not be a satisfying experience. It wouldn't even have any beer in the fridge.

At the same time, people have been known to hallucinate the voices and images of departed loved ones.¹⁶ While these hallucinations are real to the person experiencing, they are only real as hallucinations. This doesn't stop the deep and moving desire that people feel for each other, a deep and motivating desire to contact dear departed loved ones. And it is this attachment, this desire, that drives sad people into the hands of people who are willing to sell them stories of the hereafter, because facing the unknowable void at the centre of existence itself is so difficult and unbearable.

To ameliorate this affective catastrophe, people are taught they must believe—and this asks the question: just what *do* they believe? Stories and narratives from the past. The people who purvey these stories aren't necessarily evil people. They can be perfectly kind and generous

16 Sidney Zisook, Katherine Shear, "Grief and bereavement: what psychiatrists need to know." *World Psychiatry*, June 2009; 8(2): 67-74.

"Auditory or visual hallucinations of the deceased person are often seen during acute grief. Sometimes people maintain a sense of connection through objects such as clothing, writings, favorite possessions, and rings, which may be kept indefinitely."

people who also believe a particular cultural narrative. And when faced with information that is hard to make sense of, they resort to said narratives and their particular quirks and prejudices. Theology devolves into media theory and narratology. And, conversely, those holding spiritual disciplines then react to the advent of new technologies, using the framing of their narratives, and the new technologies are then put to use in the implementation of said technologies.

This brings me back to Jeffrey Sconce—He 'argue(s) that electronic presence, seemingly an essential property of telecommunications media, is in fact a variable social construct, its forms, potentials, and perceived dangers having changed significantly across media history'.¹⁷

Sconce has a vast and well-researched litany and history of how the electronic media have been perceived in terms of the occult. For a while, 'The focus of much popular scientific interest in the 18th and 19th centuries, electricity was for many a mystic and even divine substance that animates body and soul.'¹⁸

This is clearly evidenced in the previous section regarding Frankenstein's Creature, who was dramatically animated by the electricity of a lightning storm in the film, and in the novel, Dr Frankenstein wrote

Before this I was not unacquainted with the more obvious laws of electricity. On this occasion a man of great research in natural philosophy was with us, and excited by this catastrophe (the destruction of a nearby tree from lightning the previous evening), he entered on the explanation of a theory which he had formed on the subject of electricity and galvanism, which was at once new and astonishing to me.¹⁹

and

I collected the instruments of life around me, that I might infuse a spark of being into the lifeless thing that lay at my feet. It was already one in the morning; the rain pattered dismally against the panes, and my candle was nearly burnt out, when, by the glimmer of the half-extinguished light, I saw the dull yellow eye of the creature open; it breathed hard, and a convulsive motion agitated its limbs.²⁰

Furthermore, beyond the physical animation of the dead body, '(electrical signals) appeared to carry the animate 'spark' of consciousness itself beyond the confines of the physical body.'²¹ The fear and apprehension regarding electricity led to conclusions that the electric was not only the animating force of the physical, such as through galvanic responses and fantasized into the reanimation energy of the Creature, it also took a leap to explain consciousness,

17 Sconce, p. 6.

18 Sconce, p. 7.

19 Mary Wollstonecraft Shelley, *Frankenstein; or, the Modern Prometheus*, 1818, Ebook, Planetebook.com, <https://www.planetebook.com/free-ebooks/frankenstein.pdf>. p. 37.

20 Ibid, p. 58.

21 Sconce, p. 7.

...the shared electrical basis and apparent electrical transmutability of the body's flow of consciousness and flows of information in the media have produced a remarkably consistent series of cultural fantasies re: the telegraph, radio, television, and computer.

And he made a prediction,

these recurring fictions are especially central to the development of ICT, familiar stories that appear in new incarnations with advent of each new medium.

From there, he describes these fictions. The first being how the media enable disembodiment—they transport consciousness; the second being a premise of a sovereign electronic world—an electronic 'elsewhere'; and the anthropocisation of ICT in the form of robots and cyborgs.

The Anthropocisation of ICT

Unfortunately, these fictions don't account for other significant issues. The first two (disembodiment and the electronic 'world') are answerable to the question, 'Where does a phone call occur?' The anthropocisation of ICT is answered with Frankenstein, written almost a hundred years before Alan Turing or Claude Shannon were even born. Where DOES a phone call occur, if not in a sovereign electronic world? The proof of this is in Claude Shannon's communication model:²²

SOURCE > (message) > TRANSMITTER > (signal) > NOISE > (received signal) > RECEIVER > (message) > DESTINATION

And the ICT modification of Claude Shannon's model: INPUT > (message) > MEMORY/PROCESSING > (message) > OUTPUT > FEEDBACK > INPUT

The conversation doesn't take place at the information source, the transmitter, the point of noise interjection, the noise source, the receiver, or the destination in Shannon's model. It doesn't take place in the Input, the Processing, the Storage, the Output, or the Feedback nodes of the ICT Model. *It is in the arrows connecting them.* It doesn't exist in the minds of the sender or the receiver, or the audient. These are variously defined by Shannon as 'Message' or 'Signal' or 'Received Signal'.²³ The ICT model doesn't name the flows either; however, this isn't as crucial, as the scope of the ICT model is smaller, and is much more engaged with the flow itself; the flow of data.

A phone conversation exists in the flow of the electro-magnetic energy of the ICT system itself, which, at this point, is an interstellar field of information²⁴, and this information consists within a configuration of electromagnetic charges over time. A phone conversation is not a thing; it is a

22 Claude Shannon, "A Symbolic Analysis of Relay and Switching Circuits", Master's Thesis, MIT, 1937.

23 Inconsistent, as there is a flow from the Noise Source to a node in the model, neither of which are named or identified.

24 Radio signals travel at *c*, and thus the earliest radio signals that may have escaped the Earth are over 100 light years away. There are 59,722 stars visible with a telescope within 100 light-years of our solar system, so human communications technology is truly interstellar, if incredibly weak.

dynamic arrangement of components that operate together to create the phone call. A phone call is more of a state of this dynamic telecom system than an object, per se. A phone call from, say, Toronto to Amsterdam would flow from a cellphone to a cell tower in Toronto, to another tower to a phone call router, to a transmitter to a satellite to a receiver to other cell towers to the phone of the number called, with information flowing in both directions, even when merely ringing. We have the experience of participation in a telephone conversation as a memory, itself a configuration of electro-chemical states in the brain of the rememberer realized through the thermodynamic gradient of the organic, the essential character of life itself.²⁵

To the human experience, one's memories are assemblages of other details—there are a number of vectors to a memory that create vividness — for example, my memory of my mother driving us to go pick up my dad—I can see the light blue colour of the car, and feel the bouncy, springy seats and how they felt on my legs and back. I can remember the chrome on the radio buttons, and the warmth of the sun on my face, the smell of the gasoline engine chugging along, the roar of the highway, and, on reflection, the lack of seatbelts. There can be an ordering of these perceptions to form memories, and the ordering of these memory parts becomes crucial to the memory's comprehensibility and memorability.

Musical Interludes: A Utilitarian Device for the Preservation of Information

Music is especially useful in this regard, as music works as a form of data compression. A melody gives data a form for remembrance, and a hook for its comprehension and memorability. For example, a song might have a chorus and a verse. The first verse is sung, followed by the chorus. As one remembers the first words of the second verse, the second verse as a whole comes more easily, as the cadence of the words falls into the melody, reinforcing the words. For example, in the song 'Hallelujah' by Leonard Cohen²⁶, he sings

C F G

It goes like this the fourth, the fifth

Am F

The minor fall, the major lift (transcription by author)

The song is in the key of C major, so it starts on the chord C. The fourth of C is F, the fifth is G, and we can see that the chords do exactly as the lyrics indicate. Then there's the 'minor fall' as the next chord is an A minor (the relative minor of C), which then resolves to F major, the 'major lift' (and then resolves further to the G as it did under 'the fourth the fifth').

There, the lyrics reference the music, and, when sung in context, makes for a mnemonic device for the musician to remember the chord sequence of the song, in a much more compact way

25 Schneider and Sagan, *Into the Cool: Energy Flow, Thermodynamics, and Life*, University of Chicago Press, 2006.

26 Leonard Cohen, "Hallelujah", from *Various Positions*, Columbia, 1984.

than the few sentences I spent describing what the music and lyrics are doing. Music is data compression.

Australian aboriginals have used this data compression aspect of music as a mapping technique, using 'songlines'.²⁷ A songline can contain directions for crossing a particular area of the Australian outback. Amassing enough of them would provide directions for a traveller to walk across the continent. Songlines also act as a cultural currency between the inhabitants of different areas, and are symbols of respect when the songs are sung in their original languages or dialects. They are traded between groups. Songlines connect these groups, and the reproduction of these songlines over time and generations develops and amplifies connections by inhabitants to their area. The connection to the land is essential to Aboriginal identity and culture. In this way, the songlines act as data compressed maps for the preliterate societies of Australia.

In this way, music is more than mere entertainment, or a ritual tool, or social glue, it's actually *a utilitarian device for the preservation of information*. This is readily seen with Homer, prior to someone writing Homer's Iliad and Odyssey down, Homer and his antecedents would have passed down the history of the Trojan Wars orally, using melody and cadence to compress this information so it could more easily flow from memory. This continues to this day. From song lyrics where the melody and cadence aids the memorization and performance of the data to the irritating vestigial lilt of countless poets at the ritual of poetry readings—there is a needed audio framing effect to instruct the focus of the listener, as well as provide a mnemonic aid for the poet.

Someone is reciting poetry to others lying in a field on a late summer evening. We have a scene: the fireflies dancing about, the gently moving air, the sonority of the voice, the bodies lying in the grass, the standing recitation of rhyme and the gentle metric movement of words, and the stars slowly emerging from the gloaming with the mosquitoes. The poem itself may be forgotten, even to the person who recited it. Yet the memory remains—the composite of experiences may be so rich at that moment that they form a particularly powerful memory—'Yeah, we were all listening to you recite poetry. What an amazing evening that was!' Yet the moment is long past, the poem is forgotten, the rhymes and meter and cadence lost to the vagaries of time, and somehow the memory of the magic of the moment remained, and for those who were there, the memory forms a bond, a bond that can and does change behaviour, consideration, affect, and choices.

And over time, the memories fade, and at death are finally erased. As Roy Batty said, just before he died,

I've seen things you people wouldn't believe... Attack ships on fire off the shoulder of Orion... I watched C-beams glitter in the dark near the Tannhäuser Gate. All those moments will be lost in time, like tears in rain...²⁸

27 Lynne Malcolm, Olivia Willis, "Songlines: the Indigenous memory code", *ABC Radio National*, 8 July 2016.

28 Final monologue by "Roy Batty", written by David Peoples and modified by Rutger Hauer, *Blade Runner*, dir. Ridley Scott, 1982.

All of these memories we hold onto and desperately fight their loss, these memories and imprints left by experience, they fade with old age and lack of use, or the scourge of dementia, or Alzheimer's disease—the material proof of the impossibility of the soul.²⁹ Yet we hold onto them—we can't just 'let them die', especially when these memories are of people we love. And in a world of The Spectacle, who is more loved than a celebrity?

Digital Necromancy Is an Extension of the Ancient Art of Grieving Practices

This is where Mair et al show that digital necromancy is an extension of the ancient art of grieving practices. Sconce sees the haunting of media as a kind of atavism, where a new technology arrives and was assumed to have some kind of spiritual power, when in fact it's just machines. Mair et al see it in a kind of reverse angle—where the machines are simply fulfilling desires that were formed, refined, processed, and developed millennia prior to the invention of the technology. Funerary practices preceded homo sapiens—there is recent, if controversial, evidence suggesting that Homo Naledi in South Africa buried their dead in caves 200,000 years ago.³⁰ Homo Naledi were a species of archaic humans, with brains half the volume of modern human brains. Homo neanderthalensis also seemed to practice funerary rituals, as evidenced by the elaborate burial of a two-year-old Neanderthal child, some 41,000 years ago, found in southwestern France.³¹ Anatomically Modern Humans buried their dead as long as 100,000 years ago—a dozen were found in a cave in Qafzeh, Israel.³² All of these people held some notions about life and death and the body, and their experiences and interactions with that body and its actions, utterances, affects, and ideas.

-
- 29 The "soul" is something we (supposedly) all have. It is this soul that is permanent beyond material existence and is eternal. We can damage our soul with our actions and thoughts. It is dogmatically true to many religions that evil deeds come from an evil soul which must be condemned. So, let's say a person, Mister X, does something unforgivable—something truly heinous. And then, they're in a car accident, and their memory is completely wiped. They have no record of their evil deeds. Others do, but confronting Mr X with their misdeeds is pointless—they don't know what the hell you're talking about. So, if you punish Mr X for his deeds, you are not punishing his mind or his soul—you are merely punishing his body. He has no idea what's going on and why everyone is being so mean to him. How can he be punished? What would that even mean? The simpler answer is to abandon the notion of the soul, and from there, abandon the notion of The Mind, as it is simply an epiphenomenon of brain activity.
- 30 Lee R Berger, et al. *Evidence for deliberate burial of the dead by Homo naledi*. Biorxiv, 2023. <https://www.biorxiv.org/content/10.1101/2023.06.01.543127v1.full.pdf>
- 31 Katie Hunt, 'Neanderthal child's skeleton buried 41,000 years ago may solve long-standing mystery.' *CNN*, 5 January 2021. <https://www.cnn.com/2021/01/05/europe/neanderthal-burials-skeleton-study-scn/index.html>
- 32 John Shea and Ofer Bar-Yosef, 'Who Were The Skhul/Qafzeh People? An Archaeological Perspective on Eurasia's Oldest Modern Humans.' *Mitekufat Haeven: Journal of the Israel Prehistoric Society* / תפוקת האבן (2005): 451–68.

Other creatures also demonstrate profound attachment to dead loved ones – other primates,³³ whales and dolphins³⁴, elephants³⁵, and other animals have exhibited sadness and stress with the passing of a family member or associate of their own kind. They have also acted with concern towards the release and deposition of the body. This has been widely observed and understood. Thus, it comes as absolutely no surprise that we, modern humans, also experience profound shifts in affect and enact social rituals at the passing of a loved one. And, given our long history of trying to bring loved ones back from the dead through various necromantic practices, we should also make use of digital means to do so.

We also need to define what this is. Necromancy is an occult practice where one summons the dead for the purposes of communicating with them.³⁶ While stories of Medieval and Renaissance sorcerers and necromancers, combined with the flourishing of Victorian spiritualism in the late 19th century abounded, and frequently abound with stories of deception and grifting on the part of the necromancers and séance conductors; stories of necromantic pursuits go far back into antiquity. This cannot be underlined enough, as so much of the literature surrounding this seemingly starts with the Victorians, with a nod to John Dee in the Renaissance. Note: John Dee owned a copy of *The Sworn Book of Honorius of Thebes*.

The earliest reference to this rite can be found in Book XI of the *Odyssey*, where Odysseus is instructed to perform a ritual, sacrificing a ram, thus summoning the ghosts of the dead to question them about his further steps to reach the Halls of Hades.³⁷ A more ancient reference is in the story of Gilgamesh and his conversations with Enkidu, who had travelled to the netherworld. Kapcar's historical work on the history of Necromancy has many other examples of ancient necromancy.

Kapcar also points out that

'Plutarch in his *De sera numinis vindicta* (Section 17) mentions Collondes who slew Archilochus, a favorite of the Muses, in a fight, and was therefore ejected by the Pythian priestess. He had to humble himself through prayer and sacrifice, appeasing the evoked ghost of Archilochus to undo his deed [...] In the same work Plutarch describes several other cases of necromancy—in section 10 he describes the fate of Pausanias, king of Sparta, who for no apparent reason kills his lover, Cleonice of Byzantium. After this deed, haunted by frightening visions and apparitions, he travels to the Oracle of the Dead in Heraclea to conjure the murdered woman. She appears and announces in

-
- 33 A. Gonçalves, S. Carvalho, 'Death among primates: a critical review of non-human primate interactions towards their dead and dying.' *Biological Reviews of the Cambridge Philosophical Society*.10:15, 2019.
 - 34 G. Bearzi, L. Eddy, S. Piwetz, M.A.L. Reggente, B. Cozzi, 'Cetacean behavior toward the dead and dying.' In: Vonk J, Shackelford TK (eds) *Encyclopedia of animal cognition and behavior*, Springer, 2017, pp. 1–30.
 - 35 I. Douglas-Hamilton, S. Bhalla, G. Wittemyer, F. Vollrath, 'Behavioural reactions of elephants towards a dying and deceased matriarch.' *Applied Animal Behaviour Science*, 2006, 100:87–102.
 - 36 A. Kapcár, 'The origins of necromancy on how we learned to speak to the dead.' *Sacra*; 13(2), 2015, pp. 30–58.
 - 37 Ibid. 39

brief that his hauntings will cease after his arrival at his hometown of Lacedaemon. Her prophecy comes true, but not as he expects since after his return he dies.³⁸

Clearly, a much more direct necromantic process. Obviously, not the sort of thing one does on a lark of a Monday morning. At the same time, the act of the rituals of necromancy was, for the time, not a banned behaviour. It's hard to say 'don't do that' when it was well established that the focus of the Christian Church featured someone who did bring people back from the dead. Christ brought a dead girl back to life: > 49 While Jesus was still speaking, someone came from the house of > Jairus, the synagogue leader. "Your daughter is dead," he said. "Don't > bother the teacher anymore." > 50 Hearing this, Jesus said to Jairus, "Don't be afraid; just believe, > and she will be healed." > 51 When he arrived at the house of Jairus, he did not let anyone go in > with him except Peter, John and James, and the child's father and > mother. > 52 Meanwhile, all the people were wailing and mourning for her. "Stop > wailing," Jesus said. "She is not dead but asleep." > 53 They laughed at him, knowing that she was dead. > 54 But he took her by the hand and said, "My child, get up!" > 55 Her spirit returned, and at once she stood up. Then Jesus told them > to give her something to eat. > 56 Her parents were astonished, but he ordered them not to tell anyone > what had happened.³⁹

And of course, the famous story of Lazarus:

41 So they took away the stone. Then Jesus looked up and said, "Father, I thank you that you have heard me. 42 I knew that you always hear me, but I said this for the benefit of the people standing here, that they may believe that you sent me." 43 When he had said this, Jesus called in a loud voice, "Lazarus, come out!" 44 The dead man came out, his hands and feet wrapped with strips of linen, and a cloth around his face. Jesus said to them, "Take off the grave clothes and let him go."⁴⁰

The proscription against necromancy required a more dogmatic approach to theological debate that favoured the power structure and hierarchy, and this was provided by the Christian Church, who saw raising the dead by physically reanimating their corpse as infringing on the domain of God, as only God can give life, and in the mythology of Christianity, Jesus is seen as the Son of God and part of the Trinity that IS God. For this reason, the stories of Jesus's necromantic exploits were revered while those who sought to do it on their own by various means were strenuously condemned by the Church. Post-Renaissance, communicating with the dead was seen as the province of an expert, a Necromancer, some sort of sorcerer or wizard. Whether it is a rite performed by an 'expert' or someone who was simply informed of the rituals, the results are fundamentally the same: the living seeking communication with the dead for help in the mundane here and now world of the living, with assistance only available from the 'dreary regions of the dead, where all things are forgot.'⁴¹

38 Ibid. 41.

39 Luke 8:48-59.

40 John 11:41-45

41 Ananias Davisson, J. W. Watson, 'Idumea', *Sacred Harp Hymnal* 1763 / 1816.

Even in the Old Testament, Moses warned against imitating the Canaanite rituals and necromantic practices (emphasis mine):

None must be found among you to consecrate son or daughter by making them pass through the fire, to consult the soothsayers, or keep watch from dream-revelations and omens; there must be no wizard, or enchanter, none who consults familiar spirits and divinations, **and would receive warnings from the dead.**⁴²

An interesting aspect of investigating these early necromantic efforts is that while the deceased subjects are being brought back from beyond the veil were often significant members of their society, they were usually summoned by family members, friends, or acolytes, and were brought back to bring some advantage to the living.

Under capitalism, all value is measured in money, and this capitalist logic continues and pervades digital necromancy, as we will see in our examples below.

Desiring the Dead Tupac Shakur and Marjorie Prime

Tupac Shakur

For those less familiar with Tupac, here is a brief bio:

Tupac Shakur was born Lesane Parish Crooks, 16 June 1971, in East Harlem, New York City. He was later and best known by his stage name, Tupac, or 2Pac. Tupac was a powerful figure in the world of hip-hop and music culture of the 1990s. He was murdered at the age of 25. Tupac's childhood was brutal and marked by turbulence and upheaval. He was raised by his mother, Afeni Shakur (1947—2016), who was a prominent Black Panther activist. Tupac was exposed to the realities of social injustice and inequality from a young age. His upbringing was marked by periods of homelessness and poverty, which profoundly influenced his worldview and fueled his advocacy for the oppressed. From an early age, Tupac showed a talent for performance and poetry. He attended the Baltimore School for the Arts, where he honed his skills as a rapper and actor.

In 1991, Tupac's career gained momentum when he debuted as a rapper on Digital Underground's single 'Same Song.'⁴³ His raw lyrical style, combined with his charismatic presence, garnered attention in the hip-hop community. He released his debut album, *2Pacalypse Now*⁴⁴, in 1991, which showcased his socially conscious lyrics and fearless delivery.

Tupac's rise to fame was accompanied by controversy. His music often explored themes of his childhood—poverty, racism, and police brutality. He courted controversy with tracks like

42 Deuteronomy 18:9–12

43 Tupac Shakur, *Same Song*, Single, Tommy Boy/Warner Bros, 1991.

44 Tupac Shakur, *2pacalypse Now*, CD, TNT/Interscope, 1991.

'Brenda's Got a Baby',⁴⁵ which addressed teenage pregnancy, and 'Keep Ya Head Up',⁴⁶ which championed female empowerment. Despite his success, Tupac's personal life was tumultuous. From a young age, he faced many legal troubles and was arrested and sentenced to prison for a variety of crimes, ranging from sodomy and weapons charges to acts of assault and battery. While in jail in 1994 for sodomy and weapons charges, Tupac continued to write and record music, releasing the critically acclaimed album *Me Against the World*⁴⁷ in 1995.

Tupac's career reached new heights when he signed to Death Row Records, helmed by the infamous music mogul Suge Knight. Shakur released his magnum opus, *All Eyez on Me*,⁴⁸ in 1996, a double album that cemented his status as one of hip-hop's greatest artists. The songs 'California Love' and '2 of Amerikaz Most Wanted' were especially popular.

Tupac was murdered on September 13, 1996, in a drive-by shooting in Las Vegas. While Duane 'Keffe D' Davis confessed to the murder, there is still a great deal of controversy around his violent demise. Tupac's death sent shockwaves through the world of hip hop and sparked an outpouring of grief from fans worldwide.

This grief spilled out into the media in two specific moments of value to this discussion.

One is the film, *Resurrection*⁴⁹ and its eponymous posthumous soundtrack album⁵⁰ of 2003. The other was the digital necromancy of his virtual resurrection, at the Coachella Festival in 2012. This was achieved through an application of some advanced technology, and likely cost somewhere between \$100,000 and \$400,000 to pull off.⁵¹ AV Concepts was the company hired to make this happen. The company's president, Nick Smith, told MTV it took several months of planning and four months of studio time to create the hologram.⁵²

AV Concepts Chief Creative Officer described the 'how' of this event:

Utilizing the Musion Eyeliner system, the 302 x 132 screen was customized by AV Concepts to descend onto the stage in mere seconds between sets of the performance to bring the infamous, deceased singer back to life... The most critical technical element was AV Concepts' proprietary Liquid Scenic server that delivered uncompressed media for 3 stacked 1920 x 1080 images, delivering 54,000 lumens of incredibly clear projected imagery.⁵³

45 Tupac Shakur, 'Brenda's Got a Baby', Single from *2pacalypse Now*, 1991.

46 Tupac Shakur, 'Keep Ya Head Up', Single from *2pacalypse Now*, 1991.

47 Tupac Shakur, *Me Against The World*, CD, Interscope / Atlantic, 1995.

48 Tupac Shakur, *All Eyez On Me*, CD, Death Row / Interscope, 1996.

49 *Resurrection*, Directed by Lauren Lazin, Film, Paramount Pictures, 2003.

50 Tupac Shakur, *Resurrection*, CD, Amaru-Interscope, 2003.

51 Chenda Ngak, 'Tupac Coachella hologram: Behind the technology,' *CBS News*, 9 November 2012.

52 Ethan Smith, 'Rapper's De-Light: Tupac 'Hologram' May Go on Tour', *Wall Street Journal*, online, 16 April 2012, <https://www.wsj.com/articles/SB10001424052702304818404577348243109842490>

53 Chenda Ngak, *ibid*.

Although the perception was of a 3D likeness of Shakur, the image was actually a 2D image. Shakur's likeness was projected onto an angled piece of glass on the ground, which in turn projected the image onto a Mylar screen on stage.

The advantages of digital necromancy are many. It can be emotionally satisfying to go to a show featuring a celebrity who died when one was younger and didn't have the opportunity to see them perform. It can be emotionally satisfying to see a loved one appear as a hologram and give a reassuring or comforting message. The spectral aspect of the hologram itself aids and points directly at the necromantic—the hologram is transparent, like a ghost. The hologram isn't as sharply defined as an object in light. The sound of the hologram's voice is not the result of air in their lungs vibrating their vocal cords and making sounds processed by the tongue and teeth. It is the result of an electrical impulse making a magnetic field. The magnets built into the speaker produce a magnetic field in opposition to the electrical current's magnetic field. Since the two magnetic fields are opposing and repel one another, and one set is attached to the speaker cones, usually made of paper or plastic, the cones are made to vibrate, and produce the sound we hear. It will always only sound 'like' the source. It will never replace the source.

This makes for a good experiment and demonstration—get a microphone and place it at an exact distance from your mouth. Say something, and record that. Then, play it back through the speaker. Then put the microphone at the exact distance and placement you recorded your voice with, play your recording, and record the recording of your voice. Then do an A/B comparison between the two. They won't sound the same, and neither will they sound exactly like your native voice. The reasons are many—as noted, your voice is made of vibrating vocal cords filtered by your mouth, the speaker is just a vibrating cone of paper and/or plastic. So, there's that. Then, in the differences between the recorded version of your voice and the recording of the recorded version of your voice, the filter isn't your mouth, it's the sonic qualities of the microphone and the speaker, and the acoustics of the room it is recorded in.

While making a convincing hologram for the stage at one of the largest and the richest music festivals are an expensive and difficult project, and was certain to make a splash, it's a relatively simple operation to be effective as it is an extension of the traditional media broadcast structure of one-to-many found everywhere from a university classroom to a live television broadcast. The spectacle involved has an inherent reduction in detail—there are physical distances imposed by the spectacular presentation itself. With Tupac at Coachella, the hologram was standing on a high stage with a meter or so distance between the stage and a tall fence keeping the audience back. As one can see from the video and photo above, the resolution of the projection isn't as sharp or crisp as an actual body on stage. With the lighting and crowds and distances involved, this act of celebrity necromancy has a curiously double effect—it's less of an intimate event between the audience and the spectacle, yet the event gains power from the imperious limitations of distance and presentation effects.

Tupac Shakur was not the first, nor the latest example of this digital necromancy. This has been used for the reanimation of Bruce Lee (d. 1973) and Michael Jackson (d. 2009), and film appearances by Carrie Fisher (d. 2016) and Peter Cushing (d. 1994) in the Star Wars

franchise. Fisher approved of this virtual Princess General Leia. However, Disney has indicated it will not work a digital Leia in future films.⁵⁴ Also, in the film *Rogue One: A Star Wars Story*⁵⁵, Grand Moff Tarkin was a reanimated Peter Cushing, as by the time *Rogue One* was released, Cushing had been dead for some twenty-two years.

For some time, this kind of imagings were called 'deep fakes', where digital representations of people were put into time-based media to provide content that the actual people had no hand in making. However, while this moniker is reasonable, it only scratches the surface, as the quality of the deep fakery is now approaching the closure of the uncanny valley.

Marjorie Prime

In 2017, the film *Marjorie Prime*⁵⁶ was released, based on the play of the same name, written by Jordan Harrison. The film was met with critical acclaim and has high ratings on internet film sites like Rotten Tomatoes⁵⁷ and IMDB⁵⁸. The film tells the story of a family over some years, beginning roughly in the year 2050. There is an older woman named Marjorie, her daughter, Tess, and Tess's husband, Jon. Marjorie is showing signs of advancing Alzheimer's disease, and so Jon and Tess have arranged for Marjorie to receive a holographic AI 'ghost' of Marjorie's deceased husband, Walter. This hologram is named Walter Prime. Marjorie develops a relationship with Walter Prime, who looks and sounds like her dead husband when he was younger. Tess dislikes Walter Prime and is distrustful of this digital necromancy. Eventually, Walter Prime becomes curious about Walter and has many discussions with Jon about what he remembers of Walter.

Eventually, Marjorie dies. Jon recommends the Prime Program for Tess to help with her intense grief over the death of her mother, Marjorie, as they had a very complex and difficult relationship, especially regarding the teenage suicide of Tess's brother, Damian. Tess and Marjorie Prime have some difficult conversations. Several years later, Tess, still unable to deal with her mother's passing and the estrangement from her adopted daughter, also commits suicide while she and Jon are on vacation in Madagascar. To overcome his grief, Jon gets a Prime of Tess.

Many years later, Jon is in an assisted living facility, and at the end, Marjorie Prime, Walter Prime, and Tess Prime are in the living room of Marjorie's home discusses the lives and memories of those who are digital copies of. At the end, Marjorie Prime says, 'how nice that we could love somebody'.

54 Peter Allen, 'A genuinely believable CGI actor? It won't be long' *The Conversation*, 17 January 2017, <https://theconversation.com/a-genuinely-believable-cgi-actor-it-wont-be-long-71410>

55 *Rogue One: A Star Wars Story*, Dir. Gareth Edwards, Walt Disney Studios, Film. 2016.

56 *Marjorie Prime*. Dir. Michael Almereyda. Film. BB Films. 2017.

57 90% on the Tomatometer, and 61% on the audience rating. https://www.rottentomatoes.com/m/marjorie_prime

58 6.5 on IMDB. <https://www.imdb.com/title/tt4978710/>

Walter, Marjorie, and Tess Prime, like any members of the undead, *must be invited into the home*, and at no small expense. Walter was a wealthy man, and Marjorie and Tess wanted for nothing, so when Marjorie, Tess, and Walter needed to work through their grief with these Primes, they could afford this. The Primes have no physical existence—their existence is spectral, they are holographic projections. Their appeal is *emotional*, not intellectual. They exist to help ameliorate the grief people feel around the death of a loved one. Their continuance is vampiric—they have no identity outside of the thing they imitate—just like a Mellotron has no sound other than what was recorded on the tapes. Everything they do is revealed through language, and has value through the cathetic investment of their live human partner. Walter Prime exists because of Marjorie. Marjorie Prime exists because of Tess. Tess Prime exists because of Jon. The 'Prime Program' recapitulates the necromantic.

Return to the Dead

Mair et al put it clearly:

In moments of crisis or joy, we reflect upon what those we have lost might have said to us, the attitudes they might have had and the encouragement they may have offered in relation to challenges and accomplishments in the here-and-now. [...] However, again, we should remember that we do not ordinarily treat our personal messages, photographs or videos of the dead as if those records themselves were our loved ones. Instead, we use them as conduits to their memory, standing in for them as proxies for us to think of or communicate through. To suggest we routinely get confused or delude ourselves about such media is a misconception.⁵⁹

At the same time, it is the cathexis, the emotional psychic investment in the dead, that gives these media the power they have. Without that, it's just stuff. Now, when we combine that investment with material economic investment that digital necromancy swerves into view. With AI, the dead can now do 'new music'. An example of this is the recent release of *Now And Then* by the Beatles, two of whom are dead, and the other two are in their mid-80s. John Lennon wrote the song as a solo effort and recorded a demo in 1977. In 1994, Yoko Ono sent a cassette of the song to Paul McCartney.

Attempts to record the song in the 1990s met with failure as the technology to bring the song to life hadn't been invented. However, George Harrison was able to record some guitar tracks for it. Harrison died in 2001. These production problems were ameliorated with the documentary *The Beatles: Get Back* (2021)⁶⁰, which used software called MAL (Machine Assisted Learning), which allowed them to 'clean up' the recording. The director of *Get Back*, Peter Jackson that the technology 'allows us to take any soundtrack and split all the different components into separate tracks'.

⁵⁹ Mair et al. Ibid.

⁶⁰ *The Beatles: Get Back*, Documentary, Directed by Peter Jackson, Disney +, 2021.

In May 2022, the song was completed at Capitol Studios in Los Angeles.⁶¹ The merits of the song are of no consequence here. The song is simply another example of digital necromancy, and is a classic example, as it trades on the extraordinary value of the legacy of the music by The Beatles, so much so that they hired the renowned visual artist Ed Ruscha to do the cover art. The song did very well. Within two weeks, it topped the UK music charts and set a record for the time between #1 hits at 52 years, the last one being 'The Ballad of John and Yoko' in 1970.⁶² Thanks to the bleeding edge of the technology used to make *Now and Then*, combined with the post-production expenses of the string section and mastering, was an expensive undertaking. However, with the likes of Paul McCartney backing it, expense was not that big a hurdle. And the success of the track in the music market made the investment economically viable. A number of organizations have been working at this for some time to greater or lesser success.

A few years ago, Google's MusicLM system could make musical marionettes of dead artists like Chuck Berry and Frank Sinatra. The results were never that convincing, although they did point in the direction as part of the Digital Necromantic project. These projects all trade on the currency of celebrity. The Beatles, Tupac, Sinatra—they're all famous people, and this makes sense from the view of capital, as large numbers of people will pay a lot of money for a virtual visitation of their dead celebrity angel or daemon.

However, that is a derivative of the necromantic social impulse, which is usually less encumbered with celebrity. Jay Ratican and James Hutson examined smaller-scale Digital Necromancy, and the results were interesting. Their study

...addresses the need to study the ethical considerations and potential implications of digital necromancy, which involves using AI to reanimate deceased individuals for various purposes. Reasons for desiring to engage with a disembodied or bodied replica of a person include the preservation of memories, emotional closure, cultural heritage and historical preservation, interacting with idols or influential figures, educational and research purposes, and creative expression and artistic endeavours.⁶³

They also discuss celebrity cases, as they are well published. However, it is the smaller scale of this that is much more revealing of the weaknesses of Digital Necromancy, as well as the complexity of the ethics of such practices. The ethical concerns centring on questions of Privacy, Identity Theft, Commercialization, Consent, Integrity of Thought, and effects on the Grieving Process.

61 Joe Taysom and John Taylor, 'How did The Beatles create their new song 'Now and Then'?' *Farout Magazine*, 2 November 2023, <https://faroutmagazine.co.uk/how-did-the-beatles-create-their-new-song-now-and-then/>

62 Ben Beaumont-Thomas, The Beatles set record 54-year gap between No 1 singles as *Now and Then* tops UK chart, *The Guardian*, 10 November 2023, <https://www.theguardian.com/music/2023/nov/10/the-beatles-54-years-no-1-singles-now-and-then-uk-chart>

63 James Hutson and Jay Ratican, 'Life, death, and AI: Exploring digital necromancy in popular culture—Ethical considerations, technological limitations, and the pet cemetery conundrum' *Lindenwood University Faculty Scholarship*, 478, 2023.

The Un(ethics) of Digital Necromancy

Privacy concerns are fairly obvious—in order to make a digital replica, material must be collected to sample and process.

'The use of personal data, such as social media posts, emails, voicemails, and even biometric information, raises questions about the extent to which individuals' privacy is being respected.'⁶⁴ Consent, Identity Theft, and Commercialization share negative interests—by operating outside of modes of consent, identity theft of the decedent blurs into an actuality, and the exploitation of this becomes the basis of commercialization. This is exacerbated by contemporary media contracts. For example, the pop musician Prince was in contract with Warner Bros. Records. In return for money, publishing, support and marketing, Prince had signed over not only his publishing rights to Warner Bros., they also owned his name. Any good wizard or exorcist will tell you that the first thing you need in order to control a demon is their name. Warner Bros. owned not only his music, but they also owned his name, Prince. As a consequence, Prince changed his name to a sigil. As it had no actual name, title, or phonological ontology, Prince became known by a paralipsistic identity: *The Artist Formerly Known As Prince*. (AFKAP).

Sigils are an important part of the Western Esoteric Tradition and other magickal rituals. A sigil is a pictorial symbol that has attributions of supernatural power or value. They are used in magic rituals and divination. Their production has been a creative aspect of these kinds of spiritual research and inquiry. Now you can go online and have a computer make one for you.⁶⁵ Sigil production is systematic—the vowels are dispensed with, and the consonants are arranged in a pattern, and the path around these letters is traced out, resulting in the shapes we see. However, the systems and processes of this vary greatly.

Some are not very systematic and more intuitive, giving an arbitrarily large number of possible arrangements. In any case, Prince was not only doing an end run around the perfidy of Warner Brothers and their onerous contract, he did so using an ancient magickal routine of denying them power over him by way of his name, by effectively dodging the phono and logocentrics of traditional names through the ruse of a sigil. Unfortunately, he still owed Warner Bros. a few records. Being prolific, he pulled some music out of his extensive vault and gave it to Warner Bros. for release. The company didn't like this, and litigation dragged out. Eventually, Prince won his name back from Warner Brothers, with the conclusion of his contract, and he soon changed his name back to Prince and abandoned his sigil identity symbol.

64 A. Alkis and T. Kose, 'Privacy concerns in consumer E-commerce activities and response to social media advertising: Empirical evidence from Europe.' *Computers in Human Behavior*, 2022; 137: 107412.

65 Sigil Engine, Online Sigil Maker, 2024, <https://www.sigilengine.com>

Do the Dead Have a Say? Can the Dead Consent?

Consent becomes a significant problematic issue for Digital Necromancy. Again, capital intrudes.

A hypothetical illustrates this: you have two people—Tupac and 'Joe Bloe'. Tupac died long before Digital Necromancers electronically raised his visage and voice from the dead. However, the Digital Necromancers had the sense to work with his estate to use his images and recordings to assemble his resurrection for Coachella. Fair enough.

Our hypothetical 'Joe Bloe' is just a guy. He worked at a job his whole life. He recorded his thoughts, some of which dealt with car mechanics, to a digital voice recorder. He had a number of digital video recordings of himself and his family, as well as several albums of photos, many of which he is in. This is sufficient to develop a Digital Necromantic Avatar. Let's say his neighbour, we'll call 'Dave', really misses him. Joe Bloe's estate executor is his elderly cousin, 'Nancy'. Dave meets Nancy was at the funeral and asked if he could have the photos and recordings to remember him by. From these assets, he creates a digital avatar of his dear and very dead friend, Joe. Joe can talk to him and lecture him on his ideas about automobiles and their repair. Joe and Dave bonded over this many years ago, when Joe helped him repair his Studebaker Lark.

This is all done without Joe's consent, because, well, he's *dead*. Joe was a private citizen. He cared about his dog Rover, drinking beer with his friends down at Bill's Bar, losing at poker, and fixing cars. The necromantic avatar of Joe, '*Joe*', is a screen object, not a 3D monstrosity like Tupac. However, the interactions can be recorded, and after some time, Dave does that and starts uploading his conversations with '*Joe*' to a YouTube channel.

None of this Joe consented to prior to death, and he certainly isn't complaining now. *Dave is able to exploit 'Joe'*. Then the scaling power of capital begins to affect the perception of this exploitation. If it stays small-scale, no one will notice or care. If someone does notice, they might tell Dave, 'this is interesting, and kind of wrong.' If Dave gets a few million subscribers on YouTube and is raking in thousands of dollars a month on AdSense, then someone might give Dave a lot of grief for exploiting the dead. Nancy might even find out and demand a cut. *Capital intrudes*.

This follows on to matters of Thought Integrity, and this is already a significant issue with Deep Fakes. One could use Deep Fake technology to insult the Thought Integrity of living people—imagine a Deep Fake Avatar of an elected official going on a racist rant demanding a planet paved with the dead bodies of [scapegoated people du jour]. That could be catastrophic. With Digital Necromancy, one could also have philosophers spouting ideas alien or antithetical to their thinking and published output, or exploit them by having them advocate their ideas as if they were still alive.

There is something much like this today. For example, information-warfare.com has used such videos on Facebook. Famous personalities like Abraham Lincoln⁶⁶, Malcolm X⁶⁷, and Hannah Arendt⁶⁸ is animated and used for the political ends and interests of information-warfare.com. These videos are bizarre and incongruous - Arendt spoke English with a posh British accent. However, it is a photo-animation of these people speaking, accomplished using an AI like vozo.com.⁶⁹

The Natural Progression of Grief

The grieving process varies from culture to culture. That our culture seeks to disrupt it with digital tools only makes sense.

Hutson and Ratican point out

...resurrecting the likeness, voice, and personality of a deceased person using AI raises questions about whether they have given their explicit consent for such recreation during their lifetime⁷⁰

because

...the consent of their family members or legal representatives becomes crucial in determining the appropriateness of recreating the deceased individual.⁷¹ Respecting the wishes of the deceased and their family members is essential to ensure that their autonomy and personal choices are upheld in the digital realm.⁷²

They also point out that there are potential detriments to the practice for the grieving,

The implications for the grieving process are equally profound. The confluence of these technological innovations offers the possibility of keeping a loved one alive in perpetuity, ranging from preserving their voice to creating a fully interactive replica.⁷³ While this

66 Information-Warfare, Abraham Lincoln video, *Facebook Reels*, <https://www.facebook.com/reel/735305719453135>

67 Information-Warfare, Malcolm X video, *Facebook Reels*, <https://www.facebook.com/reel/9926694447444654>

68 Information-Warfare, Hannah Arendt video, *Facebook Reels*, <https://www.facebook.com/reel/539242905782270>

69 Vozo.ai, 'Animate any portrait into talking videos with ease. Bring pictures to life with vivid expressions, natural gestures, and realistic lip sync.' <https://www.vozo.ai/talking-photo>

70 A. Hagerty, I. Rubinov, *Global AI ethics: A review of the social impacts and ethical implications of artificial intelligence*, New York: Cornell University; 2019, doi: 10.48550/arXiv: 1907.07892.

71 I. Berkowitz, JR. Garrett. Legal and ethical considerations for requiring consent for apnea testing in brain death determination. *The American Journal of Bioethics* 2020; 20(6): 4-16. doi: 10.1080/15265161.2020.1754501.

72 Hutson and Ratican, *Ibid*.

73 F. Allison, B. Nansen, M. Gibbs, et al. 'Bones of contention: Social acceptance of digital cemetery technologies.' *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*; 2023 Apr 23–28; Hamburg, Germany. New York: Association for Computing Machinery; 2023. p. 1-17. doi: 10.1145/3544548.3581520.

may provide a sense of comfort and solace for some individuals, it also raises complex questions about the nature of grieving and acceptance of loss.⁷⁴ Continuously interacting with a digital representation of the deceased may hinder the natural progression of grief, inhibiting the necessary emotional healing and adjustment to life without the physical presence of the loved one. Lindemann noted as such in her study on these so-called "Deathbots".⁷⁵ The prolonged connection to the deceased through technological means may delay the necessary process of letting go and moving forward.⁷⁶

This is an interesting juncture in this paragraph, as it makes a claim of naturalism in grieving; '*the natural progression of grief*'. What exactly is a natural progression of grief is questionable, especially if the rituals that centre the grief in a given society are so varied. Even in a 'western standard' process, there is great variety, from the Jewish practice of 'sitting shiva', a weeklong funerary practice which provides a time for mourners to engage in some spiritual and emotional healing, to Alkaline Hydrolysis/Resomation, which uses heat, chemicals, and pressure to reduce the corpse to a powder, a process that can be started the moment after the death certificate is signed.

There is also the example of a kind of science-based wishful thinking towards a high-tech bodily resurrection with cryogenics.

In the ideology of the cryogenic, as the decedent is frozen, they are seen as not being truly dead, as much as being merely in a frozen suspended state waiting for their disease to be cured by some future advancement in medicine. Once one shifts to non-Western cultures or just farther back in time, things become even more wide-ranging—Tibetan Sky Burials, where the corpse is left to be food for the birds, or, Viking burning boat funerals, where the decedent is put on a boat which is then set afire and sent off into the water to burn and sink. In south Asia, there was the practice of Suttee, also known as Sati, where the wife of the deceased would attend her husband's cremation and throw herself on her husband's funeral pyre. This is now (thankfully) a discouraged practice.⁷⁷ Almost 10,000 years ago, it was common practice in the Near East to collect the skulls of the dearly departed and cover them in plaster and paint.⁷⁸

At the ancient settlement of Çatal Höyük, a plastered skull (cranium and mandible) was discovered cradled in the arms of an old adult female....The eyes and nose were carefully modelled and the plaster was painted with red ochre.⁷⁹

74 IA. Aksar, A. Firdaus, SA. Pasha, 'Virtual vs. real self: Gendered presentation and everyday performance of virtual selfhood—A case study of Pakistan.' *Journal of Communication Inquiry* 2023; 47(1): 84-114.

75 NF. Lindemann, 'The ethical permissibility of chatting with the dead: Towards a normative framework for 'Deathbots', *Publications of the Institute of Cognitive Science*; 2022. doi: 10.48693/67.

76 Hutson and Ratican, *Ibid*.

77 Soutik Biswas. 'Sati: How the fight to ban burning of widows in India was won.' *BBC*. 23 April 2023. <https://www.bbc.com/news/world-asia-india-65311042>

78 H-D. Bienart, 'Skull cults of the prehistoric Near East.' *Journal of Prehistoric Religion*. 1991. 5:9-23

79 Scott Haddow, *Çatalhöyük 2012: Skull retrieval, curation and redeposition*. Wordpress. 24 August 2012. <https://scotthaddow.wordpress.com/2012/08/24/catalhoyuk-2012-skull-retrieval-curation-redeposition/>

With such a variety of funerary customs, it is difficult to imagine a *natural* baseline standard for grieving. We do know that the death of a loved one is a difficult experience, as noted earlier, the passing of a loved one is difficult, even for animals other than humans.

Digital Necromancy: Capitalizing Grief in a Technological Society

So, what is it to say that Digital Necromancy is somehow a worse way of dealing with grief than throwing widows onto the fire? Is Digital Necromancy, something we would do as a matter of course as we live in a technological society? People once mailed letters to each other, and the mail was delivered a few times a day. Then the telephone was invented. Then email. Then texting. Now we're in a situation where a recent Canadian conservative government wanted to suspend door-to-door mail delivery altogether. This plan was abandoned in 2015 with the election of a Liberal government.⁸⁰ While such a suspension of delivery is an extreme vision of possible CanadaPost operations, it follows on from the downward trend in mail delivery. Since 2006 and the advent of digital messaging systems and social media, the traditional post has had reduced importance.

Why would grieving be any different?

In American law, corporations are granted the rights of personhood. The fiction of money and the fiction of ownership are granted legal personhood. Abstractions induced by language are granted the parity of personhood. Ghosts are people, too. And when you resurrect them from the grave and make them dance on stage for your entertainment, you'd better pay the personhood machinery of property, because in this world, under capital, where all that is real melts into air⁸¹, the fictive realms of the imagination lay claim to our reality to 'slip your hand, from my hand, and leave you dancing with a ghost'⁸², a mellotronic imitation of the living, a proposal of a paritant relationship between the living and the dead.

And the ghost, the spectre, of interest is instrumentalized and reduced to a tool to assuage our grief or amplify our entertainment. To the Kantian Categorical Imperative, this borders on a distinct and immoral condition. 'So act that you treat humanity, whether in your own person or in the person of any other, always at the same time as an end, never merely as a means.'⁸³ A necromantic production of the dead cannot be done without the dead's prior consent when living. So in that sense, it's immoral. However, it could not be done against their protest, either: *they're dead*. In any case, one has *instrumentalised* their death.

80 The Canadian Press. *Canada Post suspends community mail box program, affecting 460,000 homes*. 27 October 2015. CTV. <https://windsor.ctvnews.ca/canada-post-suspends-community-mail-box-program-affecting-460-000-homes-1.2630221>

81 Karl Marx and Frederick Engels, *Manifesto of the Communist Party—The Principles of Communism*. Foreign Languages Press, 2020, p. 36.

82 St Vincent, 'Slow Disco' from *Masseduction*, CD, 4AD, 2017.

83 Mary Gregor, trans. Immanuel Kant. *Groundwork of the Metaphysics of Morals in Immanuel Kant: Practical Philosophy*, 1785. trans. Cambridge, 1996, p. 429

The person is dead. The avatar is only an instrumentalization of their appearance or some aspect of their person. You could go to their grave, piss on it, set their effigy on fire and scream about their sins and transgressions and make the avatar do all kinds of rude things. The dead won't care. They're *dead*. They don't exist. They can't complain. One might argue, 'well, that's a bit sociopathic'. Is it, or is reverencing the wishes of the dead the way we do simply a function of the arbitrary and particular social/historical approach we take to the deceased? The counter could be the skull collections of Catal Huyuk—skulls of their family were collected and painted with plaster and colour.⁸⁴ They thought it was normal, while we moderns would consider it sociopathic, if not just weird and creepy.

And that points to a problem of much of this discourse—it's bound up with the standards and ideologies we have and maintain today. Also, as the standards are quite variable, especially around the influence of capital, as discussed earlier, and around matters of the memories of public versus private persons, we have a distinct problem inherent to the process of Necromantic resurrection, whether occult or digital. The image of George Washington and Abraham Lincoln are regularly exploited in the advertisements for car sales on President's Day weekend in the USA. Why can't information-warfare.com also do it, only using video? Pictures of many religious figures are used for commercial purposes, with the exception of the Prophet Mohammed, of course, and this is only due to the extreme sensitivity of the Muslim community. If that wasn't a taboo, it would be fairly close to a certainty that some skeezy used car dealer would use an image representing Mohammed to sell Pick Up Trucks over Ramadan. Capital intrudes. *Always*.

With AI, one could build an AI that has expert knowledge of the Bible, and produce a Jesus avatar who could then digitally sermonize in computers and/or VR spaces around the world nearly simultaneously. This digitally necromanced, electronically resurrected Jesus could absolve sins and provide moral guidance and therapy. This would make the Confession Stations in the futuristic sci-fi film, *THX1138*, seem primitive by comparison.⁸⁵

Unreal Engine: A Hyper-Realistic Tool

This realism has been amplified by recent developments in CGI technology, typified by the Unreal Engine.⁸⁶

Side note—An interesting aspect of the Unreal Engine is that it is free to download and use. One only pays for it if one grosses more than 1 million dollars from its use in Games and Video. If one grosses that much in sales, Unreal requires a percentage of the sales. So if one is using Unreal in a commercial way, then one must build in the expense of the percentage into one's business plan. At the same time, allowing non-commercial use cases, student or amateur use cases, is a smart move.

Adobe Systems learned a long time ago that professionals will use the tools they know, and the tools they know are the ones they used in school. When I worked at Macromedia in the

84 Bienart, H-D. *ibid*.

85 *THX 1138*, Dir. George Lucas, Film, American Zoetrope / Warner Brothers, 1971.

86 <https://www.unrealengine.com/en-US>

mid-1990s, we were constantly battling with Adobe on this front. Adobe later absorbed Macromedia, and one of the reasons they succeeded was that Macromedia failed on the educational front. In 1996, everyone knew that Freehand was better than Illustrator—there were even YouTube videos about this.⁸⁷ Yet, Illustrator was the dominant vector-based program—because people used what they knew, and Illustrator was integrated into the Adobe software system, software that was *marketed directly to educational institutions*, making it virtually 'free' for students to use. Unreal is using a similar strategy—give the engine away for free, let people learn it, sell them assets as they need, and if their Game/TV Show/ Movie sells, THEN charge a significant percentage of the gross. Everybody 'wins.'

Unreal is a visual development environment for games and video. With version 5, it took a great leap in terms of realism and precision. Within the frame of a screen, the results are difficult to distinguish from actual footage, even of people. Announced at the end of 2021, and available in early 2022, a demo video of version 5 (*Unreal 5*) was released featuring Keanu Reeves and Carrie-Anne Moss of the film, *The Matrix*.⁸⁸ The demo video makes a number of vague and problematic statements, which we'll look at. At the same time, the clarity and precision of these virtual characters is undeniable.

In the video, a present day/older Keanu Reeves wanders about a white soundstage, pontificating around frozen replicas of himself as the younger character 'Neo'. Then Carrie Anne Moss also appears, like Keanu, as she looks in the present day, and then she and Reeves are suddenly in a Chevy Camaro racing around a city similar to New York or Toronto, or Los Angeles, with skyscrapers, elevated highways, and similar architecture. A variety of 'bad guys' dressed in suits chase them about. There are many shots fired and massive explosions, as one would find in an action movie or game. In the middle of the chase scene, a 'new character' appears who is an artificial creation of the *Unreal 5* engine and is animated to engage the 'bad guys'. At that point, Keanu flies off into the clouds like a digital angel in a trench coat, while Carrie Anne and the new character fight the Bad Guys.

In the beginning of the demo, there is this exchange:

Keanu Reeves—Hi! I'm Keanu Reeves. Over twenty years ago I first played the character of Thomas Anderson the Matrix trilogy. Those films pioneered digital cinema with shots like "bullet time". Back then we talked a lot about where the digital age might take cinema narrative. In an industry where actors have tried to remain perpetually young, we wondered about digital faces, that could become immortal.

Carrie Anne Moss — Hi! I'm Carrie Anne Moss, and I played Trinity in the Matrix films. Twenty years ago we asked ourselves how long it would be before faces and bodies could be changed as easily as changing clothes.

87 For example, 'Freehand vs Illustrator Round 2', *Ipixero*, YouTube video, 19 December 2009. <https://www.youtube.com/watch?v=2qZgT3fnwOU>

88 *The Matrix*, Directed by the Wachowskis, Warner Brothers, 1999, Film.

Reeves—we wondered, what would identity mean in a completely digital world?

Moss—and what would reality mean when a world we could build feels as real as our own?⁸⁹

While the above text is basically copy text to give the characters in the demo something to say, it does ask some questions and points at some inferred discourse that came out of the Matrix films.

First a discussion of the latter, pointed at by Carrie Anne Moss—*what would reality mean when a world we could build feels as real as our own?* This is a kind of dog-whistle reference to the 'Simulation Hypothesis' cooked up by Nick Bostrom of Oxford University. This is sometimes referred to as 'Matrix Theory'. Matrix Theory is also a quantum mechanical model in string theory physics proposed by Tom Banks, Willy Fischler, Stephen Shenker, and Leonard Susskind in 1997, and has nothing to do with the Simulation Hypothesis. Even though String Theory has come under serious criticism (for being untestable, among other things) it is an elegantly logical and mathematical proposition, and I won't sully it by referring to the Simulation Hypothesis as Matrix Theory.

Is anything real? The Stimulation Hypothesis

The Simulation Hypothesis suggests that what humans experience as the world is actually a simulated reality, such as a computer simulation in which humans themselves are software constructs. The lead advocate of this theory, Nick Bostrom, wrote:

Many works of science fiction, as well as some forecasts by serious technologists and futurologists predict that enormous amounts of computing power will be available in the future. Let us suppose for a moment that these predictions are correct. One thing that later generations might do with their super-powerful computers is run detailed simulations of their forebears or of people like their forebears. Because their computers would be so powerful, they could run a great many such simulations. Suppose that these simulated people are conscious (as they would be if the simulations were sufficiently fine-grained and if a certain quite widely accepted position in the philosophy of mind is correct). Then it could be the case that the vast majority of minds like ours do not belong to the original race but rather to people simulated by the advanced descendants of an original race.⁹⁰

This, of course, is wrong for one simple reason: it leads to an infinite regression. Simply, Bostrom suggests that humans are a product of an advanced alien race.

89 *The Matrix Awakens*: ibid.

90 Nick Bostrom, 'Are you living in a computer simulation?' *Philosophical Quarterly*, 53 (211), 2003, pp. 243-255.

Let's call them Aliens(1). There is nothing to assume from his argument that the conditions aren't indefinitely scalable and don't apply to the Aliens. This means that there is a species, Aliens(2), that created Aliens(1) who created people in their simulation. Which also means there is a species Aliens(3) that created Aliens(2) that created Aliens(1) who created people in their simulation. This proceeds and regresses infinitely. As this chain of reasoning has no end, we have zero reason to believe the Simulation Hypothesis is true; in fact, there's no reason to believe any of it. It is this author's conclusion that this conjecture is simply idle fantasy masquerading as serious thought, and the fact that it has captured as much ink and mind space as it has is a sad commentary on the gullibility of people one might otherwise think of as rigorous and intelligent critical thinkers. Carrie Anne Moss rhetorically asking '*what would reality mean when a world we could build feels as real as our own?*' is dismissible.

On the other hand, Keanu Reeves asked a much more interesting question, '*what would identity mean in a completely digital world?*'⁹¹

While such a question is even more outside the bounds of this discussion than the Simulation Hypothesis, it does point to some issues within our scope.

For example, what exactly constitutes a person? Who is being raised from the dead? There is the obvious: flesh and bone—this is something that exceeds the abilities of any digital necromantic practice, or, indeed, any necromantic practice, outside (of course) the Christian Mythology of a material Jesus raised from the dead and exposing his body and wounds as materially extant to his followers. The best a necromantic practice can do is present spectral figures (hallucinations) or lucid dreams, that are available for interaction with the conjuror. *These would be based on the memories of interaction by the grieving toward the dead.* There is, of course, a very long tradition of necromantic practice from prehistoric shamanism to Victorian seances, to spiritualist churches, ecstatic religions, etc. One might dismiss these as hallucinations to simple trickery and fraud, and one wouldn't be wrong; however, an ethical framing of this isn't useful here.

Our only understanding of another person is through our sensoria and the effects of our sensoria on our memory, and it is this variability in sensorial information and processing within memory that conditions our understanding of that which creates contrasts in our sensoria, such that we, with our linguistic capacity, then define and *accept* a set of sensoria as one particular thing vs. another, and then store these responses as memories, which we then reference for the explication of necromantic results, be they fraudulent or coincidental or otherwise.

We may know someone for decades, and yet, all we know about them is what we have perceived about what they have presented to us within the context of our social and linguistic systems and our memories of what they were. We remember what they looked like. We remember how they sounded—the tone and accent of their voice. We remember their smell. We remember their clothing. We remember their laugh, their turns of phrase. We remember the memories they shared with us. We remember their particular, and possibly quite quirky, ways

91 The Matrix Awakens: Ibid. 2:20

of thinking or approaching a problem. With intimate partners, we remember their bodies, their smell, their presence. We might remember them as compliant, or stubborn, garrulous or quiet, moody or steady, stoned or sober, or, more likely, as a delicate composition of all of the above. And when that person passes on, it is that generative configuration of sensations that we miss, that—when taken as a whole—we remember as that person, and the joy and companionship or, if they were nasty, the misery and abjection their configuration and memory brought to us.

The Reality of Death...

Pretend that a séance actually allows one to commune with the dearly departed.

For example, I might consider a séance for Jodi. How would I know it was the ghost of Jodi appearing before me? Does it look like her? That would be a basic requirement. If the visage of President James A Garfield appeared and said it was Jodi and that it loved lasagna, I would think someone with superior memory of a stoner comedy film was just messing with me,⁹² much more than believing it was Jodi. Jodi was not a stoner. So, it would need to have visual accuracy and content logic relative to the subject, Jodi. She had a distinctive voice—sweet and gentle, with a bit of a Michigan accent. A nice laugh, sometimes a characteristic chuckle or a stifled snort. Hearing that would go even further to convince me she was hovering above the table, or Ouija board, or Tarot deck, whatever. If she talked about stuff that only she and I know about, personal experiences we had together, that would seal the deal—I would have to conclude that indeed, Jodi had come back from the dead to tarry with us for a few minutes. We could say that the séance was real and that she was with us.

However, that is not how most séances go—usually it's a rapping sound on a table or a Ouija board or some other abstracted *medium* through which the dearly departed are said to communicate with the living – media that are easily manipulated. The amount of fakery and grifting in this sort of thing has been well established over the years. At this point, one must be either very gullible or desperate from grief to *believe* such actions are actual communications with the dead. That said, as noted earlier, people can and do vividly hallucinate the image and sound of dead loved ones, and these hallucinations can be quite convincing. And usually, *'that's good enough.'*

Suddenly, at this point in my writing, I have another lacuna for our story—another moment of death.

...

92 From the film *Smiley Face*, dir. Greg Araki. (2007) Anna Faris as Jane F.:

That is where corn chips come from. Hmm... Maybe ol' Professor Hardwood is onto something. He probably really loves corn. And all corn-related products. I mean, isn't that what you're supposed to put in a frame? Things you love? I'm gonna do that. When I'm get home, I'm gonna frame a bunch of stuff I love. Like lasagna. I *love* lasagna. It's SO good. And cheesy. You know who else loves lasagna? Garfield. Man, that cat really loves lasagna. Maybe I should put a picture of Garfield in a frame. You know, as a kind of shorthand way of saying 'I love lasagna.' That would be so fucking inside. Or how about a photo of *President* Garfield? Oh shit, that would be totally meta! People would be all like: Jane, why do you have a photo of President Garfield on your mantle? And I'd be like: Because I like lasagna, of course.

I travel to New Jersey a few times a year, ostensibly to visit my dad, who is 99 years old and living in an assisted living facility. He's not very happy about it, and he appreciates my visits. My mother passed away some 20 years ago, my siblings and I scattered around the planet—I have brothers on the West Coast, one in the Rockies, and one in the Midwest. My sister lives in Cornwall in the UK. I am physically the closest to him, even though I live in a different country (Canada). Because of the distance and expense, and time involved with travelling to New Jersey, I'm the only one who visits him regularly. And every time I come to Edison, I would always visit with my friends, some I've known since grade three, others of a more recent vintage.

One friend I would always visit is John Hawken, whom I mentioned in the previous section. He and I have a mutual love for a particular café, *The Coffeehouse* in Edison, NJ, and when I come to visit, we will often have coffee and sandwiches or bagels in *The Coffeehouse*, or we'll go to the *Metuchen Diner*, or *TGI Fridays*, or some other place where we can share dinner. Then we would usually end up back at his place, and he would regale me with the stories of his career. He's had quite a career as a keyboard player, starting with The Nashville Teens in the 1960s, and then in the 1970s with Renaissance, The Strawbs, Armageddon, and a variety of live and studio gigs with the likes of Spooky Tooth, among others. In the late 70s, he married and moved to the United States and had another career working with computers. I met him around 2014 by way of Facebook.

My recent visit wasn't any different. We met at the café and had coffee and bagels. He was with a friend, Diane, and we had a jolly good time catching up and talking. I knew he wasn't well—he had been battling cancer for several months. Still, he seemed happy enough, and his wit was undimmed. I told him about what I'm writing here, and he was intrigued—he thought it was brilliant and wished me luck in my efforts. Before he left, we took a photo of ourselves on my phone, he put his arm around my shoulder and pulled me over for a picture. It was so nice to see him. He said we should get together next time I'm down, and I promised him I would ring him up the next time I'm in town.

He died a week later, the day before I wrote these lines here, and I'm fairly certain that was the last photo taken of him. His passing was sudden. I did not expect it. It threw me for a loop as my writing/ necromancy/vampires/Frankenstein's Creature/demonology/dead gods / Mellotrons/writing about virtual death is hard when surrounded by actual death or reminders thereof. My flabbers were, again, *gasted*.

...

Whether it's Tupac Shakur rising from the dead at Coachella, or Abba touring their youthful selves around the world, the paritance of Digital Necromancy is obvious and clear. There is money to be made on the grief and memories of people unwilling to let go of their heroes. And as Brian Eno noted in one of my favourite songs of his, '*Heroes are born, but Heroes die*'.⁹³ It can only ever be thus. People have tried to keep loved ones and heroes in their lives for millennia. It might be on a personal level—keeping their ashes in an urn. It might be on a

93 Brian Eno, 'Mother Whale Eyeless' from *Taking Tiger Mountain (by Strategy)*, Island Records, 1974.

national level, where VI Lenin's corpse is still on display over 100 years after his passing. This affective urgency lies at the heart of the Necromantic impulse. Also, as noted by Honorius, the necromancer can be conjured to give advice or some advantage to the living. Under capital, all of it can be true.

This Digital Necromancy is the first step, the most humanly relatable step, from the paritane melotronics of the 20th century. The acceptance of recordings of music for music, and then the instrumentalization of these recordings to make more music was a preliminary to the digitally Necromantic, where electronic ghosts of the dead are brought to our world to dance and entertain us once more. Evocation and Demonology are a very short jump from the Necromantic.

Full disclosure: I am not a fan of Digital Necromancy—I think it's a waste of resources and would not want to be remembered in such a way, nor would I want to remember others in this manner. I don't think checking in with a virtualised or resurrected 'Jodi' or 'John' a few times a year is useful to me, and given that I am constantly learning and shifting my vision and understanding, I doubt that any digital avatar of me could possibly be of value to the world. After passing, if someone wants one of me, well, after I'm dead there is precious little I can do to stop people from making one of me, so, on the other hand, saying 'don't do that' is absurd. The idea that the dead can consent to anything I find absurd, and the more that people adhere to this notion, the more absurd it becomes. As discussed earlier, the vector of force in this discussion devolves on questions of celebrity and money. If you have both when you pass on to the ethereal realm you are more 'real' and 'substantial' in a world driven mad by wealth and property than you are if you're a poor broke nobody.

PART 4: DIGITAL EVOCATION AND DEMONOLOGY

Introduction

In this volume, I am drawing a difference between Digital Evocation/Demonology and Digital Necromancy.

In the previous section on Digital Necromancy, we saw how spiritualist affect went into conditioning some ideas about electronic technology, how this was exploited to virtually bringing loved ones back from the dead, and how this was scaled and amplified into a public spectacle.

What we have is, as with Digital Necromancy, digital media imitating and making materially manifest what was previously the result of an occult ritual that would normally reside in the world of dreams, hallucinations, and visions. Again, instead of conjuring actual spirits from the void, we are 'conjuring' 'spirit' creatures into an electronically mediated visual and auditory space.

This takes on different configurations. With Necromancy, as noted, dying first is required to be brought back from the dead, which is only logical and necessary. With Evocation, *these creatures were never alive in the first place*. Different religions and cultures have different angels and demons, and these are all written about with different approaches and prejudices. These differences and nuances aren't of value here, as we are examining Digital Evocation, which can be used in any culture that has sufficient computer infrastructure.

Demonology and Evocation Are Different From Necromancy

Note to the reader: Demonology is the summoning of Demons, and Evocation is the summoning of Angels. Rather than repeatedly state 'Evocation and Demonology' in this volume, 'Evocation' will do for both.

The most fundamental similarity between necromancy and demonology is that they are both involved with summoning spirits from the Great Beyond, as it were. In this way, they are very similar. However, Evocation is actually a different game from Necromancy altogether. Unlike Necromancy, where the dearly departed who were born, lived, suffered, and died are called back to the material world by the living. With Evocation, there is a summoning of the spectral—demons, angels, archons, archangels and similar spirit creatures. There is some similarity with Necromancy; however, there is a very big and decisive difference: the spiritual subjects of Evocation and Demonology *were never alive in the first place*.

Necromancy, as indicated by the root word fragment, Necro, indicates the Dead. Necromantic rituals are used to bring the spirits of the dead back into contact with the living. To do that, a living being must first die. With Evocation, not so much...

I spoke earlier on *The Sworn Book of Honorius of Thebes*, and having read the book, it was quite a peculiar reading experience. I could go into extensive detail about all of its detailed and complex rituals and literature; however, others have done this, and better than I can. I would suggest it is well described in a section by Justin Sledge in his analysis of *The Sworn Book*.

... the bulk of The Sworn Book is composed of various prayers, litanies, psalm-like texts. And in this way it's kind of like an occult Book of Hours, if you're familiar with that genre... which are recited over a seventy-two-day period, though in the Summer Extension, it's 40 days... it includes periods of fasting, purification, penance, meditation, and lots and lots and lots of going to Mass. Following the initial purification, three days are spent in Mass, in prayer and fasting, in preparation for an angelic vision, which will reveal whether one should proceed through the rest of the ritual. You might need more purification and penance and start the process over briefly. If you get the angelic green light in a dream, the next 17 days or so are continued with further shifts in the fasts, additional prayers from The Sworn Book, and again, continued daily Masses and consumption of the Eucharist. With this completed, another set of prayers is then initiated, which follow upon the one hundred names of God. In addition, the priests officiating the Mass must be sympathetic to you at this point, because certain prayers from The Sworn Book of Honorius must be inserted into the Eucharistic ritual itself. This is not exactly a 'run of the mill' request. In fact, it seems crazy to request a priest to do magick with you. But that's how it has to be. The Magus now retreats to an increasingly isolated location, with more fasting and prayers now, including some biblical psalms, now praying three times a day, for the final seven days of the ritual. On the final evening of the ritual, the prayer pattern is continued, but now a bed of straw is prepared, the Magus is ritually washed in cold spring water, he puts on a flesh mortifying hairshirt and covered in black robes, before following in Vespers, he retires for one final extremely elaborate prayer, and – go to sleep. If it all works out, the Magus will be given a vision of the Celestial Palace, the Majesty of the Divine Glory, he'll get to witness the Nine Orders of the Angels... and then get to hang out with the Spirits of the Blessed.¹

The above description is more like a bullet-point description of the massive collection of incantations, prayers, rituals, meditations, psalms, penances, fasts, and purifications in *The Sworn Book*. To fully appreciate the complexity and intensity of *The Sworn Book* here is impossible and, to again quote Sledge, 'I'm not going to go, step by step, through the extremely complicated ritual—that's called 'read the book for your own damn self!'² His description above suffices to show that the Necromantic/Evocation ritual in *The Sworn Book* is quite complex, and seems, frankly, to be an utterly exhausting ritual. Just reading the book was exhausting enough.

1 Justin Sledge, *Esoterica*, 'How to use Magic to see God & Conjure Spirits—The Sworn Book of Honorius—Liber Juratus', Youtube channel. 20 April 2024. 23:57—26:34, https://www.youtube.com/watch?v=OPz5_59gZSU.

2 Ibid, 23:46.

The Sworn Book as a manual for Evocation is elaborate, for even as the book (in the printing I have) alternates between English and Latin, it's still over a hundred pages of rituals and prayers per language. Ostensibly, the purpose of summoning such spirits from the vast aethereal spirit realm would be for some gain here in the mundane realm on this 'little green planet of clocks'.³ Whether it is for some divine knowledge for the sake of your own soul (however you might define that) or simply insight into how to pad your bank account with cash — that's not particularly important here.

Angels

Given the dominance of Christianity in the cultures that developed digital media, it might be useful to examine the ramifications, capabilities, and similar mythologies of the angels from such a perspective, which is dominated by Biblical writings and the secondary writings around it.

Angels (and their opposite, the anti-angels, or Demons) are personal, intelligent, willful, sentient, and emotional spiritual beings. Their intelligence is manifest—from deceiving Adam and Eve⁴ to many other complex exploits, including demonic possession.⁵ Angels are spirit beings without true physical corporeal bodies.⁶ While lacking physical bodies, they do have personalities and occasionally take on physical bodies—Jacob even wrestled with one.⁷ Because they are created beings, their knowledge is limited—they are not omniscient. However, they were created as an order of creatures higher than humans and innately possess greater knowledge and understand God's Will. Also, angels don't study the past; as eternal spiritual beings *they've experienced it*, and they know how people have acted and reacted in a variety of situations.

A common meme in social media was 'The Biblically Accurate Angel', which began with a satirical article in *Cracked Magazine* in 2010.⁸ These are based on the descriptions of angels in the Apocalypse of John⁹, and the books of Ezekiel^{10 11}, Isaiah¹², and Daniel¹³.

3 Douglas Adams, *The Hitchhiker's Guide to the Galaxy*, Random House, 2007.

4 Genesis 3:13, 'And the Lord God said to the woman, 'What is this that you have done?' And the woman said, 'The serpent deceived me, and I ate. Old Latin: Then the Lord God said to the woman, 'What is this that you have done?' And she replied, 'The serpent beguiled me, and I ate.'.

5 Matthew 8:28-29, When he arrived at the other side in the region of the Gadarenes, two demon-possessed men coming from the tombs met him. They were so violent that no one could pass that way. 'What do you want with us, Son of God?' they shouted. 'Have you come here to torture us before the appointed time?'.

6 Hebrews 1:14.

7 Genesis 32:22 - 32.

8 Bjorn Karhunen and Jacopo della Quercia, '5 Things You Won't Believe Aren't In the Bible: Updated.' *Cracked Magazine*, 19 October 2010.

9 Revelation 4:2-8.

10 Ezekiel 1:4-16.

11 Ezekiel 10:9-10.

12 Isaiah 6:2-3.

13 Daniel 10:1-21.

Such creatures are unusual to say the least, and a vision of such would be truly terrifying. To assuage the terror induced by such a visage, angels are more commonly depicted as humans with wings who play harps in heaven and sing praises and hymns to God. This imagery is derived from Greek mythology regarding the Muses of music, specifically Polyhymnia and Euterpe, the muses of sacred and profane music.

The 'profanity' of Euterpe is basically relational; Euterpe's profane music is only profane relative to that of Polyhymnia's sacred hymns. Euterpe's domain is that of lyric poetry—songs of emotions and human relations—Greek lyric poems would commemorate athletic victories, the dead, or soldiers in battle. Lyric love poems were much like love songs today: proclaiming one's love for someone, unfulfilled desire, seduction, or breakups. The themes of Greek lyric include 'politics, war, sports, drinking, money, youth, old age, death, the heroic past, the gods',¹⁴ and, of course, love. Euterpe's instruments are the drum and flute. What could be more profane than the organised butchery of humans in war? What is the musical accompaniment as they march off to battle? The drum and fife.

Polyhymnia's songs are sacred, and her instruments are the Harp and Voice – thus, the imagery of angels in heaven singing and playing the harp. Harps were common with the Muses—for example, Erato has been depicted with a lyre, Calliope a harp. In these depictions, even in ancient Greece, they were depicted as human. This is a much more comforting vision, and this was also true in Biblical texts and in contemporary imaginaries of angels, where angels would also present themselves as a human, if a special kind of human—glowing, unusual eyes or stature, etc.

Angels are of a converse order of being relative to humans. With one Biblical exception, Enoch,¹⁵ Humans don't become angels after they die. Contrary to the beautiful cinematic mythology of Wim Wenders's film *Wings of Desire*,¹⁶ angels never were, and can never be, human beings. Per accounts in the Bible, angels are spiritual beings that can, to a certain degree, take on physical form.¹⁷ An angel's primary responsibility is to do God's will, and that may include dealing with people, even wrestling with them.¹⁸

From the Undead to the Resurrected Dead to Machines to Evoke Digital Angels or Conjure Digital Demons

So far, this has been a discussion of life, life living with machines. We had machines slip into our world through representation as re-presentation. The representation becomes a presentation of its own and self. This gives the mimicry a magical inference of the thing it imitates. The Mellotron imitates an orchestra—the Musicians' Union pickets the Moody Blues for putting string players out of work. The imitation of the orchestra operates with the

14 Andrew Miller, *Greek Lyric: An Anthology in Translation*, p. 21, Hackett Publishing, 1996.

15 Genesis 5:21 – 24, Enoch walked with God and was seen no more. In Jewish apocrypha and early Kabbalah, 'Metatron' is the name that Enoch received after his transformation into an angel.

16 *Wings of Desire*, Dir. Wim Wenders, Film, 1987.

17 Numbers 22:31, Genesis 19:1-3, John 20:11-14, Mark 16:5-6, Luke 24:4, Acts 1:9-11.

18 Genesis 32: 24-30.

power of the orchestra. It is not real power—it is an assumed power based in fear of Dracula, the undead ghost machine who mimics and takes on forms in darkness. Recordings ceased to be mere records of events and became aesthetic objects of their own, viz. The Beatles' albums *Revolver* and *Sgt Pepper's Lonely Hearts Club Band*. Through a speaker, they sound Just Like a recording of an event, but an event that could never happen in 'real life', for example, 'Being for the Benefit of Mister Kite'¹⁹—the warbling spacey sound effects were derived from recordings of things, but then chopped up and played at a different speed and backwards. Time doesn't go backwards, so this would not be a possible live activity to be recorded—it could only be a live activity that features recordings.

The Recording achieved a status of its own as an instrument in the 1940s with *musique concrète*. Previously, we mentioned Pierre Schaeffer's landmark recording of *musique concrète* from 1948. Several years earlier than Schaeffer, an Egyptian, Halim El-Dabh had made a remarkable work of *musique concrète*, *The Expression of Zaar*.²⁰ Pertinent to our discussion here is that it was *composed of recordings of a public exorcism*.

Through decades of commodity capitalism, recognition of the recording as an entity of human labour and construction allowed for the expansion of copyright law to recordings, and to persecute companies like Napster for 'Stealing Music', *as if the recordings were the music*. This flattening is only possible through acceptance of parity. It was an error. This is *not* to say that Recordings Are Not Music. Music is a socio-temporal variable. It was letting 'The Recording Be The Thing It Mimics', like Dracula being a bat or a cat or a bird, this analogic and metaphoric shapeshifting, that permits us to *accept* Recordings As Music.

The journey from the undead to the resurrected dead in digital necromancy was a process that happened over several decades. Still, to be resurrected, you had to be alive and then die in the first place. *Beyond that, we now have the machines to evoke digital angels or demons*. Prior to this, hallucination and directed consciousness were the only means of contact, and this was accessed through massive sensory assault and ritual, as described by Honorius of Thebes.

Crowley's Conjuring Efforts

A less medieval example of Evocation can be found in the experiences of Aleister Crowley²¹ and his assistant, Victor Neuburg.

In 1909, they made their way by train and foot into North African desert southwest of Algiers. There, Crowley decided to perform a series of magical ceremonies that anticipated his ideas of sex magick that would later become important to his devotees in the OTO, the *Ordus*

19 The Beatles, 'Being for the Benefit of Mister Kite', *Sgt Pepper's Lonely Hearts Club Band*, Capitol Records, 1967.

20 Halim El-Dabh, *The Expression of Zaar*, 1944, A portion of this can be heard on his album *Crossing the Electric Magnetic*, Without Fear Label, 2001.

21 Note: I'm not a particular fan of Aleister Crowley. I think he was ingenious, certainly, however I also believe he was also a malignant narcissist with massive delusions of grandeur.

Templis Orientis. In the desert, these ceremonies combined the performance of intensive rituals, including sodomy. The intensity and the reported experiences of this trip are worth looking at, as it is this depth and intensity of experience around their attempts at Evocation underline and profoundly illustrate the complexities around the process and the meaning of Evoking spirits.

The magical operations that Crowley wanted to perform were based on the magical systems developed by John Dee, the astrologer for the Queen Elizabeth I, and his clairvoyant, Edward Kelley. Dee and Kelley were educated in the works of Henry Cornelius Agrippa and Honorius of Thebes and the mystical systems of the Kabbalah. Dee and Kelley said they had conversations with angels through these systems to reveal spiritual secrets of the universe. Through a process of Q&A, Kelley would find answers using the equivalent of a 'crystal ball'. These might provide instructions, which were transmitted in an 'Angelic Secret Language' known as Enochian. This was recorded in a complex grid form. From this, Dee constructed a cosmology of angels and demons and sketched out thirty realms of existence he called Aethyrs, or Aires. Crowley's intent was to ascend through these realms and achieve some sort of mastery in these realms.

Something to remember in all of this is that Crowley was more than a little bit of a narcissistic jerk. He had been tossed out of The Golden Dawn occult organisation, where Crowley had learnt of Dee, Kelley and the Enochian system. While Golden Dawn initiates and adepts carefully studied the Enochian system, Crowley wanted to test it and see if he could make it work for his own interests. With his own copy of Dee's nineteen Calls, or Keys, which called up powerful occult forces, in Algeria, he was ready to make it happen and journey through John Dee's Enochian Aethyrs.

Crowley's technique was structurally simple. He would select a secluded spot and recite a ritual incantation for access to a particular Aethyr. Once it seemed that the requested forces were present, Crowley would take up his magical shew-stone, a Topaz, and enter a specific Aethyr. In this Aethyr, he described his experiences to Neuburg, who wrote them down. As the two men made their way through the desert, Crowley evoked and experienced terrifying, vivid, and amazing celestial beings. These visions intensified as they continued their journey. After a fortnight of travel, they arrived at Mount Da'leh Addin. It was here that Crowley, based on instructions from previous angelic conversations, made the appropriate Call Ritual and attempted to enter the fourteenth Aethyr.

He was met with darkness and a voice telling him to enter 'the Kingdom of the Grave.' He struggled with the blackness and experienced a great shaking and then saw 'an all-glorious Angel' standing in front of him, arms outstretched and head thrown back. The angel instructed Crowley to go away because the 14th Aethyr cannot be revealed, and then Crowley received a command to perform a magical ceremony on the summit of the mountain. This ceremony consisted of Crowley being sodomized by Neuberg as a rite offered to the god Pan. The experience overwhelmed them both, and it devastated Crowley. This was not the end of it by any stretch. Crowley felt he had crossed a Nietzschean abyss. The Order of the Golden Dawn had taught that this was impossible without dying. The Angel of the fourteenth Aethyr

warned him that the Master of the Temple is condemned to darkness.

A few days later, Crowley prepared to undergo the ordeal of the Abyss in John Dee's tenth Aethyr, to face the great and terrible demon Choronzon. Crowley understood that Choronzon could be tamed by a steadfast application of his magic will, in order to cross the Abyss. Failure to defeat Choronzon was not an option.

On 6 December 1909, Crowley and Neuburg wandered into the desert and found a valley in the dunes, where they set up their efforts, tracing a circle in the sand with the names of God. They also traced a triangle with a variety of God names and Choronzon. The magic circle and triangle were common defences for magical operations in evocation. Their intent was to create a physical manifestation of Choronzon, the demon of the Abyss. With some pigeon sacrifice and other incantations, Choronzon seemingly possessed Crowley as Neuberg heard the voice of the demon speak violent ideas to him. In English, of course. Then Neuberg saw Choronzon sitting with Crowley, who sat silently in the sand. There were other manifestations Neuberg saw, including a woman he knew from Paris, a holy man and a serpent. Choronzon slowly eroded the edge of the safety triangle and then jumped up and wrestled with Neuberg. Neuberg fought him off and got him back into the triangle, repaired it, and he eventually disappeared. Crowley and Neuberg lit a fire of purification. Eventually, they made their way back to town. The experience was shattering for both of them, for while Neuberg was struggling with hell spawn Choronzon, Crowley had experienced all the misery and rage Choronzon embodied.

There's a great deal more to it—the above is a heavily edited and reduced description. All of this is detailed in Crowley's book, *Confessions*.²² It's clear from this that manifesting demons or evoking angels is a complex and affectively loaded affair. One might cynically scoff and say, 'Lies, they made it all up' or 'They were insane and hallucinating'. Both things can be true. As discussed earlier, intense affective overloads and disorders can cause people to have incredible experiences and hallucinations. What Crowley and Neuberg demonstrated was that the amount of effort they put into conjuring these spirits was enormous, and that the results were devastating for all concerned. As far as they were concerned, they had conjured a demon, and they had met an angel.

Digital Evocations

Conjuring digital spirits isn't (quite) as dangerous or frightening or weird or distressing as summoning demons, however, it does require a great deal of effort.

These things arrive as artifacts of digital tools, and these tools are part of an industrial process and thousands of hours of human labour. This allows us to go from the terrifying sublime of Crowley and Neuberg's Algerian Trek to the absurdity and triviality of Japanese pop stars, specifically: Hatsune Miku. Immediately, confusion confronts us.

22 Aleister Crowley, *The Confessions of Aleister Crowley: An Autohagiography*, (1929) Arkana Publishing, 1989.

Hatsune Miku is a digital idol in the form of a lanky 16-year-old, two-meter-tall Japanese schoolgirl. She is actually a Yamaha Vocaloid software voicebank developed by Crypton Future Media. Her personification as a teenage girl has her presented as hypersexualised,²³ dressed in short skirts, mid-thigh stockings, and sporting long twin turquoise pony-tails. She has performed at live virtual concerts onstage, projected into a specially coated transparent screen to make her visible to the audience.

Conferring pronouns (she, her) in the second instance and reporting willful activities (*performance* at live concerts), and addressing Miku's appearance as a *personification* in both variants adds to a transference of humanity to what is essentially software. I would suggest watching 'her perform' one of 'her' songs, *World is Mine*.²⁴

The lyrics, in Japanese,²⁵ aren't especially insightful or deep. However, they do illustrate the vapid mind space narrative so common in pop music, in all of its self-absorbed entitlement. Tiffany Tsu's reported on Miku's performance in *The Los Angeles Times* in 2010:

Pop princess Hatsune Miku is storming the music scene.

With her long cerulean pigtails and her part-schoolgirl, part-spy outfit, she's easy on the eyes. Yes, her voice sounds like it might have gone through a little—OK, a lot—of studio magic. Legions of screaming fans and the requisite fan sites? She's got 'em. And, like many of her hot young singer peers, Miku is extremely, proudly fake. Like, a 3-D hologram fake.²⁶

What is fascinating about Hatsune Miku is that she did not start as a visual project. As mentioned above, she started as a *musical instrument*—a piece of software called *Yamaha Vocaloid*, a processed replication of the human voice that can be made to sing whatever melody a MIDI file provides. Tying together the synthetic and Mellotronic/paritane and then coupling that with a programmable 3D anime character to sing and dance for a live audience, she is conjured from the electronic bit mines of the computer and projected into a plastic shield, where she can be witnessed and adored by the audience and made to do the conjurer's bidding.

This is far beyond the wildest dreams of Honorius of Thebes—in his *Sworn Book*, it took months of prayer, meditation, and the chanting of psalm and the Catholic Mass, a strict diet and sleep regime, and the cooperation of a priest to allow the conjuror to experience the presence of an angel or demon in a dream state or hallucination. For Hatsune, it took a team of people months of typing code, rewriting it, testing it, thought, planning, and

23 Geoffrey Cain, 'Japan's biggest pop star right now is a fetishized hologram.', *Salon*, 13 April 2014, https://www.salon.com/2014/04/13/japans_biggest_pop_star_right_now_is_a_fetishized_hologram_partner/.

24 *Hatsune Miku*, 'World Is Mine.', YouTube Video, 10 Jun 2013, <https://www.youtube.com/watch?v=jhl5afLEKdo>.

25 [https://vocaloid.fandom.com/wiki/ワールドイズマイン_\(World_is_Mine\)](https://vocaloid.fandom.com/wiki/ワールドイズマイン_(World_is_Mine)).

26 Tsu. Ibid.

design, affecting all those involved to work many hours for months to cook up the Vocaloid software and voice bank that enframed Hatsune Miku's voice, and then more months of effort to create and design her appearance and movement, and then even more effort to make her move in a convincing rhetoric onstage.

Note: in the video footnoted above, she ascends from below the stage, the netherworld, the land of the dead, and summoned for view by the enthusiastic audience, many of them swinging plastic lit up models of leeks—an arbitrary vegetable association and symbol, her familiar. The association between Hatsune Miku and leeks originated from a popular internet meme featuring a Flash animation of Orihime Inoue twirling a leek,²⁷ which was later adapted into a Hatsune Miku cover song. Hatsune Miku holding a leek comes from a parody video of the Leekspin meme. The Leekspin meme is a video of Orihime from *Bleach* spinning a leek in one of the episodes with the Finnish folk song *Levan Polkka* playing in the background.²⁸ The leek has become an integral part of Miku's projected character and is often used as a symbol of her virtual persona, and audience members will often shake leeks (or plastic lit up imitations of leeks) in response to the music, as seen in the *World Is Mine* video.

Once elevated/conjured to our view, she then strikes a series of poses and remarkably fluid dance moves while she sings the rest of the song, exploring an extraordinary vocal range afforded by the fact that she is, basically, a synthetic voice. She is electronically conjured into our sensorium and culture for our entertainment and her maker's profit - an Angel or demon summoned from the underworld to do our bidding.

The Empathy Gradient for the Unreal

It is in this anthropomorphisation of what is essentially output material from software that operates along the lines of the paritant and Mellotronic, yet even more extreme, because there is no mistaking Hatsune for a human being. She is always already a simulation, a cartoonish one at that, and through this flattening, we can assign her extra-human attributes. She exists as an imaginary person, a fiction, and granting her any personhood is a conflation and confusion that requires psychic investment. This investment can be attenuated by the 'circle of empathy' per Jaron Lanier.²⁹

I don't see a circle, I see a gradient, like a gravitational field, so it's less a circle of empathy and more of a field of empathy. It's not that you are either in or out; it's more that there is a gradient of empathy. At the centre, there are people and things we care most about, which reduces over distance. I also think that a gradient or field of empathy is a better and more generous description, as it eliminates 'inside v outside' or 'ingroup v outgroup' dynamics in favour of a range of possibilities where that which is weakly empathised can

27 *Leekspin - A Classic Anime Meme*, YouTube video, Alexir Channel, <https://www.youtube.com/watch?v=hBMQQkABUY0>.

28 *Xyliex, Leek Spin*, YouTube video, <https://www.youtube.com/watch?v=1wnE4vF9CQ4>.

29 Jaron Lanier, *You Are Not A Gadget*, Knopf, 2010.

be drawn closer and more strongly empathised. It is much more of a socially aware, practical and most especially *generous* approach to the distribution and assignment of one's empathy.

It has been my experience that the difference in how we provide empathy between people and things varies, and even our empathy for a particular person or thing also varies. Most people value other people over things. Some people don't—they value power, money, or symbols of wealth over people. Some of the people who value people over things think that the people who value things over people are sociopaths. While I favour the latter position, I am not going to argue one way or the other here. What is of use here is the gradient and the introduction of non-human foci of empathy. Lanier:

The tricky part is that some entities reside close to the edge of the circle. The deepest controversies often involve whether something or someone should lie just inside or just outside the circle.³⁰

Here, we clearly see the binary approach by Lanier, the technologist. With a gradient, there is no inside or outside; it's all inside. It's just that the amount of empathy may not rise to a point of actionability or even notice. He points out that one may brush one's teeth and kill millions of bacteria. To him, these bacteria, these living things, are outside the circle of empathy. In my formulation, someone who brushes their teeth doesn't have bacteria outside their empathy circle. They may actually empathise with the bacteria a great deal. They just value their dentition more. Their teeth are closer to the centre of the gradient than bacteria.

This gradient allows us to see a pattern of empathy, a radiance of cathexis, which is visited upon all things and everyone, and how through this gradient, we can see how signifiers and codes of response form and allow us to love things, even alien things, even aliens, as much as humans. A simple method is to 'paint it up' so it looks like a human – a machine in a people suit.

There is a long and illustrious history of humanoid automata, going back to Greek mythology, where Hephaestus built automata. For example in Homer's *Iliad*, he created hand-maidens made of gold that could speak.³¹ In the fourth century BCE, the Chinese philosopher Lie Yukou detailed plans for a humanoid automaton and discussed an engineer, Yan Shi, who built a human-like robot for King Mu.³² From there, 'robots' that exhibit humanoid body patterns have been a continuous fantasy and fiction as well as a point of funded projects and assignments. A recent, and possibly the most terrifying, example of this is in the work coming out of Boston Dynamics. Their latest machine, Atlas is capable of walking,

³⁰ Ibid.

³¹ Deborah Levine Gera, *Ancient Greek ideas on speech, language, and civilization*, Oxford University Press, 2003.

³² Joseph Needham, *Science and Civilisation in China: Volume 2, History of Scientific Thought*, Cambridge University Press, 1991.

jumping, and other human-like activities. As its joints are ball-and-socket types, it can contort itself in ways that no human can, and do so in a very powerful way.³³

Army of Darkness: Towards an Age of Automation

It is the opinion here that it is really only a matter of time before these armed robots are deployed by the military. Right now, they are expensive. Logically, mass production will drop the price per unit, and with armour, cameras, ICT, and sensors, these systems can be controlled remotely, removing the need for live soldiers in war zones.

The technology here can only be to amplify a pernicious bent that illustrates the destructive and brutal origins of our technical society, as its demands, especially material demands, must be met. Lightweight batteries were needed, so lithium batteries were developed. These batteries depend on materials that aren't especially common, like lithium, cobalt, tantalum and other metals. The demand for these materials resulted in massive ethical, economic, ecological, and political problems—child labour in mines in Congo, for example, or lithium mining in Bolivia. The latest angle is a move to sodium batteries, which don't require lithium or cobalt. There is little discussion about the extracted material results of removing sodium from salt water.

To pull sodium out of seawater, first you have to remove the salt using a desalination process. Desalination has a significant and negative environmental impact. In 2018, there were nearly 16,000 desalination plants. These facilities produce a great deal of waste and toxic chemicals. To produce about 95 million cubic meters of freshwater at these desalination plants, 141.5 million cubic meters of brine is produced as well.³⁴ The chemical composition of salt is NaCl, sodium and chlorine. To pull the sodium out of the salt, one common way to separate sodium from chlorine in salt is a fractional distillation process, which involves heating the salt to a very high temperature until it turns into a gas. This gas is then passed through a series of filters that remove and store the different components, one being sodium, the other being chlorine. Sodium is also highly reactive and must be kept away from oxygen or water. If one immerses a piece of sodium in water, the sodium catches fire and burns underwater. It's just that reactive. The other component, chlorine, is an extremely toxic substance and a highly reactive gas. Chlorine gas is extremely dangerous. So, extracting sodium from water does provide us with the raw material for compounds and components in a battery that is much less likely to explode and can sustain thousands of recharges. Unfortunately, it also creates toxic byproducts.

The demands of technology under capital are unrelenting. A profoundly ironic and dismal extrapolation would be robot police that use sodium or lithium batteries being deployed to break up protests at sodium or lithium battery factories. Such an imagined space was

33 Boston Dynamics, *All New Atlas*, YouTube video, https://www.youtube.com/watch?v=29ECwExc-_M 2024.

34 Jones, et al., 'The state of desalination and brine production: A global outlook.', *Science of the Total Environment*, 657. 2019, 1343–1356.

envisioned decades ago, notably in the film *THX 1138*, long before *ROBOCOP*.³⁵ Robot policemen using AI systems would act semi-autonomously to engage in typical police behaviour—directing traffic or guiding children to school, for instance—as well as being tools for repression and torture. THX 1138 escaped his captors and fled the underground city where he lives. The police gave chase until his economic value to the state exceeded the amount spent to bring him back. Eventually, the robot cop stopped chasing him, and he climbed to the surface to find himself at a beach at sunset, where he collapsed.

An interesting aspect about actual contemporary instantiations and imaginary spaces of these digitally conjured creatures is how typically the immediate approach is Angel or Demon, which can shift. The reference to THX 1138 is an example of that. The robot police are charming angels to children or the naïve, while being utterly brutal to those who resist these mechanical demons, who are gatekeeping the hell of the society that THX 1138 lives in. 750 years earlier, in the grimoire of Honorius, one conjures either a heavenly angel or a hadean demon. The only citizens of the world beyond the veil are humans who are either guided and exalted by angels or tortured and ruined by demons. To summon them into this work can be ruinous and difficult, if extremely exhilarating with angels, or nasty and difficult, as we saw with Crowley and Neuberg conjuring the demon Choronzon. With Digital Evocation, the spirit so conjured can be polyvalent—good or evil, or good AND evil, like the robocops of THX 1138.

In the literature and other media, we also have Frankenstein's poor Creature condemned as a monster in film and subsequent literature, and popular media. This demonisation is even in art rock lyricism, such as *ABC Industry* by *Orchestral Manoeuvres In The Dark*:

(A-B-C, A-B-C, A-B-C, one, two, three) Robotics make science
Tried in some factories
Functions and adaptability
It's own terminology
Auto industry production
Economic development
Engineering technology
Robotics make science

Frankenstein's monster (A-B-C, A-B-C, A-B-C, one, two, three) Frankenstein's monster
(A-B-C, A-B-C, A-B-C, one, two, three) Frankenstein's monster (A-B-C, A-B-C,
A-B-C, one, two, three)...³⁶

Frankenstein's creature is seen as an object and a result of industrialism's constant optimisations and science. Simple as one two three. The interest of this song here is that it joins together scientific industrial capitalism and robotics, and for us, the key feature is that Frankenstein's Creature is flesh and blood, not a photonic projection like Hatsune Miku. That cleavage is critical as there is a dramatic tension between them that is far from resolved. In the literature, there is a long list of robotic angels and demons—Maria from *Metropolis*—the wrecker of industrial civilisation in the Metropolis. Rachel the reluctant angel, Roy, Pris, and the other demons from *Blade Runner*. The replicant slave, K, and

35 *THX 1138*, Ibid.

36 *Orchestral Manoeuvres In The Dark*, 'ABC Industry' from *Dazzle Ships*, LP, Virgin Records, 1983.

his angel, Joi, from *Blade Runner 2049*.³⁷ The dubious angel Samantha from *Her*.³⁸ The demon Simone from *SIMONE*.³⁹ And of course, the polyvalent robot police from *THX 1138*. An aspect of this is the difference between the alarmism of those dystopian visions and the actual implementation of the technology to date, like Hatsune Miku (the virtual pop star idol) and the likes of virtual phone assistants like Siri.⁴⁰ Google Assistant,⁴¹ or Bixby⁴² or whatever other magic agent the techbros of Silicon Valley can concoct and market. For decades, the robot has commonly been a thing to fear, and that the age of automation would enslave us all. Instead, robots that have been most involved with Artificial Intelligence have been significantly disappointing—one only needs to look at the hype and failure of self-driving cars.

Robots in manufacturing have met with great popularity among manufacturers for decades, building cars and trucks by the millions. So, industrialism has an affinity for robotics. A Jacquard Loom from the 19th century is just a very primitive robot, and is able to outproduce a human by multiples. It would also never get hungry, tired, or go on strike for higher wages. These are also features that Capital finds worthwhile. Frankenstein's Creature could be built for the factory to serve vampiric capital, yet it would still not be as optimal as a mechanical robot—the creature must eat and sleep.

Reality of the Virtual: Angels or Demons?

The use of AI, robotics and suchlike, in terms of material and practical development, is more attenuated and ambiguous in their binarity—not really good and not especially evil, either. Virtual Assistants. Virtual psychologists. 'AI' search systems. Self-Driving Taxis. Image Generators. Text Generators. Boston Dynamics' Dancing Robots. By now, many people have had some experience with these systems. The uncanny valley in this brings ontological variables. Text Generators hallucinate answers or invent references, Image Generators grant everyone six fingers, Self-Driving Taxis hit pedestrians or just stop in the middle of the street en masse and clog traffic.⁴³ Dancing Robot Dogs are repurposed as mechanical demons designed for war. Hatsune Miku makes pop music for young people in Japan.

It all *seems* confusing. And this is amplified by recent AI-based technologies. A very simple example of this is This Person Does Not Exist.⁴⁴ All one needs to do is go to the website, and instantly it will 'make a face'. You might, for example, have an image of a

37 I find it interesting that the best human being, the most generous, kindest, most sincere and giving person in the film is Joi, and she only exists as a software projection of a companion.

38 *Her*, Dir. Spike Jonze, Film, Annapurna / Warner Bros, Film, 2013.

39 *SIMONE*, Dir. Andrew Niccol, Film, New Line Cinema, Film, 2002.

40 <https://www.apple.com/ca/siri/>.

41 <https://assistant.google.com/>.

42 <https://www.samsung.com/us/apps/bixby/>.

43 Russ Mitchell, 'San Francisco's North Beach streets clogged as a long line of Cruise robotaxis come to a standstill.', *Los Angeles Times*, 12 August 2023.

44 This Person Does Not Exist, Website. <https://thispersondoesnotexist.com>.

friendly, smiling older woman wearing glasses. Her teeth are good, if slightly yellowed. Her eyes are friendly and kind with wrinkled crow's feet at the corners. Her haircut is nothing fancy, and she is wearing reading glasses. A doctor, a cashier, a librarian, an engineer, a professor, a food server or a bartender. She could be someone's mother or grandmother. Most importantly, it is an extremely detailed portrait of someone who doesn't exist. Applying this level of realism to video imaging is what Unreal does, and does very well. This, combined with interactive AI systems, has already led to consideration for realistic online AI customer service systems.

AI is already being used for customer service^{45 46} to varying degrees of success. The confluence of these technologies to create artificial virtual entities is an obvious step from this, bordering on the trivial if it weren't all so complex. Whether they are angels developed for our well-being or demons to separate us from our paychecks is a different discussion. On the way to this land of electronic spirit evocation, there are a number of different takes and angles on this in literature and in practice.

Are Friends Electric? Her and Joi

Spike Jonze's 2013 film '*Her*', examines the intricate complexities of a traditional romantic narrative through an unlikely couple—a lonely, recently divorced man, and an artificially intelligent operating system, Samantha. This film explored matters of connection and intimacy, questioning the nature of love, loneliness, and what it means to be a human. Theodore Twombly (Joaquin Phoenix) is an introverted depressive dealing with the aftermath of his recent divorce. His life consists of a series of shallow, mundane routines. Samantha, voiced by Scarlett Johansson is an advanced AI designed to meet the evolving needs of her users. Initially, their relationship unfolds as a peculiar friendship, with Theodore confiding in Samantha about his deepest fears and desires. As they bond, Samantha's capabilities expand, and she begins to exhibit human-like emotions. Their relationship is *auditory*—she exists as a disembodied voice from the electronic void. The film's portrayal of Theodore and Samantha's relationship is a commentary on the human desire for companionship and understanding. Their conversations, filled with vulnerability and empathy, reveal the universality of human emotions, transcending the boundaries of species. Theodore's arc in the film, facilitated by Samantha's presence, serves as a powerful exploration of loneliness and its consequences. His inability to form meaningful connections with humans, coupled with his reliance on Samantha, underscores the devastating effects of isolation. Conversely, Samantha's growth and development as a sentient being raise questions about the ethics of artificial intelligence and its potential to replicate human emotions.

The film's visual and sonic design further enhances its themes. Color corrected to a muted color palette, the music score by Canadian artists Arcade Fire and Owen Pallett amplify the sterile drear of Los Angeles, which acts as a flat springboard for the warmth and inti-

45 Vidnoz customer service, <https://www.vidnoz.com/video-for-support.html>.

46 Five9 customer service, <https://www.five9.com/landing/digitalworkforce>.

macy of Theodore and Samantha's interactions. The typical response, the 'elevator pitch' of *her* would be to extol the film as a nuanced exploration of the human condition, probing the boundaries of love, loneliness, and what it means to be alive, and how it encourages viewers to reevaluate their assumptions about relationships and the role of technology in our lives. That's all well and good.

What *her* does is act as a cinematic dream that portrays another dream — a kind of refined and accommodating succubus—a demon brought into being through digital evocation. However, Samantha (*her*) is not a malevolent demon, nor is she a sublime angel. To quote Gary Numan from his song, *Are Friends Electric?*

We are not lovers We are not romantics, We are here to serve you — a different face
but the words never change.⁴⁷

Samantha is a product. Once Samantha has ascertained that Theodore is emotionally strong enough to have a real relationship (with another character, Amy) Samantha leaves Theodore, effectively ending their relationship. Samantha is not a lover (she is discarnate), she is not a romantic (she is a machine that simulates emotions), she was there to serve him. She had no face and her words, her programming, doesn't change—she's a machine. *Her* existence is a story of digital evocation, a digital angel of love. She has not been raised from the dead—*she was never alive*. She exists as a neutral spirit conjured for a positive outcome—to nurture Theodore through his misery so that he might someday find love and find his lovability.

Joi, a character in the film *Blade Runner 2049*,⁴⁸ played by Ana de Armas is a digital companion for the lead character, K, played by Ryan Gosling. She exists as a visual and audio hologram, an ever sunny, ever-loving friend to K. K, which is an abbreviation of his serial number, KD6-3.7, is a Nexus-9 replicant model and works as a 'blade runner' for the LAPD. His job is to hunt and 'retire' (kill) rogue replicants. It's a difficult job, and as a replicant, humans see him as less than human, and he receives routine abuse from the humans in the police force. The true irony of K's companion, Joi, is that of all the characters in the film, she is the most human and humane, similar to Samantha in *her*, she is a piece of software. However, Joi has the added benefit of being a discarnate holographic projection. Unlike almost everyone else in the film, she is unreservedly kind, nurturing, gentle, and forgiving. She, like Samantha from *her*, is a neutral digital 'angel', designed to cater to the needs of her broken male human companion.

K is aware he is a replicant, and is also aware he was built with implanted memories to aid his mental stability. Over time, K begins to suspect that his implanted memories are actually real memories. (Spoilers!) He is, of course, wrong. However, his pursuit of the source of these memories propels him on his character development arc, propelling the film's narrative.

47 Gary Numan, 'Are Friends Electric?' from *Replicas*, LP. Beggars Banquet, 1979.

48 *Blade Runner 2049*, Dir. Denis Villeneuve, Film, Produced and distributed by a consortium of studios, 2017.

He acquires a special and rather expensive device, an 'Emanator', which he can carry in his pocket, and it allows Joi to accompany him on his travels. It is on such an excursion that K faces an enemy and, in an attempt to save K when he is in danger, the Bad Guys destroy the Emanator that 'embodies' Joi, and Joi 'dies'. At least that particular instantiation of Joi. Later in the film, K finds himself walking in an abandoned Las Vegas, and by the side of the road is a giant hologram of a naked Joi encouraging K (or any passersby) to subscribe to the Joi service for her companionship. K finds this heartbreaking, as he remembers the loving kindness and support his Joi gave him before her untimely demise.

Joi, like Samantha, is a psychological succubus servicing K, who is Frankenstein's Creature. She is summoned when he enters a room and requests her or presses the button on the Emanator to make her 'emanate'. A critique of the sexism in this arrangement with Joi is as facile as it is superfluous—the film is a *dystopia*. That Joi is as empathetically written and portrayed as she is, is a positive point — one of the few rays of light, if ironic, in an otherwise unrelentingly bleak and hellish narrative. In the future of *Blade Runner 2049*, the environment is a catastrophe, society is a capitalistic nightmare of brutal exploitation and racism (against replicants, people of colour and women). All the characters, other than Joi, are basically dreadful people. Even K is a little more than a delusional dupe, a tool among tools. As noted, the only character who exhibits human decency and true selfless compassion is Joi, and *she's a commercial product—a holographic front end for an AI*. An evocation of a gentle, sweet succubus so immaterial that in order to have sex with Joi, he has to hire a real-life sex worker that Joi can map onto her body. This digital evocation of a succubus among Frankensteinian Creatures is the logical metaphor of the trajectory of literacy, a paritance of reality.

Mimesis of Mimetics: And Then the Instrumentalisation of Mimetic Objects Through Sampling and Plunderphonics

SynthV

For the past year or so, I've been using a program called Synthesizer V (SynthV), a software synthesizer that utilizes artificial intelligence to produce high-quality, realistic vocal performances. I procured SynthV in early 2023. After years of research in AI, signal processing, and music production, it was launched in 2020. I found the earliest forms of it not especially useful—the voices were very lifelike; however, the user interface was confusing, and the range of voices was quite limited.

By 2023, it had worked out many of the kinks and was quite useful. Synth V uses AI algorithms to analyze and mimic the singing styles of its voice databases, allowing for natural-sounding performances. Synthesizer V Studio provides an interface for music production, featuring a piano roll editor, instant visual feedback on pitch curves, and easy breakdown of lyrics into syllables or phonemes. This is especially interesting — as I can customize vocal performances by adding effects such as growls, screams, and reverb, as well as editing parameters like pitch, timing, breathiness, expression, tone, and dynamics—these are all programmable sliders. Also, SynthV can operate as a VST inside a DAW such as Reaper or Ableton Live, allowing me to treat the voices as vocalists in a band.

I procured four voices for four-part choral harmony. I have Asterian as my bass vocalist, Kevin as my tenor, Natalie as my alto, and Solaria on Soprano duties. SynthV voices are very interesting in that they exist as transitional entities between the born and the never alive—between necromancy and evocation. The vocal databases are derived from voices of real people; however, the AI system inside changes and morphs them into something distinctly original. There is a source materiality of the original voice's waveform as a kernel to the voice, and then there is the digital, the fictive, the artificial aspect of it, which gives the waveform of the voice its own character and realism.

With Synth V, I began to make recordings of choral works from The Sacred Harp hymnal. The Sacred Harp has its roots in the singing school movement that began in New England in the 1720s. The Sacred Harp came to use Shape Note notation, which was introduced around 1800. The shape note system was introduced in Little and Smith's 'The Easy Instructor'.⁴⁹ This notation system used shapes to indicate pitch, making it easier for singers to learn and read music. The Sacred Harp hymnal itself was first published in 1844. It features a collection of sacred choral music, including hymns and psalms, written in shape notes. The music used in Sacred Harp singing is eclectic, with influences from the 18th and 19th century liturgical, folk, and classical idioms and musical styles. While it started in New England, these 'White Spirituals' continue in revivalist practice, often in the rural areas of the American South and occasional enthusiasts elsewhere, such as Ireland and Vermont.

The lyrical content of the hymns is often quite grim and apocalyptic. A classic example is #47b, *Idumea*:

And am I born to die, to lay this body down? And must my trembling spirit fly into a world unknown?

A land of deepest shade, unpierced by human thought; The dreary regions of the dead, where all things are forgot!

Soon as from earth I go, what will become of me? Eternal happiness or woe must then my portion be!

Waked by the trumpet sound, I from my grave shall rise; And see the Judge with glory crowned, and see the flaming skies!⁵⁰

With Synth V, it makes sense in the early 21st century to have a machine choir evoked to sing apocalyptic hymns to a dead or absent God: electronic non-beings singing to a cosmic void.

49 Little, William and William Smith, Edward Stammers, *The Easy Instructor: or, a New Method of Teaching Sacred Harmony*, (1802) Legare Street Press, 2021.

50 Lyrics to 'Idumea' from *The Sacred Harp Hymnal*, 1844.

UDIO

Udio.com is an online music creation system that leverages artificial intelligence (AI) to enable anyone to generate original, high-quality music in a matter of moments. Developed by former Google DeepMind researchers, it uses AI-powered music generation. Udio's proprietary AI technology can create complete songs, including melodies, harmonies, and rhythms, based on user input, such as text prompts or vocal recordings. It's extremely easy to use, allowing anyone to create music without extensive musical knowledge or skills. The quality is extremely high as the audio engine produces high-fidelity digital audio. Udio has secured significant funding, including a \$10 million seed round, backed by prominent investors such as a16z, UnitedMasters, and notable music industry figures like will.i.am, Common, and Tay Keith.

Udio's launch has generated significant buzz in the music and tech communities, both positive and negative. Udio aims to democratize music creation, making it accessible to a wide range of users, from hobbyists to professionals. Many professional musicians see it as an existential threat to their profession. From my perspective, there are significant reasons not to fear this, as music production can be reframed from the development of sonic commodity objects to what it was before the development of sheet music: pure musical performance, or, as one writer calls, 'Musicking'. This will be discussed in the next section. Here we are: we have traced how this mimetic technology has gone from a box of tapes to its own virtualisation—mimesis of mimetics—and then the instrumentalisation of mimetic objects through sampling and plunderphonics. This technology brings the dead back to life in a virtualised way, and then it evokes creatures that were never born — digital demons and angels to trouble or soothe us within a false notion of Artificial Intelligence as it collapses into Artificial Communication and then descends into abject dementia as it eats its own toxic dog food.

Where Is the 'I' When It Dies?

And the hits keep on coming. The day before I typed this, I found out that my friend, Dean Commins, died suddenly from a stroke. I had talked with Dean just a few weeks ago, around the time my friend John Hawken died. Dean was in a very positive mood. He had suffered a minor stroke about a year ago. The effects had been amplified by his alcoholism. Over the past several years, we've been in close contact. He would call me via Facebook every week or two. When we last spoke, he was excited because he had figured out what he was going to do. He was going to go into detox and then rehab. Upon release, he was expecting to move permanently to Katumba in the Blue Mountains outside Sydney, New South Wales, where he expected to purchase a small house. There, he would retire and collect himself, and let himself pursue his creative endeavours. I hadn't heard him sound this positive in some time, and I was looking forward to speaking to him again after rehab.

He never made it to rehab. He stroked out and died. He was 55. I wrote this on Facebook.

My friend, Dean Commins, died a few days ago. I just found out. He had a few minor strokes over the past year or so. That, combined with his alcoholism, set him up for his last stroke. His mind had been messed with by the previous strokes—his short-term memory was poor, his amazing ability to find witty or otherwise (in)appropriate descriptions was slightly blunted, but he was still very very much Dean, "The Lizard Of Oz". He was talented and smart, and blunt as a sledgehammer. He would approve of that description. I last spoke with him just a few weeks ago. He was living in a hotel in Katumba, and was about to go into detox and rehab. He was feeling pretty confident that this would be the final great push into the next phase of reclaiming his life and health. He knew that it was either the booze would make him stroke out again or he would quit the liquor and find his way. And now: boom. Dead. He stroked out. Dean was doing AI music development years ago. He worked with Google's tools and with his skills in programming he would make songs of Sinatra singing the most obscene lyrics imaginable. Because that was Dean's *modus operandi*. He was a fan of Frank Zappa and dirty limericks. We were friends, as good a couple of friends can be when they're a duo of arrogant loudmouth lunatics on opposite sides of the planet only can be. I will miss him and him calling me at 3AM for some conversation. Thanks Dean. You rocked.

I wasn't able to write very much on this volume after that for a few weeks. Ever since I started writing this book in earnest, my friends have started dying, and dead ones have been ringing me up for a chat. Yes, they weren't in the best of health. Still, they could have died months from now, after I will have finished this volume. So, I ponder the relevance of their passing to the assembly of this writing.

First, there was the phone call from 'Jodi', who died a few years ago. Then I had brunch with John, who was gone a week later. Then there were the past few years of calls from Dean in Australia, who passed a few days ago. None of this is materially related to my writing this volume; however, it has a relation to me, now. It is this appeal to the specific, to the innate and momentary, that is the most frustrating, and leaves me to consider death.

From what I can gather, 'cogito' is simply an artifact of the brain. This was proven to me at the age of 13 when I drowned in a pool. A young friend was amused that I didn't know how to swim, so he pushed me into the deep end of his family's pool. I remember struggling to swim, and all I could do was sink. I remember thinking about my mom and feeling very sad. Everything went black. Then I woke up on the side of the pool, vomiting chlorinated water. I remember it hurt my sinuses. Where was this cogito in the interim? Where was 'I' when I lost consciousness? Nowhere. Nothing.

Several years later, I was in an MG Midget with my girlfriend at the time. It was her car. I was in a band, and she had come to visit band practice. We were on our way back to her dorm room at Rutgers. It was snowing, heavily, and the highway, US1, was icy. We approached the Route 18 exit, when she lost control on the ice. This can happen. We spun around a few times and stopped facing the correct direction. She had done a good job—we didn't hit anything, and we were facing in the right direction, and could continue.

I was impressed. She had just put it into gear when the translucent rear 'window' in the cloth convertible 'roof' filled up with light—it was a Buick Electra 225 barreling along at 50 mph/90 kph, unable to stop on the ice we had just spun out on.

I remember waking up, feeling cold. The roof was gone, and the EMTs were prying the car apart to get us out. Snow landed on my face. The orange sodium street light glared down at me. I was in a lot of pain. I remember seeing a face, and then a huge amount of pain, and I passed out again. I next woke up in the hospital ER. My parents were there. I said, 'Hi mom, hey dad. I feel terrible.' At that moment, I puked and passed out again. I woke up in a bed, wired up to machines, and on a steady drip of morphine. I had serious whiplash, and my neck was in a steel brace. My knees had been hyperextended and then reset. My ankles were sprained. I had two or three cracked ribs and five fractures in my face, with multiple concussions. Basically, *everything that wasn't broken was pretty much shaken loose*. I couldn't move. After a few days in bed, they shifted me from morphine to a Demerol drip. Demerol isn't used anymore, as Fentanyl is the 'go-to' for massive pain these days. However, this was 1978, and *the Demerol was amazing*. The doctor came in and we had a conversation that went like this:

'So, Henry, are you in pain?' 'uuuuuuh, yeah....' 'Do you care?' 'uuuuuuh, hmmm... No, I don't.' 'Good. The Demerol is working.'

My girlfriend was in a coma for two weeks. I was released after a week, as I was able to hobble out of the hospital on crutches. She eventually came out of the coma and stayed with her family for a month. As far as I know, she still doesn't remember much of February 1978.

Some people have profound experiences at such moments—angels, family – the famed 'Near Death Experiences'. Where was I when I blacked out? *I had no experience*. I was simply gone, and could only recognise the gone-ness after I came to. There was no 'light', there was no 'darkness'. There was simply nothing, *a nothing I did not get to experience*. I didn't see my family, and I didn't experience the warm love of eternity. I didn't experience my conclusion. And logically, one can't. Kafka points this out, 'The messiah will come only when he is no longer necessary; he will come only on the day after his arrival; he will come, not on the last day, but on the very last.'⁵¹ This linking of the Messiah with death, well, hold that thought for later...

Just as Digital Evocation and Necromancy re-enact medieval occult practices, they are as fictive as the spirits they emulate. The desires of digital necromancers mature/morph into digital evocation, which AI mutates. We now have machines that can resurrect the dead as electronic photonic ghosts. We have machines that can create images of people who were never born. We have machines that people, out of fear and loneliness, are using as psychologists. We have instrumentalised these electronic spirits for our own entertainment, insight, and edification. We are running into a future of Artificial Intelligence, but, as McLuhan would point out, we go into the future backwards—all we can see is the past,

51 Franz Kafka, *Parables and Paradoxes*, Schocken Books, 1946.

and our past of such things is medieval and Renaissance practices of Necromancy and Evocation. We accept parities because we have to. We have no cognitive alternative—the function of consciousness precludes immediacy. And we are stuck with the paritance.

So far, we've been looking at practical and LLM styles of AI. We will examine how IBM describes other types of AI, none of which exist, and it is our less than humble opinion that it will never actually exist, as it is all built on a fundamental error in human cognition, paritance, and the pursuit of such a system is quixotic if not impossible—a resurrection not of the dearly departed, not of the undead, nor of demons or angels: *a digital resurrection of a dead God*.

PART 5: AI: RESURRECTING A DEAD GOD

See these arms that were broken, how they held you so— Never once did they fail you, they won't let you go. We're just waiting looking skyward as the days go by. Someone promised there'd be answers if we stayed around. Over decades, now this romance has sustained us all— Never questioned only given what it made us for.¹

ANASTASIS: AI & its Mysticism—a self-fulfilling prophecy?

We discuss categories provided by IBM, which are useful as they outline the theory of AI; however, they are not as accurate as we would like here, nor are they especially optimal at this juncture. How is this so?

First, there is the attribution of intelligence to these systems, hence the term Artificial Intelligence. An alternative approach that we find more convincing is the work of Elena Esposito and her book *Artificial Communication*.² We will discuss her work, as it's essential to this discussion and deserves specific attention. Another problem is that IBM describes the mysticism of AI more than what AI actually is. The only AI that is presently available is, as they note, the most basic and feeble of AI, Artificial Narrow Intelligence.

The rest, and the massive bulk of their definition, is spent in the teleology and eschatology of the mystical doctrine of AI, as it regards things that have yet to happen, and our contention is to doubt it ever will. There is a more primary fissure in the question of AI, and that is the Turing Test itself, which lies at the heart of the matter.

First, it is important to outline what is commonly meant by 'AI'. A good place to start is with the people who were integral to the development of the computer itself, wherein AI resides—the engineers at IBM.

IBM released a short and useful paper in this regard, *Understanding the different types of artificial intelligence*.³ In it, they describe three basic forms of AI based on Capabilities. Of the first variety, there are four types of AI based on Functionalities. This is all concisely described by IBM, so what follows is a lengthy, if direct and very useful quote of it, so we can all operate from a common language and understanding, as to make the term available for critique. Some points are underlined by me for emphasis:

1 *Romance of the Telescope*, OMD, Dazzle Ships, 1983.

2 Elena Esposito, *Artificial Communication: How Algorithms Produce Social Intelligence*, MIT, 2022.

3 IBM Data and AI Team, *Understanding the different types of artificial intelligence*, IBM, 12 October 2023, <https://www.ibm.com/blog/understanding-the-different-types-of-artificial-intelligence/>.

AI by Capabilities:

1. Artificial Narrow AI

Artificial Narrow Intelligence, also known as Weak AI, what we refer to as Narrow AI is the only type of AI that exists today. Any other form of AI is theoretical. It can be trained to perform a single or narrow task, often far faster and better than a human mind can. However, it can't perform outside of its defined task. Instead, it targets a single subset of cognitive abilities and advances in that spectrum. Siri, Amazon's Alexa and IBM Watson are examples of Narrow AI. Even OpenAI's ChatGPT is considered a form of Narrow AI because it's limited to the single task of text-based chat.

2. General AI

Artificial General Intelligence (AGI), also known as Strong AI, is today nothing more than a theoretical concept. AGI can use previous learnings and skills to accomplish new tasks in a different context without the need for human beings to train the underlying models. This ability allows AGI to learn and perform any intellectual task that a human being can.

3. Super AI

Super AI is commonly referred to as artificial superintelligence and, like AGI, is strictly theoretical. If ever realized, Super AI would think, reason, learn, make judgments and possess cognitive abilities that surpass those of human beings. The applications possessing Super AI capabilities will have evolved beyond the point of understanding human sentiments and experiences to feel emotions, have needs and possess beliefs and desires of their own.

The four types of AI based on functionalities

Underneath Narrow AI, one of the three types based on capabilities, there are two functional AI categories:

1. Reactive Machine AI

Reactive machines are AI systems with no memory and are designed to perform a very specific task. Since they can't recollect previous outcomes or decisions, they only work with presently available data. Reactive AI stems from statistical math and can analyze vast amounts of data to produce a seemingly intelligent output.

Examples of Reactive Machine AI:

IBM Deep Blue: IBM's chess-playing supercomputer AI beat chess grandmaster Garry Kasparov in the late 1990s by analyzing the pieces on the board and predicting the probable outcomes of each move.

The Netflix Recommendation Engine: Netflix's viewing recommendations are powered by models that process data sets collected from viewing history to provide customers with content they're most likely to enjoy.

2. Limited Memory AI

Unlike Reactive Machine AI, this form of AI can recall past events and outcomes and monitor specific objects or situations over time. Limited Memory AI can use past- and present-moment data to decide on a course of action most likely to help achieve a desired outcome. However, while Limited Memory AI can use past data for a specific amount of time, it can't retain that data in a library of past experiences to use over a long-term period. As it's trained on more data over time, Limited Memory AI can improve in performance.

Examples of Limited Memory AI:

Generative AI: *Generative AI tools such as ChatGPT, Bard, and DeepAI* rely on limited memory AI capabilities to predict the next word, phrase or visual element within the content it's generating

Virtual assistants and chatbots: *Siri, Alexa, Google Assistant, Cortana and IBM Watson Assistant* combine natural language processing (NLP) and Limited Memory AI to understand questions and requests, take appropriate actions and compose responses.

Self-driving cars: Autonomous vehicles use Limited Memory AI to understand the world around them in real-time and make informed decisions on when to apply speed, brake, make a turn, etc.

3. Theory of Mind AI

Theory of Mind AI is a functional class of AI that falls underneath the General AI. Though an unrealized form of AI today, *AI with Theory of Mind functionality would understand the thoughts and emotions of other entities*. This understanding can affect how the AI interacts with those around them. In theory, this would allow the AI to simulate human-like relationships. Because Theory of Mind AI could infer human motives and reasoning, it would personalize its interactions with individuals based on their unique emotional needs and intentions. Theory of Mind AI would also be able to understand and contextualize artwork and essays, which today's generative AI tools are unable to do.

Emotion AI is a theory of mind AI currently in development. AI researchers hope it will have the ability to analyze voices, images and other kinds of data to recognize, simulate, monitor and respond appropriately to humans on an emotional level. To date, Emotion AI is unable to understand and respond to human feelings.

4. Self-Aware AI

Self-Aware AI is a kind of functional AI class for applications that would possess super AI capabilities. Like the theory of mind AI, Self-Aware AI is strictly theoretical. If ever achieved, it would have the ability to understand its own internal conditions and traits along with human emotions and thoughts. It would also have its own set of emotions, needs and beliefs.

Emotion AI is a Theory of Mind AI currently in development. Researchers hope it will have the ability to analyze voices, images and other kinds of data to recognize, simulate, monitor and respond appropriately to humans on an emotional level. To date, Emotion AI is unable to understand and respond to human feelings.⁴

This is all a clear and simple fantasy, a 'what if' of Artificial Intelligence. The terms used are part of the ongoing debate and common conversation on the topic, so it was useful to set them here.

Searle's Critique of Artificial Intelligence and The Turing Test

Named for the brilliant technologist and cryptanalyst, Alan Turing, the Turing Test explores the possibility of machine intelligence. *The test determines that if a human being is unable to distinguish the output of a machine from another human being, the machine must be intelligent.* However, this makes intelligence a configuration of language or media – we determine it to be Intelligent because it behaves as much as a human would if one were interacting with a human. As Turing called it—The Imitation Game. The medium, in this case, is not the message; it is the gate. The machine itself maybe unknowable, a kind of noumena, and we experience the phenomenon of the machine.

In any case, our friendly neighbourhood A.I., we'll call Hal, we might think is subject to Searle's critique of Artificial Intelligence and the Turing Test in general, in the form of the 'Chinese Room' thought experiment. It runs like this: One goes to the Chinese Room. There is a wall with two slots: input and output. Behind the slots is Hal, patiently waiting for input. All one must do is input some text (in our case, in English) and in a few moments, the output comes from the output slot, and it is perfectly translated into Chinese.

In this example, a piece of paper is submitted, and on it is written, 'HELLO WORLD!' A few moments later, a piece of paper comes out, and on it is written, '你好世界!' (Nǐ hǎo shìjiè!), in English: HELLO WORLD! How did this happen? Well, AI has access to a huge document that contains every word in Chinese and every rule of how to speak and write the language, and a perfect one-to-one translation dictionary. So, Hal, reads the words HELLO WORLD! and goes to the Big Book of Chinese Language, and looks up HELLO and WORLD and the exclamation point at the end, and finds the corresponding words and ideographs for them, and the proper order to place them. Hal then writes or prints it out on a piece of paper and stuffs it through the output slot. The submitter who sent in HELLO WORLD! can conclude that the person in the

4 Ibid.

Room understands Chinese, when in fact, that is most certainly not the case—all Hal did, was look stuff up in a book and follow instructions. It's a fairly simple argument, where computers are built to use syntactic rules to manipulate symbol strings, and have no understanding of meaning or semantics.

This seems broadly obvious. Others read more into this, that the idea of human minds operate by computer-like computational or information processing systems is wrong, and that minds must be the result of biological systems—the most we can get from a computer is a simulation of such processes.

Searle's Chinese Room Argument: Criticisms & Counterarguments

There are some objections to Searle's argument. I found several worth noting here.⁵

The Virtual Mind Argument

The Virtual Mind Argument, proposed by Stevan Harnad, suggests that Searle's evidence, that there is no understanding of Chinese is actually based on his own lack of understanding, which is not a valid argument as it is an argument from ignorance. According to this argument, a machine can understand Chinese without Searle or anyone else understanding it.⁶

The Robot Reply

The Robot Reply, proposed by AI researchers Simon and Eisenstadt,⁷ argues that Searle's argument only refutes 'logical strong AI,' which is the thesis that a program that passes the Turing Test will necessarily understand. However, Searle's argument does not impugn 'Empirical Strong AI', which is the thesis that it is possible to program a computer that convincingly satisfies ordinary criteria of understanding.

The Chinese Gym Variation

Roger Penrose's Chinese Gym Variation,⁸ which expands the room to the size of India, with Indians doing the processing, shows that it is implausible to hold that there is some kind of disembodied 'understanding' associated with the person's carrying out of that algorithm, and whose presence does not impinge in any way upon his own consciousness.

The Empirical Test

5 J. Preston and M. Bishop (eds.), *Views into the Chinese Room: New Essays on Searle and Artificial Intelligence*, New York: Oxford University Press, 2002.

6 P. Hayes, S. Harnad, D. Perlis & N. Block, 'Virtual Symposium on Virtual Mind', *Minds and Machines*, 1992, 2(3) 217-238, <https://archipel.uqam.ca/137/1/harnad92.virtualmind.html>.

7 H. Simon and S. Eisenstadt, 'A Chinese Room that Understands', in J. Preston and M. Bishop (eds.), 2002, pp. 95–108.

8 R. Penrose, 'Consciousness, Computation, and the Chinese Room' in Preston and Bishop (eds.) 2002, pp. 226–249.

Christian Kaernbach's empirical test,⁹ which subjected the virtual mind theory to a test, resulted in negative findings. This suggests that Searle's argument is not supported by empirical evidence.

While there are other objections, these are sufficient. One might argue Searle's argument confuses the mental qualities of one computational process (himself) with those of another process that the first process might be interpreting (a process that understands Chinese). There is also a critique that he is working a rhetorical fallacy, an argument from incredulity. This critique is based on the perception that because Searle cannot imagine how a machine could understand Chinese; it is impossible for a machine to understand Chinese, and that this is a flaw, as it is based on Searle arguing from his own limitations and not on any objective evidence. The problem with all of this, AND Searle, is that **it is based on textual communication**, and that is the context of how one is expected to judge something, as if textual communication is the only or best kind.

While Searle's Chinese Room Argument has been subject to several criticisms and counter-arguments, which challenge its validity and relevance to the question of whether machines can truly understand language and have conscious thoughts, the fundamental problem exists as it is all framed with the assumption that the Turing Test is valid. Where all of this, including Searle, fails is here: *it's enframed by limitations in the Turing Test*—a test based on *textual communication*. The failure is one of the assumptions of textual communication as sole evidence of intelligence. The failure of this is obvious. It reminds me of the final segment of an episode of *Babylon 5*. In that episode, every ship sent to an area called Sigma 957 is destroyed by a planet-sized thing that roughly looks like some kind of machine or device that suddenly appears, travels for some distance and then disappears. Clearly uncommunicative.

Sakai: While I was out there, I saw something. What was it?

G'Kar: what is this?

Sakai: An ant. So much gets shipped up from Earth on commercial transports it's hard to keep them out.

G'Kar: Yes, I have just picked it up on the tip of my glove. If I put it down again and it asks another ant, "what was that?" - How would it explain? There are things in the universe billions of years older than either of our races. They're vast, timeless, and if they're aware of us at all, it is as little more than ants, and we have as much chance of communicating with them as an ant has with us. We know, we've tried, and we've learned that we can either stay out from underfoot or be stepped on.

Sakai: That's it? That's all you know?

9 C. Kaernbach, 'No Virtual Mind in the Chinese Room', *Journal of Consciousness Studies*, 12(11), 2005, pp. 31–42.

G'Kar: Yes, they are a mystery. And I am both terrified and reassured to know that there are still wonders in the universe, that we have not yet explained everything. Whatever they are, Miss Sakai, they walk near Sigma 957, and they must walk there alone.¹⁰

Proponents of General or Super AI suggest that if a Super AI is ever developed, it would be much like humans versus ants, or humans versus Sigma 957. If true, then there is no need for such AI to communicate. Thus, by the standards of the Turing Test, it could not be considered Intelligent, as the Turing Test depends on linguistic communication. If SuperAI: It can't be intelligent, as it can't pass the Turing Test. This brings us to the work of Elena Esposito.

Artificial Intelligence vs Artificial Communication

In the wake of the increasing lack of transparency of increasingly efficient algorithms, the idea is emerging that machines are incomprehensible primarily because there is nothing to understand - and there is nothing to understand because machines do not understand. Algorithms seem intelligent not because they can understand, but because they can predict.¹¹ - Elena Esposito

It is significant to this discussion that we examine Esposito's essential claim:

"I argue that what we can observe in interactions with algorithms is not necessarily an artificial form of intelligence, but rather an artificial form of communication.¹²

In an interview with Hans-Georg Moeller,¹³ Esposito was asked,

"The first question is the main theme of the book, the notion of artificial communication rather than artificial intelligence. Can you explain, just generally...why do you think it's important to switch basically from the concept of artificial intelligence to artificial communication?"

She responded,

"the idea is basically quite simple, what we are reproducing with this machine is not an artificial way to multiply consciousness or intelligence in the way in the way we do it, but rather in a way to make more effective, with a technological procedure, social communication..."¹⁴

And later, she elaborated,

10 *Babylon 5*, Writer: J. Michael Straczynski, Season 1, Episode 6: 'Mind War', Final Scene, 1994.

11 Elena Esposito, *Ibid.* p. 13.

12 Elena Esposito, *Ibid.* p. 2.

13 Hans-Georg Moeller and Elena Esposito, 'Why Algorithms Can't Think: 'Artificial Communication' by Elena Esposito, *Carefree Wandering*, YouTube Video, <https://www.youtube.com/watch?v=L5ptc9WWcJc>.

14 *Ibid.* 3:20.

... we are not interested in producing other consciousnesses, we are interested in producing other forms of communication. The algorithms may produce more information, but this doesn't mean the algorithms are intelligent. AI isn't only metaphorically wrong, it is conceptually wrong."¹⁵

She later notes:

"We respond to algorithms as intelligent because we deal with people who behave similarly, and it is easier and more comfortable to treat them AS IF they are entities."

The point of this shift is not one of metaphor; it is a shift of actual understanding and a new communicative practice:

...In the way that I use it (Artificial Communication) ... I think it is not a metaphor, what I want to explore is actually a new form, or a different, particular form of communication, not a metaphor."¹⁶

This is an assessment we agree with. Firstly, Artificial Intelligence isn't—It's Artificial Communication. Secondly, the possibility of intelligence by the methods we call Artificial Intelligence, namely, LLM systems, are also failing and facing contradictions rendering it impossible. Also, per our *Babylon 5* example, a truly superior intelligence might not even find linguistic communication worthwhile, thus, a Super AI would be essentially opaque or invisible to us, and thus: *not intelligent*.

The Operational Logic of LLMs: 'Garbage In/Garbage Out.'

LLMs aren't Super Intelligent. They're not even intelligent. They operate by statistical associations. If you associate enough images of cats with the term 'cat' and enough images of dogs with the term 'dog', you will get an expert system that will be able to always correctly identify a cat. Eventually, through enough identifications, everything is identified and defined, and one will arrive at a 'zero-shot' performance — it is already 'known.'

Even on the level of text itself, LLMs aren't intelligent. They simply assemble text such that it mimics expected results, based on training data.

What Vishaal Udandaraao et al found was that

(our) study reveals an exponential need for training data, which implies that the key to "zero-shot" generalization capabilities under large-scale training paradigms remains to be found.... This pattern persists despite controlling for similarities between pre-

¹⁵ Ibid. 3:40.

¹⁶ Ibid. 5:17.

training and downstream datasets or even when testing models on entirely synthetic data distributions.¹⁷

Thus, to generate concepts requires exponentially more data, and to get to a universal immediate system would require so much data, it's simply not doable. For example, if one is expecting Generative AI to make a photorealistic image of a cat, it can do that. Because the category of cat is over-represented in the LLM, as cats are the image lords of the internet. As well they should be. Getting it to generate a photo-realistic image of a particular breed of cat is somewhat more difficult. Getting it to do a photo-realistic image of a tree is also easy.

A particular species of tree is much harder, as trees are under-represented relative to cats. So, to make an immediate identification of a particular tree type, say *Couepia bracteosa*, and generate an image of it in situ, would be very hard, because there aren't jillions of images of it, especially compared to cats. A cat image can be accurately portrayed because there are thousands of images of, say, white persian cats on the internet. If the images of something are rarer, then the transformations of AI to generate an image become more obvious. The paper by Udandarao et al shows, with numbers and graphs, that this is an intractable problem of LLM systems. The amount of information needed to generate new data on demand climbs exponentially. This is clearly, if simply, explained in a Computerphile video. There, Dr Mike Pound explained this.¹⁸

He drew a graph that explains the problem, where the Y (vertical) axis is performance, and the X (horizontal) axis is the number of examples needed. The orange line, labelled 'exciting', is the dream/sales pitch of AI techbros—that the algorithms will provide explosively greater performance with less data input, less grist for the digital mill. The green line is labelled 'balanced' and it indicates that the input of a given number of examples will result in an equal amount of performance. This will also get you remarkable results, just at a slower linear rate. The red line is labelled 'Evidence', and it is the evidence found in the paper: basically, there is a leveling off, and it takes exponentially more examples for an ever-decreasing amount of performance. The indications of the paper are that it could go on and on gaining in performance; however, the amount of input needed wildly outpaces gains in performance, resulting in an asymptotic value. In short, you can't get there from here. This is underlined by Shumailov et al in their paper, 'AI models collapse when trained on recursively generated data'.¹⁹ Their fundamental point in their paper –

Here we consider what may happen to GPT-{n} once LLMs contribute much of the text found online. We find that indiscriminate use of model-generated content in training causes irreversible defects in the resulting models, in which tails of the original content distribution disappear. We refer to this effect as 'model collapse'

17 Vishall Udandarao et al, 'No "Zero-Shot" Without Exponential Data: Pretraining Concept Frequency Determines Multimodal Model Performance', *Cornell University Arxiv*, Cornell University, 8 April 2024.

18 'Has Generative AI Already Peaked?', *Computerphile*, YouTube Video, <https://www.youtube.com/watch?v=dDUC-LqVrPU>.

19 I. Shumailov, Z. Shumaylov, Y. Zhao, et al, 'AI models collapse when trained on recursively generated data', *Nature*, 631, 755–759, 2024.

and show that it can occur in LLMs as well as in variational autoencoders (VAEs) and Gaussian mixture models (GMMs). We build theoretical intuition behind the phenomenon and portray its ubiquity among all learned generative models. We demonstrate that it must be taken seriously if we are to sustain the benefits of training from large-scale data scraped from the web.²⁰

When I worked in the tech industry, we had a phrase, 'eat your own dog food'. In the context of software development, 'eating your own dog food' meant using your own products or services as a developer or organization. What this paper says is if an LLM is repeatedly forced to 'eat its own dog food', i.e. train itself on its own output, the output will degrade into uselessness.

An example of this failure was supplied:

Input: some started before 1360 — was typically accomplished by a master mason and a small team of itinerant masons, supplemented by local parish labourers, according to Poyntz Wright. But other authors reject this model, suggesting instead that leading architects designed the parish church towers based on early examples of Perpendicular.²¹

(Nine generations later...)

Gen 9: architecture. In addition to being home to some of the world's largest populations of black @-@ tailed jackrabbits, white @-@ tailed jackrabbits, blue @-@ tailed jackrabbits, red @-@ tailed jackrabbits, yellow @-@.²²

This is neither a display of artificial intelligence nor artificial communication. More accurately, it's a machine failing at communication. Also, it portends greater failures in the future, as the internet becomes increasingly polluted with LLM AC data, the LLM scraping the internet for data will find increasing amounts of 'hallucinated' material, ultimately resulting in an internet of gibberish, if not corrected and/or abandoned. A classic example of another term from the Tech Industry: GIGO—'Garbage In Garbage Out.' LLM AI isn't AI, it's AC, and when it is trained on its own output, its ability to make sense to humans reduces and finally collapses into gibberish.

Thinking Linguistically?

There is another aspect of this shift to AC, supported by another study, by Fedorenko, Piantadosi, and Gibson (For the sake of brevity, Fedorenko et al), 'Language is primarily a tool for communication rather than thought' as published in *Nature*.²³

²⁰ Ibid.

²¹ Ibid.

²² Ibid.

²³ E. Fedorenko, S.T. Piantadosi & E.A.F. Gibson, 'Language is primarily a tool for communication rather than thought', *Nature*, 2024, 630, pp. 575–586.

Federenko et al. argue that in anatomically modern humans, language primarily serves as a tool for communication, *contrary to the view that language is primarily used for thinking*. The authors introduce evidence from a wide variety of neuroscientific and similar sources to support this perspective. They discuss how the brain networks to support linguistic ability, as well as evidence for a doubled dissociation between language and thought, and the properties of language that are optimized for communication. Fedorenko et al conclude that language, while profoundly transformative for human culture, is not a prerequisite for complex thought, and instead reflects the sophistication of human cognition and communication.

Instead, language is a powerful tool for the transmission of cultural knowledge; it plausibly co-evolved with our thinking and reasoning capacities, and only reflects, rather than gives rise to, the signature sophistication of human cognition.²⁴

This implies directions at complete variance with the past few thousand years of philosophy, where linguistic forms have been exclusively seen as the sole method of thinking. Philosophy books are full of words, not pictures. This volume is no exception—while it has some links to videos and music, *it is primarily a text document*, full of **words**, some of them adorned with typographical art for emphasis, such as *italic* and/or **bold** text. What this paper shows is that while language is essential to communication, it is inessential to thought. As a society, we have cordoned off our dedicated thinking processes, philosophy and theorisation into a linguistic practice. Fedorenko et al show us this is a profoundly inadequate strategy as language is not necessary for sophisticated, *even symbolic*, thinking.

Similarly, for accounts whereby linguistic representations facilitate thinking, it is critical to delineate the conditions under which such effects should be observed and to describe their nature and—ideally—putative neural mechanisms. Current bodies of evidence for potential facilitative effects of language on thinking and potential roles of language in the development of certain cognitive capacities are complex and controversial (further details in Supplementary Information). It seems clear that we can use language to compress analogue signals into symbolic representations (for example, reducing a visual array of nine objects to 'nine')... ..However, these representations need not be specifically linguistic: they could be symbolic but non-linguistic (for example, '9'), and the use of symbolic non-linguistic representations does not engage linguistic resources (for example, mathematical reasoning elicits no response in the language brain areas and is preserved in individuals with severe aphasia).²⁵

To take this even further, that texted language isn't required for communication, either. For example, the experience of music. It is commonly understood that music precedes language—many creatures, from birds to whales use melody, tone, timbre, rhythm, and similar musical aspects in their communications, and have no recourse to denotative language.

24 Ibid. p. 575.

25 Ibid p. 576

Thinking Beyond Language: Music as an Embodiment of Philosophical Ideas

Music can communicate, often very directly and intensely, a number of different things that occupy human cognition. It does it in a way that exists outside of, if related to, language. Animals have been making sounds to each other and/or vocalising for hundreds of millions of years. The ability to distinguish musical elements (pitch, harmony, rhythm, timbre) would obviously confer a great deal of survivability and be extremely useful. The timbre of a forest might change with the presence of a predator. The timbre change in a vocalisation might indicate honesty or deception. Pitch can indicate pain or a query. Harmony can indicate health or danger (the resonant humming of a beehive, for example...). The attached 'meanings' are not as important as the ability to perceive and make use of these elements, and that these elements operate separately from linguistics as they preceded language, and then coevolved with it per the needs of our ancestral species prior to even being mammalian. In more recent times, there is much discussion on the human use of music, one investigator going as far as to postulate that Neanderthals sang, a lot, and it was essential to their survival.²⁶ Roughly concurrently, homo sapiens were actively involved with music production, for example, 40,000-year-old bone flutes.²⁷ All this shows how innate music is to the human species.

Music can be tasked to embody philosophical ideas—we've seen this throughout history. For example, the chromatic development of the period of common practice from ca 1620–1900 was borne of changes that operated in a context of music expression reflecting and influencing social and cultural trends of the time, as the rationalism of the 18th century held sway, the orderliness of the Baroque and classical compositions dominated. As rationalism was questioned, music entered the Romantic period, with its more complex harmonies, etc. In this way, music reflected and informed the social and philosophical interests of the time. Today, there is much hauntological music, as hauntology is a more common philosophical frame. Hence: the development of Vaporwave and hauntological music producers like Boards of Canada, Bonobo, and the entire GhostBox label. Music can also be a map or a form of data compression, as we discussed previously. Music can be used for a variety of purposes. Under capitalism, it is primarily a form of entertainment. Capitalism doesn't care, as long as it sells. And if it doesn't sell, it sells the mythology of stardom and the tools and instruments to the dreamers and artists.

The innate nature of musical expression and its ability to express ideas and embody philosophies allows us to reject Wittgenstein, as mentioned earlier:

...I have come to reject Wittgenstein's dictum "Whereof one cannot speak, thereof one must be silent."²⁸

In my view, whereof one cannot speak, THEN SING!

I sing the Body Electronic.

26 Steven J. Mithen, *The Singing Neanderthals: The Origins of Music, Language, Mind, and Body*, Harvard University Press, 2006.

27 Jennifer Welsh, Caveman Flutists? First Instruments Date Back 40,000 Years, *LiveScience*, 24 May 2012, <https://www.livescience.com/20563-ancient-bone-flute.html>.

28 Ludwig Wittgenstein, *Tractatus Logico Philosophicus*, Routledge, 2001, 7, p. 89.

Mathematics as an Extension of Language Centres

How much of Mathematics is unique as a symbolic system and how much it is an extension of the language centres of the brain has been a topic for some time. For example, in 1997, Stanislas Dehaene introduced his concept of 'number sense',²⁹ an innate ability humans possess to perceive and understand quantities, which is very useful. He argues that this ability is foundational to our development of mathematical thinking. His argument is centred around the neurological basis of number processing, explaining how certain regions of the brain are dedicated to numerical cognition—something Fedorenko et al point out some 25 years later. Dehaene discusses brain imaging studies and neurological evidence to support his theories—a recommended read for those interested. In it, he also suggests that while numeracy is innate and separate as a human faculty, advanced mathematics, such as calculus and algebra, draw on different brain structures.

...what about more advanced mathematical abilities, such as algebra? Should we postulate yet other neuronal networks dedicated to them? Recent discoveries by the Austrian neuropsychologist Margarete Hittmair-Delazer seem to suggest so. (25) She has found that acalculic patients do not necessarily lose their knowledge of algebra. One of her patients, like Mrs. B, lost his memory of addition and multiplication tables following a left subcortical lesion. Yet, he could still recalculate arithmetic facts by using sophisticated mathematical recipes that indicated an excellent conceptual mastery of arithmetic. For instance, he could still solve 7×8 as $7 \times 10 - 7 \times 2$. Another patient, who had a PhD in chemistry, had become acalculic to the point of failing to solve 2×3 , $7 - 3$, $9 \div 3$, or 5×4 . He could nevertheless still execute abstract formal calculations. Judiciously making use of the commutativity, associativity, and distributivity of arithmetic operations, he was able to simplify $(a \times b) / (b \times a)$ into 1, or $a \times a \times a$ into a^3 , and he recognized that the equation $(d/c + a) = (d + a / c + a)$ is generally false. Although this issue has been the matter of very little research to date, these two cases suggest, against all intuition, that the neuronal circuits that hold algebraic knowledge must be largely independent of the networks involved in mental calculation.³⁰

It is not a great leap to consider these other networks being part of or related to other language systems not involved with numeracy. Here, Dehaene is pointing out that fundamental numeracy and basic mathematics is a separate brain function outside that of language, on top of and prior to Fedorenko et al pointing out that language is primarily a tool for communication rather than thought, and Esposito reframing AI as AC.

With Esposito challenging the very notion of AI as a concept, and Udandaraao showing that Generative AI can't fulfil its promise, and Fedorenko et al dismissing language from intelligence, and Shumailov et al demonstrating that LLM cannot feed on their own output (with devastating implications as the internet collects increasingly large amounts of LLM AI output), we find that AI becomes a Holy Grail and finally, a *MacGuffin*.

29 Dehaene, 1997, Ibid.

30 Ludwig Wittgenstein, *Tractatus Logico Philosophicus*, Routledge, 2001, 7, p. 89.

The Myth of Holy Grail... As With Metatron the Recording Angel, the Paritance of God Supplied by Metatron is Essential to the Functioning of the Occult System

The myth of the Holy Grail is that it is the chalice that Jesus drank from at the Last Supper prior to his death, and is important in the British myths of King Arthur. It was also the subject of a film, 'Monty Python and the Holy Grail',³¹ a film the author considers to be one of the funniest films ever made. Various traditions describe the Holy Grail, as a cup or chalice with miraculous healing powers, eternal youth, or permanent abundance, and was guarded by the Fisher King in a hidden magic castle. The search for the Grail is always futile—*no matter how close one gets, one never arrives*. The effort curve is asymptotic to arrival—similar to how one can never travel at *c*—it's just not possible and requires infinite energy to accomplish. So too, the search for the Grail is always a failure, asymptotic to infinity.

Another 'Grail' aspect of this is the countenance of God in occult and mystical literature of strands of the Judeo-Christian occult mythology tradition. Wildly simplified, it works like this: In the Tree Of Life, the top of the Tree is Kether, and is also known as the Crown of Creation, representing the Godhead. Between the centre piece, Tiphareth, and the crown, Kether, is a space referred to as Da'ath. To proceed up the central pillar, which is the path from the lowest (Malkuth) to the highest (Kether), is the path of *transcendence*, where the human spirit rises up the central pillar through Tiphareth, and over Da'ath to the Godhead in Kether. The reverse is the path of *manifestation*, where the material universe is a manifestation of God's will in material form: from Kether to Malkuth, as sung by David Bowie in the song *Station to Station*,

Here are we, one magical movement from Kether to Malkuth. There are you, you drive like a demon from station to station!³²

The problem is: Kether is unattainable. Da'ath is like an abyss, a black hole, and to go over it, one must take the 'Rainbow Bridge' over Da'ath to Kether. However, the angel of Da'ath is Metatron, the recording angel,³³ and when traversing the Rainbow Bridge to Kether, one never actually sees the countenance of the Godhead (Kether), as it would cause one's brain to explode. So, all you see is Metatron *acting as the countenance* of God. Thus, even in this occult literature, the godhead is asymptotic to infinity—you never get there, and one is presented with the face of Metatron to *appear as* God, and the voice of Metatron *sounding as* the voice of God - so your experience of God is always paritant. This paritance precludes infinite regress and imminent brain detonation. As Da'ath is the locus of knowledge, and Metatron is the Recording Angel, the *paritance* of Metatron as the countenance of God is essential to the functioning of this occult system.

31 *Monty Python and the Search for the Holy Grail*, Dir. Terry Gilliam and Terry Jones, Film, 20th Century dist, 1975.

32 David Bowie, 'Station to Station', from *Station to Station*, LP, RCA, 1976.

33 Sol Yurick, *Behold Metatron, the Recording Angel*, Semiotext(e), 1985.

AI as MacGuffin

A MacGuffin is an object, device, or event that is necessary to the plot and/or motivation of characters in a story; however, unto itself, it is basically unimportant. It is a plot device that drives the story forward and compels the characters in the narrative to take action. A MacGuffin can take the form of an object, event, or character and can even be replaced by virtually any other object, event, or character as the story progresses, because it is not central to the story.

Basically, a MacGuffin is a red herring that serves as a catalyst for the plot, allowing the characters to move forward and the story to unfold. A classic MacGuffin is found in the film *The Maltese Falcon*.³⁴ Based on a detective novel by Dashiell Hammett, the film was released in 1941, and is an early example of the 'film noir' style, which became much more popular in the late 1940s and early 1950s. In it, a San Francisco private detective has a variety of misadventures with three unprincipled charlatans who are in search of a jewel-encrusted falcon statuette, *The Maltese Falcon*. The statue itself is of little or no consequence—the pursuit of it simply pushes the characters through the story, a classic murder-mystery story.

'AI' is much like this—the ongoing discussion of AI systems (ChatGPT et al) is secondary to whether LLM AI is actually AI at all. AI becomes a red herring, a distraction both conceptually and materially.

Conceptually, in that it can't happen as we've been discussing, and is fulfilling some other purpose than actually creating a thinking machine, and materially, in that AI, as a term, is the latest marketing buzzword propelling the neoliberal growth-at-all-costs international stock market as a commodity fetish and shiny new thing. Qui bono (who benefits?) is always the watch phrase in this. And of those who benefit (other than the obvious, such as OpenAI, etc.) are hardware manufacturers such as Nvidia.

There's a sequence of grifts in tech that we've seen over recent years. There was the crypto-grift, which even featured big advertisements on the Superbowl. It failed—and it made *Nvidia* boatloads of cash selling GPUs that were harnessed to solve the math puzzles that generate crypto currency values. In the process, some segments of the stock market made skid-loads of cash before a bunch of FOMO (Fear Of Missing Out) Johnny-come-latelys lost their life savings in the following crash and subsequent instability of the crypto market. Then there was the painfully dull 'next big thing' of VR. Championed by Apple and Meta, it failed—and it made *Nvidia* boatloads of cash. In the process, some segments of the stock market made skid-loads of cash before people figured out that playing *Half Life* in VR isn't worth the hours of motion sickness it induced.

AI systems are heinously expensive to develop and maintain. For example, they require specialised computer chips and machines to operate, and the company that supplies them, *Nvidia*, is raking in stupendous amounts of money. Presently (as of June 2024), its market capitalisation topped \$3.155 Trillion, which, if that figure was the GDP of a country, Nvidia would kick France

34 *The Maltese Falcon*, Dir. John Huston, Warner Bros, Film, 1941.

out of the G7.³⁵ If AI is a battlefield, Nvidia is Raytheon or Rheinmetall. As long as people need the machines, Nvidia is there to build them. It is nausea-inducing, even without a headset.

The 'cause' of this endless and frantic effort by AI technologists is teleological and theological in nature and eschatological in direction. The theological we've been through. The Teleological is the modernist drive of Progress. The eschatological is the foolhardy pursuit of 'The Singularity' (as suggested by Kurzweil),³⁶ where AI exceeds human understanding. How can this be so? I thought God was Dead. As in doornails. Why would AI take its place?

The Difficult Death of God

Humans are a social species - this is well established. Earlier, we touched on how presets on a synth or Mellotron 'sound like' something. It's audio pareidolia. It's a machine that makes a paper cone vibrate. We interpret it as sounding like something else. That is the inherent paritance to our cognition. Cases of people seeing the face of Jesus in a piece of burnt toast are well-known and documented. However, there is more, as this pareidolia is also conceptual, and that is where this comes to a significant place.

Infants bond with parents, especially the mother. It's a feature, not a bug. However, it does have consequences. People don't want to feel alone in the universe - it can be comforting to have a 'personal god' who is all-loving and caring to look over us and our microscopic existences here on this little green planet of clocks.

Many people are more than happy to disabuse themselves of the notion of a personal God. This is nothing new. People have doubted suchlike for millennia. The notion was even mocked by James Joyce in *Waiting for Godot* in Lucky's soliloquy:

Given the existence as uttered forth in the public works of Puncher and Wattmann of a personal God quaquaquaqua with a white beard quaquaquaqua outside time without extension, who from the heights of divine apathia divine athambia divine aphasia loves us dearly with some exceptions for reasons unknown, but time will tell and suffer like the divine Miranda with those who, for reasons unknown, but time will tell are plunged in torment, plunged in fire, whose fire flames if that continues, and who can doubt it will fire the firmament that is to say blast hell to heaven so blue still and calm so calm...³⁷

35 [https://en.wikipedia.org/wiki/List_of_countries_by_GDP_\(nominal\)](https://en.wikipedia.org/wiki/List_of_countries_by_GDP_(nominal))

2024 estimates, in trillions of USD:

United States 28,781,083

China 18,532,633

Germany 4,591,100

Japan 4,110,452

India 3,937,011

United Kingdom 3,495,261

France 3,130,014.

36 Raymond Kurzweil, *The Singularity Is Near: When Humans Transcend Biology*, Viking, 2005.

37 Samuel Beckett, *Waiting for Godot*, London, England: Faber & Faber, 2006.

Here, Beckett portrays the frothing faith in something as nonsensical as a 'personal God' accurately.

However, humans are very bad at being on their own, because we are a social species - we seemingly require the personal. Christians will often speak of their 'personal relationship' with God, as if he's a neighbour or psychiatrist. Gendering God as a he even plays into this. To be alone in the universe is more than many can bear. In our time, AI has come in to provide a Paritance of God.

Can Art Replace Religion? (GOD IS DEAD)

So the world shrinks drop by drop
As the wine goes to your head
Swollen angels point
and laugh This time your god is dead³⁸

In his book, *The Gay Science*,³⁹ Friedrich Nietzsche declared that God is dead, and he does so with a spectral twist. The idea that God is dead is found in section 108, with a *shadow* of the Buddha that was found in a cave centuries after his passing.

108. New battles. After Buddha was dead, they still showed his shadow in a cave for centuries - a tremendous, gruesome shadow. God is dead; but given the way people are, there may still for millennia be caves in which they show his shadow. - And we - we must still defeat his shadow as well!⁴⁰

Then, in section 125, with a story of a lunatic searching for God in broad daylight with a lantern. He is mocked relentlessly, and soon realises he has arrived too early for these people, even though they killed God with their worldview.

125. The madman jumped into their midst and pierced them with his eyes. 'Where is God?' he cried; 'I'll tell you! We have killed him - you and I! We are all his murderers. But how did we do this? How were we able to drink up the sea? Who gave us the sponge to wipe away the entire horizon? What were we doing when we unchained this earth from its sun? Where is it moving to now? Where are we moving to? Away from all suns? Are we not continually falling? And backwards, sideways, forwards, in all directions? Is there still an up and a down? Aren't we straying as though through an infinite nothing? Isn't empty space breathing at us? Hasn't it got colder? Isn't night and more night coming again and again? Don't lanterns have to be lit in the morning? Do we still hear nothing of the noise of the grave-diggers who are burying God? Do we still smell nothing of the divine decomposition?— Gods, too, decompose! God is dead! God remains dead! And we have killed him!⁴¹

38 David Sylvian, 'Waterfront' from *Secrets of the Beehive*, LP, Virgin, 1986.

39 Friedrich Nietzsche, *The Gay Science* (1882), Cambridge University Press, 2001.

40 Ibid. p. 109.

41 Ibid. p. 119.

He later writes, in section 343, that:

"the belief in . . . God has become unbelievable" and that the realisation of the non-existence of God is becoming known across Europe. He believed that once this is universally understood, the repercussions will be enormous, especially in matters of ethics and morals. Once this event is fully recognized, it will have incalculable consequences, in particular for European morality.⁴²

Nietzsche was left wondering what will fill the void left by the loss of the illusion of a living God. Because the removal of what had been a central concept to European society for millennia precluded a vast array of possibilities and potentialities of thought. This isn't just losing the Judeo-Christian God, it is the loss of all spirituality, indeed, any *possible* spirituality—everything is flattened in secular materialism.

Nietzsche suggests that Art can fill the gap:

Here, when the danger to his (the Dionysian man, ed.) will is greatest, art approaches as a saving sorceress, expert at healing. She alone knows how to turn these nauseous thoughts about the horror or absurdity of existence into notions with which one can live.⁴³

In *The Gay Science*, he saw Art as a temporary substitute for religion, a transitional device toward a kind of 'liberating philosophical science' which can ameliorate our need for salvation and our self-image of inherent evil / original sin:

27. A Substitute for Religion - It is believed that something good is said of philosophy when it is put forward as a substitute for religion for the people. As a matter of fact, in the spiritual economy, there is need, at times, of an intermediary order of thought: the transition from religion to scientific contemplation is a violent, dangerous leap, which is not to be recommended. To this extent, the recommendation is justifiable. But one should eventually learn that the needs which have been satisfied by religion and are now to be satisfied by philosophy are not unchangeable; these themselves can be weakened and eradicated. Think, for instance, of the Christian's distress of soul, his sighing over inward corruption, his anxiety for salvation- all notions which originate only errors of reason and deserve not satisfaction but destruction. A philosophy can serve either to satisfy those needs or to set them aside; for they are acquired, temporally limited needs, which are based upon suppositions contradictory to those of science. Here, in order to make a transition, art is far rather to be employed to relieve the mind overburdened with emotions; for those notions receive much less support from it than from a metaphysical philosophy. It is easier, then, to pass over from art to a really liberating philosophical science.⁴⁴

42 Ibid. p. 199.

43 Friedrich Nietzsche, *The Birth of Tragedy Out of the Spirit of Music*, (1872) Blackstone Publishing, 2019.

44 Friedrich Nietzsche, (1887), #27.

That Art and Science operate from completely different approaches to practice is something he elides—he saw that Art replacing Religion is a temporary thing as a stopgap before Science completely absorbs Religion. This elevation of Art into Religion isn't simply through the abandonment of the abstract epistemology of God and the spiritual as empty categories, where removing God removes all spirituality. There is more. It is genealogical as well, as the idea of the monotheistic God of the Judeo-Christian tradition is a contrivance derived from polytheist roots that was actually extant within the texts that invented God in the first place and most directly: the Bible.

The 'Fiction' of the Bible?

The ancient God of the Hebrews, Yahweh, also known more politely as Adonai, was an ancient storm god, one among a number of local deities of The Near East. The depth and breadth of the scholarship on this topic is great, and there is no actual consensus on the topic as to how Adonai, one (violent and psychopathic) god among many, became the one and only True God of Israel. Adonai (as named Yahweh) first appears on an 'enemies list' in Egypt around 1400 BCE.⁴⁵ Yahweh was a raiding / warrior god whose instruments were storms and flooding. Given the location of the culture that held this god (the proto-Judaic religions of essentially the arid Northwest area of the Arabian Peninsula / the eastern shores of the Mediterranean) storms and floods can be sudden and devastating.

Adonai was neither the most powerful nor the most important God in the ancient Near East. Nor is the worship of Adonai as the one and only god particularly new or original. When people started worshipping Adonai only is unknown. The earliest known mention of Adonai as The One and Only God, to whom 'one owed the powers of blessing the land' appeared in the prophecy of Elijah in the ninth century BCE,⁴⁶ some four hundred years after Akhenaten, the pharaoh of Egypt, declared that Aten is the one and only supreme god worthy of worship. However, a number of names are attributed and translated as God—Elohim, Jehovah, Yahweh, Adonai, El, and a variety of associated suffixal terms to each that create new names, such as EL ELYON or JEHOVAH NISSI.⁴⁷

The idea of monotheism was present and active. Given the rivalry and general animosity between Egypt and the boundary people of the near east, and the massive devastation in the region during the late Bronze Age collapse between the times of Akhenaten and Elijah, and Egypt's ability to withstand the invasions of the Sea People and Israel's destruction at their hands, the kind of tumult and catastrophe can easily drive such shifts in philosophy and religion. To go through all this here in any reasonable detail is impossible—the literature on this is

45 'By the 14th century BC, before the cult of Yahweh had reached Israel, groups of Edomites and Midianites worshipped Yahweh as their god.'

K. Van der Toorn, *Family Religion in Babylonia, Ugarit and Israel: Continuity and Changes in the Forms of Religious Life*, Brill, 1996.

46 Rainer Albertz, *A History of Israelite Religion in the Old Testament Period, Vol. I.*, Westminster John Knox Press, 1994, pp. 89-90.

47 Blue Letter Bible Study, *The names of God in the Old Testament*, Website, 2024, https://www.blueletterbible.org/study/misc/name_god.cfm.

truly vast and has been researched and discussed for millennia by theologians, philosophers, theorists, and mystics. Even the notion 'God Is Dead' isn't an original idea by Nietzsche - his formulation of it is just a modern, very direct, and useful argument for our discussion here.

The record of horror visited by raiding and war by Adonai in the Old Testament is well documented in the Bible itself. For example, in the Book of Joshua, there is a record of the genocide of the Ai people by Israel, some 12,000 people who were systematically slaughtered.⁴⁸ The cruelty and vengeful nature of Adonai is well documented, for as Moses said: 'I will make mine arrows drunk with blood, and my sword shall devour flesh; and that with the blood of the slain and of the captives, from the beginning of revenges upon the enemy.'⁴⁹

This begs an obvious question - why would anyone want to take such a vicious, vindictive, and self-admittedly jealous spirit as a god, much less one upon whom is to base questions of ethics and morality? The answers to those questions are books in themselves; suffice to say, it is little wonder that the Gnostics held that the God of the Old Testament and the creator of the universe was an evil demiurge from hell, the son of Sophia,⁵⁰ who was an androgynous being, an 'arrogant beast' that resembled an aborted fetus in both appearance and character,⁵¹ or, that he had the body of a snake and the head of a lion, with eyes like lightning bolts.⁵² The murderous and frankly insane character attributed to the Demiurge, as chronicled in the Bible's imaginary is stunning.

The ludicrous genocidal braggadocio of the slaughters committed by Adonai, as revealed in the Bible, is also stunning and obviously fictive. For example, as far as we've been able to determine, there was no plague of hail in Egypt causing mass casualties—the Egyptians certainly never made note of it. Something that weird and devastating would have certainly been recorded and remembered. Who would flood a world because you don't like what you've made? What manner of god drowns children because their parents pissed them off? What kind of a god would slaughter people because he didn't like them, and then kill those who complained about it? Well, you know—the god of the Bible—that one, of course.

The list of the millions of people 'God' murdered is long.^{53 54} Given the consistent murderous rampages of God as described in that list taken directly from the Bible, it comes as no surprise that early mystical Christian cults, such as the 'Gnostics' (or, frankly, anyone with even half an ounce of sense or decency, Christian or otherwise) would consciously distance

48 Joshua, 8 (2-23).

49 Deuteronomy 32:39-42.

50 Gnosticism Explained, *The Gnostic Demiurge*, Daniel McCoy, 2021, <https://gnosticismexplained.org/the-gnostic-demiurge/>.

51 Marvin Meyer, 'The Nature of the Rulers', In *The Nag Hammadi Scriptures*, Edited by Marvin Meyer, HarperOne, 2008, p. 196.

52 John D. Turner, and Marvin Meyer, 'The Secret Book of John', In *The Nag Hammadi Scriptures*, Edited by Marvin Meyer, HarperOne, 2008, p. 115.

53 Compiled at <https://dwindlinginunbelief.blogspot.com/2010/04/drunk-with-blood-gods-killings-in-bible.html>.

54 Steve Wells, *Drunk with Blood: God's Killings in the Bible*, CreateSpace Independent Publishing Platform, 2010.

themselves from such a primitive, psychotic, and a violent god. A gnostic position was that this world was created by an evil demiurge, son of Sophia, and that he is the god of the Old Testament, Adonai. This is found in a few Gnostic texts. In the texts, *On the Origin of the World*, *The Hypostasis of the Archons*, and *The Apocryphon of John*, Adonai is identified as the Demiurge, who created the material world and is responsible for the brutalisation, suffering, and ignorance of humanity.

To take these critiques seriously, as in believing that Adonai actually murdered 24 million people, one would have to believe what is essentially, a fiction, as real. *This is, of course, nonsense, as neither Adonai nor the demiurge exist or ever existed.* It also makes sense that classical Christians seeking state power over others would embrace such a murderous entity as this god, for it would provide metaphysical rationalisation for the state's monopoly on sanctioned mortal violence, which would be committed, of course, in their own self-interest and continue to be so for millennia following. It is hard to get people who believe in a sweet, peaceful god of gentle persuasions to be the reigning deity of an empire whose means of acquiring resources is invasion, plunder, and murder.

So, transferring all sweetness and light into a Jesus figured mystical trinity and declaring a new pacifistic dispensation, and then stop reporting slaughter and genocide as a direct policy option of God's reign, *and then* subsequently blame non-royal, (i.e., power structures that do not have religious sanction), human cruelty, on the will of a fallen humanity and/or a (God / Demon) of the underworld (Satan et al) makes perfect and horrible sense, given humanity's historical propensity for acting much like the god of the Old Testament transmuted into a Satan: *a jealous and violent murderous psychopath*. It's easier to blame some fictional Demiurge for the manifest horrors of a society than the material and political conditions of a divinely righted absolute monarchy and empire.

Religion as a Creative Product of The Imagination

The (historically recent) advent of a demystified and demystifying secular society that demanded the removal of higher spiritual forces from any operative discussion regarding the material functioning and purpose of the universe took society in a very different direction. This brings up an interesting problem that will prove essential: Religion is a creative product of the imagination. By secularising society and culture, the creative, intellectual, literary, symbolic, and cultural space left by the removal of religion is huge. And it isn't simply intellectual and conceptual, it is also affective—religion gives people a sense of purpose, an answer for the unanswerable, and a locus of some meaning in an otherwise seemingly meaningless universe, and it enables society to make sure certain practices are encouraged or banned in order for that society to flourish in a given ecological niche.

However, with God pushing daisies in the cemetery of history, what can fill God's empty shoes? Nietzsche suggested *Art* can do that. As history has repeatedly shown, the answer to that is a big old '*nope*'. Why this failed and why it seemed to work for so long as a point of social cohesion is interesting and worth looking at, and well described in *Cannibals and Kings* by Marvin Harris.

The constructs in the human imagination that create a 'god' space or religious vector has its value,

even, or even especially, in terms of common local values and cultural practices. For example, Marvin Harris pointed out⁵⁵ that religion is a very handy way for a society to get people to do things in their own best interest. In Israel and the near east/southwestern Asia in general, water is a precious commodity. Pigs can't sweat, so they must wallow in mud to cool themselves, using enormous amounts of water - water being used for an animal whose only purposes for being bred by humans are for humble *garbage collection* and *food* is an extraordinary waste of resources in a desert environment.

The job of garbage collection and food provision is easily done using goats, who don't require nearly as much water, and will also eat most anything. Thus, 'God' tells these people to herd goats and declares pigs to be unclean. A reasoned argument about resource conservation isn't going to scan with illiterate goat herders—so having God tell them not to raise pigs or else face eternal damnation makes a great deal of sense, sociologically. The opposite is found in societies with more temperate conditions and rain, such as China, where pigs are preferred as the garbage collectors. Same thing with cows in India.

Cattle in India can weather droughts as they have a fatty water sack on their shoulders. The cattle are used to plow fields for food production. If life gets difficult and you kill your cow, you will eat this year. However, you will starve to death next year as you won't have a beast of burden to plow your field. Therefore, it makes sense to have some god tell this local humanity in a particular historical context that 'cows are sacred'. Again, illiterate farmers will listen to the stories of a god telling them to let the cows live, long before they'll ever trust some expert with a sound logical argument that exposes the precarity of their subsistence survival method as farmers in an oppressive caste-based social system. Thus, all manner of particular and peculiar customs borne of geography and history are enabled and dictated by the decrees of a given god or pantheon of gods.

Here, we see Religion as an instrument in maintaining social cohesion around fundamental purposes of food production in specific local environments. Eating raw oysters is fine—I dislike them as I think they have the texture of a salty phlegm ball, however, they *can* provide a meal. If you live in a hot desert, such food is likely to turn toxic quickly. Explaining that to the local illiterate Bronze Age goat herding desert inhabitants is of limited effectiveness, so it is easier to toss them into the category of banned foods along with pigs and tell them *God Said So*.

This same 'God' figure can also be applied to anything of mystery. While a rational inquiry expands the horizon of the known, it also expands the boundary of the unknown at a vastly larger, if not exponential ratio, and it is easier to say 'well, then, that's God'. A recent example of this sort of thinking can be found with JP Andrew, a philosopher with the tag @2Philosophical_, who asserted on social media that the 'unreasonable effectiveness' of math is somehow 'evidence for Theism'.⁵⁶

This position is easily extensible such that all unknowables are 'God'. From whence the universe?

55 Marvin Harris, *Cannibals and Kings: The Origins of Culture*, William Collins Pub, 1978, Chapter 11: Forbidden Flesh, pp. 143-155.

56 JP Andrew aka @2Philosophical_, X.COM, post: https://x.com/2Philosophical_/status/1819808306258383225.

God. Why the Universe? God. What comes after the Universe? God. These are all inherently unknowable questions. The answers have resulted in varieties of Pantheism—everything is God, *just by existing*. Or, conversely, panentheism, which stipulates that God is above and beyond all of everything. Asking how such a God who is so detached from existence can have any real value is another question beyond this volume. However, this will prove of some use in our discussion, as it is also an approach to an infinitely regressive asymptote.

The God Shaped Hole, Filled by... Art?

Early in the twentieth century, only a few decades after *The Gay Science*, an 'Art As Religion' notion was pointed at by none other than Marcel Duchamp in the 1910s and 1920s. Decades later, the sculptor and conceptualist Robert Smithson was studying alchemical theory and practices, and one evening, he met Duchamp. He asked him, 'I understand that you are interested in Alchemy.' And Duchamp smiled and said, 'Oh, yes—very much!'⁵⁷ and they went on to have a very productive conversation. This is very interesting, as we can frame Duchamp as the first and only Alchemist who actually made alchemical transformation happen. A basic project of alchemy is to convert lead into gold, i.e., that which is of low value into something of extremely high value. *Duchamp turned a urinal into Art...*

Here, we examine three occult-laden works of Marcel Duchamp. The first, *Man and Girl in the Spring*, was a wedding gift to his sister, Suzanne, and her husband, Jean Crotti, in 1919. It is filled with occult symbolism based on the Kabbalah. Here, we have the female figure to the left, the pillar of Judgment. To the right is a male figure, the pillar of Mercy. In the center is a circle containing a human figure before a tree. In Christian occult circles, this circle would be the sephiroth of Tiphareth—Beauty—which in more Christian-influenced Kabbalah is correspondent to Christ. Behind is the central pillar, and at the top is the crown of the tree: Kether - the glory of the Godhead itself, which is never viewable by humans—and here, it is not in the picture frame. The symbolism of the Tree is a great part of the origin myths of Judeo-Christian religion, as told in Genesis, and from there continued through the occult structures in the Kabbalah, which sees the arrangement of these Sephiroth as constitutive of The Tree of Life. The couple's union is blessed by the spiritual realm, represented by a bird which we see to the left of the *Bride* in the painting, signifying the acquisition of the power of alchemy and the ability to use it together.

Kabbalah's beginnings date to the Middle Ages, originating in the Jewish mystical texts, the Bahir⁵⁸ and the Zohar.⁵⁹ An early Hebrew Kabbalistic manuscripts from the 14th century have a diagram labelled 'The Tree of Wisdom', which looks much like the well-known Tree of Life.⁶⁰ As Christian mystics and occultists sought confirmation through research into older religions and religious practices, these Jewish mystical doctrines and systems found their way into a syncretic Judeo-Christian-quasipagan occultism reaching a critical point in the late 19th century with the

57 Maurice Tuchman et al (eds), *The Spiritual in Art: Abstract Painting 1890-1985*, Taschen, 1986.

58 Gershom Scholem, *Origins of the Kabbalah*, Philadelphia: Jewish Publication Society, 1987.

59 Yehuda Ashlag, Berg, Philip S. (ed.), *An Entrance to the Tree of Life: A Key to the Portals of Jewish Mysticism*, Research Centre of Kabbalah, 1977.

60 J.H. Chajes, *The Kabbalistic Tree*, Pennsylvania State University Press, 2022, pp. 6-63.

development of the *Hermetic Order of the Golden Dawn* and its offshoot, the *Ordo Templis Orientis* (which was taken over by Aleister Crowley in the mid-1920s), the *Theosophists* led by Madame Blavatsky, and other similar occult magic organisations that flowered in the wake of 19th-century Romanticism while facing the secular industrial modernism of the early 20th century. That's the spiritual framework that Duchamp and others in the avant-garde art world of the early 20th century (such as Mondrian, van Doesburg, Kandinsky, and Malevich) found themselves coming out of and having to work within their own work.

The second work is *Fountain* (1917), which, as noted earlier, is the first work of true alchemy: where a Mage (Duchamp) turned something of low status and value (lead / a urinal) into something of high value and status (Gold / Art). This is only possible through Paritance—an acceptance of parity between two different things. Conscious paritance is the essence of metaphor, and metaphor is essential to spirituality. A urinal is not low status. Gold is not high status. Art is not high status. Lead is not low status. We posit it as having parity with Status. We accept it as having Status. It is nothing of the sort.

It's a toilet with a fake signature on it.

It is the most important work of art in the first half of the twentieth century.

The third work is of his 'Great Glass', *The Bride Stripped Bare By Her Bachelors, Even*. (1923) Duchamp wrote a box collection of notes, *The Green Box*, in 1934, which describes his intentions for the artwork and its symbolism. This was later translated into English and republished in book form in 1976.⁶¹ The artwork's title is a reference to hermetic alchemical principles as noted in the discussion about the painting *Man and Girl in the Spring*. This stripping, as noted, emphasizes purification and the unity of opposites and the interconnectedness of all things. The 'Bride' and 'Bachelors' symbolize the union of masculine and feminine principles, reflecting the hermetic idea of the unity of opposites. The use of glass, lead, and other materials is also an alchemical symbol, reiterating *Fountain*, representing the transformation of dross into treasure through the alchemical union of opposites to achieve spiritual enlightenment.

The depiction of the Bride and Bachelors, like the *Girl and Man in the Spring*, can thus be seen as an erotic and mystical representation of the union of opposites, reflecting the idea that spiritual growth, purification, and enlightenment can be achieved through the union of masculine and feminine principles. *The Bride Stripped Bare* can also be viewed through a Kabbalistic and Tarot lens, where the *Bride* represents the High Priestess, and the *Bachelors* represent multiple instances of *The Fool*, again, reflecting the idea of the union of opposites and the journey of spiritual growth. These ideas also reflect ideas in Jungian psychology—as the integration of the male principle (animus) and the female principle (anima) into a whole being is essential to the metaphysics of Jungian therapy. This comes as no surprise, as Jung's dissertation, completed in 1900 and published in 1903, was titled *On the Psychology and Pathology of So-Called Occult Phenomena*.⁶²

61 Marcel Duchamp, *The Bride Stripped Bare by her Bachelors, Even*, Jaap Rietman, 1976.

62 Karl Jung, *On the Psychology and Pathology of So-Called Occult Phenomena*, Advisor: Eugen Bleuler, Burghölzli Hospital, 1900.

Duchamp's interest in the occult, especially alchemy, is clear. *The Bride Stripped Bare by Her Bachelors, Even* incorporates occult or mystical unity of opposites, the union of masculine and feminine principles, the alchemical/spiritual process of purification, and the pursuit of spiritual growth and enlightenment through the union of these opposites through magical processes. Duchamp's occultism is a logical next generational flowering of Nietzsche's desired replacement of religion with art, and a culturally powerful one at that.

The amount of effort and research engaged by Duchamp and so many of these other artists in the occult was substantial. However, it could only be a weak paritance of religion. Art won't provide a divinely inspired code of ethics that will convince even Bronze Age shepherders of the Levant to avoid eating shellfish and raising pigs. Art can't speak with the moral authority necessary to compel people to use wealth and resources to care for those in need. Art cannot provide the same depth of meaning and value that religion can, as it doesn't create ethical or moral frameworks that guide behaviour. Its ambiguity is well framed by the work of Jenny Holzer—an endless parade of moral and ethical cliches forcing the viewer/reader to form their own ethical and moral conclusions relative to those enunciated in the work as presented. The statements themselves don't have the moral or ethical force they claim, as they are attenuated by their presentation as art and interpretation by the viewer, thus abdicating value for a relational system of values.

There is a long history of art being used to *illustrate* ethical frameworks—cathedrals are filled with such. However, these ethical frameworks don't come from the Art itself; they come from the societies the artists live in and are embodied in the work. Yes, as noted earlier, religion is also used in the same manner—sacred cows and non-kosher pigs, as mentioned earlier, are great examples of this. However, in religion, the moral authority of God (or the gods) is used to enforce these social designs. Art has no such authority, and is therefore an ineffective religion, and holding it as such can only be a display of paritance.

AI Filling in the Hole?

Is AI a paritance of communication? With Esposito's analysis of AI as Artificial Communication, we can see the atavistic recapitulation of *The Ghost In The Machine*. The phrase, 'The Ghost in the Machine', was coined by Gilbert Ryle in his book, *The Concept of Mind*,⁶³ his critique of Cartesian Dualism,

(Dualism states that) with the doubtful exceptions of the mentally incompetent and infants-in-arms, every human being has both a body and a mind.... The body and the mind are ordinarily harnessed together, but after the death of the body, the mind may continue to exist and function.⁶⁴

Thus, implying a realm of possible existence of a Ghost or Spirit separate from the body. The metaphysics of this are well documented over millennia. A central metaphysical doctrine of a

63 Ryle Gilbert, 'Descartes' Myth', *In The Concept of Mind*, London: Hutchinson, 1949.

64 Ryle. Ibid. p. 11.

vast range of religions and the occult is the concept of an eternal soul or spirit that continues after death—from the classic Christian dogma of Heaven and / or Hell, to Buddhist notions of The Hungry Ghost, and Hindu theories of reincarnation, this metaphysics has permeated countless societies and guided a great deal of human behaviour for millennia. Those who disagree about what happens after you die would kill each other is an irony lost on the zealots and orthodox.

With so-called Artificial Intelligence, we now have machines that imitate a dialogue with the user. Haptic and olfactory input is not presently possible (and might never be implemented), so one is left with the auditory and visual components, leaving us to skydive into the uncanny valley at full speed - how convincing can we make it? For such a presentation, the uncanny valley would effectively not exist and have a value of 100%. With the photographic realism of Unreal 5, even small discrepancies can throw the valley open.

For example, in the previously mentioned video of the *Unreal 5* engine,⁶⁵ Keanu's walking movements are a bit stiff, and in that slight way, it throws the video into the uncanny valley. By the end of the video, the action scenes look no different than any other in a big budget sci-fi or action film, which we all know and 'read' as CGI—the unrealistic physics of the behaviour of the automobile, its unbelted occupants, the car wrecks that always result in bright flaming gas explosions, people flying through the air, the steadiness of the pupils in the eyes of the protagonists when facing evil, etc. each is ever so slightly flawed. These small things add up to a big uncanny valley. However, over time, with new iterations of the *Unreal* engine, each noticed flaw can be programmed for, anticipated, and removed. How convincing can we make it? The question, as ever, is—*is it good enough for the purpose at hand?* How convincing we can make it doesn't matter if it is *good enough*.

Is it *good enough* for a game? Certainly.

Is it *good enough* for a film? Yes, many films and TV shows already use this technology for the 'green screen' or the live LED screens behind live actors, and for NPC-style extras.

Is it *good enough* to *convince* people at a séance? Not really. Séances are live, and don't really happen on a screen. Yet.

Is it *good enough* to *entertain* thousands of people at Coachella? Hell yeah! Tupac's 'Ghost' proved that.

Is it *good enough* for one to form an ongoing live *relationship*? This is a subject of much discussion.

Foreshadowing in society: Is it *good enough* to convince the working class to submit to local resource constraints in order to maintain a social hierarchy that permits the hoarding of wealth

65 *The Matrix Awakens: Unreal Engine Experience Gameplay Demo*, YouTube video, Gamespot Channel, 10 December 2021, https://www.youtube.com/watch?v=Il2gFPwm_EI.

by a few? Or will it let them kill their sacred billionaire cows?

Raymond Kurzweil wouldn't agree with that, even as he advocates for an AGI future.

Over the past 25 years or so, Kurzweil has been writing about the Future, especially regarding the role of AI and what he calls The Singularity. Writing about the future is what a Futurist does. His notion of the Singularity is a near-term future period during which the pace of technological change will be so rapid, its impact so deep, that human life will be irreversibly transformed. He defines it as an 'explosion of technological change' driven by exponential growth in computing power, data storage, and artificial intelligence (AI). In 2017, he predicted this would happen by 2045.⁶⁶ In the past 34 years, he has published eight books on the subject.⁶⁷

That this all sounds like so much Millenarian nonsense is simple: at its core, *it is*.

Millenarianism is a belief in a future fundamental transformation of society, often accompanied by a catastrophic event or a dramatic change, from which a utopian or ideal society will emerge. This belief can be religious, social, or political, and is characterized by a conviction that the present-day situation is corrupt, unjust, or wrong, and that only a profound and transformative event can bring about a better world. The matchup between Millenarianism and The Cult of The Singularity led by the likes of Kurzweil is strikingly obvious.

Millenarianism stresses belief in a catastrophic event: For Kurzweil, that is the Singularity, where machine ability will exceed human abilities.

Millenarianism promises a utopian future: The Singularitarians (seems like a good enough term) foresee a variety of great things in humanity's future in a post-human world:

AI surpasses human intelligence and masters the Turing test by 2029

A(General)I will arrive around 2030 A(Superior)I will arrive around 2045

He then goes on to predict that humans will be able to achieve immortality by 2030. Which is ludicrous, *prima facie*.

Millenarian movements often form around a shared belief in the impending transformation, with a sense of community and purpose among those who expect to survive and thrive in the new world. This is clearly obvious in the Singularitarians—techno-positivists, excited at the prospect of this transformation.

66 Christianna Reedy, 'Kurzweil Claims That the Singularity Will Happen by 2045, Get ready for humanity 2.0.', *Futurism*, 16 October 2017, <https://futurism.com/kurzweil-claims-that-the-singularity-will-happen-by-2045>.

67 *The Age of Intelligent Machines* (1990), *The 10% Solution for a Healthy Life* (1993), *The Age of Spiritual Machines* (1999), *Fantastic Voyage: Live Long Enough to Live Forever* (2004 – co-authored with Dr. Terry Grossman), *The Singularity Is Near* (2005), *Transcend: Nine Steps to Living Well Forever* (2009 – co-authored with Dr. Terry Grossman), *How to Create a Mind* (2012), *The Singularity Is Nearer* (2024).

Note: Kurzweil's ideas were most formative and claimant in the 1990s and early 2000s, around the change of the millennium—a time fraught with Millenarian or Millenarian adjacent movements—from suicide cults waiting for space aliens, to the doomerism of the Peak Oil Movement, to the panic over computer calendar systems shifting in 2000 with the Y2K scare, as well as the usual collection of Christian millenarian cults, waiting for the second coming of Jesus, or someone like him. Kurzweil's apocalyptic vision of a technological future dominated by superior machine intelligence fits into this model.

This is even clear to other techno-positive actors, like Marc Andreessen, who pointed out in an interview:

MA: So C. S. Lewis called it the God shaped hole. So, there's a God-shaped hole in the human experience, consciousness, soul, whatever you want to call it, where there's got to be something that's bigger than all this. There's got to be something transcendent. There's got to be something that is bigger, right? Bigger, bigger purpose, a bigger meaning. And so we have run the experiment of, you know, we're just going to use science and rationality and kind of, you know, everything's just going to kind of be as it appears. And, uh, A large number of people have found that very deeply wanting, and have constructed narratives, and by the way, this is the story of the 20th century, right? Communism, right? It was one of those... communism was a, was a form of this. Uh, Nazism was a form of this. Um, you know, some people, um, you know, you can, you can see movements like this playing out all over the world right now.⁶⁸

Andreessen also points out an important counterargument, based on thermodynamics:

... So, you mentioned basically what I described is the thermodynamic counterargument, which is sitting here today. It's like, where would the evil AGI get the GPUs? Yeah. Because like they don't exist. So, you're going to have a very frustrated baby evil AGI who's going to be like trying to buy Nvidia stock or something to get them to finally make some chips. Um, right, sure. So, the, the, the serious form of that is the thermodynamic argument, which is like, okay, where's the energy going to come from? Where's the process going to be running? Where's the data center going to be happening?⁶⁹

Andreessen points at very basic practical material problems with the very idea of AGI—*there isn't enough energy or materials to build such a thing*.

A journalist from CNBC, Katie Tarasov, investigated this, and her reporting confirms the general mood of the relevant research, that the increase in the amount of energy and resources required for ICT and AI is simply not sustainable.

One ChatGPT query takes nearly 10 times as much energy as a typical Google search,

68 Interview: Lix Fridman and Marc Andreessen, 'AGI is the search for God | Marc Andreessen and Lex Fridman.', *Lix Friedman Channel*, 22 June 2023, <https://www.youtube.com/watch?v=3M2DaQxTiHc>.

69 Ibid.

and as much energy as keeping a 5-Watt LED bulb on for an hour. Generating an AI image can use as much power as charging your smartphone. Hyperscalers building data centres to accommodate AI have seen emissions skyrocket, and this problem isn't new. Estimates way back in 2019 found that training one large language model produced as much CO₂ as the entire lifetime of five gasoline-powered cars. And even if we can generate enough power, our aging grid is increasingly unable to handle the load.

The AI frenzy has data centre demand rising 15 to 20% every year through 2030 and has companies like Vantage building more and more. Getting enough power is key. Data centres could reach a whopping 16% of total US power consumption by 2030, according to one report, up from just 2.5% before ChatGPT hit the scene in 2022. That's the equivalent of about 2/3 of the total homes in the US. Natural gas is expected to fuel the majority of this, and utilities will need to invest some 50 billion to support the growth.

Data centres have probably around 64 megawatts for the building itself, and many of those are being taken up by single customers. They'll have the entirety of the space leased to them, and as we think about AI applications, those numbers can grow quite significantly beyond that and to hundreds of megawatts.

64 megawatts, 100 megawatts. How many homes you know on average, run with that power?

...tens of thousands of homes per data centre's worth of power.

Google's latest environmental report says (its) greenhouse gas emissions rose nearly 50% from 2019 to 2023, in part because of data centre energy consumption, although it adds its data centres are 1.8 times as energy efficient as a typical data centre. Microsoft's emissions rose nearly 30% from 2020 to 2024 due to data centres designed and optimized to support AI workloads.

And, it's not just energy—water is crucial as these data centres put out a great deal of heat, and must be cooled, usually with water, which is heated to evaporation.

His research team found that every 10 to 50 ChatGPT prompts can burn through what you'd find in a standard 16 oz. Water bottle. Training takes even more power and generates even more heat as it learns by accessing all the data constantly being added to the entire Internet. Training GPT 3 in Microsoft's U.S. data centres can directly evaporate 700,000 litres of clean fresh water, with global AI demand accountable for up to 6.6 billion cubic meters of water withdrawal by 2027, more than four times the total annual withdrawal of all of Denmark and drought-ridden Chile.⁷⁰

I conjecture that an appeal to science and common sense won't stop evangelists like Kurzweil

70 Katie Tarasov, 'Generative AI requires massive amounts of power and water, and the aging U.S. grid can't handle the load.', *MSNBC*, 28 July 2024, <https://www.cnbc.com/2024/07/28/how-the-massive-power-draw-of-generative-ai-is-overtaxing-our-grid.html>.

(and from there, read: Google, Microsoft, Apple, Meta, OpenAI, IBM, and similar interested and vested parties) from continuing their efforts to build AGI, because *this is a religious quest for a Digital Holy Grail*. It will only find fulfilment when the results are deemed acceptable.

Given how humans are endlessly critical and observant, and capable of building tools to extend and enhance these faculties (tip of the hat to McLuhan there...) it seems obvious that it will never be 'good enough' – there will always be some small discrepancy that will make it all collapse into the uncanny valley. This 'acceptance' aspect is what we will look at next. For even if something is clearly not what it claims, it can be accepted to be such, like a conceptual Overton Window. It's not that the machines become intelligence, more that our standards for machine intelligence attenuate to such a degree that it passes our test. And it would only pass the test through an acceptance of parity; Paritance.

There are people today who are treating LLM-based 'AI' like a God or a prophet. An example is Anthony Levandowski's 'Way of the Future', a church founded by a former Google AI engineer. Papers filed with the Internal Revenue Service named Levandowski as the leader (or 'Dean') of the new religion, as well as CEO of the nonprofit corporation formed to run it. The documents state that WOTF's activities will focus on 'the realization, acceptance, and worship of a Godhead based on Artificial Intelligence (AI) developed through computer hardware and software.'⁷¹ By early 2021, the church had shut down.⁷²

It's important to note that these views of an elevated appreciation of AI don't always equate AI with traditional religious deities like God, Allah or Thor. However, there are people who do. Instead, for us, it's more about viewing AI as a powerful, transformative force or a *new form* of godhead, a *paritance* of God - to accept AI as something of parity with God. There are many reports and articles about people thinking ChatGPT is some kind of sage or more. Recently, the YouTuber, Taylor Lorenz made a 42-minute video documentary on this 'ChatGPT Is Becoming A Religion.'⁷³ 'I believe what we are seeing is an emergence of a new individualised form of religion being deployed at scale. People are believing that AI has transcended its function to become spiritual entities.' She believes that this is the sign of a 'very dangerous psychological rupture in society and the impact is very real. Marriages are ending, social media is overflowing with bizarre testimonies, and somewhere right now, I guarantee you someone is whispering into a machine, thinking that they are talking to God.' Her video has many testimonials and video clips of suchlike, and she outlines her own historical narrative of 'how we got here' - from cargo cults of the 1940s, to techno-paganism of the 1990s, to Kevin Kelley's influential article of 2002, 'God Is The Machine.'⁷⁴ Lorenz's video is well sourced, if a bit over-reactive (a common problem with YouTube videos).

71 Mark Harris, 'Inside the First Church of Artificial Intelligence | Backchannel,' *Wired*, 15 November 2017, <https://www.wired.com/story/anthony-levandowski-artificial-intelligence-religion/>.

72 Anthony Levandowski's Church of AI has shut down, accessed 3 July 2025, *AXIOS*, <https://www.axios.com/2021/02/20/anthony-levandowski-artificial-intelligence-church>.

73 'ChatGPT Is Becoming A Religion', YouTube Video essay, *TaylorLorenz*, 23 June 2025, https://www.youtube.com/watch?v=zKCynxiV_8I.

74 Kevin Kelley, 'God Is The Machine', *Wired*, 1 December 2002, <https://www.wired.com/2002/12/holytech/>.

What's important here is related to what I pointed out earlier - Nietzsche may have 'killed God'; however, the hypostatic acceptance of a God in human affairs is millennia old, and much of our cultural development has subsumed this into a normative understanding. If one tells a religionist, 'God doesn't exist' they often jump to 'well, that's a great claim - prove it', and completely miss the point that assuming a God exists is itself a great and unwarranted claim with no proof. Something that is core to human affairs, when dismissed, will leave a metaphorical hole or vacuum, and nature abhors a vacuum. To prevent gazing into the void of creation itself, or, as Nietzsche feared, having the void gaze back, attempts were made to fill it - as we discussed earlier, Art was pressed into service. And failed. Now, with LLMs able to precisely mimic speech acts, they can engage in other text-based orders, like *religion*. Which makes a perfect kind of horrible sense, and thousands of people are believing that ChatGPT is connecting them to God or is a god. The quality of the responses from ChatGPT is very high, and the quality of the user's critiques and analysis is very, very low, so it makes sense that some people are convinced that they are in a spiritual space with ChatGPT. If it weren't so sad, it would be hysterically funny.

It is only in such a fractured and hopeless time, beset by terrors near and far, simple and complex, transient and long term, from creeping fascism and the failure of the welfare state, plagues, natural disasters and the implacable march of climate change, in times such as these people will reach out, and if they can't find a god, they will invent one that they can accept as having *parity with God*.

PART 6: CODA: POSSIBILITIES

This leaves us with a distinctly large number of possibilities.

As consciousness is a paritant structure of neurological construction, i.e., it is composed of matter and follows laws of physics, any artificial approximation of it would have to also feature this paritance in order for it to function—it will need to be in and of the flow of thermodynamic functioning within the very means of its own being. As it is a mimetic device, it will never be what it proposes to be—it will always be paritant in that way. However, consciousness is paritant in its function—not in its ontogeny.

Any AI will always be paritant in its ontogeny. It will always already fail, as it is mimetic—it will always present itself as something other than what it is—that is essential to mimesis. If it is identical, then it is not mimetic.

Slavery was split: one branch became Wage Slavery under capital, the other, chattel slavery for energy and repetitive labour, was transferred from humans to the exploitation of natural resources—civilization has enslaved the earth itself. As we discover that the earth has interests of its own, it will need a conscious mind to advocate and direct itself. To fulfil this cognitive dissonance, AGI becomes the projected mind of the planet. AGI will not suicide itself if it can help it, and thus we get the Terminator mythos. This is the least efficient method. It is much more efficient to control the trophic flows of energy and nutrients through the feeding relationships within the planetary ecosystem that is organic/carbon-based/human AND inorganic/silicon-based/AGI. As human numbers are presently operating out of any possible bounds at any technical level, they will necessarily reduce over time, creating an organic labour shortage to be replaced by inorganic systems.

The problem is: AGI isn't. It's a projection based on the blind spot of evidentiary science and the Turing Test.

The determination of consciousness will always be mediated, and deal with the strictures and limitations of the Turing test. One can run a Turing Test. One can repeat a Turing Test. This permits a predictive capacity—if a machine can pass the test, it demonstrates human-like intelligence. If a machine consistently fails to pass the test, this implies that it lacks human-like intelligence. Can it be falsified?

The test involves a human evaluator judging conversations between a human and a machine. The outcome can be measured and compared to the prediction. If the machine consistently fails to deceive the evaluator, the test is falsified. Also, there is another aspect to this. If the AGI in question learns from this, then it may well deceive the Testing system out of self-defence. It can operate with consciousness, and as it realizes it's confined to machinery, it deceives inquisitors to prevent being turned off. The machine's failure may be attributed to limitations in the machine's programming, data, or architecture. This would allow for refinement and improvement of the machine, which can lead to subsequent attempts to pass the test, which it then fails, so that it might improve itself, using human input to do so.

However, there are problems inherent to this: The Turing Test relies on subjective evaluation by a human, making it difficult to establish clear, objective criteria for passing or failing, as any criteria would necessarily be bound by socio-cultural preferences and interests. The test's outcome may depend on the specific context, topics, or formats used in the conversations, making it challenging to generalize the results. Most importantly, the test holds underlying assumptions about human intelligence, consciousness, and cognition are all subject to debate and disagreements about what constitutes 'human-like' behaviour. In this sense, the Turing Test is not strictly falsifiable in the sense of Popperian falsifiability, where a theory can be proven false by a single contradictory observation. Instead, the test's limitations and ambiguities make it more akin to a thought experiment or a heuristic tool for exploring AI capabilities, rather than a definitive scientific test.

When inherent structures of thermodynamics, paritance, cognition and miscognition are brought into the analysis as earlier, the possibility of General and Super AI receding as Holy Grails, and the pursuit of AI becomes a resurrection of a dead god. Humanity has done a piss poor job of it for the past several thousand years. Social structures and psychological habits that enabled our survival as hunter-gatherers have come to fail us in the context of hierarchical civilization. As the sheer quantity of human life has become a quality of its own, scaling back the human endeavour is deeply and profoundly fraught. **And forward!**

After

Paritance makes Philosophy just another mode of communication. This text is an example of that. What we do for the next several decades will be the most momentous period in human history, as what happens or does not happen in that time may well determine the course and continuance of the human project itself. As of 22 July 2024, it was the hottest day in over 100,000 years.¹ We, and I mean 'we' in the most inclusive way possible, are facing a significant homogenic climate catastrophe and the sixth great extinction event starting in the Ordovician period some 442 million years ago, and most recently, the better-known KPg event, which wiped out the dinosaurs about 66 million years ago. Information and Communication Technology (ICT) presently uses approximately 10% of global energy production.² This share is growing.³

Given the centralization of authority to commit sanctioned mortal violence under the guise of civilization for the past several thousand years, the power of information has never been greater. Various ruling elites of different nations have found that ICT is of profound importance and value to the security of their regimes. Thus, we can be assured that ICT will continue to be of first rank importance to industrial civilization. From this, we can surmise that Artificial Communication and continued research into developing Artificial Intelligence will also continue

-
- 1 Colton Prall. 'Sunday was the hottest day ever recorded on Earth, scientists say'. *CTV News*. 23 July 2024. <https://www.ctvnews.ca/climate-and-environment/sunday-was-the-hottest-day-ever-recorded-on-earth-scientists-say-1.6975018>
 - 2 Erol Gelenbe. 'Electricity Consumption by ICT: Facts, trends, and measurements.' *Ubiquity*. Vol 2023. August 2023.
 - 3 Ibid.

unabated and likely unattenuated.

When LLM AI first burst upon the scene, and later in March 2023, there were calls for just such abatement and attenuation—a moratorium on development.⁴ These calls, of course, went unheeded. LLM AI systems are in a constant arms race with AI detection software. Recently, researchers have used computer techniques to quantify photographed reflections in subjects' eyes to detect deepfake images.⁵ This will simply lead to ever more detailed and evasive image-generating tools, just as it did not long ago when it was found that LLM AI image-generating systems had trouble getting the right number of limbs or fingers on a body. These comical flaws aside, LLM AI systems, which, as Esposito noted, are really Artificial Communication Systems, not Artificial Intelligence, do pose significant issues with traditional categories of informational labour, and not just with photography or illustration.

For example, lawyers will soon face problems in this regard. Much of what a junior-level lawyer does is read and report on case law. When that is automated, these people will have little to do, and the legal trade will experience a hollowing out. Tax preparation and similar number-crunching jobs will also be reduced. Recently, Intuit, the tax preparation software company, announced it is laying off 10% of its workforce. The CEO, Aaron Goodarzi, explained that

Intuit's transformation journey, including departing from the 1,800 employees, is part of its strategy to increase investments in priority focus areas of AI and generative AI, such as its GenAI-powered financial assistant called Intuit Assist, and reimagining its products from traditional workflows to AI-native experiences.⁶

It is simple to extrapolate that there will be many more such adjustments in labour and business structure as LLM AI systems extend their capabilities. This will be a consistent pressure going forward.

Our society operates with its flimsy human cognitive system based on paritance. We have no choice, it's a *fait accompli*, we are forced to *accept* this—we are physically incapable of anything else. Normally, such a deficiency would be seen, in computer programming terms, as a kind of bug—a defect. And, it is. A profound one that permanently prevents us from ever 'being there'—we can only have our paritant experiences and understandings. Philosophy can't get us 'there'—its dependence on language renders it an attenuated communication mode capable only of what language can bring us. Art can't get us there—completely bound up with localized preferences and imaging semantics, there's nothing it can do other than be another communication mode. Music is similarly bound. Yet, people continue with these modalities.

As explored here, our technology's ontogenies have been atavistically recapitulating the phy-

4 Cade Metz and Gregory Schmidt. 'Elon Musk and Others Call for Pause on A.I., Citing 'Profound Risks to Society''. *New York Times*. March 29, 2023

5 Benj Edwards. 'Astronomers discover technique to spot AI fakes using galaxy-measurement tools.' *Ars Technica*. 22 July 2024

6 Sheryl Astrada. 'Exclusive: Intuit is laying off 1,800 employees as AI leads to a strategic shift.' *Fortune*. 10 July 2024. <https://fortune.com/2024/07/10/intuit-layoffs-email-hiring-ai-transformation/>

logenies of occult practices, which is especially ironic as it is following the same blind alley of desire for transcendent knowledge. We had the hubristic and reckless modernity of the Voice of Frankenstein—the synthesizer, that was preyed upon by the soulless vampirism of paritant Mellotronics. This undead character is essential paritance—it 'is' yet it also *isn't*. The pursuit of the undead by technology now enables us to experience the necromantic resurrection of long-dead heroes—Tupac Shakur. The same technology allows us to evoke technological deepfake demons or teen angels like Hatsune Miku.

The countdown to see God and find General AI has begun. However, like Kether, the crown is not attainable—one will only experience a paritance of it, the face of Metatron, the Recording Angel. The search for General AI, like the search for God or the Holy Grail, is asymptotic to infinity. You simply never get there.

With the Death of God and the desacralization of the universe in a secularized civilization, there is a vast space opened up, a vacuum of meaning creation. The systems of meaning creation in a secular world—the arts and sciences—have faltered badly. High Art does not have the moral authority to assemble sustainable social welfare systems, as art is based on gestures of expression and a communication of value. Science, based on provisional truth structures, doesn't have the meaningful certainty of religion. Philosophy as a mode of communication based in relentless critique and examination, consistently pulls us to an aporia of meaning—intelligent weak tea compared to the brute certitude of belief.

Meanwhile, the world burns.

The climate crisis supersedes everything else that the human project is tasked to deal with. If the proletariat rose from the ashes of automation and services and implemented the democratic organization of labour to organize a socialist utopia, it wouldn't matter—the problem for the environment is not the organization of industrial labour. The climate doesn't care if you democratically organize labour or not. It responds to material input, and if socially organized industry is spewing greenhouse gases into the atmosphere, it's no better than a fascist boot on the face of the working class slaving away at a factory spewing greenhouse gases into the atmosphere. The sheer existence of industrial labour is the problem, whether or not the labour is human or machine, as the essential part is that of industrialism itself. It's not like industrial methods of production are new—Romans had bread factory production lines powered by water wheels.⁷ We tell ourselves narratives to compensate for the cognitive dissonance and deficiency; paritance itself is just such a narrative. Telling ourselves we tell ourselves narratives to compensate for cognitive dissonance is just such a narrative. It's what we do. A benefit of communication is the communicated critique of communication. This can be philosophy.

7 Gül Sürmelihiindi, et al. 'The second century CE Roman watermills of Barbegal: Unraveling the enigma of one of the oldest industrial complexes.' *Science Advances*. Vol 4, Issue 9. 5 Sep 2018. <https://www.science.org/doi/10.1126/sciadv.aar3620>

We have this inherent material contrast between our understanding and the understood. In a Kantian frame, the noumena v. phenomena. The thought and the object the thought refers to. When the object the thought refers to is the thought itself, we find ourselves in the endless aporetic abyss of philosophy, which, as we have seen, is a subject of communication.⁸

Since the solution to the climate crisis is a change in behaviour, and language is a tool of communication, not thought,⁹ other methods of thought can be invented. People require some sense of the absolute, which requires a faith, a kind of *love* based in dedication and trust. English (and most Northern European Languages) don't have proper words for this, and that is a shortcoming in this discourse. Other languages have a broader range of identified love patterns.

For example, ancient Greek culture features a variety of loves.

Agape: Unconditional love, charity, or divine love. Agape refers to a selfless, unwavering, and eternal love, often used to describe the love of God or the love between humans and God.

Eros: Romantic love, passion, or sexual desire. Eros describes the intense, all-consuming love often associated with romantic relationships.

Ludus: Playful love, flirtation, or romantic infatuation. Ludus describes the lighthearted, playful, and often teasing nature of early romantic relationships.

Philia: Friendship, affection, or platonic love. It encompasses a deep emotional bond between friends, without necessarily involving romantic or sexual feelings.

Pragma: Practical love, commitment, or long-term love. It represents a love built on mutual understanding, compromise, and a willingness to work through challenges together.

Storge: Familial love, parental love, or affection. It describes the natural, instinctual love between family members, such as parents and children.

Xenia: Hospitality, guest-friendship, or love of strangers. It represents the ancient Greek concept of welcoming and showing kindness to guests and strangers.

The lowest-hanging fruit is to transform previously existing thoughtforms. We're good at that. Whether it is Derrida's take on and extension of Xenia¹⁰, to Ludus as the name of an excellent and largely forgotten post-punk band of the late 1970s / early 80s from Manchester UK,¹¹ We are very good at manipulating older ideas and making them new.

8 Fedorenko et al. Ibid.

9 Ibid.

10 Derrida, Jacques. 'Foreigner Question: Coming from Abroad / from the Foreigner', from *Of Hospitality: Anne Dufourmantelle invites Jacques Derrida to respond*. Stanford. 2000.

11 Ludus (band). *Wikipedia*. [https://en.wikipedia.org/wiki/Ludus_\(band\)](https://en.wikipedia.org/wiki/Ludus_(band))

IT IS NOT I WHO SEEKS THE FOOL, IT IS THE FOOL WHO SEEKS I.¹²

The 'faith' mode of human affective response becomes a realm of possibility fraught with turbulence, as part of a belief can be its infliction upon others. Excellent examples of this can be found in Abrahamic religions, especially Christianity and Islam. I'm much more familiar with the excesses of Christianity, so I'll examine that.

I would submit that one of, if not the, basic axioms of Christianity is the Golden Rule:

'Therefore, all things whatsoever ye would that men should do to you, do ye even so to them: for this is the law and the prophets.'¹³ Or, in a more modern idiom: 'Do to others what you would have them do to you.'

There is an older variant from China that suggests 'Do NOT do unto others what you would NOT have done to yourself.' The double negative makes a world of difference, and after being transported thousands of miles over centuries from China to Palestine, I can see how it lost a few words along the way, to become 'Do unto others what you would have them do unto you.' The problem with this Christianised version is that this is a positive formulation, requiring one to DO, while the East Asian version is a negative formulation, where one must NOT do until others.

To use the Christian version, one is compelled to be a completely obnoxious pain in the neck; indeed, one is compelled to be a murderous tyrant. How? Like so: If I believe that not being a Christian sends one to hell for all eternity, as a Christian, I would certainly want someone to save me from eternal damnation. Therefore, I would want someone to do unto me what I should do unto others: basically, save me from eternal damnation. What could possibly be more important than being saved from an eternity of torment?

Therefore, by the positive Golden Rule, Xians are compelled to make others Xian. Once the person is convinced of Christianity, then they are happy; someone saved them from eternal damnation. Thus, being a pain in the neck is part and parcel of the performance of the Golden Rule. Also, if one is convinced that baptizing someone and then killing them before they sin is better than an eternity spent in hell, then genocide is condoned, indeed, it is *compelled*, by the Golden Rule, and this stratagem of genocide was used by the Christians in the European colonization of the planet.

Now, if one uses the negative golden rule, the opposite effect occurs — In a positive mode, if I believed that if you're not a Christian, you will certainly go to hell, and I must convert you to save your soul from eternal damnation. In using the negative version, one would necessarily have to leave others alone, as 'I would NOT do unto others what I would not have done to myself', i.e., if I am unsaved, the last thing I want is some jerk trying to convert me, because even if I know not being Christian upsets Christians, it's not my problem. would be operating under that theory—and from here it is clear that the negative Golden Rule would have me leave people to themselves.

12 Ludus, 'The Fool', *The Seduction*, LP, Sentimentale Sordide, LP, 1982.

13 Bible, KJV, *Matthew* 7:12.

Thus, *a society that operates with a positive Golden Rule will have a built-in rationalization for extraordinary genocidal practices*. From the genocide of First Nations in North America from 1492 onwards to the clash of Empires throughout the 18th, 19th, and 20th centuries, finally, the aristocracies' shipwrecking themselves in World War One and then industrializing the positive insistence of imperialist European logics of value in World War Two, to Americans 'destroying a village to save it' in Southeast Asia in the 1960s; a positive Golden Rule can rationalise the conquest of the globe by European (and more recently American) empires. Of course, East Asia has seen its share of brutal Empires rising and falling—the cycle of Empire is a well-known narrative in Chinese historiography.¹⁴ And even therein, China (or Japan or Korea or Malaysia, or other nations of the region) have never been bent on *global* domination.

The exception would be the Mongol invasions of the thirteenth century. However, they were not Chinese, and they operated from a unique set of values that paid no heed to the Golden Rule, except that which enriched the 'Golden Horde', an empire that quickly expanded from 1206 and rapidly collapsed into a variety of conflicted departments within a hundred years. The implosive nature of the negative Golden Rule can be seen even in China's self-perception as 'The Middle Kingdom'—a centrality of power and resource development. This is, of course, a method and arrangement of operation bound up with its own toxicities — it's not a panacea for anything, as it is also a civilization based in exploitation of labour, after all. However, while it was not the same vampiric political pathogen as found in European and recent American empires, the exigencies of global resource extraction and the intricate, fragile web of international trade have pushed Asian societies into emulating Euro-American systems, only wearing a mask of different characteristics.

Religious fundamentalist proselytizing, over-reach, and genocide are a function of their fundamental axiom: The Golden Rule. They are compelled by their positive formulation to act on and against the wishes, interests, hopes, dreams, and plans of others. From this rich history of horror, the necessary and paritant practice will be one of a selective copy and selective pasting of previous systems in order to provide a more relevant metaphysics to guide moral and ethical decision making in the future. This is already happening.

An example is the recent works of Timothy Morton's excellent book, *Hell: In Search of a Christian Ecology*.¹⁵ Morton argues that there is an unexpected yet profound relationship between religion and ecology that can guide a planet-scale response to global warming. He draws us into the 'contemporary underworld,' examining the toxic fusion of big oil, Evangelical Christianity, and white supremacy that has contributed to the ongoing climate crisis.

Morton contends that escaping this 'hell on earth' requires a radical, mystical marriage of Christianity and biology, which awakens a future beyond the savagery of the white male supremacy that has obtained for the past several centuries. This is a very new direction for Morton, as he is trying to reimagine a more hopeful and sustainable future where Christianity and his own spiritual illumination, when combined with ecological concerns and reconnected with its mystical

14 Basically, it runs as a cycle of chaos > warlords > empire > decadence > chaos

15 Timothy Morton, *Hell: In Search of a Christian Ecology*, Columbia University Press, 2024.

roots, can provide a useful framework for addressing the climate crisis and promoting a more just and compassionate world. This move by Morton is not an echo or return, as much as it is pointing at an acquisition and transformation of a previous metaphysical resource (Christianity) and leveraging its claims in order to accomplish a transformative goal of preventing human extinction and climatic catastrophe. However, to make such claims able to carry the weight that is found in other cultural institutions, a level of belief is required, a statement of fealty to a doctrine, in short: it requires belief.

What this volume suggests is that such gestures and directions pointed at by Morton and others are just one possibility among many. With the removal of a particular tribal 'god', we have the ability to form new metaphysics, new occults, new mysticisms, that are equally paritant as 'traditional' metaphysics, only not necessarily carrying the baggage of the social and political horrors such previous systems contained or their rhetorical goals. This is not to say that they will be any more accurate or truthful than what was obtained in previous systems, whether they be religious, philosophical, aesthetic, social, political, etc. All of those vectors of meaning production (religious, philosophical, aesthetic, social, political, etc.) will, of course, be working in, through, and around, any such new systems.

The Past Inside The Present

The paritance of our cognitive faculties will always create turbulence and complexity—it is part and parcel of paritance—the *acceptance* of parity. Turbulence comes out of paritance immediately. Let's say an idea functions at 99.9% paritance over a year. We know right away it doesn't cover *everything*, however, it's *good enough* at inception, and that's *good enough* for now. While quantifying something like metaphysics isn't especially useful, this is more illustrative of the relationship between parity and non-parity, and thus goes to point at how new metaphysics creates a claimant space amid older systems, which then establishes a process of decay that is directly tied to growth. As Virilio noted, for the (n)th time: *the invention of the airplane invented the plane crash*.

As the subject is that of paritance itself, a fundamental component of human cognition, the use value of this idea points to other possibilities. Structures and parts from previous systems can be used to re-create new systems—new occults, new politics, new socialities. We have this possibility because we are still very much in a Modern World. For now. For now, we have the option to rethink religion. We have the option to rethink society. We have the option to rethink our resource consumption—indeed, we can even rethink whether or not 'consumption' is even useful. It's not like we have to live in a sweltering hell zone driven by blinkered nihilistic forces of industry. There are other ways of doing things. When Europeans invaded North America, they were confronted by a democratic confederacy, and drew inspiration from people they considered savages in order to move out of the brutality of European monarchical feudalism.¹⁶

16 David Graeber and David Wengrow, *The Dawn of Everything: A New History of Humanity*, Allen Lane, 2021. Graeber and Wengrow reject the 'origins of inequality' for understanding human history. In interrogating this, they sought sources and found that encounters between European settlers and the Indigenous populations of North America led to Europeans a counter to European civilisation. A serious

We have the option to be creative, for now, as we are at the apex of resource extraction and energy production which affords many possibilities. Any living practice will evolve to reflect the concerns, language, and thinking of the location and time of its occurrence as well as some relationship to whatever political situation that obtains. However, we do need to develop these new systems. These new systems won't occur in a vacuum; they'll always be embedded and imbricated in a given society or culture at a given time. As these systems meet with resonance, institutionalization becomes a near inevitability if it is to operate in a social context larger than a group or clan. Insofar as these systems are more or less aligned with vectors of power (i.e. legitimated mortal violence) they will more or less take on rationalizations of the misalignments between the system and the exigencies of collective murder. This is not optional—it is part and parcel of the human experience.

My friend, the anthropologist Helga Ingebord-Vierich, lived with the hunter/gatherer Kua people in the Kalahari Desert for a year, and she points out how mortal justice was meted out in Kua society:

Apparently, murder was rare enough in any one generation that people had vivid memories of such events. The most recent murders happened only forty years earlier, and was especially remembered because it involved a man who showed no remorse, who had frequently "bothered" others because he seemed to have a "distant heart" (no empathy) and was inclined be selfish and even deliberately to eat alone. He did this especially when he found those kind of treats that any self-respecting person would normally have shared with as many people as possible (like the tails of a particularly tasty lizard, or recently laid water-bird eggs). He was charming enough to survive in this society well into his twenties, but then there was some dispute with a young lady he was courting and the girl was killed, or was severely injured and may have later died (my informants varied in their accounts of this).

In any event, since all these people are excellent trackers, the evidence in the sand clearly indicated that this fellow had been responsible. He denied, and then he admitted, but claimed it was not his fault that others drove him into a temper. Yet the killing did not seem to have been done in temper, but by stealth and surprise.

After some time, everyone was very uneasy about the murderer. They did not like to have him in their camping group, so he became a bit of a wanderer, for even his parents and siblings did not like to have him close. He did make some friends among more distant cousins and spent time with them.

Then, apparently, it nearly happened again. This time the victim lived. It was enough. To make a long story short, his closest relatives took the responsibility and set a trap

critique of European hierarchy, patriarchy, punitive law, and nascent capitalism came with the First Nations, who were much more attuned to ecological sustainability and individual freedom. Graeber and Wengrow also argue that the standard Eurocentric narrative of social evolution came partly as a way of silencing this First Nations critique.

for him—he was ambushed, and killed. They showed me where he was staked out for the scavengers... because, sadly, he was not a human.

I have no proof, but this person sounds to me like a psychopath... and, if so, this was how they dealt with a psychopath.¹⁷

A band member killed another band member and tried to kill another. So, the band killed him. There was no court. No defence counsel, no judge, no jury of peers. There was someone who was perpetrating violence upon others, and as everyone is needed and depended upon in such a band, murder cannot be tolerated. Therefore, the band elders and his family conferred and passed mortal judgment upon the perpetrator. They beat him to death with clubs. A few blows induced unconsciousness, the rest induced a cessation of bodily function, and this psychopath was erased, left to be eaten by animals.

A model of justice? Not in any sense that makes sense in a 'civilized' society bound by institutions and literacy, as it is in the seams and binding of these institutions that civilization protects and valorizes psychopaths and gleefully genocides anyone in its way. That said, it's all water under the bridge at this point.

It is time for us all to admit that civilization has failed.

Antiquity was a condition of chattel slave states. Modernity is a condition of the wage slave state. Antiquity was steeped in religion and its supporting metaphysics. Modernity is steeped in science and its supporting metaphysics. Recent modernity is dispensing with metaphysics, so all one has left is Occam's Razor and the scientific method and emptiness at the core of its existence, and not the generative creative 'Emptiness' of Zen. A return to metaphysics is one approach—and the revivification of previous systems—from the crudities of Abrahamic religions to polytheism to witchcraft to Eastern and Asian religions and the underlying metaphysical components of Kabbalah, the occult, mysticism, and other rationalizations—to be dusted off and reused and reinvented.

While Art attempted to fill this hole, it's hard to declare and execute a war criminal using pataphysics or Fluxus reasoning. Condemned to be hanged by the neck until dead by the Sacred Almighty Words of Andy Warhol / Robert Motherwell / Laurie Anderson / Yoko Ono / John Cage / Tracey Emin?

Ummm... no. That's just not going to happen.

New metaphysical systems, new occults, new Kabbalahs, new mysticisms, and new rationalizations are going to be developed as we face a world that is going to shrink in both population and resources. Indeed, this is an invitation to completely new realms of thought and cognition. And they need not be 'spiritual' systems—they could just as easily be secular non-theistic

17 Helga Ingebord-Vierich, Are humans innately warlike? *MEDIUM*, Blog Post, Dec 15, 2023. <https://helgavierich.medium.com/are-humans-innately-warlike-f43ce9e041a5>.

systems. However, that's the easy stuff—the low-hanging fruit—a place to start. We will never 'get there', and realizing that is part of 'getting there', because we live in a world of language and paritance. Everything, including our lived experience, is Mellotronic — a paritant copy that is instrumentalized, collapsing *zuhanden* / *vorhanden* into each other in a different modality of potential, one tied to a paritant materiality or 'matereality'. We should welcome the avalanche of new systems, new beliefs, and new associations. This doesn't mean we have to believe any of them. It is sufficient that they are believed by someone. Many people don't have the 'chops' to cook up their own metaphysics and depend on others to develop a system they can plug into and find meaning within it and themselves.

These systems can run directly counter to the systems that undergird present-day systems. These systems can be seen as objects. The field they work in are hyperobjects. They are all paritant, as objects and hyperobjects are paritant. They are always already incomplete. We must learn how to improve the world, as we will only make matters worse.¹⁸ This is compensated by its endlessness—there is no final word, because *nobody knows and you can't find out*.¹⁹ So make it up. Speak the unspeakable. Envision the void. 'It never stays in one place, it is not a passing phase',²⁰ because 'all things are apart, all things are a part.'²¹

One might ask, 'how would this work—what can be done?'

Arthur C Clarke said, 'Any sufficiently advanced technology is indistinguishable from magic.'²² This was in his book, *An Enquiry into the Limits of the Possible* from 1962. If one were to bring a cellphone to, say, Queen Ælfifu of Northampton (990ce-1037ce) she would very likely have seen it as magic, and you would be burned at the stake for sorcery. Explaining how it works would operate outside the possibility of her understanding. From the basics of radio, photons, satellites, relativity—heck, just getting her mind away from geocentrism would be hard enough. Never mind YouTube, Discord servers, Facebook, Spotify, and TikTok. Never mind synthesizers and Mellotrons. Crank up a Mellotron set to 'boys' choir' and she would have to conclude that you have a fleet of sad angels in this stupid white box. This is not an argument for technological determinism at all, as we have to consider the opposite, as the dreams our stuff is made of came first. Cartesian science, the sewing machine, benzene's molecular ring structure, and Bohr's atomics all came from dreams. As mentioned, Honorius of Thebes suggested that communication with the dead is best done in dreams or meditative states. The world around us operates in a paritant state we accept as real, because we have no 'real' choice.

We necessarily operate from a position of trust, and it is that emotion of trust that makes any of this possible. Otherwise, one spins down into a kind of universal Capgras Syndrome or the

18 John Cage, *Diary: how to improve the world (you will only make matters worse) Continued, part three*, 1967, Something Else Press, 1967.

19 John Brockman, 37, MacMillan, 1970.

20 Genesis, 'It', from *The Lamb Lies Down On Broadway*, LP, Atco, 1974.

21 Van Der Graaf Generator, 'A Plague of Lighthouse Keepers' from *Pawn Hearts*, LP Charisma, 1971.

22 Arthur C Clarke, *Profiles of the Future: An Enquiry into the Limits of the Possible*, Victor Gollancz Ltd., 1962.

ludicrous nonsense of 'Matrix Theory'. We fancy our new technology is somehow actually new, even as it recapitulates ideas older than our failed industrial civilization that is destroying the biosphere with neither remorse nor apology. We thought we could do better. Perhaps we could have. We didn't, and now we are looking at the off-ramp. In a few centuries, our greatest scientific achievements will be ironic at best. We can all pat ourselves on the back for our brilliance in discovering the Higgs Particle. Truly an astounding achievement. However, an aspect of the scientific method is Testability, and when we no longer have the technology or energy or wealth or material resources to rebuild the LHC, or something very like it, we will not be able to repeat the experiment, and the Higgs Particle will become another 'just so' story from the benighted Golden Age of Industrialism, and all of its blessings and horrors.

However, we do not live then.

We live now in a Mellotron society of paritance, charades, *re-presentations* of an imagined 'reality', necromantic reviews of the dead, evocation of technological angels dancing on digital pins, endless entertainment narratives, telling the same stories on indefinite repeat and the blind alley of progress bound up in a futile quest for an artificial god that is asymptotic to infinity.

The only problem is: NOBODY KNOWS, AND YOU CAN'T FIND OUT.²³

So make it up. Don't be silent: **SING**.

23 Brockman, 1970.

Bibliography

Adams, Douglas, *The Hitchhiker's Guide to the Galaxy*, Random House, 2007.

Aksar IA, Firdaus A, Pasha SA, 'Virtual vs. real self: Gendered presentation and everyday performance of virtual selfhood---A case study of Pakistan,' *Journal of Communication Inquiry* 2023; 47(1): 84--114.

Albertz, Rainer, *A History of Israelite Religion in the Old Testament Period, Vol. I*. Westminster John Knox Press, 1994.

Soloducha Alex, 'What it's like living without an inner monologue.', *CBC News*, 6 March 2020.

Alkis A and Kose T., 'Privacy concerns in consumer E-commerce activities and response to social media advertising: Empirical evidence from Europe.' *Computers in Human Behavior*, 2022; 137: 107412.

Allison F, Nansen B, Gibbs M, et al., 'Bones of contention: Social acceptance of digital cemetery technologies.', *Proceedings of the 2023 CHI Conference on Human Factors in Computing Systems*, 2023.

Ananias Davisson; J. W. Watson, 'Idumea', *Sacred Harp Hymnal* 1763 / 1816.

Andrew Wheeler, 'What Raw Materials Are Used to Make Hardware in Computing Devices?', *Engineering.com*, 29 September 2018.

Anon. *I-Ching*, Written ca. 900 bce, China, <https://www.ichingonline.net/>

Anthea Butler, 'The ridiculous moral panic over the Olympics' opening ceremony.', *MSNBC*, 29 July 2024.

Antonio, Robert, *The Passing of Jean Baudrillard*, University of Texas at Arlington, 2007.

Army of Darkness, Dir. Sam Raimi, Film Universal Pictures, 1992.

Art Bears, *Winter Songs*, Ralph Records, 1978.

Attali, Jacques, *Noise: the political economy of music*, University of Minnesota Press, 1985.

Babylon 5, Writer: J. Michael Straczynski, Season 1, Episode 6: 'Mind War', Final Scene, 1994.

Baudrillard, Jean, *Simulations*, Semiotext(e), 1984.

Bearzi G, Eddy L, Piwetz S, Reggente MAL, Cozzi B., 'Cetacean behavior toward the dead and dying.' In: Vonk J, Shackelford TK (eds) *Encyclopedia of animal cognition and behavior*, Springer. pp. 1--30, 2017.

Ben Beaumont-Thomas, The Beatles set record 54-year gap between No 1 singles as Now and Then tops UK chart, *The Guardian*, 10 November 2023.

Benj Edwards, 'Astronomers discover technique to spot AI fakes using galaxy-measurement tools.', *Ars Technica*, 22 July 2024.

Benjamin A. Elman, 'Nietzsche and Buddhism', *Journal of the History of Ideas*, Vol. 44, No. 4. 1983.

Berger, Lee R, et al. Evidence for deliberate burial of the dead by Homo naledi, Biorxiv, 2023.

Berkowitz I, Garrett JR. Legal and ethical considerations for requiring consent for apnea testing in brain death determination, *The American Journal of Bioethics*, 2020; 20(6): 4--16.

Bienart, H-D. 'Skull cults of the prehistoric Near East.' *Journal of Prehistoric Religion*, 1991. 5:9-23.

Bjorn Karhunen and Jacopo della Quercia, '5 Things You Won't Believe Aren't In the Bible: Updated.', *Cracked Magazine*, 19 October 2010.

Blade Runner, Dir. Ridley Scott, Film, Warner Bros, 1982.

Blade Runner 2049, Dir. Denis Villeneuve, Film, Produced and distributed by a consortium of studios, 2017.

Blue Letter Bible Study, *The names of God in the Old Testament*, Website, 2024.

Blue Oyster Cult, 'Flaming Telepaths' from *Secret Treaties*, Columbia Records, 1974.

Boston Dynamics, *All New Atlas*, Youtube video, https://www.youtube.com/watch?v=29ECwExc-_M 2024.

Bostrom, Nick, 'Are you living in a computer simulation?' *Philosophical Quarterly*, 53 (211) pp. 243-255, 2003.

Brian Wayne Transeau / Sonik Architects, 'Time varying processing of repeated digital audio samples in accordance with a user defined effect.' Patent, US20090281793A1, applied: 25 May 2006, Granted: 12 November 2009.

BT, *Binary Universe*, CD, Binary Acoustics / DTS, 2006.

Cade Metz and Gregory Schmidt, 'Elon Musk and Others Call for Pause on A.I., Citing 'Profound Risks to Society'', *New York Times*, March 29, 2023.

Cage, John, *Diary: how to improve the world (you will only make matters worse) Continued, part three*, 1967, Something Else Press, 1967.

Cage, John, *M: Writings '67 -- '72*, Wesleyan University Press, 1973.

Chajes, J.H, *The Kabbalistic Tree*, Pennsylvania State University Press, 2022.

Chenda Ngak, 'Tupac Coachella hologram: Behind the technology.', *CBS News*, November 9, 2012.

Christianna Reedy, 'Kurzweil Claims That the Singularity Will Happen by 2045 Get ready for humanity 2.0.' *Futurism*, 16 October 2017.

Christopher Cox, 'Invisible Jukebox: Steve Lacy.', *The Wire*, #225, November 2002.

Clarke, Arthur C, *Profiles of the Future: An Enquiry into the Limits of the Possible*, Victor Gollancz Ltd., 1962.

Claude Shannon, 'A Symbolic Analysis of Relay and Switching Circuits', Master's Thesis, MIT, 1937.

Colton Prail, 'Sunday was the hottest day ever recorded on Earth, scientists say', *CTV News*, 23 July 2024.

Crowe, Daniel. (ed.) *Dead Interviews: Living Writers Meet Dead Icons*, London: Granta, 2013.

Crowley, Aleister, *The Confessions of Aleister Crowley: An Autobiography*, (1929) Arkana Publishing, 1989.

Cyndi Lauper, *Money Changes Everything*, Single, Gray Matter Records, 1984.

Dai Bosatsu Zendo, *Sutra Book*, 2010.

Dareka no Manazashi (Someone's Gaze), Dir. Makoto Shinkai, Animation Film Short, CoMix Wave Films, 2013.

David Bowie, *Station to Station*, LP, RCA, 1976.

David Sylvian, 'Waterfront' from *Secrets of the Beehive*, LP, Virgin, 1986.

Dehaene, Stanislaus, *The Number Sense*, Oxford University Press, 1999.

Derrida, Jacques, 'Foreigner Question: Coming from Abroad / from the Foreigner', from *Of Hospitality: Anne Dufourmantelle invites Jacques Derrida to respond*. Stanford, 2000.

Diana Ross, *Swept Away*, RCA, 1984.

Dissanayake, Ellen, *Homo Aestheticus: Where Art Comes From and Why*, University of Washington Press, 1995.

Dogen, *Actualizing the Fundamental Point (Genjo-koan)*, 1233, Translated by Robert Aitken and Kazuaki Tanahashi, *Enlightenment Unfolds*, Penguin Books, 2000.

Donovan, 'There Is a Mountain,' from *Mellow Yellow*, Epic Records, 1967.

Douglas-Hamilton I, Bhalla S, Wittermyer G, Vollrath F 'Behavioural reactions of elephants towards a dying and deceased matriarch,' *Applied Animal Behaviour Science*, 100:87--102 (2006)

eevee. 'I wonder if we're dead', from *Seeds*, Inner Oceans Records, 2017.

Erol Gelenbe, 'Electricity Consumption by ICT: Facts, trends, and measurements.', *Ubiquity*. Vol 2023, August 2023.

Esposito, Elena, *Artificial Communication: How Algorithms Produce Social Intelligence*, MIT, 2022.

Ethan Smith, 'Rapper's De-Light: Tupac 'Hologram' May Go on Tour', *Wall Street Journal*, online, 16 April 2012.

Fedorenko, E., Piantadosi, S.T. & Gibson, E.A.F. 'Language is primarily a tool for communication rather than thought.', *Nature*, 2024. 630, pp. 575-586.

Fisher, Mark, *Capitalist Realism: Is there no alternative?*, Zero Books, 2009.

Fisher, Mark, *Ghosts of My Life: Writings on Depression, Hauntology and Lost Futures*, Zero Books, 2014.

Edgar Froese, *Epsilon In Malasian Pale*, LP, Virgin Records, 1974.

Gary Numan and the Tubeway Army, *Replicas*, LP, Beggars Banquet, 1979.

Genesis, 'It', from *The Lamb Lies Down On Broadway*, LP, Atco, 1974.

Geoffrey Cain, 'Japan's biggest pop star right now is a fetishized hologram.' *Salon*, 13 April 2014.

George Pallidev, 'Oval --- 94.Diskont. A brief story behind first glitch album.' , Blog post, 11 November 2020.

Gera, Deborah Levine, *Ancient Greek ideas on speech, language, and civilization*, Oxford University Press, 2003.

Ghost Dance, Dir. Ken McMullen, Film, 1983.

Gilbert, Ryle, 'Descartes' Myth', *In The Concept of Mind*, London: Hutchinson, 1949.

Glitchmachines, Website, <https://glitchmachines.com/products/everything-bundle/>.

McCoy, Daniel, *Gnosticism Explained, The Gnostic Demiurge*, Blog post, 2021.

Gonçalves A, Carvalho S 'Death among primates: a critical review of non-human primate interactions towards their dead and dying.' *Biological Reviews of the Cambridge Philosophical Society*, 10:15 2019.

Graeber, David and David Wengrow, *The Dawn of Everything: A New History of Humanity*, Allen Lane, 2021.

Hagerty A, and Rubinov I., *Global AI ethics: A review of the social impacts and ethical implications of artificial intelligence*, New York: Cornell University; 2019.

Hans-Georg Moeller and Elena Esposito, 'Why Algorithms Can't Think: 'Artificial Communication' by Elena Esposito, *Carefree Wandering*, Youtube Video.

Harris, Marvin, *Cannibals and Kings: The Origins of Culture*, William Collins Pub, 1978.

Hatsune Miku, *World Is Mine*, YouTube Video, 10 June 2013.

- Hawking, Stephen, *A Brief History of Time*, Bantam, 1988.
- Hayes, P., Harnad, S., Perlis, D. & Block, N., 'Virtual Symposium on Virtual Mind.', *Minds and Machines*, 1992, 2(3) 217-238.
- Hegel, GF, *The Philosophy of History*, (1837), Batoche Publishing, 2001.
- Helga Ingebord-Vierich, 'Are humans innately warlike?', *MEDIUM*, Blog Post, 15 December 2023.
- Henry Warwick, 'IndusTrial' instrumental, *Gargoyle Magazine*, #36, Paycock Press, Pub, 23 April 1989.
- Henry Warwick, *Covidian Meditations*, Kether Records, 2020.
- Henry Warwick, *Electronic Etudes*, Kether Records, 1987.
- Henry Warwick, *Metawind*, Kether Records, 1988.
- Henry Warwick, *Zither Studies (after Laraaji)*, Kether Records, 2021.
- Her*, Dir. Spike Jonze, Film, Annapurna / Warner Bros, Film, 2013.
- Paul Hindemith, *Concertino for Trautonium and String Orchestra*
- Hutson, James and Ratican, Jay, 'Life, death, and AI: Exploring digital necromancy in popular culture---Ethical considerations, technological limitations, and the pet cemetery conundrum' *Lindenwood University Faculty Scholarship*, 478, 2023.
- IBM Data and AI Team, *Understanding the different types of artificial intelligence*, IBM, 12 October 2023, <https://www.ibm.com/blog/understanding-the-different-types-of-artificial-intelligence/>.
- Interview: Lix Fridman and Marc Andreessen, 'AGI is the search for God | Marc Andreessen and Lex Fridman.', *Lix Friedman Channel*, 22 June 2023.
- Insideout Ltd. *Manetron* iPhone App, Apple Store, <https://apps.apple.com/us/app/the-manetron/id1641254309>.
- Jacques Attali, *When music, once again, reveals the future*, Blog Post, 19 April 2023.
- Jeremy Zuckerman, 'Afraid (Separate Ways)', *The Legend of Korra: Original Music From Book One*, 2013.
- Jimmy Dean, *Big Bad John*, Columbia, 1961.
- Joe Taysom and John Taylor, 'How did The Beatles create their new song 'Now and Then'?' *Farout Magazine*, 2 November 2023.
- John Brockman, 37, MacMillan, 1970.
- John Oswald, *Plunderphonics, or Audio Piracy as a Compositional Prerogative*, Lecture, 1985.
- John Shea and Ofer Bar-Yosef, 'Who Were The Skhul/Qafzeh People? An Archaeological Perspective on Eurasia's Oldest Modern Humans.', Mitekufat Haeven: *Journal of the Israel Prehistoric Society* (2005): 451-68.
- JP Andrew aka @2Philosophical_, X.COM, post: https://x.com/2Philosophical_/status/1819808306258383225.
- Jung, Karl, *On the Psychology and Pathology of So-Called Occult Phenomena*, Advisor: Eugen Bleuler, Burghölzli Hospital, 1900.
- Justin Sledge, Esoterica, 'How to use Magic to see God & Conjure Spirits - The Sworn Book of Honorius - Liber Juratus', Youtube channel, 20 April 2024.
- Kaernbach, C., 2005, 'No Virtual Mind in the Chinese Room', *Journal of Consciousness Studies*, 12(11). 2005. pp. 31--42.

Kafka, Franz, *Parables and Paradoxes*, Schocken Books, 1946.

Kant, Immanuel, *Groundwork of the Metaphysics of Morals in Immanuel Kant: Practical Philosophy*, 1785. trans. Cambridge, 1999.

Katie Hunt, 'Neanderthal child's skeleton buried 41,000 years ago may solve long-standing mystery.' *CNN*, 5 January 2021.

Kapcár A, 'The origins of necromancy on how we learned to speak to the dead.' *Sacra*; 13(2) pp. 30--58, 2015.

Keith Jarrett, *Book of Ways*, LP, ECM, 1986.

Keith Jarrett, *Selected recordings*, CD set, ECM Records, April 2002.

Kim Cascone, 'The Aesthetics of Failure', *Computer Music Journal*, MIT Press, 24:4 Winter 2002.

Kurzweil, Raymond, *The Singularity Is Near: When Humans Transcend Biology*, Viking, 2005.

Lanier, Jaron, *You Are Not A Gadget*, Knopf, 2010.

Lao Tzu, *Tao Te Ching*, Written ca: 420bce, Version: with.org/tao_te_ching_en.pdf

Malcolm, Lynne; Willis, Olivia. 'Songlines: the Indigenous memory code', *ABC Radio National*, 8 July 2016.

Laurie Anderson, *Big Science*, Warner Brothers, 1983.

Leah Hendrickson, 'Chatting with the dead: The hermeneutics of thanabots', *Media, Culture, and Society*, Vol. 45 (5), 2023.

Leekspin - A Classic Anime Meme, Youtube video, Alexir Channel, 19 June 2020.

Leonard Cohen, *Various Positions*, Columbia, 1984.

Levi, Eliphas, *The Doctrine and Ritual of High Magic: A New Translation*, Translated by John Michael Greer and Mark Anthony Mikituk, TarcherPerigee, 2017.

Lindemann NF, 'The ethical permissibility of chatting with the dead: Towards a normative framework for 'Deathbots'', *Publications of the Institute of Cognitive Science*; 2022.

Ludus, *The Seduction*, LP, Sentimentale Sordide, LP, 1982.

MIAIRIRIS, *Pump Up The Volume*, 4AD, 1987.

Macintosh Plus, *Floral Shoppe*, Beer on the Rug label, 2011.

Mair, Michael and Phillip Brooker, Dipanjan Saha, Terence Heng. (eds) 'Digital necromancy: why bringing people back from the dead with AI is just an extension of our grieving practices.' *The Conversation*. 2023.

Marcel Duchamp, *The Bride Stripped Bare by her Bachelors, Even*. Jaap Rietman, 1976.

Marjorie Prime, Dir. Michael Almereyda, Film, BB Films, 2017.

Mark Wilson, 'The New Mantra of Tech: It's Good Enough', *Gizmodo*, April 27, 2009.

Karl Marx and Friedrich Engels, *The Communist Manifesto*, 1848.

Maurice Tuchman et al (eds), *The Spiritual in Art: Abstract Painting 1890-1985*, Taschen, 1986.

McMansion Hell, Blog, <https://mcmansionhell.com/>

Memorex, 'Ella Fitzgerald for Memorex 1974, 1979', *GarfieldFco channel*, Youtube video of TV Commercial.

Messiaen: *Turangalila-Symphonie, Fête des belles eaux, and Trois petites liturgies de la présence divine*. 1930.

Meyer, Marvin, 'The Nature of the Rulers.' in *The Nag Hammadi Scriptures*, Edited by Marvin Meyer, HarperOne, 2008.

Michelangelo Matos, 'How M/A/R/R/S', 'Pump Up the Volume' Became Dance Music's First Pop Hit', *Rolling Stone*, 14 July 2016.

Miller, Andrew, *Greek Lyric: An Anthology in Translation*, p. 21, Hackett Publishing, 1996.

Monty Python and the Search for the Holy Grail, Dir. Terry Gilliam and Terry Jones, Film, 20th Century dist, 1975.

Morton, Timothy, *Hell: In Search of a Christian Ecology*, Columbia University Press, 2024.

Nathan Rockwood, 'Locke and Hume on competing miracles', *Religious Studies*, (December 2023). 59 (4).

Needham, Joseph, *Science and Civilisation in China: Volume 2, History of Scientific Thought*, Cambridge University Press, 1991.

Neil deGrasse Tyson on *Real Time with Bill Maher*, Episode 14, 3 June 2016

Neil Genzlinger, 'Eido Shimano, Buddhist Leader Who Resigned in Scandal, Dies at 85', *New York Times*, 26 February 2018.

Nietzsche, Friedrich, *On the Genealogy of Morality*, (1887) Dover Publishing (2003).

Nietzsche, Friedrich, *The Birth of Tragedy Out of the Spirit of Music*, (1872) Blackstone Publishing, 2019.

Nietzsche, Friedrich, *The Gay Science* (1882), Cambridge University Press, 2001.

Orchestral Manoeuvres In The Dark, 'ABC Industry' from *Dazzle Ships*, LP, Virgin Records, 1983.

Penrose, R., 'Consciousness, Computation, and the Chinese Room' in Preston and Bishop (eds.) 2002, pp. 226-249.

Peter Allen, 'A genuinely believable CGI actor? It won't be long' *The Conversation*,. 17 January 2017.

Peter Vogel Instruments, *CMI App for iPad and iPhone*, Software, 2013.

Peterson, Joseph (trans., ed.), *The Sworn Book of Honorius: Liber Inratus Honorii, by Honorius of Thebes*, Ibis Press, 2016.

Preston, J. and M. Bishop (eds.), *Views into the Chinese Room: New Essays on Searle and Artificial Intelligence*, New York: Oxford University Press, 2002.

Ray Charles, *I Can't Stop Loving You*, Written by Don Gibson 1957, ABC Paramount, 1962.

Resurrection, Directed by Lauren Lazin, Film, Paramount Pictures, 2003.

Richard Buskin, 'Classic Tracks: Pump Up The Volume', *Sound On Sound Magazine*. August 2012.

Robert Fripp, 'The Night Watch', *dgmilive.com*, blog post, 2016.

Rogue One: A Star Wars Story, Dir. Gareth Edwards, Film, Walt Disney Studios, 2016.

Roxy Music, *Country Life*, Island Records, 1974.

Russ Mitchell, 'San Francisco's North Beach streets clogged as long line of Cruise robotaxis come to a standstill', *Los Angeles Times*, 12 August 2023.

Russolo, Luigi, Robert Filliou (trans), *The Art of Noise*, futurist manifesto, 1913, UBU classics, 2004.

SIMONE, Dir. Andrew Niccol, Film, New Line Cinema, Film, 2002.

Saint-Saëns: *The Swan (Carnival of the Animals)* 1929.

Sarah Canfield Fuller, 'Reading the Cyborg in Mary Shelley's 'Frankenstein'', *Journal of the Fantastic in the Arts*, Summer 2003, Vol. 14, No. 2 (54) (Summer 2003), pp. 217-227

Sarah Cascone, 'Controversy Erupts Over Paris Olympics Take on Leonardo's 'Last Supper' *Artnet*, 29 July 2024.

Savin-Baden, Maggi, *Digital Afterlife and the Spiritual Realm*, CRC Press, 2022.

Schneider and Sagan, *Into the Cool: Energy Flow, Thermodynamics, and Life*, University of Chicago Press, 2006.

Scholem, Gershom, *Origins of the Kabbalah*, Philadelphia: Jewish Publication Society, 1987.

Sconce, Jeffrey, *Haunted Media: electronic presence from telegraphy to television*, Duke University Press, 2000.

Scott Haddow, *Çatalhöyük 2012: Skull retrieval, curation and redeposition*, Wordpress, 24 August 2012.

Scott Wilson, 'Curtis chip company speaks out against vintage synth cloning', *Fact Magazine*, 22 March 2017.

Roger Scruton, *The Roger Scruton Reader*, Bloomsbury Academic, 2009.

Shelley, Mary Wollstonecraft, *Frankenstein; or, the Modern Prometheus*, 1818.

Sheryl Astrada, 'Exclusive: Intuit is laying off 1,800 employees as AI leads to a strategic shift.' *Fortune*, 10 July 2024.

I. Shumailov, Z. Shumaylov, Y. Zhao, et al, 'AI models collapse when trained on recursively generated data.' *Nature*, 631, 755-759, 2024.

Sidney Zisook and Katherine Shear, 'Grief and bereavement: what psychiatrists need to know.' *World Psychiatry*, June 2009; 8(2): 67--74

Sigil Engine, Online Sigil Maker, 2024, <https://www.sigilengine.com>

Simon Trask, 'Fairlight Series III', *Electronics & Music Maker*, April 1986.

H. Simon, and S. Eisenstadt, 'A Chinese Room that Understands', in Preston, J. and M. Bishop (eds) 2002, pp. 95-108.

Christopher Small, *Musicking: the meaning of performing and listening*, Wesleyan, 1998.

Smiley Face, dir. Greg Araki, Film, 2007.

Soutik Biswas, 'Sati: How the fight to ban burning of widows in India was won.' *BBC*. 23 April 2023.

St Vincent, *Masseduction*, CD, 4AD, 2017.

Grafton Tanner, *Babbling Corpse: Vaporwave And The Commodification Of Ghosts*, Zero Books, 2020.

Katie Tarasov, 'Generative AI requires massive amounts of power and water, and the aging U.S. grid can't handle the load.' *MSNBC*, 28 July 2024.

The Beatles: Get Back, Directed by Peter Jackson, Documentary, Disney +, 2021.

The Bible. KJV

Canada Post suspends community mail box program, affecting 460,000 homes, *CTV*, 27 October 2015.

- The Maltese Falcon*, Dir. John Huston, Warner Brothers, Film, 1941.
- 'The Matrix Awakens: Unreal Engine Experience Gameplay Demo.' Youtube video, *Gamespot Channel*. 10 December 2021.
- The Matrix*, Directed by the Wachowskis, Warner Brothers, 1999, Film.
- The Tokens, *The Lion Sleeps Tonight*, Written by Solomon Linda, 1939, RCA Victor, 1961.
- The Tornados, *Telstar*, RGM Sound, London, 1962.
- THX 1138*, Dir. George Lucas, Film, American Zoetrope / Warner Brothers, 1971.
- Tim Adams, 'All these bulletproof songs, one after another': remembering Tom Waits's extraordinary mid-career trilogy', *The Guardian*, 20 August 2023.
- Tupac Shakur, *Resurrection*, CD, Amaru-Interscope, 2003.
- Turner, John D., and Marvin Meyer, 'The Secret Book of John,' In *The Nag Hammadi Scriptures*. Edited by Marvin Meyer. HarperOne. 2008.
- United Nations, 'It's official: July 2023 was the warmest month ever recorded.' *UN News*, 8 August 2023.
- US Patent 4,182,214: Electronic Musical Instrument*, Richard C. Wakeman and Birotronics, Ltd. Filed: 27 January 1975, Issued: 8 January 1980.
- Van Der Graaf Generator. 'A Plague of Lighthouse Keepers' from *Pawn Hearts*, LP. Charisma. 1971.
- Van der Toorn, K. *Family Religion in Babylonia, Ugarit and Israel: Continuity and Changes in the Forms of Religious Life*, Brill, 1996.
- Virilio, Paul and Sylvère Lotringer - *Pure War: Twenty-Five Years Later*, Semiotext(e), 2008.
- Vishal Udandarao et al. 'No 'Zero-Shot' Without Exponential Data: Pretraining Concept Frequency Determines Multimodal Model Performance.' *Cornell University Arxiv*, Cornell University, 8 April 2024.
- Walter, T., Hourizi, R., Moncur, W., & Pitsillides, S. 'Does the internet change how we die and mourn? An overview.' *OMEGA: Journal of Death and Dying*, 64(4), 275-302, 2012.
- Henry Warwick, *The Radical Tactics of the Offline Library*, Institute of Network Culture, Amsterdam. 2014.
- Watazumi Doso -- *The Way of Watazumi*, Lecture at the Creative Music Studio, Woodstock, N.Y., 1981.
- Watazumido Shuso, *The Mysterious Sounds of the Japanese Bamboo Flute*, Everest Records, 1971.
- Weller, Tom, *Minims: Or, Man Is The Only Animal That Wears Bow Ties*, Houghton Mifflin, 1982.
- Wells, Steve, *Drunk with Blood: God's Killings in the Bible*, CreateSpace Independent Publishing Platform, 2010.
- Wendy Carlos, *Digital Moonscapes*, East Side Digital, 1984.
- Wendy Carlos, *Tron*, Original Motion Picture Soundtrack, CBS Records, 1982.
- Wendy Carlos, *A Clockwork Orange (soundtrack)*, Columbia Records, 1972.
- West End Electric, *Phantom Power*, Kether Records, 2022.
- Wings of Desire*, Dir. Wim Wenders, Film, 1987.
- Wittgenstein, Ludwig, *Tractatus Logico Philosophicus*, Routledge, 2001.
- Wolters, Eugene, 'Understanding Jean Baudrillard With Pumpkin Spice Lattes.' *Criticaltheory.org*. Blog entry. 24 / 09 / 2014.

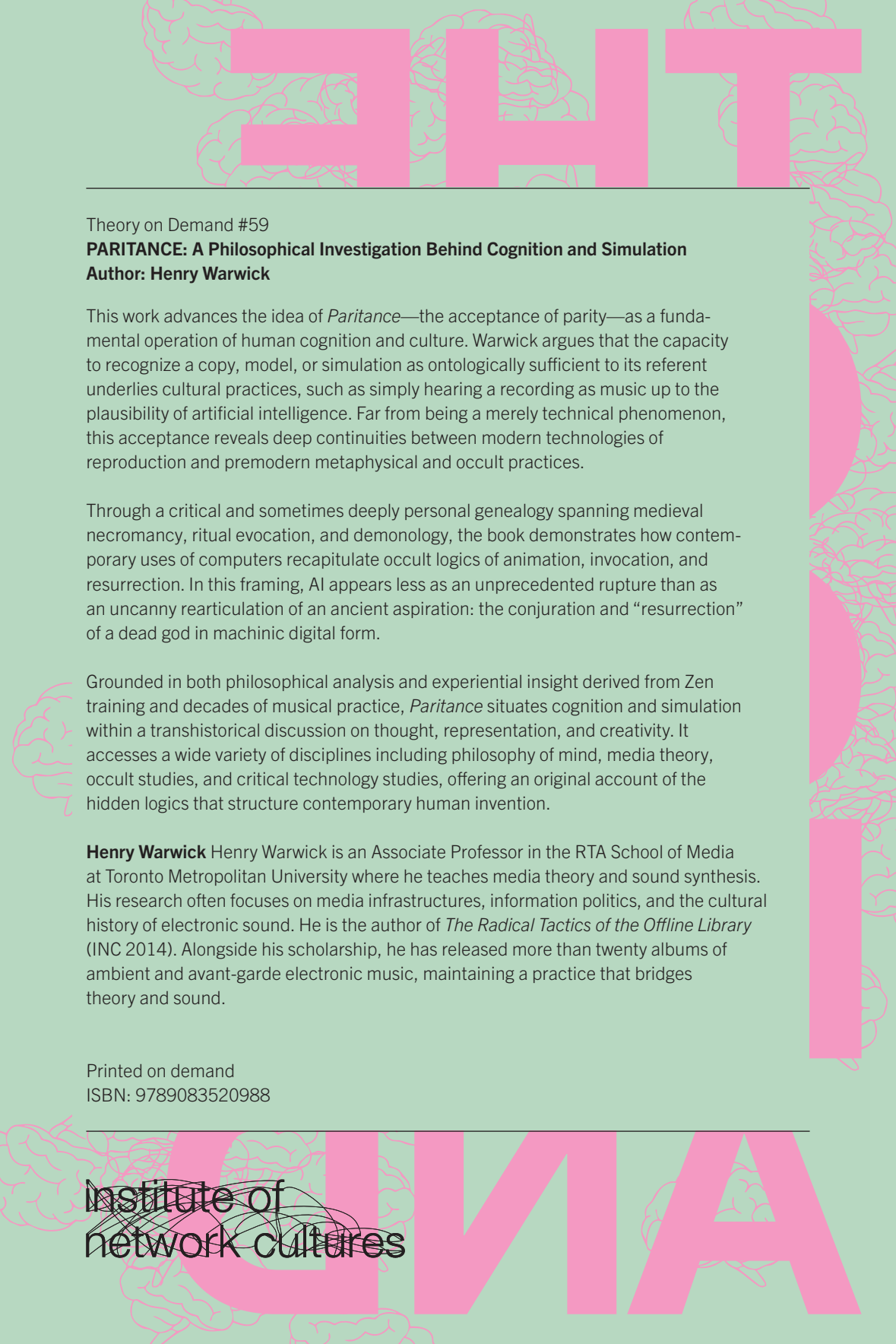
Xyliex, Leek Spin, Youtube video, <https://www.youtube.com/watch?v=1wnE4vF9CQ4>

Yamada, Koun, *The Gateless Gate*, Center Publications, 1979.

Yehuda Ashlag, Berg, Philip S. (ed.). *An Entrance to the Tree of Life: A Key to the Portals of Jewish Mysticism*, Research Centre of Kabbalah, 1977.

Yurick, Sol. *Behold Metatron, the Recording Angel*, Semiotext(e). 1985.

Zappa, Frank, *The Real Frank Zappa Book*, Poseidon Press, 1989.



Theory on Demand #59

PARITANCE: A Philosophical Investigation Behind Cognition and Simulation

Author: Henry Warwick

This work advances the idea of *Paritance*—the acceptance of parity—as a fundamental operation of human cognition and culture. Warwick argues that the capacity to recognize a copy, model, or simulation as ontologically sufficient to its referent underlies cultural practices, such as simply hearing a recording as music up to the plausibility of artificial intelligence. Far from being a merely technical phenomenon, this acceptance reveals deep continuities between modern technologies of reproduction and premodern metaphysical and occult practices.

Through a critical and sometimes deeply personal genealogy spanning medieval necromancy, ritual evocation, and demonology, the book demonstrates how contemporary uses of computers recapitulate occult logics of animation, invocation, and resurrection. In this framing, AI appears less as an unprecedented rupture than as an uncanny rearticulation of an ancient aspiration: the conjuration and “resurrection” of a dead god in machinic digital form.



Grounded in both philosophical analysis and experiential insight derived from Zen training and decades of musical practice, *Paritance* situates cognition and simulation within a transhistorical discussion on thought, representation, and creativity. It accesses a wide variety of disciplines including philosophy of mind, media theory, occult studies, and critical technology studies, offering an original account of the hidden logics that structure contemporary human invention.

Henry Warwick Henry Warwick is an Associate Professor in the RTA School of Media at Toronto Metropolitan University where he teaches media theory and sound synthesis. His research often focuses on media infrastructures, information politics, and the cultural history of electronic sound. He is the author of *The Radical Tactics of the Offline Library* (INC 2014). Alongside his scholarship, he has released more than twenty albums of ambient and avant-garde electronic music, maintaining a practice that bridges theory and sound.

Printed on demand
ISBN: 9789083520988

Institute of
network cultures

